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Essays on Empirical Economics of Public Procurement: Drivers of Efficiency and Effectiveness

Marly Tatiana Celis Galvez

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Essays on Empirical Economics of Public Procurement: Drivers of Efficiency and Effectiveness

Essays over de Empirische Economie van Publieke Inkoop: Drijfveren van Efficiëntie en Effectiviteit

(met een samenvatting in het Nederlands)

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Marly Tatiana Celis Galvez

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Promotor: Prof. dr. ir. F. Schotanus
Copromotor: Dr. V. Titl

Beoordelingscommissie:

Prof. mr. dr. E. Manunza
Prof. dr. S. Murtinu
Prof. dr. B.M.J. Rijkers
Prof. dr. S. Rosenkranz
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Contents

Acknowledgements vii

List of Tables ix

List of Figures xi

List of Abbreviations xiii

1 Introduction **1**
1.1 Motivation 1
1.2 Organizing framework of the dissertation 4
1.3 Institutional context 6
1.4 Methodology and data sources 7
1.5 Results and contributions 9
1.6 Roadmap 14

2 Discretion and Political Favoritism: Evidence from Two Reforms in Public Procurement **15**
2.1 Introduction 15
2.2 Setting and data 20
2.3 Empirical strategy 29
2.4 Results 37
2.5 Conclusions 44
2.A Appendix section 47
2.B Robustness checks estimates excess mass 52

3 Quality of Public Projects and Procurement Officers Value Added **65**
3.1 Introduction 65
3.2 Institutional context 70
3.3 Data 71
3.4 Empirical approach 73
3.5 Results 75
3.6 Discussion and conclusion 83
3.A Appendix section 87

4 Competition, Entry, and Firm Dynamics in Public Procurement 99
4.1 Introduction . . . 99
4.2 Institutional setting . . . 105
4.3 Empirical framework . . . 108
4.4 Results . . . 112
4.5 Mechanisms . . . 122
4.6 Discussion . . . 127
4.7 Conclusions . . . 128
4.A Details public procurement data . . . 129
4.B Details firm's balance sheet information . . . 133
4.C Matching algorithm . . . 134
4.D First contracts test . . . 137
4.E Estimation tables . . . 138
4.F Robustness checks . . . 160

5 Discussion 167
5.1 Summary and main findings . . . 168
5.2 Channels for effective and efficient public procurement . . . 169
5.3 Limitations and external validity . . . 171
5.4 Future research . . . 172
5.5 Policy implications . . . 173

Bibliography . . . 177

Nederlandstalige Samenvatting 191

Curriculum Vitae 193

U.S.E. Dissertation Series 195

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Without her you would not have set out.
She has nothing left to give you now.
— from "Ithaka" by Constantine P. Cavafy

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List of Tables

1.1 Overview of thesis chapters and corresponding research questions. . . 3

2.1 Summary discretionary thresholds reforms and range for discretion . 22
2.2 Regression estimates of excess mass below the thresholds 27
2.3 Estimates of excess mass below discretionary threshold. Reform 2012 - Restricted bureaucratic discretion. 39
2.4 Estimates of excess mass below discretionary threshold. Reform 2016 - Broader bureaucratic discretion. 41
2.5 Regression discontinuity design estimations for normalized price, number of bids and competitiveness 43
2.A.1 Descriptive statistics - public procurement contracts Czech Republic 48
2.A.2 Number of contracts and number of firms per year 50
2.B.1 Estimates excess mass below thresholds by winning firms' likelihood of corruption. 52
2.B.2 Estimates excess mass below thresholds by winning firms' joined likelihood of corruption. 52
2.B.3 Estimates excess mass below thresholds - alternative estimation in equation 2.B.1 Reform 2012 54
2.B.4 Estimates excess mass below thresholds - alternative estimation in equation 2.B.1 Reform 2016 55
2.B.5 Estimates excess mass below thresholds - alternative estimation in equation 2.B.2 Reform 2012 55
2.B.6 Estimates excess mass below thresholds - alternative estimation in equation 2.B.2 Reform 2016 55
2.B.7 Density discontinuity tests - McCrary manipulation testing 59
2.B.8 RDD estimations for *normalized price* - Reform 2012 61
2.B.9 RDD estimations for *normalized price* - Reform 2016 61
2.B.10 RDD estimations for *number of bids* - Reform 2012 62
2.B.11 RDD estimations for *number of bids* - Reform 2016 62
2.B.12 RDD estimations for *productivity* - Reform 2012 63
2.B.13 RDD estimations for *productivity* - Reform 2016 63

3.1 Participation regression . . . 79
3.2 Exploring officers value added: legal engagement and education . . . 81
3.A.1 Description of survey questionnaire . . . 87
3.A.2 Descriptive statistics of survey measures . . . 88
3.A.3 Contract-level email and completed-survey coverage . . . 89
3.A.4 Survey sampling and response . . . 90
3.A.5 Balance respondents versus non-respondents . . . 91
3.A.6 Officers' value added: legal engagement and education categorized legal engagement . . . 94
3.A.7 Officers' value added: full results . . . 95
3.A.8 Exploring Officers Value Added: Legal Engagement and Education . 96
3.A.9 Exploring officers value added: legal engagement and education 3-year quality measure . . . 98

4.1 Descriptive statistics . . . 107
4.1 Market staggered entry . . . 113
4.2 Event study estimates financial performance . . . 117
4.1 Relative bid reduced-form regression . . . 124
4.2 Number of bidders, entry events and type of firms . . . 126
4.A.1 Descriptive statistics for all contracts . . . 133
4.A.2 Sectoral distribution of contracts . . . 134
4.C.1 Sample restrictions and staggered events . . . 136
4.D.1 Test relevance of the first contract . . . 137
4.E.1 Event Study estimates financial performance (part 2) . . . 139
4.E.2 Overall coefficients . . . 142
4.E.3 Firm Heterogeneity Characteristics . . . 143
4.E.4 Share of procurement revenue in total firm revenue . . . 144
4.E.5 Event study estimates financial performance established participants - part 1 . . . 146
4.E.6 Event Study estimates financial performance established participants - part 2 . . . 147
4.E.7 Event Study estimates financial performance capacity constrained firms - part 1 . . . 148
4.E.8 Event Study estimates financial performance capacity constrained firms - part 2 . . . 149
4.E.9 Event Study estimates financial performance short-term participants - part 1 . . . 150
4.E.10 Event Study estimates financial performance short-term participants - part 2 . . . 151
4.F.1 Event Study estimates financial performance not yet suppliers (part 1) 161
4.F.2 Event Study estimates financial performance not yet suppliers (part 2) 162
4.F.3 Relative bid reduced-form regression - alternative specification . . . 165
4.F.4 Contracts characteristics . . . 166

List of Figures

1.1 The economy and the components of public procurement markets. 4

2.1 Bunching mass estimation around re-centered anticipated contract value . . . 26
2.2 Distribution of contract anticipated value by reform . . . 38
2.A.1 Areas for levels of discretion during different public procurement acts. 2006 - 2021 . . . 47
2.A.2 Total number of bids per year and total number of contracts . . . 49
2.B.1 Distribution of contracts anticipated value by winning firms' likelihood of corruption . . . 53
2.B.2 Representation of the sample split for the two reforms and the fuzzy RDD design . . . 56
2.B.3 Intention to treat and compliance of new regulatory threshold . . . 56
2.B.4 McCrary density continuity plot . . . 57
2.B.5 Contracts' date of award distribution with respect to date reform 57
2.B.6 Contracts' date of award distribution with respect to date reform, by size of contract whether above or below . . . 58
2.B.7 Contracts' date of award distribution with respect to date reform-Days . . . 58
2.B.8 Multiscored running variable Local Polynomial Density Estimation 59
2.B.9 Price over cost ratio after the discretionary reforms by type of flagged favoritism . . . 59
2.B.10 Average number of bids . . . 60
2.B.11 Competitiveness (Lagged productivity) . . . 60

3.1 Public projects' quality . . . 77
3.2 Procurement officers' value added . . . 78
3.3 Role of legal engagement and legal education . . . 82
3.A.1 Examples of contact people identified in the contract records . . . 89
3.A.2 Share of survey respondents by region . . . 91
3.A.3 Procurement officers' value added correlation of primary and alternative measures . . . 93

3.A.4 Officers' value added and legal engagement 96
3.A.5 Procurement officer value added and survey participation 97
3.A.6 Procurement officers' value added 3-year quality measure 98

4.1 Event Study Plots Financial Performance 116
4.2 Event Study plots financial performance: assets growth 121
4.1 Distribution of relative bids by bidder type 123
4.A.1 Misreporting of contract values of public procurement contracts . 132
4.E.1 Event Study plots financial performance - part 2 140
4.E.2 Event Study plots financial performance - part 3 141
4.E.3 Event Study plots financial performance: assets thousand Euro . . 152
4.E.4 Event Study plots financial performance: current ratio 153
4.E.5 Event Study plots financial performance: liabilities thousand Euro 154
4.E.6 Event Study plots financial performance: liabilities growth 155
4.E.7 Event Study plots financial performance: shareholders funds . . . 156
4.E.8 Event Study plots financial performance: solvency 157
4.E.9 Event Study plots financial performance: number of employees . . 158
4.E.10 Event Study plots financial performance: employee growth 159
4.F.1 Event Study plots financial performance 163
4.F.2 Event Study plots financial performance assets growth by firms heterogeneity 164

5.1 Channels for effective and efficient public procurement 169
5.1 Economics of Public Procurement: Drivers of Efficiency and Effectiveness 174

List of Abbreviations

API	Application Programming Interface
ATT	Average treatment effect on the treated
CBS	Statistics Netherlands
CPV	Common Procurement Vocabulary
CZK	Czech koruna
ECA	European Court of Auditors
EU	European Union
EUR	Euro
EZ	Ministry of Economic Affairs - *Ministerie van Economische Zaken*
FAR	Federal Acquisition Regulation
FE	Fixed effects
FPA	First-price auction - Lowest Price
GDP	Gross domestic product
GPPD	Global Public Procurement Dataset
LPM	Linear Probability Model
MEAT	Most Economically Advantageous Tender
NACE	Statistical Classification of Economic Activities
OECD	Organization for Economic Co-operation and Development
OLS	Ordinary least squares
RDD	Regression discontinuity design
RSIN	Legal Entity Partnership Information Number
RVO	Netherlands Enterprise Agency *Rijksdienst voor Ondernemend Nederland*
SD	Standard deviation
SME	Small and medium-sized enterprise
TFP	Total factor productivity
VAT	Value-added tax
WTO	World Trade Organization

CHAPTER 1

Introduction

1.1 Motivation

This dissertation studies the economics of public procurement and the channels that shape its efficiency and effectiveness. Public procurement refers to the award of public contracts by or on behalf of governments to external suppliers for the acquisition of works, supplies, and services necessary for the functioning of the public sector. Its efficiency is understood as the balance between costs and benefits, whereas its effectiveness refers to the degree to which objectives are achieved (European Commission, 2025). Public procurement markets are sizable, accounting on average for 13% of GDP in OECD countries (OECD, 2025). These markets are also economically and politically consequential, since public procurement uses of public funds, from taxpayers' money, to allocate contracts to firms in the private sector. These markets are also fundamental for governments across the world to fulfill their mission and can create economic incentives that align with societal goals. Public procurement can boost competition while promoting sustainable and inclusive growth (World Bank, 2026; European Commission, 2026b; OECD, 2026). Nonetheless, governments face limited resources and fiscal pressures while ensuring that public services function properly and that citizens can trust this to happen OECD (2025).[1] Furthermore, the nature of the market is vulnerable to perverse incentives that may encourage corruption (Bandiera et al., 2021b) and cartelization among firms (Chassang and Ortner, 2019). These objectives and constraints raise a central question: What factors shape the efficiency and effectiveness of public procurement markets, and through which institutional, buyer-side, and supplier-side channels do they operate? Answers to these questions can inform procurement

[1] See for instance the press release '*Governments should strengthen public trust by improving efficiency, citizen engagement and access to public services*' of the report 'Government at a Glance 2025' (OECD, 2025a)

design and ensure efficient public spending.

Public procurement markets vary substantially across countries. Differences in legal frameworks, procurement procedures adapted to national needs and technological capabilities, social norms, and economic environments shape how governments purchase goods and services and prioritize sectoral and policy objectives (OECD, 2025). Countries also differ in how they define and organize their procurement workforce. While some OECD countries recognize public procurement as a standalone profession within the civil service, in most countries procurement responsibilities remain embedded within broader administrative functions (OECD, 2025, pp. 149-150). National public procurement laws also differ in their definitions of thresholds, procedures, and the degree of discretion granted to buyers (The World Bank, 2021). These institutional differences are associated with important variation in procurement performance, suggesting that the design of procurement systems is itself a determinant of efficiency and effectiveness (Bosio et al., 2022) and that people and organizations matter (Best et al., 2023).

Despite institutional differences across countries, one stylized fact is consistent across public procurement markets: competition is often limited and has declined over time. Public procurement tenders receive, on average, 4.4 bid submissions, a global average from 2018 to 2022[2]. In Europe, the average number of bidders per procedure fell from 5.7 in 2011 to 3.2 in 2021 (ECA, 2023). One particular manifestation of this trend is the prevalence of single bidding. This concerns contracts allocated in tenders with only one bidder, implying that firms face no or limited competition. Reduced competition increases procurement costs and weakens incentives for efficiency. For example, Titl (2025) shows that prohibiting the award of single-bid contracts is beneficial; a strict restriction reduces contract prices by about 10%. A low number of competitors poses challenges for public buyers, who may search for additional bidders (Titl, 2025) or prioritize the extraction of informational rents from existing suppliers (Kang and Miller, 2022). More broadly, firm participation and matching frictions link the number of competitors to market transaction costs (MacKay, 2022). The decline in competition is driven by factors related to market behavior, buyers' capacity, and institutional frameworks (Schotanus, 2025). The study of competition and firm participation is central to understanding public procurement markets, as participation decisions determine the intensity of competition, which in turn shapes prices and contract performance (Carril et al., 2026). Consequently, the market environment and firms' capabilities play a key role in shaping the efficiency and effectiveness of public procurement markets[3]. Understanding why firms participate in procurement markets is therefore central to understanding procurement efficiency.

Research on the field has emerged from theoretical foundations to empirically grounded analysis of how procurement markets function and perform (Thai, 2001;

[2]Calculation based on the data available for 54 countries at the Global Public Procurement Database

[3]The dynamics of firm participation are also informative for research on anti-competitive behavior (Bajari and Ye, 2003; Chassang et al., 2022) and corruption risks taking place in public procurement markets (Fazekas et al., 2016; Procurement Integrity Network, 2025)

Kelman, 1990). In 1947, William Baumol stated '[public] *...procurement is assuming larger proportions than ever before with the constantly increasing size and scope of governments.*' (Baumol, 1947, page 1). In the post-war period, the economics literature on public procurement focused on the theoretical aspects of principal-agent relations and contract theory (Laffont and Tirole, 1993), delineated the scope of governments' purchasing decisions (Hart et al., 1997), and proposed early definitions of the field: '*[...] procurement when the firm supplies a good to the government and regulation when it supplies a good to consumers on behalf of the government*' (Laffont and Tirole, 1993, page 9). More recently, the field of public procurement has leveraged the growing availability of detailed datasets to design promising research avenues using empirical approaches, including the drivers of market efficiency and the role of discretion and corruption[4] (Fazekas et al., 2024; Decarolis and Srhoj, 2025; Mazzucato et al., 2025). Yet, despite these theoretical foundations and the growing body of research agendas (Chabrost, 2019; Chen, 2024; Tkachenko, 2022; Cavalieri, 2024; Jääskeläinen, 2025), the factors that shape the efficiency and effectiveness of discretionary reforms, public buyer skills, and participation of new suppliers in public procurement markets remain insufficiently understood.

Table 1.1 Overview of thesis chapters and corresponding research questions.

Chapter	Research Questions
Discretion and Political Favoritism: Evidence from Two Reforms in Public Procurement	How do political favoritism and corruptible channels deepen the misuse of discretion? To what extent do corrupt political channels affect procurement market outcomes after changes in the discretion granted to procurement officers?
Value Added of Procurement Officers on Quality of Public Projects	What is the value added of procurement officers on the quality of public procurement projects? What skills, knowledge, and individual characteristics explain the officers' value added on procurement quality?
Competition, Entry, and Firm Dynamics in Public Procurement	How does entry into public procurement markets affect firms' financial performance? To what extent does the competitive environment at the time of entry shape the gains?

Note: The chapter-level research questions correspond to three drivers of efficiency and effectiveness in public procurement markets: regulatory and institutional design, public administration capacity, and firm participation and competition. These analyses contribute to answering the dissertation's overarching research question: what factors shape the efficiency and effectiveness of public procurement markets? Chapter 5 offers an integrated discussion of these findings.

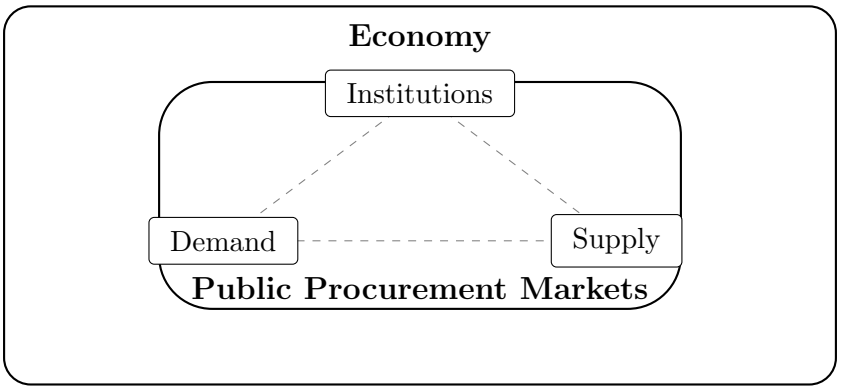
This dissertation aims to fill in this research gap by studying three drivers of efficiency and effectiveness in public procurement markets. Each driver is examined in a separate empirical chapter guided by its own research questions. Table 1.1 summarizes the questions that guide this dissertation.

[4]For instance, Decarolis et al. (2025) link procurement data with prosecuted officers and discuss observed corruption as a source of inefficiency. Fazekas et al. (2024) introduce the Global Public Procurement Dataset (GPPD) and illustrate how these data can be linked with other data for the study of market dynamics, ownership networks, among other aspects

1.2 Organizing framework of the dissertation

This section provides an organizing overview of the public procurement market, drawing on insights from Bandiera et al. (2021a), Bosio et al. (2022), Decarolis and Srhoj (2025), and Carril et al. (2026). Here I describe the elements that structure the analysis in this dissertation and situate it within the broader literature. I focus on three elements of the market: institutions, buyers, and suppliers. Figure 1.1 summarizes this organizing framework and illustrates the relationship between the three core components of public procurement markets within the broader economy. It illustrates the position of public procurement markets within the broader economy. Public procurement markets are represented as a specific market environment embedded in the economy, where demand, supply, and institutions interact[5]. Demand is generated by public buyers, who initiate procurement procedures to acquire goods, services, or works. Supply is provided by firms that decide whether and how to participate in tenders. Institutions shape both sides of the market by defining the legal rules, administrative procedures, and constraints under which public buyers and firms interact. The figure highlights that procurement outcomes are not determined by demand and supply alone, but also by the institutional framework that structures their interaction.

Figure 1.1 The economy and the components of public procurement markets.



The first element is institutions, which refer to regulations and laws that govern public contracting. Empirical work often proxies institutions with central government or congress decisions (Laffont and Tirole, 1991; Bosio et al., 2022; Carril, 2021)[6]. Institutions in public procurement markets aim to increase, among other things, efficiency, transparency, and competition. These aims are constrained by, among other things, economic cycles, corruption, and market inefficiencies. In-

[5]The definition of market used in this dissertation is modest. It aligns with the conception of markets presented by Rosenbaum (2000) as exchange characterized by voluntariness, specificity, regularity, and competition. This perspective is compatible with recent work in related fields of purchasing and public procurement that calls for greater attention to markets as objects of procurement policy, practice, and research (Knight, 2023; Borobia et al., 2026).

[6]In related public administration literature, institutions are also conceptualized as social norms (EL Bizri et al., 2023) or as actions of legitimacy (Nicolas and Schotanus, 2025).

stitutions are present in different hierarchical layers[7] and shape decision-making in public procurement markets, such as how to select suppliers, which awarding mechanisms to implement, and other procedural choices. In the context of this dissertation, the European Commission and the European Union (EU) public procurement directives are relevant institutions because these set rules for Member States, influencing even procurement activities that are not directly regulated by these rules, such as under-threshold procurement (Schotanus, 2025). Specifically, in Chapter 2, I focus on discretionary thresholds below the EU public procurement thresholds, and in Chapter 4, I study contracts tendered above the EU thresholds.

The second element of the market is the public buyer. I define the public buyer as the specific unit within a governmental agency responsible for designing public tenders. Public buyers can be understood broadly as an entire public entity or narrowly as an individual public servant (Rejeb et al., 2023). In the literature, public buyers are defined as contracting authorities (Bandiera et al., 2009; Allende et al., 2024; Bucciol et al., 2020), procurement teams (Decarolis et al., 2020), or individual procurement officers (Bandiera et al., 2021a). They demand discretion (Szucs, 2024), search for suppliers (Titl, 2025), and contribute to better prices (Best et al., 2023). Public buyers manage public budgets funded by taxpayers' money (Liebman and Mahoney, 2017; Rasul and Rogger, 2018) and make trade-offs within institutional rules that influence both the price offered by firms and the quality and sustainability of what is procured (Bosio et al., 2022). Public buyers' decisions are subject to their preferences (Tukiainen et al., 2024), career incentives (Bertrand et al., 2020), political incentives (Coviello and Gagliarducci, 2017; Spenkuch et al., 2023; Titl et al., 2021), and mission preferences and intrinsic motivations (Besley and Ghatak, 2005). Chapter 3 defines public procurement officers as buyers, positions them as agents who affect the quality of the execution of contracts, and explores skills and professionalization constraints.

The third element of the market is the firm. Firms, also defined as suppliers of the market, are in charge of delivering goods, services, or works; some of which are directly used by consumers (e.g., public roads) and some by civil servants (e.g., public office supplies). The study of the firm within public procurement markets involves the notion that most firms are driven by profit maximization, which materializes through their participation decisions and bidding strategies in competitive environments. In addition, public procurement is a demand shock that affects growth (Ferraz et al., 2015), survival (Cappelletti et al., 2024), investment (Hebous and Zimmermann, 2021; di Giovanni et al., 2024), and sales (Hoekman et al., 2024; Lee, 2021). Firms operate within the constraints imposed by the market's institutional design, while also responding to political incentives (Baranek and Titl, 2024; Titl et al., 2021; Titl and Geys, 2019). Moreover, firms are constrained by the market structure (Carril and Duggan, 2020; Carril et al., 2026) and economic cycles (Gugler et al., 2015; De Silva and Rosa, 2024). Chapter 2 explores how the institutional design affects firms' participation. Chapter 4 studies how competitive

[7]Institutional analyses have focused on how organizations operate in decentralized versus centralized public procurement markets (Dimitri et al., 2006a; Lotti et al., 2024)

entry affects procurement demand shocks on firms' financial growth.

The main hypothesis of this dissertation is that institutional discretion, procurement officer capacity, and firm entry conditions are fundamental drivers of the efficiency and effectiveness of public procurement markets, as reflected in procurement prices, project quality, competition, and supplier performance. The three empirical chapters examine these three components of public procurement markets in turn, leading to the following hypotheses. First, greater discretion in procurement rules can improve procurement efficiency by allowing buyers to adapt procedures to specific circumstances, but it also creates scope for political favoritism in the allocation of public contracts. Second, individual procurement officers differ systematically in their ability to produce high-quality procurement outcomes, implying that administrative capacity is partly shaped by officer-level value added. Third, entry into public procurement markets improves firm performance, with competition and sustained participation shaping these gains.

This organizing framework positions the contributions of this dissertation within the broader literature. This dissertation bridges three strands of the economics literature: political economy, public economics, and industrial organization. Chapters 2 and 3 focus on the demand side of public procurement markets, providing empirical evidence on institutional design and administrative capacity, and their effects on price and quality. Chapter 4 shifts to the supply side, analyzing firm entry and competition. As such, the framework should be interpreted as a guiding lens that structures the discussion and connects the individual chapters, rather than as a comprehensive model of procurement markets.

1.3 Institutional context

The empirical setting of this dissertation consists of Czechia and the Netherlands, two EU Member States that share a supranational procurement framework but differ in their national regulatory trajectories and administrative practices. This contrast is central to the dissertation's empirical approaches. Czechia offers variation in formal procurement rules, as successive reforms changed the scope for discretion. The Netherlands provides a comparatively stable institutional setting, making it possible to study firm participation and competition without major regulatory shocks.

Within this institutional context, public buyers are allowed limited discretion in their decisions for (i) the choice of procurement procedure (e.g., open vs. negotiated), (ii) the design of award criteria (price vs. multidimensional scoring), (iii) the use of bidder selection and qualification requirements, and (iv) the evaluation of bids when there are non-price criteria. Following an initial external demand shock from a tender notice, firms respond strategically to leverage the shock. They submit bids based on, among other things, expected profitability, which depends on the level of competition, the transparency of the process, and the perceived fairness of evaluation. In environments where discretion is high and favoritism is likely, entry may be discouraged, particularly among less connected firms.

General principles such as transparency, non-discrimination, and equal treatment are established in the EU Treaties. Building on these principles, the 2014 EU public procurement directives[8] provide detailed regulatory framework, specifying procurement procedures while allowing contracting authorities discretion in how tenders are designed and awarded (Janssen, 2023). As such, public procurement constitutes a central pillar of the European Single Market (European Commission, 2026b) and EU competitiveness (Draghi, 2024). It has also been positioned as a strategic policy instrument, extending its role beyond the efficient allocation of public funds, for instance, for green public procurement (Nicolas et al., 2025).

Although these policy priorities are defined at the EU level, their implementation depends on national legal frameworks and administrative practices. As a result, European procurement policies *trickle down* in heterogeneous ways across Member States, shaping public procurement markets differently depending on the institutional context. This interaction between a common supranational framework and diverse national implementations is instrumental to this dissertation. The Czech Republic provides a setting with regulatory variation since the establishment of Act no. 137/2006 Coll. on public contracts. A sequence of reforms altered public procurement rules along key dimensions related to discretion and transparency. These changes generate quasi-experimental variation in the institutional environment. In contrast, the Netherlands offers a relatively stable institutional environment following the Public Procurement Act 2012.

The comparison is not intended to treat the two countries as institutionally equivalent. Rather, the dissertation uses their differences analytically: Czechia speaks primarily to the consequences of regulatory change, while the Netherlands speaks to firm behavior within a stable procurement regime. This distinction is important for interpreting the findings and for assessing their external validity across EU procurement markets.

The institutional context highlights a central trade-off: while discretion allows public procurement officers to incorporate quality and policy objectives, it also creates scope for favoritism and reduces transparency (Bosio et al., 2022; Fazio, 2025). This trade-off has implications for firm entry, competitive intensity, and the allocation of contracts. The empirical analysis in this dissertation exploits variation in discretion, firm entry, and institutional environments to study how these forces shape public procurement outcomes.

1.4 Methodology and data sources

This dissertation adopts a set of standard applied econometric methods to quantify how discretionary thresholds distort the allocation of public resources, the contribution of individual procurement officers' on the quality of public construc-

[8]Directive 2014/24/EU, which sets out generally applicable procurement rules, Directive 2014/25/EU for entities operating in the water, energy, transport, and postal services sectors, and Directive 2014/23/EU on concession contracts (European Parliament and Council of the European Union, 2014).

tion tenders; and firms' entry into the public procurement market. In particular, I use regression discontinuity designs (RDD), fixed effects models, propensity score matching, and staggered difference in differences, each tailored to the institutional and data features of the respective research questions. In this dissertation, I aim to follow best practices in applied econometric research. The empirical frameworks are proposed to address distinct challenges related to identification, measurement, and mechanisms of different drivers of efficiency, and to provide policy-relevant evidence.

The empirical approaches are based on the construction and integration of multiple complementary datasets from the Czech Republic and the Netherlands, designed to enable credible quantitative approaches. I combine comprehensive administrative procurement data with firm-level financial information from Orbis (formerly Bureau van Dijk, now Moody's), with additional primary, and secondary data. For the study of changes in regulation and public procurement officers' value added, I exploit the universe of public procurement tenders in the Czech Republic, collected by data.lab through manual scraping of the Czech Gazette of Public Procurement. Covering the period 2006 to 2022, these data provide a uniquely comprehensive basis for analyzing regulatory changes and public procurement outcomes over time. Identification in the analysis of political favoritism is strengthened by merging the tender dataset with external datasets identifying firms with political connections, anonymous ownership structures, and political campaign donations, extending earlier work in the literature (see Titl and Geys, 2019; Titl et al., 2021; Baranek and Titl, 2024). To more directly measure procurement officers' value added and decision-making constraints, I additionally rely on original survey data collected from public procurement officers via Qualtrics. The survey, designed jointly with my co-authors, showcases the effort of harmonizing heterogeneous data sources and linking them at the tender, firm, and public procurement-officer level, thereby enabling the measurement of outcomes related to prices, quality, and institutional performance that are typically not jointly observable.

For the Netherlands, the empirical approach relies on combining several public procurement data sources to address the absence of a fully integrated national procurement data space (Titl and Schotanus, 2025). These include public tender data from spreadsheets hosted by the platform TenderNed. Additionally, I use tender data accessed through the API embedded for tenders in the Netherlands. I also made use of additional public contracts from a large public utility company that includes bidding information. Across chapters, the analyses demonstrate how leveraging the complementarity of administrative, financial, and survey data improves measurement and enables research designs that isolate institutional mechanisms in public procurement.

1.5 Results and contributions

1.5.1 Discretion and Political Favoritism: Evidence from Two Reforms in Public Procurement

Chapter 2 studies discretion and political favoritism. The research developed together with Vitezslav Titl and Fredo Schotanus, examines the misuse of bureaucratic discretion in public procurement. Discretion in public procurement arises from the degree of autonomy that contracting authorities have within a given institutional framework, including the design of procedures, selection of suppliers, and contract implementation. One important regulatory instrument shaping this discretion is the use of thresholds that determine when simplified procedures may substitute for open competition. These thresholds affect the level of discretion by altering the balance between flexibility and oversight. While greater discretion can enhance responsiveness and reduce administrative frictions, such as during the COVID-19 pandemic (Bandiera et al., 2021b), it also introduces risks associated with reduced competition and weaker accountability (Bosio et al., 2022). The empirical evidence regarding the effects of allocating discretion to procurement officers has remained ambiguous. Some studies document adverse effects, including contract manipulation (Palguta and Pertold, 2017), favoritism (Gallego et al., 2021), and higher prices (Szucs, 2024; Carril, 2021), while others suggest that discretion may improve procurement outcomes by enabling higher-quality purchasing and better post-award performance (Fazio, 2025; Decarolis et al., 2025; Arozamena et al., 2025). In our research, we provide new evidence by leveraging two reforms to Czech public procurement law that changed discretionary thresholds in opposite directions. The first reform in 2012, reduced the threshold for simplified procedures (with high discretion). The second reform in 2016 raised the threshold for simplified procedures. This implies that after 2012, public buyers had less discretionary room compared to the preceding period, while after 2016 discretionary room increased again.

We use comprehensive administrative data on public contracts (2010 to 2018); detailed firm-level data on three potential sources of political favoritism, such as political connections, donations to political parties and firms with anonymous ownership, and firms' financial information from the Orbis database. Our empirical strategy relies on two approaches. First, we use bunching regressions to detect manipulation of contract values around thresholds. Second, we use Regression Discontinuity Design to estimate causal effects of discretion on prices and competition.

The main contribution is causal evidence that the effects of discretion are asymmetric and depend on political favoritism. First, we find that public contracts are substantially clustered at the simplified procedure thresholds, regardless of the threshold value. We observe less clustering of contracts at the highest threshold. This suggests that procurement officers manipulate contract values to remain below discretionary thresholds. The two reforms also indicate that bunching shifts from the old thresholds to the new thresholds, indicating that policy officers actively respond to threshold changes.

Second, based on the estimations from the Regression Discontinuity, we find that the first reform led to a significant reduction in contract prices of 0.13 percentage points, indicating increased efficiency. However, when discretion increased with the 2016 reform, final prices were 0.06 percentage points lower, but this reduction is not statistically significant. Thus, our estimations suggest asymmetric effects of discretion on efficiency, an aspect not studied in the literature, which has primarily focused on studying expansions of discretionary (Palguta and Pertold, 2017; Coviello et al., 2018; Castro et al., 2022; Carril, 2021).

Third, we find that the effects of discretion are mediated by political favoritism. Particularly, efficiency gains from restricting discretion are concentrated among non-connected firms, while politically connected firms do not exhibit similar improvements. Our findings suggest that political favoritism hinders market outcome improvements from stricter regulations. Overall, we find suggestive evidence that public contracts just below the threshold are overpriced, especially for contracts delivered by firms with political connections, firms donating to political parties, and anonymously owned firms.

This research contributes to two streams of literature. First, it contributes to the literature on discretion in public procurement. Previous research has documented contract manipulation after the introduction or expansion of discretionary thresholds and found that it leads to a higher share of contracted politically connected firms (Palguta and Pertold, 2017; Baltrunaite et al., 2021; Szucs, 2024; Titl et al., 2021). Our findings provide new causal evidence using reforms that both tighten and relax discretion. Second, we contribute to the literature on political connections in public procurement by showing that political favoritism limits the effectiveness of institutional rules designed to improve efficiency (Coviello et al., 2018, 2022; Szucs, 2024; Carril, 2021; Titl and Geys, 2019; Titl et al., 2021).

1.5.2 Quality of Public Projects and Procurement Officers Value Added

This chapter studies the value added of public procurement officers on the quality of public procurement projects. The research developed together with Michal Soltes, Vitezslav Titl, and Fredo Schotanus focuses on the role of individual public procurement officers in shaping public procurement outcomes, an aspect that remains poorly understood in the economics literature due to challenges in measuring outcomes other than prices (Best et al., 2023) or due to the methodological challenges for isolating officer-level contributions (Decarolis et al., 2020; Bucciol et al., 2020; Allende et al., 2024). Additionally, prior evidence suggests relatively homogeneous preferences among officers within countries (Tukiainen et al., 2024). Consequently, there remains limited empirical evidence on how individual skills translate into realized public procurement outcomes. This gap is particularly relevant given persistent policy concerns about skill mismatches and insufficient technical expertise among public buyers (Saussier and Tirole, 2015; OECD, 2021).

Our research quantifies the contributions of public procurement officers to the quality of public procurement projects. We assemble a unique dataset combining the universe of public procurement contracts in the Czech Republic with original

survey data on public procurement officials. Our analysis focuses on construction contracts and introduces a novel measure of procurement quality based on the value of subsequent repair contracts required after project completion (Baranek and Titl, 2024; De Jaegher et al., 2025). This approach captures economically meaningful dimensions of quality while being less susceptible to manipulation, as repair needs typically materialize only years after contract award and are unlikely to be directly targeted by contemporaneous incentive schemes.

Our empirical strategy proceeds in two steps. First, we build on the value-added literature (McCaffrey et al., 2004; Koedel and Rockoff, 2015; Chetty et al., 2014a,b; Bau and Das, 2020; Bertrand et al., 2020; Limodio, 2021) to estimate public procurement officers' value added by exploiting variation in project quality across contracts, controlling for observable contract characteristics. Second, we analyze the determinants of this value added by regressing it on officers' skills and preferences.

We document substantial heterogeneity in officer performance. Moving from the 10th to the 90th percentile of the procurement officer value-added distribution is associated with a 4.5 percentage point improvement in project quality. For a median-sized construction contract (CZK 11.5 million; EUR 465,000), this corresponds to a reduction in expected five year repair costs of approximately CZK 520,000 (EUR 21,000), highlighting the quantitatively important role of procurement officers in determining the long-run efficiency of public procurement. While these estimates are not causal, they indicate economically meaningful variation in officer performance.

We find that officers with a legal education are associated with lower-quality projects on average. Particularly, we find that conditional on level of legal engagement, these officers are associated with higher expected repair costs of approximately 1.3%. This pattern suggests a potential trade-off between legal compliance orientation and effective project execution.

Our research contributes to the literature studying determinants of public procurement outcomes. Existing work relies primarily on proxies such as cost overruns or delays (Coviello et al., 2018; Decarolis, 2014; Best et al., 2023). By contrast, we provide an outcome-based measure of realized project quality and are, to our knowledge, the first to estimate individual-level value added using such a measure in a complex contracting environment. Our findings highlight that individual officers explain economically meaningful variation in non-price outcomes, particularly in settings with non-standardized projects.

Finally, we contribute to the broader literature on bureaucratic capacity and state effectiveness (Bertrand et al., 2020; Besley et al., 2022). Our results suggest that differences in skills, training, and professional orientation translate into persistent differences in performance, consistent with models of heterogeneous bureaucrats with varying mission preferences (Besley and Ghatak, 2005; Ashraf et al., 2020; Barfort et al., 2019). By documenting substantial individual heterogeneity, we extend the worker value-added framework to public procurement and show that bureaucrats, like teachers, managers, and private-sector workers, exhibit meaningful productivity differences that aggregate into state capacity.

1.5.3 Competition, Entry, and Firm Dynamics in Public Procurement

In Chapter 4, I study to what extent entry into public procurement markets affects firms' financial performance, and how the competitive environment at the time of entry shapes the size and persistence of those gains. Public procurement constitutes an important source of demand for private firms. However, bidder participation in public tenders has declined steadily; by 2024 the average number of bids per tender in the EU stood at 3.2, two fewer than a decade earlier (ECA, 2023). This decline in participation is puzzling given growing evidence that winning public contracts improves firm outcomes, including productivity (Chiappinelli et al., 2025; Ferraz et al., 2015), access to credit (di Giovanni et al., 2024), and sales (Hoekman et al., 2024). At the same time, policymakers increasingly rely on public procurement to promote innovation, green growth, and broader societal objectives.

The analysis focuses on Dutch public procurement above EU thresholds over 2016–2024. The Netherlands is one of the largest public buyers in the OECD (procurement amounts to roughly 21% of GDP), making it an informative setting. The paper combines contract-level data from TenderNed, a Dutch national procurement platform, with firm-level balance sheet information from Orbis (Moody's Analytics), yielding a firm-year panel that tracks financial outcomes before and after a firm's first public contract. Complementary bid-level data from ProRail, a large Dutch public buyer, are used to study bidding behavior and mechanisms.

Entry is defined as the year a firm first wins a public contract above EU thresholds. Two definitions are distinguished: *general entry* (any first contract; 4,235 events) and *competitive entry* (first contract awarded in an open procedure with at least two bidders; 1,095 events). Because entry occurs in a staggered fashion, I implement the Callaway and Sant'Anna (2021) group-time estimator, grouping firms into cohorts by entry year and comparing each cohort to a matched sample of never-supplier firms with similar pre-entry age, size, and sector. Dynamic event-study coefficients are then aggregated across cohorts at each relative event time. As a robustness check, not-yet-supplier firms are used as an alternative comparison group; results are qualitatively consistent.

The findings suggest that entry into public procurement leads to improvements in firms' financial outcomes, particularly capital accumulation. General entry is followed by a statistically significant increase in asset growth of approximately 9.9%, while the effect on total assets in levels (EUR 28,065) is positive but imprecisely estimated. Competitive entry yields weaker evidence on asset growth (2.5%, not statistically significant), and effects on asset levels are not sustained beyond the entry year. Both entry definitions are associated with higher liabilities and declining liquidity. Under general entry, total liabilities rise by approximately EUR 97,975 and liability growth increases by 22.3%; the current ratio declines by roughly 0.24 points at entry and continues falling, reaching a reduction of about 1.5 points five years later. Under competitive entry, liabilities increase by approximately EUR 71,887 (liability growth of 13.5%) and the current ratio falls by around 1.0 point by year five. Employment effects are statistically insignificant across both entry definitions, consistent with mixed evidence in the broader procurement literature.

Total assets increase by approximately 8% at entry and reach around 11% within four years, with persistent effects on equity. However, these average effects hide substantial heterogeneity. First, firms entering through competitive procedures do not experience sustained gains: while assets increase in the short run, these effects dissipate and become statistically insignificant over time. Second, the benefits of entry differ across firm types. Less experienced firms exhibit temporary increases in assets that fade in subsequent years, whereas more established firms experience sustained and statistically significant growth.

Motivated by the concern that entry may be correlated with observed and unobserved firm characteristics (Krasnokutskaya, 2011; De Silva et al., 2009), I compare the three types of firms to decompose the average post-entry effect by procurement trajectory and document how financial outcomes differ across firms with different degrees of market engagement. Entrants are classified into three groups: *established firms* (continuously active after entry), *capacity-constrained firms* (intermittent subsequent participation), and *short-term participants* (no further contracts after entry). Established firms exhibit the largest and most persistent asset growth, about 16% under general entry, while competitive entry effects attenuate over time even for this group. Capacity-constrained firms show moderate and less persistent gains (approximately 10% under general entry). Short-term participants experience the smallest and most transient improvements (6-7%), which dissipate within two years. These patterns hold broadly across other financial outcomes: increases in assets and liabilities are concentrated among established and capacity-constrained firms, while short-term participants show no significant changes in assets or liabilities under competitive entry.

Additionally, I explore two mechanisms. First, using bid-level data from ProRail, I document that one-time and short-term bidders submit bids roughly 35 percentage points below the engineer's cost estimate (conditional on contract and firm characteristics), consistent with a winner's curse and learning-by-doing dynamics. Second, using a Poisson pseudo-maximum-likelihood regression, the paper shows that short-term participants and capacity-constrained firms tend to enter contracts with 8-10% fewer bidders than those won by established firms. Observable contract characteristics do not explain this sorting, suggesting that unobserved contract attributes or informational advantages of incumbents play a role. Together, these mechanisms help reconcile why short-term entrants win contracts but fail to translate those wins into sustained financial improvement.

This chapter contributes to a growing research agenda that studies who contracts with the government and why (di Giovanni et al., 2024; Ferraz et al., 2015; Cappelletti et al., 2024; MacKay, 2022), by showing that entry into public procurement improves firm outcomes on average, but these gains are not driven by competitive entry and are concentrated among firms that remain active and have the capacity to scale. I also contribute to the literature that studies competition in public procurement and procurement auctions (Carril et al., 2026; Carril and Duggan, 2020; Titl, 2025; De Silva and Rosa, 2024), by discussing that inexperienced firms win less attractive tenders, enter aggressively and may suffer from a winner's curse, which limits their gains. The findings in this paper speak to the lit-

erature that documents that young producers charge lower prices than incumbents (Foster et al., 2008) and to the literature using event studies to measure effects of market entry (Alfaro-Urena et al., 2022; Amiti et al., 2024). All in all, this chapter highlights that in public procurement markets, entry and permanence are central to competition outcomes. The implications of these results are informative to public entities and international agencies that promote public procurement as a channel through which governments can achieve innovation, green growth, and societal goals (OECD, 2026; World Bank, 2026). Furthermore, studying competition in public procurement markets provides evidence to enlighten salient discussions on European competitiveness (Draghi, 2024).

1.6 Roadmap

In answering the question of what factors shape the efficiency and effectiveness of public procurement markets, this dissertation discusses three central channels: political favoritism, public capacity, and market structure (see 5 and Figure 5.1). Political favoritism shapes whether bureaucratic discretion is used to improve procurement outcomes or to manipulate contract allocation. Public capacity matters because procurement officers' skills, preferences, and professional backgrounds influence the quality of tender design and contract outcomes. Market structure affects procurement performance because firms' entry and bidding decisions determine competitive intensity, supplier viability, and the allocation of contracts.

This dissertation is organized as follows. The next chapter (Chapter 2) presents the paper Discretion and Political Favoritism: Evidence from Two Reforms in Public Procurement, published in the Southern Economic Journal. It examines how changes in discretionary power affect the allocation of public contracts and the extent of political favoritism. The following chapter (Chapter 3), Quality of Public Projects and Procurement Officers' Value Added, shifts the focus to the role of individual bureaucrats, analyzing how procurement officers influence project outcomes and quality. The subsequent chapter (Chapter 4), Competition, Entry, and Firm Dynamics in Public Procurement, studies how entering public procurement markets affects firms' financial performance. The dissertation concludes with a final chapter (Chapter 5) that synthesizes the main findings, discusses channels that mediate efficiency, broader implications for public procurement policy, and outlines directions for future research.

CHAPTER 2

Discretion and Political Favoritism: Evidence from Two Reforms in Public Procurement

The misuse of bureaucratic discretion in public procurement risks political favoritism and corruption. Discretionary thresholds regulate this, with lenient rules below and strict oversight above these thresholds. We examine the impact of changes in these thresholds in the Czech Republic in 2012 and 2016 on discretion misuse and market competition, using bunching estimators, regression discontinuity, and comprehensive data on construction contracts, political ties, and firm productivity. Our findings show a concentration of contracts just below thresholds, both pre and post-reforms. Reforms reallocating contract values to new thresholds reveal that limiting discretion lowers final contract prices, indicating increased efficiency. However, when discretion increases, final prices are unaffected. Efficiency gains are not seen in contracts awarded to politically connected firms, suggesting that political favoritism hinders market outcome improvements from stricter regulations.

Keywords: Procurement, Discretion, Political Favoritism, Bunching.
JEL Codes: D73, D86, H57, K23, L74

2.1 Introduction

The public procurement market accounts for approximately 12% of the GDP in Organization for Economic Co-operation and Development (OECD) countries (OECD, 2017). One of the critical questions about organizing this market is how much discretion procurement officers should have. During the Covid-19 pandemic, several

This chapter is joint work with Fredo Schotanus and Vitezslav Titl and was published in the *Southern Economic Journal* (Volume 91, Issue 3, 15 January 2025, pp. 915–968).

academic voices (Bandiera et al., 2021b) supported the need for increased discretion in public procurement because the pandemic was a critical moment that required flexibility, expedited public contracting, and prompt responses given the emergency. Allowing more discretion requires changing the law because the level of discretion granted to public procurement officers is regulated by comprehensive legal acts in which so-called discretionary thresholds are established. Below these thresholds procurement officers are entitled to preselect potential bidders for a contract, and thus are granted more autonomy in awarding procurement contracts. In the United States, modifying these thresholds involves adjustments in the Federal Acquisition Regulation (FAR), as illustrated by Carril (2021). In the case of European countries, modifying levels of discretion requires changes in national public procurement laws, as illustrated by Palguta and Pertold (2017); Castro et al. (2022); Coviello et al. (2018); Szucs (2024). Furthermore, in the European Union (EU), the Directives of public procurement exert regulatory references for member states. Organizing procurement markets through regulatory changes is lengthy because of the administrative burden embedded in those changes.

Increasing discretion, even in difficult times, carries risks of inefficiency and corruption, particularly in countries with a low capacity public sector (Bosio et al., 2022; Gallego et al., 2021). The misuse of discretionary power often materializes in contract value manipulation (Palguta and Pertold, 2017) and procurement that is driven by political favoritism (Szucs, 2024; Baranek and Titl, 2024). While adapting the law to address extraordinary circumstances is acceptable and aligns the interest of governments and the general public, introducing or adjusting discretionary thresholds in the law raises questions about optimal policies and calls for evidence. The empirical evidence on how much discretion procurement officers should have remains ambiguous. There is evidence suggesting that more autonomy leads to contract manipulation, higher prices, selection of politically connected firms (Szucs, 2024), and sorting of contracts below thresholds i.e. "bunching" (Palguta and Pertold, 2017; Carril, 2021). On the other hand, other studies find that higher prices are due to the acquisition of higher-quality products (Fazio, 2025), and that more discretion leads to better post tendering performance (Decarolis et al., 2025).

This paper investigates two research questions. First, how do political favoritism and corruptible channels deepen the misuse of discretion? Second, to what extent do corrupt political channels affect procurement market outcomes after changes in the discretion granted to procurement officers? We focus on specific outcomes such as the number of competitors, contract price, and the previous year's productivity of winning suppliers.

We study two reforms of the public procurement law in the Czech Republic that changed the discretionary thresholds for public procurement contracts. The first reform, enacted in 2012, amended the procurement law Act no. 137/2006 Coll. and reduced the threshold for simplified procedures to 10 million Czech crowns (CZK) ($470k[1]) from 20 million CZK ($940 thousand (k)). Therefore, the 2012 Reform

[1]Using the exchange rate from 14th July 2023 – $1 = 21.2 Czech crowns or Czech korunas (CZK).

left less room for discretion. The second reform in 2016 raised the threshold for simplified procedures to 50 million CZK ($2.36 million (M)). Therefore, the 2016 Reform broadened the range for procurement officers to use simplified procedures.

The reforms provide two unique quasi-experimental designs. These two shocks exogenously created rules that differentiated contracts awarded with more discretion (below the threshold) from those with less discretion (above the threshold), and differentiated contracts awarded before and after each reform. Regression discontinuity design provides all the elements for a structured empirical strategy. With defined cutoff values, we can implement an analysis focused on the discontinuity. We leverage the analysis of the distribution of the running variable, which, in our case, is the anticipated contract value. This allows us to study the extent to which discretion is misused in the form of contract value manipulation. This step is customary in regression discontinuity and is essential in our discussion of the effects of changing thresholds. We devote the first part of our analysis to studying contract value manipulation, sorting, and bunching.

We also leverage the decrease and increase in the threshold for identification. The reform in 2012 offers a policy treatment of low discretion, where contracts awarded just below the threshold before the change are a valid counterfactual to those contracts awarded just above the threshold after the change. Therefore, this policy treatment informs about the effects of the imposition of less discretion. The reform in 2016 offers a policy scenario of high discretion, where contracts awarded just above the threshold before the change are a valid counterfactual to those just below the threshold after the change. Thus, this policy scenario informs about effects of granting more discretion.

We motivate our study of these two reforms based on non obvious evidence about whether the effects of increasing discretionary thresholds would be the opposite of decreasing the thresholds. The uncertainty about what the potential outcomes would be in the two scenarios, motivates us to pursue our research design. As the literature has mostly focused on studying reforms that increase discretionary thresholds (Palguta and Pertold, 2017; Coviello et al., 2018; Castro et al., 2022; Carril, 2021), our research setup contributes to addressing this gap in the literature.

We use the population data of public procurement contracts in the Czech Republic. This dataset contains information about the characteristics of the tenders, the contracting authorities, and the procurement outcomes. We restrict the analysis to construction contracts awarded between 2010 and 2018. We combine the contract data with detailed firm-level data on three potential sources of political favoritism: political connections, donations to political parties and firms with anonymous ownership. Our source of political connections regards to people who were candidates of political parties and that were ever members of board or owners of a company. This definition is aligned with that of Baranek and Titl (2024). A second source of political favoritism is the connection of firms who have been donors to political parties. We match contract winners with records on donations to political parties. The third source of political favoritism is the record of anonymous ownership[2].

[2]This type of ownership was a legal form of representation for firms in the Czech Republic

Furthermore, we use the dataset *Organisation, Beratung und Innovatives Systemengineering* (Orbis) from Bureau van Dijk to explore contractors' productivity as a market outcome.

Political favoritism represents a significant mechanism by which discretion can be misused. It is likely that mechanisms prone to corruption might alter the effects of changes in discretionary thresholds on public procurement market outcomes. To disentangle the role of political favoritism, we estimate the effects on market outcomes for the case where the winning suppliers have political connections in a different regression specification to the cases where the suppliers are not politically connected. The extent to which procurement officers misuse the discretion granted to them may vary with the degree of discretion they actually posses; likewise, corrupt mechanisms might differ depending on whether the potential for misuse of discretion is high or low. In other words, the political favoritism channel might differ when estimating the effects of the Reform in 2012 from the effects of the Reform in 2016. For instance, less room for discretion can make the misuse of discretion more evident, while more room for discretion could hinder it. Therefore, we separately estimate the effects of increased discretion in 2012 Reform and the effects of decreased discretion in 2016.

We find that public procurement contracts are substantially clustered at the simplified procedure thresholds irrespective of their values. There is less clustering at the highest threshold of 50 million CZK ($2.36M), suggesting that higher discretionary thresholds may be preferable to reduce manipulation. In analyzing the two reforms, we document that contract value bunching shifts from old thresholds to new thresholds. This indicates that procurement officers actively respond to the thresholds changes by manipulating contract values accordingly. We also observe that contracts just below the threshold tend to be overpriced, particularly those awarded to firms with political connections, firms donating to political parties, and anonymously owned.

Our contribution to the literature is two-fold. First, we contribute to the stream of literature focusing on the effects of discretion in public procurement. We are the first to document the effects of changes following a decrease and an increase of discretionary thresholds within the same institutional environment. Most studies present evidence focused on reforms that either introduce discretionary thresholds (Szucs, 2024; Palguta and Pertold, 2017), or increased them (Coviello et al., 2018; Castro and Guccio, 2021; Carril, 2021; Baltrunaite et al., 2021). Only one study by (Coviello et al., 2022) uses a decrease in discretionary thresholds to cross-validate their analysis of the increase in discretionary thresholds. Therefore our focus on decreasing and increasing the discretionary threshold is unique.

We advance the literature by estimating effects of changes in the discretionary threshold after their introduction in the regulatory framework of public procurement. Contract value manipulation has been documented in public procurement following the introduction of discretionary thresholds in the law, first by Palguta and Pertold (2017) and recently by Szucs (2024). We find that the lowering of the

and have been linked with corrupt scandals (Palguta and Pertold, 2017)

threshold by the 2012 Reform led to adjustments in contract value manipulation, suggesting that reduced discretion continued to reflect bunching as a substantial issue. On the other hand, the increase in the threshold by the 2016 Reform resulted in a less evident adjustment in contract value manipulation. This indicates that granting more room for discretion helped alleviate bunching. Our estimations of the effects of decreasing and increasing these thresholds explore policy changes that have previously gone unstudied.

Within the same stream of literature, we contribute by exploring the relationship of manipulation and political connections. Our results are aligned with the growing body of literature showing that political connections are associated with increased prices (Baltrunaite, 2020; Titl and Geys, 2019; Baranek and Titl, 2024) and lower efficiency (Baltrunaite et al., 2021; Titl et al., 2021). Baltrunaite et al. (2021) shows that manipulation leads to a higher share of politically connected winners, which is at odds with our findings. Additionally, Baltrunaite et al. (2021) and Szucs (2024) also show that less productive firms tend to be more likely to win just below the threshold. In addition, we discuss that procurement has more efficient outcomes when discretion is more regulated and becomes less efficient when more discretion is present in the context of the Czech Republic. Our contribution touches upon procurement process (Coviello et al., 2018), cost overruns (Coviello et al., 2022), increased prices (Szucs, 2024), and bunching of contracts (Carril, 2021).

This paper contributes to the empirical understanding of agency theory in the context of public spending, particularly focusing on the misalignment issues between principals and agents, driven by the agents' preference for discretion (Bandiera et al., 2009; Bosio et al., 2022; Szucs, 2024) in an institutional context where corrupt activities are likely to occur (Decarolis et al., 2025; Baranek and Titl, 2024). We rationalize our findings under the theoretical framework of active and passive waste in public spending of Bandiera et al. (2009) and on the definition of "*demand for discretion*" proposed by Szucs (2024). According to Szucs (2024) demand for discretion can be characterized by the preference of buyers to have contracts under discretionary thresholds as a preference for less formalized and less monitored procedures. Similarly to Decarolis et al. (2025), we consider that risks of corruption and political favoritism are institutional factors that confound the preference for discretion.

According to Bandiera et al. (2009) active waste is characterized by the preference of buyers to setup the price for contracts. In relation to regulatory thresholds, active waste occurs when contracts are sorted to lie under the threshold and are awarded to politically connected firms. Thus, active waste can be identified from contract manipulation and political favoritism when the public buyers do so in order to claim bribes, gain political status, contract with insiders (Bosio et al., 2022). Those factors are unobservable for us, but observable to the procurement officer.

Passive waste refers to the preference for lower-priced contracts and a reluctance to design contracts that exceed the discretionary threshold. This form of discretion could be motivated to avoid increased monitoring and the additional tasks associated with open auctions. Additionally, passive waste can be associated with

contracting with politically connected firms. In other words, political favoritism, while a highly corruptible activity, can serve as a screening mechanism that the buyer implements to select high quality suppliers Schoenherr (2019). Thus, passive waste can emerge when political connections are used for market efficiency. We interpret our results that awarding contracts to politically connected firms can be a screening tool for identifying productive and reliable suppliers, thus reducing information asymmetries and transaction costs.

Our paper empirically contributes to the hypothesis described in Schoenherr (2019) by examining whether firms who were awarded tenders with contracting values in the sorting region are politically connected and whether those firms were more productive than the average firms at the moment of the tender award. If these firms were productive prior to being awarded a contract, the manipulation of contract values indicate passive waste - an effort to avoid unnecessary work due to sufficient information about the contractor's quality. However, if non-competitive firms with political connection receive contracts, this suggests active waste driven by discretion.

The remainder of the paper is structured as follows. Section 2 describes the institutional design of the procurement market in Czechia and the reforms. In Section 3, we describe our empirical strategy. Section 4 presents our main findings about the effect of the discretionary threshold. Section 5 concludes and lays out the policy implications of our findings.

2.2 Setting and data

Following the Velvet Revolution in 1989 and the subsequent dissolution of Czechoslovakia in 1993, the Czech Republic shifted from a communist economy to a market-based system. Despite achieving an average real gross domestic product (GDP) growth of 2.5% from 1997 to 2017, it has yet to reach the economic levels of its pre-communist era peers like Austria or Germany. The National Economic Council pinpointed corruption and weak institutions in 2010 as key obstacles to growth, likening the Czech Republic's institutional quality to that of countries like South Korea or Italy, which have high incomes but also high corruption levels.

The public procurement market in the Czech Republic is worth about 14.5% of GDP (OECD, 2021). The market is fully harmonized with the EU legislation. There are two main awarding procedures open and simplified procedures. An open procedure is an auction with lowest price awarding mechanism. When a public buyer uses an open procedure there could be an unlimited number of firms submitting bids, upon fulfilling the qualification criteria. In a simplified procedure, a public buyer is entitled to invite only specific candidates to submit a bid. In Czech public procurement, evaluation criteria include lowest-price auctions, where price is the sole determinant, and the Most Economically Advantageous Tender (MEAT), which integrates price with other factors to select the winning bid. Most public procurement contracts are awarded using lowest-price criterion 63%. Public procurement is particularly affected by these institutional weaknesses, with politicians

often influencing contract allocation and the appointment of public officials, leading to issues such as unclear contract specifications and bid rigging. The EU and anti-corruption bodies have criticized this politicization. The influence of politicians and officials varies based on the procurement process (open or restricted) and evaluation criteria (lowest price or most economically advantageous tender), affecting the fairness and efficiency of public procurement.

We study public tenders awarded in the construction sector in the Czech Republic. Public procurement in the construction sector suits the study of discretion and political favoritism because public contracts in the construction sector are often tailored by public buyers' needs and potential contractors in the private sector are not many (Bajari and Tadelis, 2001; Bajari et al., 2009). Construction tenders represent 30% of the total public procurement in the Czech Republic and refer to public works of engineering, building development, and installations. Furthermore, we focus on national contracts or contracts classified as below the European threshold for public works in procurement[3].

To enable small contracts to be awarded in a simpler and more flexible way, the Czech Republic procurement law allows for a so-called simplified procedure to contracts with estimated value up to an arbitrary legislative threshold. Public buyers are allowed to use the simplified procedure for contracts whose anticipated value lies below the legislative threshold. This simplified procedure gives more autonomy to public buyers because it entitles them to invite specific suppliers for tender proposals. Unlike an open procedure, the contracting authorities do not have to call for proposals to an unlimited number of potential suppliers. Similarly, unlike a restricted procedure the contracting authorities do not have to make known their intention to award a public contract and to invite all qualified suppliers. In the simplified procedure the contracting authority can invite no less than five qualified candidates to submit tenders. We refer to the threshold for simplified procedure as discretionary threshold as it constitutes the core of our methodological strategy to study discretion.

2.2.1 Public Procurement Acts reforms

Beginning in 2006, the Czech Republic has experienced major changes and amendments in its public procurement acts. These public procurement reforms, among other things, modified the threshold for simplified procedures. We define the reform in 2012 as the legal change that narrowed discretion and the reform in 2016 as the legal change that broadened discretion. The reform in 2012 imposed a decrease in range of discretion. Before 2012, Czech public procurement was governed by the Act no. 137/2006 Coll. on public contracts, which stipulated the discretionary threshold for simplified procedures established at 20 million CZK ($940k). Contracts with anticipated value under this limit and above small-scale limit (6 million CZK ($280k)) were eligible to simplified procedures and thus subject to

[3]The Directive 2014/24 from the European Commission on public procurement establishes the European Union (EU) thresholds for all members of the union ec.europa.eu

Table 2.1 Summary discretionary thresholds reforms and range for discretion

Period	Date Acts' modifications		Simplified[a]			National[b]				EU[c]
Reform	Effective	1st July 2006	6 <	X_i	**20** ≤	X_i	≤	132	≤	X_i
	Valid	19 April 2006		range 14'						
2012	Effective	1st April 2012	3 <	X_i	**10** ≤	X_i	≤	132	≤	X_i
	Valid	24 February 2012		range 7'						
Reform	Effective	1st January 2014	6 <	X_i	**10** ≤	X_i	≤	132	≤	X_i
	Valid	30 September 2013		range 4'						
2016	Effective	1st October 2016	6 <	X_i	**50** ≤	X_i	≤	132	≤	X_i
	Valid	29 April 2016		range 43'						

Notes: Effective date refers to the exact date when the law or amendment came into effect. Valid date refers to the date when the law or amendment was announced or enacted. We exclude contracts within this transition period. The numbers are millions of CZK, as follows: 3 CZK ($140k), 6 CZK ($280k), 10 CZK ($470k), 20 CZK ($940k), 50 CZK ($2.36M), 132 CZK ($6.23M).
[a] Indicates the range within which a simplified procedure may be utilized.
[b] Applies to all procurement processes that exceed the simplified procedure threshold but remain below the European threshold for procurement
[c] Denotes the threshold above which contracts are required to be tendered in the European market.

significantly fewer rules. The small-scale limit refers to a threshold for under which public officers have full discretion.

After 2012 and until 2014, the Amendment no. 55/2012 Coll reduced the threshold for simplified procedures to 10 million CZK ($470k). As table 2.1 presents, after 2012 the range for discretion was halved (7' million). Thus this reform is characterized for a decrease in discretion. The 2012 reform, also modified the small-scale procurement to 3 million CZK ($141k). In 2014, the Amendments no. 303/2013 Coll. and no. 341/2013 Coll modified the threshold for small-scale procurement for construction to 6 million CZK ($280k). These amendments kept the simplified procedures threshold unchanged, thus from 2014 to 2106 there was a narrower range for discretionary decisions.

From 2016 public procurement has been governed by the Act no. 134/2016 Coll[4]. This law raised the discretionary threshold to 50 million CZK ($2.36M) while small-contracts value remained unchanged. Therefore, the range for contracts eligible for simplified procedures broadened significantly. The bottom row in table 2.1 indicates a range of 43 million[5]. Similar to Szucs (2024) high discretion area is the range where simplified procedures are allowed. We use the term medium discretion instead of low discretion because contracts under the European threshold are procured in the national market with more likelihood for political favoritism.

[4]To date the Czech Republic is adopting a new regulatory framework Act No. 166/2023. The new regulation is aligned with the main research questions of this paper. One aim is to reduce the administrative burden on both contracting authorities and suppliers. Additionally, the regulation aims to provide more information to the Competition Protection Office for the tender documentation to promote competition (Open Government Partnership, 2023). The new regulatory framework does not aim to modified discretionary thresholds.

[5]In the appendix figure 2.A.1 depicts a visual area presentation of the ranges. For the purpose of this paper we focus on the simplified procedure threshold or the area for high discretion in figure 2.A.1.

2.2.2 Data sources

We use three sources of data. First, we use the (public) administrative records of all Czech procurement contracts for the 2006 to 2021 period. The data have been web scraped, processed, and manually cleaned by a company.[6] This dataset contains the anticipated and final values of procurement contracts, the number of bids, the date of award, the name of the public buyer and the winning firm, and some other few characteristics such as the classification of the main subject of the procurement Common Procurement Vocabulary code (CPV). The most relevant variables from the public procurement dataset are the final price awarded to tenders, the contract anticipated value and the number of bidders that participated in the tendering process.

From the contracts from 2006 to 2021, 74% of contracts published were awarded. To further simplify the analysis, in this paper we exclude contracts awarded under framework agreements or dynamic purchasing systems[7].

Second, we use extensive firm-level data on risks of corruption and political favoritism. This dataset includes data on personal political connections of firms, firms donations political parties, and firms with anonymous owners. All the data come from publicly available sources. To measure personal political connections, we combine data from the company registry in the Czech Republic and the lists of political candidates. We have web scraped the Czech company registry. The company registry contains about 3,200,000 records. Each record represents an appointment term of a (supervisory) board member – her/his name, date of birth, academic title, city of residence, and the beginning and end of the appointment. The dataset of political candidates contains all candidates running for office in any municipal, regional, or central government (the Parliament) elections since 2004. It contains 740,000 records. Note that these are not distinct individuals as many politicians/political candidates have run repeatedly for office at different levels of the government and time. The measure of political connections is then created by matching elections' candidate lists and the lists of board members of all Czech companies mentioned above. Our definition of political connections is based on Baranek and Titl (2024). In Czechia, to become a candidate, the person must be approved within an internal selection procedure. That is often a formal voting process of a committee of high ranked party members. Because becoming a candidate requires support from within the party this means that the person has established connections and networks with party members, therefore in our context a candidate has influential relationships with party members and thus political favoritism is not ex-ante negligible. Hence, our paper reflects the way of functioning and the structure of political parties in the Czech Republic.

The data on political donations come from the annual reports of political parties. We have manually transcribed these reports into Microsoft Excel files. More

[6]The company is called Datlab, s.r.o. and registered in Czechia.

[7]These types of procedures include complex buyer-supplier relationships where multiple tenderers can be involved for the same or multiple contracts. These features restrict tractability of the discretionary process and the mechanisms we aim to study might be different.

recent (after 2016) annual reports were available on the website of the Office for Economic Supervision of Political Parties and Political Movements. We merged these two sources of political donations and the combined dataset contains political donations to all political parties since 1995 from private citizens as well as firms. Our dataset contains in total 11,345 firms' political donations.

Finally, we identify firms who have been anonymously owned. Such firms had so call bearer shares, i.e. the ownership of these companies was transferred by transferring the shares in paper form. This dataset covers registries from 1990 until 2019.

Our data on political connections does not have exact date of when the connection was made and the years covered by political donations are not an exact match with the timeframe of our analysis, which spans from 2010 to 2018. To overcome this limitation, we define a dichotomous variable as indicative of whether a firm in our procurement data has ever been associated with political connections, has ever made donations to political parties, or has ever been anonymously owned. We also consider combinations of these characteristics and include in our analysis firms that have been politically connected and have made donations to political parties, or that have been politically connected and have been anonymous, or that have made donations to political parties and have been anonymous, or that have been politically connected, made donations and anonymously owned.

Thirdly, we use the dataset Orbis from *Bureau van Dijk* to measure the market competitiveness of suppliers. This dataset contains balance sheet information of 1,746,766 Czech firms from 1993 to 2019, although not all firms have data for all the years. We focus mostly on data from 2005 onward because we are interested in characterizing the market competitiveness of the winning firms prior to the contract award. We were able to match up to 38% contracts with Orbis. We compute the results of productivity following the methodology proposed by Gal (2013) with an application of the software package developed by Rovigatti and Mollisi (2018) to estimate total factor productivity (See Appendix 2.A.3).

Our final dataset constitutes a total of 35,731 procurement contracts awarded in the construction sector from 2010 to 2018. Our unit of analysis is contract level data. A contract is observed only once, in our case, when a procurement officer or generally a public buyer awards the contract to a winning bidder. The final dataset is the result of a trimming exercise that we do in two dimensions[8]. The first dimension is the range of the anticipated value we study. We keep contracts which anticipated value lies within a range suitable for our analysis, meaning contracts that are around the simplified procedure threshold. Having as a reference table 2.1, we have a lower limit defined by the small-scale threshold and an upper limit defined by the European threshold. For the threshold at 20 million CZK ($940k) that was the rule from 2006 until 2012, we use contracts that have surpassed the small contracts value threshold (i.e. 6 million CZK ($280k)). As mentioned, contracts below 6 million are subject to other laws. We also restrict the analysis to contracts

[8]We follow similar ideas used for this kind of administrative data previously done by Palguta and Pertold (2017) and Coviello et al. (2018).

up to no more than 40 million.

After the reform in 2012 until 2014, the limit for small contracts was 3 million, therefore the contracts of interest start from 3 million CZK ($141k). For the threshold at 10 million CZK ($470k) ruling from 2012 until 2016, we keep the upper limit of 40 million. From 2014 until 2016 the limit for small contracts was increased to 6 million and therefore contracts of our interest must be higher than 6 million ($270k). However, from 2014 to 2016 we lift the upper limit to allow for the comparison with the threshold after the 2016 reform. For the threshold at 50 million CZK ($2.36M) ruling from 2016 onward, we have less restrictions since the limit for small contracts remained 6 million ($270k). For the upper limit our restriction is the EU threshold which is higher than twice the threshold of 50 million and therefore has a similar condition we have established for the previous upper limits.

The second dimension is time defined by the contract awarding date. While it is tempting to include all contract data from 2006 until 2021, the reality is that for our goals of measuring manipulation through bunching and the effects of the discretionary reforms, more data could include more sources of biases and miss-measurements. For instance, we consider the reform implemented in 2012. We want to study this reform during the time its ruling remained on hold, which dates 2014. Hence, we include contracts awarded during the two years post-reform. To make an accurate comparison we start from two years before the introduction of the amendments in 2012. Thus, we restrict the analysis to contracts awarded from 2010 to 2012 under the 20 million ($940k) regime and contracts from 2012 to 2014 under the 10 million ($470k) regime. This is important because most of our estimations depend on densities around the described cutoffs. Carelessly including more data to one side could alter the estimations and bias the results. We also exclude contracts from 2018 onward for similar reasons. Thus, we are able to study contracts two years before and after each reform.

2.2.3 Evidence of bunching at each discretionary threshold

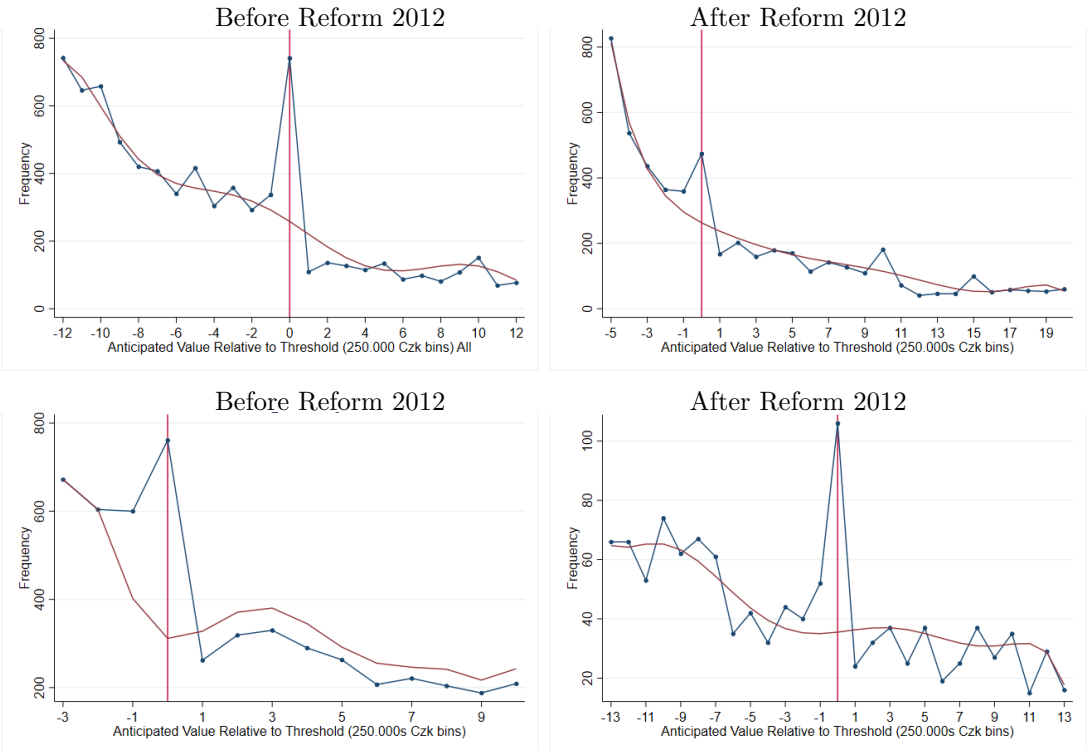
To provide further description of our setting and data we illustrate how the excess mass below the thresholds takes place in the Czech Republic. Here we follow the methodology proposed by Chetty et al. (2011)[9]. This benchmark methodology estimates the excess of mass by subtracting from the empirical distribution a counterfactual distribution where the threshold does not exist. Therefore, the excess of mass computed by Chetty et al. (2011) is static and must be calculated for each period separately. Figure 2.1 present the graphical result of this benchmark bunching estimation. These figures depict a spike below the thresholds during each period of our analysis.

Table 2.2 quantifies the excess of mass of the spikes in figure 2.1 for the reforms of 2012 and 2016. The estimations suggest that before the reform in 2012, when the threshold was defined at 20 million CZK ($940k), the bunching mass was $\hat{b} = 1.43$. This result can be interpreted as an excess of 143 percent of the average height

[9]We implement the Stata program `bunch count` available from replication package available in `rajchetty.com`

of the counterfactual distribution. After the reform in 2012 (second row in table 2.2), we find an excess of mass of 76 percent for all contracts below the post-reform threshold of 10 million CZK ($470k). This result suggests that the manipulation of contracts after restricting discretion reduced. Regarding the reform in 2016, which allowed more discretion, the estimations show an increase of bunching for all contracts (column All contracts). Table 2.2 shows before the reform in 2016 (third row) when the threshold was at 10 million ($470k) the excess mass was about 168 percent; and after the reform (fourth row), when the threshold was at 50 million ($2.36M), and the excess mass increased to 210 percent. This result suggests a direct relationship between discretion and contract value manipulation. In other words, when discretion thresholds are high, for instance before 2012 and after 2016, bunching is big; and when discretion thresholds are low, after 2012 and before 2016, bunching is low.

Figure 2.1 Bunching mass estimation around re-centered anticipated contract value



Notes: The graphs are produced following Chetty et al. (2011). The red lines represent the counterfactual distribution. The blue lines display the empirical distributions of the anticipated values. The four graphs are: top left the distribution before the 2012 reform, top right the distribution after the 2012 reform, bottom left the distribution before the 2016 reform, and bottom right the distribution after the 2016 reform. The thresholds are presented in normalized terms. The absolute value of the thresholds are as follows: Before 2012- 20 CZK ($940k); After 2012 - 10 CZK ($470k); Before 2016 - 10 CZK ($470k); After 2016 - 50 CZK ($2.36M). These graphs consider all contracts.

Table 2.2 shows the excess mass of contracts awarded to firms without political connections in column 2, firms with personal political connections in column 3, firms with history of political donations in column 4, and firms with history of

anonymous ownership in column 5.

Table 2.2 Regression estimates of excess mass below the thresholds

oprule		All contracts (1)	Non connected (2)	Connected (3)	Donated (4)	Anonymous (5)
Before 2012	$\hat{b}$	1.434 ***	1.436 ***	1.843 *	1.783 **	1.489 ***
	$se_{\hat{b}}$	(0.395)	(0.524)	(0.971)	(0.851)	(0.553)
After 2012	$\hat{b}$	0.772 **	0.558 *	0.683	0.784	0.352
	$\hat{se}_b$	(0.358)	(0.313)	(0.609)	(0.498)	(0.715)
Before 2016	$\hat{b}$	1.675 ***	2.370 **	1.996 ***	1.104	0.395
	$\hat{se}_b$	(0.348)	(0.996)	(0.392)	(1.229)	(1.993)
After 2016	$\hat{b}$	2.109 ***	3.996 **	0.200	3.106	-0.405
	$\hat{se}_b$	(0.566)	(1.526)	(0.927)	(2.023)	(0.756)

Notes: The estimations are computed following Chetty et al. (2011). The excess of mass of contracts is quantified by $\hat{b}$ in parentheses are the standard errors. The first column considers all contracts, the second column includes contracts awarded to firms with political connections, the third column includes contracts awarded to firms with records of donations to political campaigns, and the fourth column considers contracts awarded to firms with history of anonymous ownership. See complete regression tables 2.B.1 and 2.B.2 in the appendix.

The results indicate that after the reform in 2012, contractors with political connections were associated with less bunching. Regarding the reform in 2016, the estimations in column 4 for firms flagged for donations show an excess mass for donating firms of 110 percent before 2016 and 310 percent after the 2016 reform. This increase after 2016 is statistically significant and higher than that of personally connected firms or anonymous firms. These results over estimate the reforms effects on bunching, because this benchmark estimation does not take into account the previous existence of thresholds. Even though this benchmark methodology falls short for exploring which mechanisms explain this shift, we speculate that connected firms are less responsive to contracts with high anticipated value whereas donating firms could be more attentive to those expensive contracts as those could be seen as the return of the donation investment.

2.2.4 Potential outcomes and procurement reforms objectives

We are interested in studying the effects of changes in the discretionary thresholds. A primary outcome variable is contract value manipulation. As previously studied by Palguta and Pertold (2017) in the Czech Republic and documented in other cases (Szucs, 2024; Carril, 2021), the main result from restricting discretion with a limiting threshold value is the clustering of contracts below such cutoff. Such bunching, or excess mass below the threshold indicates demand for discretion, a concept introduced by Szucs (2024). This demand for discretion has also been interpreted as a preference by officers for discretionary procedures, or simplified contracting (Carril, 2021). From the broader theoretical perspective illustrated by Bandiera et al. (2009), the demand for discretion is a revealed preference for con-

tracting with lower costs. In our case, this means contracting below the threshold and it indicates positive gains derived by the procurement officers. Those gains come from (i) corrupt motives and bribes, (ii) workload and oversight avoidance, (iii) less costly screening of high quality suppliers.

Secondly, in line with related literature, we study the effects of the reforms on the cost-efficiency outcome of ratio between the final price of the contract and the anticipated contract value. We later use the term normalized price. This measure has been used in the literature to proxy different aspects of the procurement process, such as competition (Szucs, 2024), winning rebate (Coviello et al., 2018), and cost-overrun Carril (2021).

Thirdly, we define measures of market competition. We study competition in public procurement contracts measured as the number of bidders participating for a procurement contract. In public procurement the concept of competition has usually been measured by the number of bids that each contract receives. The number of bids is a good proxy for competition because it indicates how many firms were interested in providing the good or service required by the public authority. Furthermore, the number of bidders is a direct market competition outcome affected by the threshold that allows for a simplified procedure because the regulatory framework sets a lower bound of a minimum of five invited bidders. Competition can be affected by the existence of corruption between the public authority and potential firms, for example, the existence of bribes (Celentani and Ganuza, 2002; Compte et al., 2005). The potential existence of bribes is an unobservable variable for us. We use the characterization of political favoritism of winning firms as a channel through which discretion can be corruptible and affect the competition of the market. Another notion of competition has to do with how competitive the participation and awarding processes are. The competitiveness exerted by potential suppliers for a procurement contract has to do with how aggressively the participating firms bid against each other, whether they use previous knowledge to undercut competitors bids. Competition can also be constrained by the existence of cartels or bid rigging agreements. One can imagine a case of a corrupt public officer receiving bribes from one firm or we could have a case of a corrupt public officer receiving bribes from the representative of a cartel. We do not observe firms bidding behavior and studying competitive bidding goes beyond of our aim and it is material for future research.

In addition, we define a measure of competitiveness as the productivity of the year before of winning a contract. This lagged productivity indicates whether at the moment of wining a contract the firm was above or below the average productivity of its economic sector. We are interested in lagged productivity, one year prior to a winning award because this allows to test whether a procurement officer who awards a contract to a firm in a given year, might have been informed their decision by observing the productivity level of the firm in the year before. This enables us to test two hypotheses. First, an efficient, rational and aligned procurer who observes the productivity score would prefer to allocate the contract to a firm with productivity above the market average in the year of reference. The second hypothesis is that a wasteful and misaligned procurer would prefer or be indifferent to allocate a

contract to a firm with negative productivity or below the market average in the previous year. The existence of favoritism and cronyism could be flagged by low competition and low levels of productivity of the winning firms.

Finally, for favoritism and risks of corruption we classify the winning firms as non-connected to political campaigns and firms with any source of political connections. We subset our data among firms that have connections with politicians, firms that have made donations to political parties, and firms with background anonymous ownership. Due to data limitations, our study offers an alternative approach to understand the role of political connections, donations and anonymous ownership. Rather than calculating the likelihood of procurement contracts awarding conditional on political connection, or on donations to political parties, or on anonymous ownership, we explore how these three sources of political favoritism channel the use or misuse of bureaucratic discretion. The premise is that procurement officers might use their discretion to favor firms with political affiliations, such as those making donations. This favoritism could stem from two possibilities: either political connections serve as a signal of firm's productivity and capacity, or they act as a means of corrupt activities.

2.3 Empirical strategy

Our research design consists of the study of the distribution of contract estimated value and relies on regression discontinuity analysis. We separate our analysis in two. First, we estimate bunching of procurement contracts below discretionary thresholds. Second, we estimate the effects of changes in discretionary thresholds by adapting a regression discontinuity design approach for variables of cost efficiency and market competition.

Regression discontinuity design provides all the elements for a structured empirical strategy. First of all, we want to study the effects of decreasing and increasing discretionary thresholds. Having such defined cutoff values, we can implement an analysis around the discontinuity. We leverage the analysis of the distribution of the running variable, in our case the anticipated or estimated contract value to study the extent to which discretion is misused in the form of contract value manipulation. This first customary step in regression discontinuity is essential in our discussion of the effects of changing thresholds. We devote our first part of analysis to studying contract value manipulation, sorting, and bunching.

In addition, we leverage both the decrease and increase in the threshold for identification by arguing that contracts just below the threshold and before the change versus just above the threshold and after the change are almost the same as contracts just above the threshold and before the change versus just below the threshold and before the change. The only difference being that the contract happened to be awarded under a different law. Furthermore, we implement a fuzzy design because while procurement officers are allowed to use discretionary procedures they are not obliged and thus we have to account for the probability of the procurement buyers of using the simplified procedures.

Our estimation strategy consists in studying separately the reform in 2012 and the reform in 2016. Doing so facilitates the comparison between the change that narrowed discretion and the change that broadened the range for discretion. This separation also isolates other legal changes that took place in between 2012 and 2016. While the simplified threshold remained the same, the small-scale threshold changed in 2014. Thus, the range of contracts eligible for simplified procedure differs between the period in 2012-2014 to the period 2014 to 2016.

First, discretionary thresholds are defined as cutoff values of thresholds in public procurement. These thresholds set a reference amount below which procurement officers are allowed to use a procedure called simplified procedure. This procedure, only available below the threshold amount, entitles procurement officers to preselect potential bidders for a contract. This type of procedure grants autonomy to officers to award procurement contracts. Second, we define changes in such discretionary thresholds as the decrease and the increase of the cutoff values. We ask what the procurement market outcomes are if there is less discretion and what would those outcomes be if there is more discretion granted to procurement officers.

Regression discontinuity focuses on treatment that is assigned at a certain cutoff value. The idea behind implementing a regression discontinuity in our setting is to compare contracts just on either side of the discretionary threshold. Without this legal cutoff the contracts would likely be similar. Therefore, if the contracts are different, we can probably attribute the difference to whatever it is that happens at the discretionary threshold. In our setting, to one side of the cutoff, for instance above the threshold, the treatment condition procurer officers to act under more oversight and must comply with stricter rules. Only to the other side of the cutoff, or below the threshold, procurer officers have discretion and are allowed to use simplified procedures.

In this paper, we study this discontinuous allocation of rules and discretion after the Reform in 2012 that lowered discretionary thresholds and after the reform in 2016 that increased the discretionary threshold. Several papers have relied on this identification strategy (Coviello et al., 2018, 2022; Szucs, 2024), however the main issue of this approach is that procurement officers tend to sort contracts below the threshold. Therefore identification is at stake and estimations might be biased. To address the issue of contracts sorting below the threshold we introduce an additional cutoff point determined by the date when the new regulation was implemented. This temporal cutoff point is an external shock that creates an additional treatment rule. The idea here is to compare contracts just on either side of the temporal cutoff. Without the regulatory cutoff the contracts would likely be similar. If the contracts are different, we can probably attribute the difference to whatever it is that happens at the temporal cutoff. We combine these two ideas. We have two running variables, the anticipated contract value and the awarding date. We also have two cutoff points, the discretionary threshold and the date of the reform implementation.

2.3.1 Distribution and sorting around reforms implementation

In the core of our paper we motivate the analysis of contract value manipulation as a source of discretion misuse by procurement officers. We observed contract value manipulation from studying the distribution of the anticipated value of contracts and identifying the level of sorting or clustering around the discretionary thresholds by estimating bunching regressions. In other words, we study the monetary dimension of the running variable of our empirical strategy.

In the appendix we present the analysis of the temporal dimension of our running variable (see 2.B.1). We analyze sorting around the time when the Reforms of 2012 and 2016 were announced and enforced and present the results of the McCrary test on the density distributions. Overall, we do not find an excessive number of contracts awarded before or after key dates such as announcement and implementation dates. We exclude contracts awarded within the transition window from one legislation to the other. In other words, contracts awarded between the date of announcement and the date of effectiveness are excluded. In principle such contracts are covered by current valid regulations, but we excluded those to avoid ambiguities. We also exclude contracts that were designed under one legislative act and awarded under another one.

2.3.2 Bunching estimation: contract value manipulation, discretion and favoritism

We build upon the work of Palguta and Pertold (2017) and expand their method by proposing a new specification suitable for our context. Our method consists in studying the density distribution of the contracts anticipated value pre-reform and post-reform. First, we define a set of evenly spaced intervals or bins. Second, we estimate the frequency and density of contracts in each bin pre-reform and post-reform for 2012 and 2016. We use the pre-reform distribution as the counterfactual distribution that we would observe had the discretionary threshold been kept unchanged.

Our method is similar to Chetty et al. (2011), (Kleven and Waseem, 2013), Palguta and Pertold (2017) and Carril (2021) for estimating bunching, but the counterfactual assumption in our approach is fundamentally different from their assumptions. Chetty et al. (2011) assumes that the counterfactual scenario is one where no bunching exists. He assumes a smooth counterfactual distribution. We cannot argue for a smooth counterfactual distribution because we know that without the reform the distribution of the anticipated value was not smooth, there was bunching below the discretionary threshold. Palguta and Pertold (2017) use as counterfactual distribution the pre-reform distribution. Their counterfactual distribution is smooth because there was no threshold and no bunching. We cannot rely on the assumption of a smooth pre-reform distribution because the discretionary threshold was already established and there was bunching before the reforms. Instead, we use a counterfactual scenario in which the threshold would have stayed the same after the reform. Our counterfactual scenario includes the fact that the

presence of bunching affects the counterfactual distribution.

We estimate the main specification defined in equation 2.3.1. Here, our dependent variable NC_{jt} is the total number of contracts awarded within an interval or bin j at year t. Intervals are defined as one every 1 million CZK. This means that the contract value distribution is divided into intervals of one million units. Thus $NC_{jt} = \sum_j = X_{it}$. The dependent variables are interaction terms between the area formed by the bins below the pre-reform cutoff value or discretionary threshold AC^- and the area formed by the bins below the post-reform cutoff value AC^+. Thus, $AC^- = \sum_j^{C-} X_{it}$ and $AC^+ = \sum_j^{C+} X_{it}$. We identify with dummy variables Bin_j bins below the pre-reform or old threshold (AC^-) and below the post-reform or new threshold (AC^+). We include a dummy variable for years after the reforms or *Legal Modifications* $\mathbb{1}[t > Y_{LM}]$. The parameters of interest are γ and η. First, γ_i indicates the mass of contracts in the excluded area corresponding to the old threshold before the reform. Therefore, γ_i quantifies the mass of contracts after the reform but with the pre-reform threshold as reference. Second, η_i quantifies the mass of contracts after the reform at the post-reform threshold; therefore testing for adjustment of bunching.

$$NC_{jt} = \alpha_j + \alpha_t + \sum_{i=-AC_Y^-}^{C-} \gamma_i \mathbb{1}[Bin_j = i]\mathbb{1}[t > Y_{LM}] + \sum_{i=-AC_Y^+}^{C+} \eta_i \mathbb{1}[Bin_j = i]\mathbb{1}[t > Y_{LM}] + \varepsilon_{jt} \tag{2.3.1}$$

We estimate the equation 2.3.1 twice, one for the reform in 2012 and another for the reform in 2016. In 2.3.1, the two reforms are mutually exclusive, $Y_{LM=2012}$ cannot be estimated at the same time when $Y_{LM=2016}$. The regression for 2012 estimates evidence of the effects of restricting discretion with a lower discretionary threshold. The regression for 2016 estimates effects of allowing for more discretion when the thresholds were raised. These two specifications are defined in terms of Y_{LM}; where Y_{2012} represents the reform in 2012 and Y_{2016} represents the reform in 2016.

To illustrate, we use the reform of 2012 as an example. In equation 2.3.1, Y_{LM} would be Y_{2012}. In that case, $-AC_{2012}^-$ denotes the width of the excluded region below the cutoff threshold pre-reform in 2012. Similarly, $-AC_{2012}^+$ represents the width of the excluded region below the cutoff threshold post-reform of 2012. In this case, excluding a region of 1 million below the cutoff threshold before the reform in 2012 means that $AC_{2012}^- = 19$ because that is the first bin of 1 million interval under the 20 million cutoff, the discretionary threshold pre-reform 2012. Thus $i = 19$ and the indicator variable would be equal to one when $\mathbb{1}[Bin_j = 19]$. Similarly, excluding a region of 1 million below the cutoff after the reform in 2012 means that $i = -AC_{2012}^+ = 9$ because that is the first bin under the 10 million cutoff, the discretionary threshold after 2012. In principle, there should not be major differences in the sense that the number of contracts with an anticipated value of 20 million would be similar before and after the reform. Similarly, the number of

contracts with an anticipated value of 10 million would be similar before and after the reform. This example for 2012 applies the same for the reform of 2016 where the regions of bins below the limits are defined with respect to the discretionary thresholds of the 2016 period.

We formulate three hypotheses to test with our specification in equation 2.3.1.

- We test $\gamma_i = 0$ and $\eta_i = 0$. In other words, we consider there is no clustering of contracts below a region defined by an i number of bins below the pre-reform threshold $i = -AC_Y^-$ nor clustering of contracts below a region defined by an i number of bins below the post-reform threshold $i = -AC_Y^+$.

- We test for bunching after the reform within a region defined by an i number of bins below the post-reform threshold $i = -AC_Y^+$, thus, we test $\eta_i = 0$.

- We test for bunching after the reform within a region defined by an i number of bins below the pre-reform threshold $i = -AC_Y^-$, thus, $\gamma_i = 0$.

Insights of bunching theory Kleven and Waseem (2013) suggest that a new threshold would likely imply adjustments in where the clustering of contracts happens. In the post reform context, we expect a decrease in the clustering of contracts below the pre-reform cutoff and an increase in contracts below the post-reform threshold. For example, in the context of a new reform the pre-reform cutoff value losses the bunching incentives. If before the reform there was clustering of contracts at the pre-reform cutoff, there are no reasons to believe that after the reform the old-pre-reform cutoff has meaning or carries bunching incentives. In contrast, the new-post-reform threshold creates new incentives to clustering contracts at the new established cutoff. Therefore, we suspect and test for bunching adjustment. These hypotheses follow the argument that bunching is evidence of the public buyers' demand for discretion and the argument that such demand won't disseminate into a smooth distribution when facing new constraints Szucs (2024) Carril (2021). Our design provides evidence of what happens under low discretion thresholds in reform 2012 and evidence of high discretion thresholds for reform 2016.

Apart from estimating equation 2.3.1 for all contracts in our sample, we estimate it for contracts awarded to politically connected firms, firms with history of political donations, firms with anonymous ownership records, and the possible combination of these favoritism characteristics.

2.3.3 Discontinuities estimation: restricting discretion, granting discretion and favoritism

Changes in bureaucratic discretion can not only cause adjustments in bunching area, but can also cause changes in market competition outcomes. To study that plausible causation path we quantify the effects of the discontinuities created by the discretionary thresholds after the reforms, the first one that limited the scope of discretion and the second one that enabled broader discretion. Nevertheless, quantifying the effect of public procurement reforms on procurement outcomes faces several identification challenges (Szucs, 2024) because of selection biases coming from

different sources. First, the choice of awarding procedure is not chosen randomly. The procurement authority decides whether a simplified, open or other procurement procedure is selected. Second, the nature of the public contract, its economic sector, and its value, all define the level of discretion. For instance, complex construction contracts are associated with more discretion. Third, the political and institutional context also mediates in how the procurement is carried out, whether there is prevalence of favoritism and high risks of corruption.

We exploit the policy reforms, the modification changes in public procurement laws of 2012 and 2016 in the Czech Republic, as sources of exogenous variation. Particularly, the policy reform as an Intention to Treat and the simplified procedure as the compliance variable, which allows us to use the Regression Discontinuity Design framework. Our identification strategy relies on concepts from Regression Discontinuity Design models. Particularly, we use two sources of discontinuity or two variables to define our treatment and control group. We use the arbitrary discretionary threshold and the date of the reform as two conditions to define treatment and control. We define as the treatment group contracts above the post-reform thresholds where regulations restrict discretion by not allowing the use of simplified procedures. The control group consists contracts below the old thresholds where the simplified procedures are allowed and the reforms have not affected the procurement process. This identification strategy follows the principles of multiple scores and multiple cutoff designs discussed by Cattaneo et al. (2016), Cattaneo et al. (2020), and recently used by Gori et al. (2023) in the public procurement context.

Our approach makes use of the fact that we can pool the anticipated value dimension when considering the relative contract value with respect to the thresholds by recentering the anticipated value of contracts around the discretionary thresholds. We first recenter the anticipated contract value before and after the each reform. We pool the time dimension, and restrict our analysis to contracts above the new threshold and contracts below the old threshold. In the appendix we provide visual representation of the pooled design for the regression discontinuity estimation (see figure 2.B.2)

To implement our approach we rely on Multi-Score regression discontinuity design assumptions (Papay et al., 2011; Wong et al., 2013; Bertanha, 2020; Cattaneo et al., 2020). First, the two forcing variables, contract value and date of award are continuously distributed around the cutoffs. The anticipated value of a contract is continuously distributed around the new threshold for simplified procedures. The date of award of a contract is continuously distributed around the reform date. Second, the probability of being treated changes discontinuously at the simplified procedure threshold and at the reform date. Third, in the absence of a reform and a new discretionary threshold the expected outcome changes continuously around the threshold. Fourth, there is a cutoff bias but this is a constant bias across the cutoff points. Fifth, "the pooled RDD estimand is a weighted average of cutoff specific RDD treatment effects for subpopulation facing different cutoffs" (Cattaneo et al., 2016). Sixth, for a Fuzzy design one-sided noncompliance shall be assumed (Cattaneo et al., 2016). We can make the one-sided noncompliance assumption,

as we know that contracts above the threshold comply deterministically to the regulation and do not implement simplified procedures to award contracts; whereas contracts below the threshold may or may not award contracts with simplified procedures. Under these assumptions our estimation design identifies effects around the thresholds and at the moment of the change. Furthermore, for a Fuzzy Design the estimations identify local changes for compliers of the simplified procedure.

The economic intuition of our RDD approach is that estimates are obtained by comparing contracts, which in terms of normalized value are above the simplified procedure threshold and awarded after the reforms (either 2012 or 2016) versus contracts with normalized value below the simplified procedure threshold and awarded before the reform. These two groups of contracts are awarded under two distinct levels of discretion, in other words, the treatment group corresponds to contracts upon which more strict rules operate and are excluded of using simplified procedures. The control group does not follow strict rules and is observed before the reforms. Moreover, the separation between the reform of 2012 and the reform of 2016 allows us to draw conclusions from the institutional context where the scope of discretion got constrained by law and an institutional context where the law enabled broader discretion in public procurement.

We define our treatment in terms of two dimensions, namely contract value and date of award. Our design relies on the assumption that a contract belonging to treatment and control group is random upon two running variables, one for each dimension. First, in terms of the running variable time, the source of randomness comes from the fact that the reforms constitute external shocks. We argue that a contract awarded before or after either of the reforms is randomly defined by this institutional shock. Although there could be some anticipation effects, we do not find significant bunching before the reforms. One argument is that the lifetime of a procurement contract in Czechia is on average a year (Baranek and Titl, 2024), and even though procurers might want to arbitrage in time, there are capacity limits and legislative time limits that restrict them to achieve that. Additionally, in terms of the second running variable we rely on the normalized contract value is continuously distributed around the "new threshold" at the time when the reform becomes effective. This means that even though there is sorting in the before the reform threshold this sorting is not happening yet in the after the reform threshold.

We argue that discretion is an inverse function of rules. This is key because discretion is not directly observed, but rules are. Hence, we define discretion as the inverse function of rules as $D = f^{-1}(X_i)$ provided that $R = f(X_i)$; where the rule is that above the threshold there is strict compliance of non using simplified procedure. Regression Discontinuity Designs allows us to relate the anticipated contract value with rules as in equation 2.3.2. The equation indicates that certain rules apply $R_i = 1$ when the anticipated contract value X_i is greater or equal than a given threshold C_Y

$$R_i = \begin{cases} 1 \text{ if } X_i \geq C_Y \\ 0 \text{ if } X_i < C_Y \end{cases} \tag{2.3.2}$$

Equation 2.3.2 is also known as a *sharp* Regression Discontinuity Design. This is because the probability that a contract is assigned a certain set of rules happens deterministically after the defined threshold. In the Czech public procurement context the deterministic change is true for tenders surpassing the limit for small-scale, and for European tenders (see 2.1). Above the small-scale there is no other option than to comply with the national procurement law. For the European threshold there is no other option than to comply with the European public procurement guidelines. However, for contracts with an estimated value that fall under the simplified procedure threshold, the assignment to rules is not deterministically, but probabilistically related to the threshold-crossing rule. This happens because contracting authorities are not obliged to use a simplified procedure. We represent this in equation 2.3.3:

$$P(R_i|X_i) = \begin{cases} g_1(X_i) \text{ if } X_i \geq C_Y \\ g_0(X_i) \text{ if } X_i < C_Y \end{cases} \tag{2.3.3}$$

We implement a Fuzzy RD design where the probability of taking the treatment or awarding contracts with simplified procedures changes at C_Y. As specified in equation 2.3.4 we regress our outcomes of interest Y_i over D_i that refers to discretion or its inverse rules. We also include fixed effects for type of buyer α_B, whether the contract was funded with European funds α_{EUF} and a dummy for contracts whose exact value is the same as the threshold to control for mismeasurements α_{mse}.

$$Y_i = \alpha_B + \alpha_{EUF} + \alpha_{mse} + \beta(X_i - C_Y) + \lambda D_i(X_i - C_Y) + \delta D_i + \varepsilon_i \tag{2.3.4}$$

As instrument we use the uptake of simplified procedures, Z_i and its interaction with the cutoff $Z_i(X_i - C_Y)$. Such first stage regression is described in equation 2.3.5

$$D_i = \alpha_B + \alpha_{EUF} + \alpha_{mse} + \beta(X_i - C_Y) + \lambda Z_i(X_i - C_Y) + \delta Z_i + \mu_i \tag{2.3.5}$$

The effect of interest is then estimated by $\hat{\tau}_{FRD} = \frac{\delta_{Y_i}}{\delta_{D_i}}$ This design allows us to compare the outcomes of contracts awarded under high discretion versus a medium discretion regime. Moreover we are able to setup such comparison for the case when discretion was constrained by a low value discretionary threshold and for the case when discretion was granted with a higher discretionary threshold value.

We estimate equations 2.3.4 and 2.3.5 for three outcome variables the normalized price, or price over cost ratio, the number of bids, and our measure of competitiveness.

We subset our sample to estimate the effects for all contracts, for contracts awarded to firms with personal political connections, firms with financial political connection, and firms with anonymous ownership records. We also test the combination of these favoritism characteristics.

2.4 Results

This section describes the main results of our empirical strategy and contains two parts. First, we show the results of our econometric specification for estimating bunching when discretionary thresholds change. The estimations also include contract value manipulation explained by firms with a higher probability of engaging in corrupt practices. The second part presents the estimation results of fuzzy regression discontinuity design for cost efficiency and market competition variables.

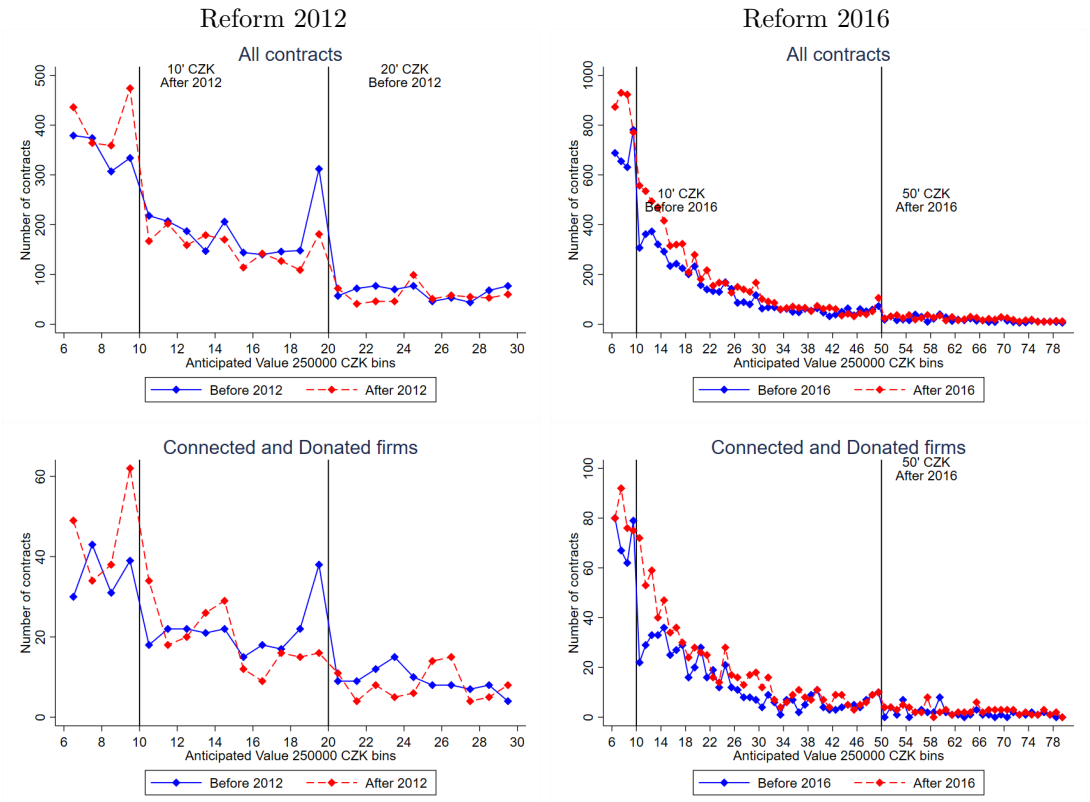
2.4.1 Contract value manipulation and political favoritism

Here we focus on the results of the specification we introduced in equation 2.3.1, where we relax the assumption of a smooth counterfactual density distribution. This assumption means that our method accounts for a pre-reform distribution characterized by bunching. Figures 2.2 illustrate this assumption. The anticipated contract value for the counterfactual is illustrated with blue solid lines by the distribution of the value before the reforms. The red dashed lines are the distributions of anticipated values after the reforms. Figures in 2.2 focus on all contracts (top figures) and contracts awarded to firms with political connection or connected firms; and firms that have donated to political parties. Companion figures are included in the appendix in 2.B.1. The distributions of the anticipated contract values have notorious spikes below the thresholds for all contracts and for contracts awarded to firms with political connections. The existence of spikes also for contracts awarded to politically connected firms suggests that favoritism and the risk of corruption are important channels of contract value manipulation.

The most salient result of these distributions is the change in the position of the spikes from the pre-reform threshold before the reforms to the post-reform threshold after the reforms. These moving of the bunching or adjustment of the contract value manipulation is more visible for the reform in 2012 (graphs on the left) than for the reform in 2016 (graphs on the right). To estimate the spikes and their adjustments after the discretionary reforms, we fit a Poisson Fixed Effects regression model to equation 2.3.1. For the estimations, we group contracts into sorted accumulated bins of 1 million CZK ($47k) per year. Therefore, our dependent variable NC_{jt} is a counting variable that indicates the number of contracts in each bin. Tables 2.3 and 2.4 present the estimates for reform 2012 and reform 2016 respectively.

Table 2.3 presents the estimates of excess mass below discretionary threshold for the Reform of 2012 when the legal modification restricted discretion. After the reform in 2012 there are no incentives to manipulate contract values to be at 20 million ($940k), the old threshold. We observe a reallocation of the misuse of discretionary power as contract manipulation moves from 20 million cutoff to 10 million cutoff (from $940k to $470k). In table 2.3, $\hat{\gamma}_1$ represents this expected result. The value of $\hat{\gamma}_1 = -0.467$ for column (1) All contracts estimates the difference in the number of contracts at the bin below the 20 million threshold ($940k) after the reform in 2012 versus before the reform, thus it is the pre-reform threshold studied post reform.

Figure 2.2 Distribution of contract anticipated value by reform



Notes: The graphs represent the distribution of number of contracts of bins of 1 million CZK ($47k). The anticipated contract value for the counterfactual is illustrated with blue solid lines by the distribution of the value before the reforms. The red dashed lines are the distributions of anticipated values after the reforms. The black solid lines represent the discretionary thresholds. Graphs on the top plot the whole sample. Graphs on the bottom plot contracts awarded to firms with political connections and who have made donations to political campaigns.

The negative sign of $\hat{\gamma}_1$ indicates that after the 2012 reform there was a decrease in the amount of contracts located in the area of 1 million below the threshold. In other words, after the reform constrained the bureaucratic autonomy less contracts had anticipated values of 19 million. This result is expected, because after the reform in 2012 there are no incentives to manipulate contract values to be 20 million ($940k). Similarly, there were significantly less contracts with anticipated value of 18 million, as $\hat{\gamma}_2 = -0.275$. The sign and magnitude of $\hat{\gamma}_1$ and $\hat{\gamma}_2$ reject the null hypothesis of $\gamma = 0$, thus we confirm that there is decreased bunching after the reform at the pre-reform threshold. $\hat{\eta}_1$ represents the interval area or bin below the post-reform threshold of 10 million ($470k) after the reform in 2012. And $\hat{\eta}_2$ represents the interval area for two bins below the post-reform cutoff. The positive sign for the column (1), All contracts, indicates an increase in the number of contracts clustered just below the post-reform threshold. This result rejects the hypothesis of $\eta = 0$ and confirms that there is an increase of bunching at the post-reform threshold.

Table 2.3 Estimates of excess mass below discretionary threshold. Reform 2012 - Restricted bureaucratic discretion.

	All contracts	Non connected	Connected	Donated	Anonymous	Connected & Donated	Connected & Anonymous	Donated & Anonymous	Connected & Donated & Anonymous
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\hat{\gamma}_1$	-0.467*	-0.556**	-0.386	-0.003	-0.497***	-0.939**	0.028	-0.898***	-0.198
	(0.253)	(0.265)	(0.333)	(0.172)	(0.149)	(0.436)	(0.478)	(0.228)	(0.434)
$\hat{\gamma}_2$	-0.275***	-0.306**	0.041	-0.462***	0.076	-0.465**	-0.738**	-0.148	-0.619
	(0.054)	(0.138)	(0.317)	(0.119)	(0.242)	(0.211)	(0.375)	(0.343)	(0.474)
$\hat{\eta}_1$	0.447***	0.475***	0.585*	0.720***	-0.004	0.419***	0.194*	0.070	0.131
	(0.070)	(0.107)	(0.302)	(0.083)	(0.179)	(0.078)	(0.112)	(0.234)	(0.210)
$\hat{\eta}_2$	0.218***	0.142	0.565**	0.243*	0.591***	0.277***	0.034	0.508*	0.288
	(0.072)	(0.097)	(0.233)	(0.135)	(0.185)	(0.087)	(0.113)	(0.292)	(0.223)
Constant	4.484***	3.923***	1.995***	2.785***	1.579***	2.185***	1.907***	1.848***	1.369***
	(0.017)	(0.020)	(0.034)	(0.027)	(0.034)	(0.030)	(0.029)	(0.030)	(0.038)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bins FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	293	293	250	290	254	290	273	281	252
ll	-888.7	-671.1	-356.9	-575.9	-373.5	-490.7	-454.6	-459.5	-340.3

Standard errors in parentheses

* p¡0.1, ** p¡0.05, *** p¡0.01

Notes: The total number of observations are number of bins 1 million CZK ($47k) per year from 2010 to 2014. Estimation from Poisson Fixed Effects regression. Reported coefficients are not exponential. We interpret coefficients as $(\exp(\gamma - 1)) * 100$ and $(\exp(\eta - 1)) * 100$

We are mostly interested in the columns with the estimations for the subset of contracts awarded to politically connected contractors; these are columns (3) to (9) in Table 3. The excess mass below the threshold for contracts awarded to firms with political connections suggests different mechanisms through which political favoritism and corruption correlate with contract value manipulation. First, firms with personal connections with politicians, column (3), do not have a significant reduction in contracts clustered at the pre-reform threshold; but they do have a significant increase of contracts below the post-reform threshold as shown by $\hat{\eta}_1 = 0.585$ and $\hat{\eta}_2 = 0.565$. Second, firms who have donated to political parties, column (4), have a significant decrease at the pre-reform threshold around the

second bin $\hat{\gamma}_2 = -0.462$. Simultaneously, there's a marked increase of clustering below the post-reform threshold as observed by $\hat{\eta}_1 = 0.720$ and $\hat{\eta}_2 = 0.243$. Thirdly, anonymous firms, column (5), decrease the bunching at the pre-reform threshold $\hat{\gamma}_1 = -0.497$, but have a timid increase at the post-reform threshold. Timid in the sense that the first bin $\hat{\eta}_1 = -0.004$ is small and non significant and the second bin $\hat{\eta}_2 = 0.591$ is significant. Lastly, when exploring the combination of political favoritism factors, we find that firms with personal political connection who have also made donations (column 6) have the biggest shift from the pre-reform threshold to the post-reform threshold, with a decrease in the excess mass in the pre-reform cutoff of $\hat{\gamma}_1 = -0.939$ and an increase in the excess mass at the post-reform cutoff of $\hat{\eta}_1 = 0.419$. This last coefficient indicates that firms with both political connections and donations to politicians had an increase of 52% for contracts awarded at the post-reform threshold of 10 million after the reform in 2012.

Using back of the envelope calculations this percentage means a total of 93 more contracts awarded to firms who made donations to political campaigns in the two year prior to the reform, or the equivalent to 839 million CZK or around 39 million dollars transferred to firms flagged for political favoritism. This evidence is suggestive on how risks of corruption move along with the reallocation of discretion even when bureaucratic discretion is restricted. The results are evidence of the relationship between political connections and manipulation of contract values in procurement. The results suggest that political favoritism in procurement deters the goals of regulations to constrain discretion. The observed bunching adjustments and the adaptability of politically connected contractors shed light on the misuse of bureaucratic discretion via contract value manipulation along with political favoritism.

After the reform in 2016 there are several incentives into play. First, there are no reasons for clustering of contracts at the old threshold of 10 million. Second, the reform expanded the scope to use or misuse discretionary power because simplified procedures are available for larger contracts. Before the reform, simplified procedures could be employed for contracts ranging from 6 million to 10 million. After the reform, this range was increased to contracts between 6 million and 50 million. This expanded range is 44 million, providing more room for discretionary decisions. Third, the reform does not necessarily imply a unavoidable incentive to design and award contracts with higher anticipated values. In other words, the mere existence of more discretion regarding contracts does not mean there will be more money available in the budget to fund additional contracts. The ability of bureaucrats to make choices doesn't automatically translate into having more financial resources to support those choices. And even if there were more resources available, the reform does not ensure that the potential suppliers have the capacity to successfully manage higher-value contracts. The private market would also require an adjustment period to effectively handle higher-valued contracts.

The reallocation of contract value manipulation dissipates when the discretionary change increases from 10 million ($470k) to 50 million ($2.36M). This time the bunching located at the threshold of 10 million decreased significantly after the 2016 reform. The estimates of $\hat{\gamma}_1$ in table 2.4 show that political donors, and

anonymously owned firms decrease the number of contracts just below the 10 million pre-reform threshold. The results from the bins below the post-reform threshold of 50 million ($2.36M), $\hat{\eta}_1$ and $\hat{\eta}_2$ are small and non statistically significant. Compared to the benchmark calculation in the descriptive statistics sections, the results here are more precise because the estimates consider non smoothness in the distribution. After controlling for pre-reform thresholds, and fixed effects for bins and years, there is a non significant clustering of contracts below the post-reform threshold.

Table 2.4 Estimates of excess mass below discretionary threshold. Reform 2016 - Broader bureaucratic discretion.

	All contracts	Non connected	Connected	Donated	Anonymous	Connected & Donated	Connected & Anonymous	Donated & Anonymous	Connected & Donated & Anonymous
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\hat{\gamma}_1$	-0.317***	-0.327***	-0.284	-0.286**	-0.347**	-0.446*	-0.319	-0.306	-0.225
	(0.065)	(0.043)	(0.224)	(0.118)	(0.162)	(0.229)	(0.322)	(0.419)	(0.593)
$\hat{\gamma}_2$	0.107	0.250*	-0.384***	-0.029	-0.219	-0.185	0.034	-0.315*	-0.436***
	(0.096)	(0.143)	(0.146)	(0.175)	(0.213)	(0.129)	(0.156)	(0.182)	(0.163)
$\hat{\eta}_1$	0.203	0.457	0.950	0.171	-0.953***	-0.288	0.475	0.149	-0.134
	(0.280)	(0.333)	(0.732)	(0.338)	(0.345)	(0.459)	(0.336)	(0.335)	(0.554)
$\hat{\eta}_2$	-0.408	-0.614	-0.554***	-0.348	0.705**	-0.317	-1.471***	-1.103	-1.520**
	(0.313)	(0.554)	(0.193)	(0.495)	(0.293)	(0.336)	(0.329)	(0.741)	(0.732)
Constant	4.488***	3.821***	2.219***	2.860***	1.611***	2.264***	2.016***	1.688***	1.329***
	(0.012)	(0.014)	(0.031)	(0.021)	(0.027)	(0.028)	(0.037)	(0.037)	(0.054)
Bins FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	413	413	398	407	403	409	401	404	370
ll	-1,273.2	-1,012.2	-563.4	-789.9	-597.8	-656.0	-650.8	-595.7	-461.5

Standard errors in parentheses
* p¡0.1, ** p¡0.05, *** p¡0.01

Notes: The total number of observations are number of bins 1 million CZK ($47k) per year from 2014 to 2018. Estimation from Poisson Fixed Effects regression. Reported coefficients are not exponential. We interpret coefficients as $(\exp(\gamma - 1)) * 100$ and $(\exp(\eta - 1)) * 100$

The estimations of γ_1 for political donors, anonymously owned firms, and firms both with political connection and link through donations to politicians show a decrease in bunching at the pre-reform threshold. The decreases are -24%, -29% and -35% respectively. A back of the envelop calculation suggests that there are potential savings of 467 million CZK, around 22 million dollars, associated to less contracts awarded to connected and donors firms. These results are also explained by the fact that there are not so many contracts valuated at 50 million ($2.36M), as figures 2.2 show. Thus, bunching below 50 million will be less frequent. Alternatively, the distributions show an increase of contracts with anticipated value between 10 ($470k) and 30 millions CZK ($1.1M) for all contracts and for politically connected firms. We estimate alternative specifications to explore this alternative explanation but the estimations do not show a significant increase. The fact that we find less bunching at the post-reform threshold after the reform in 2016 could suggest that the higher values for discretionary thresholds perform better (Carril, 2021).

2.4.2 Discontinuities in bureaucratic discretion, market outcomes and political favoritism

In this section, we present the results from our identification strategy based on regression discontinuity estimators of multiple assigning variables. We are interested in understanding how the restricting discretion and enabling discretion impact procurement market outcomes and how political favoritism is a mechanism that channels positive and negative incentives. Figures in the appendix 2.B.3 illustrate the discontinuous probability of being treated at the two thresholds. Tables 2.B.8, 2.B.9, 2.B.10, 2.B.11, 2.B.12 and 2.B.13 in the appendix present the results for the three market outcome variables of our interest, these are normalized price or price over cost ratio, number of bids, and lagged productivity. Figures 2.B.9, 2.B.10, and 2.B.10 present visual and suggestive evidence of the discontinuities in the distributions of the outcome variables; and the differences between firms with political connections, firms that have made donations to parties, and firms with previous history of being anonymously owned.

Table 2.5 presents the estimations for each of the market outcome variables for the reform of 2012 of restricted discretion and the reform of 2016 of granted discretion. The columns of $\hat{\tau}_F$ contain the estimated coefficients of Fuzzy estimations where the compliers are the contracts awarded under simplified procedures. Those coefficients represent local average treatment effects for compliers within the bandwidth of the regression discontinuity. As benchmark, we include the sharp estimations in the columns $\hat{\tau}_S$, these are columns (2), (4), (6), (8), (10), and (12). As indicated in the table, each row contains the estimations for the sub samples of political favoritism. For instance, the row for connected coefficients refers to the estimations for contracts awarded to firms with personal political connections. The row labeled as donated presents the coefficients for the sample of contracts awarded to firms that have made donations to parties. The row labeled as anonymous presents the coefficients for the sample of contracts awarded to firms with history of anonymously ownership. The row for non connected refers to the coefficients for the sample of contracts awarded to firms without political connections. Finally, the row all sample refers to all contracts without distinguishing the political favoritism status of the contract.

For the normalized price, or the ratio of final price and anticipated value, the estimations in columns (2) and (3) suggest that contracts treated by the reform of 2012 had lower final prices relative to the initial anticipated value. For Total sample, last row in table 2.5, we find that $\hat{\tau}_F = -0.13$, meaning that contracts with anticipated value above the new threshold and after the reform had a normalized price 0.13 percentual points lower than contracts awarded before the reform and below the old threshold. As a reference, the descriptive statistics in table 2.A.1 show that on average the normalized price before 2012 and below the threshold of 20 million was 0.809 and on average the normalized price after 2012 and above the threshold of 10 million was 0.743. The result of the fuzzy regression discontinuity refers specifically to compliers or contracts that would have been eligible to be awarded with simplified procedures but are not eligible anymore due to the reform. This result

Table 2.5 Regression discontinuity design estimations for normalized price, number of bids and competitiveness

	Normalized Price				Bids				Competitiveness			
	Restricted Discretion 2012		Granted Discretion 2016		Restricted Discretion 2012		Granted Discretion 2016		Restricted Discretion 2012		Granted Discretion 2016	
	τ_F (1)	τ_S (2)	τ_F (3)	τ_S (4)	τ_F (5)	τ_S (6)	τ_F (7)	τ_S (8)	τ_F (9)	τ_S (10)	τ_F (11)	τ_S (12)
Connected	-1.10	-0.26	-0.03	-0.01	7.08	3.32	-1.56	-0.90	0.03	-0.01	0.16	0.09
	(2.05)	(0.09)**	(0.15)	(0.07)	(5.81)	(1.43)*	(2.08)	(1.10)	(0.11)	(0.03)	(0.13)	(0.08)
Donated	-0.19	-0.08	-0.01	-0.00	10.84	3.93	2.79	1.28	-0.23	-0.04	0.03	0.01
	(0.12)	(0.04)*	(0.08)	(0.05)	(4.92)*	(1.20)**	(2.35)	(1.05)	(0.24)	(0.03)	(0.13)	(0.06)
Anonymous	-0.74	-0.11	-0.13	-0.06	-6.90	-2.13	3.65	1.57	0.29	-0.35	-0.20	-0.18
	(1.80)	(0.05)*	(0.18)	(0.08)	(5.84)	(1.21)	(5.58)	(1.99)	(0.10)**	(0.03)**	(0.64)	(0.45)
Non connected	0.10	0.03	-0.05	-0.03	1.70	0.49	4.28	2.63	0.11	0.03	0.13	0.05
	(0.09)	(0.03)	(0.06)	(0.04)	(2.07)	(0.59)	(1.46)**	(0.80)**	(0.12)	(0.03)	(0.10)	(0.04)
All sample	-0.13	-0.04	-0.06	-0.03	2.87	0.81	3.65	2.03	0.13	0.03	0.01	0.00
	(0.06)*	(0.02)*	(0.04)	(0.02)	(1.55)	(0.41)*	(0.92)**	(0.47)**	(0.11)	(0.02)	(0.05)	(0.03)

Standard errors in parentheses

* p¡0.1, ** p¡0.05, *** p¡0.01

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts was European level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshold or date of award.

suggests that when discretion is restricted the final prices of the contracts awarded are more efficient. Moreover, contracts awarded to firms with political connections, firms that have made donations to political parties, and firms with anonymous ownership have lower normalized prices. For the reform in 2016, column (4) shows the effect for compliers or the fuzzy estimation. The sign of the coefficients suggest a lower normalized price for All sample contracts, but also for firms with political connections. The descriptive statistics had shown a slight increase in the normalized price from 0.801 to 0.823. Nonetheless, we have also discussed that the reform in 2016 with a higher discretion threshold carries different mechanisms of potential misallocation of contracts, but also challenges of statistical power. These results suggest that a discretionary reform that grants more discretion does not increase nor decrease final prices. Potentially, in the context of the Czech Republic public procurement there are more benefits from restricting discretion and having more oversight.

In terms of competition observed by the number of bids, we find an increase in the number of bids for compliers. This result is expected because contracts that are not eligible anymore for simplified procedures should follow more competitive tendering process. For the reform of 2012, column (6), we don't find statistical significance of the increase in the numbers bids. The only exception is the estimation for contracts awarded to firms that have made donations to political parties. The coefficient of $\hat{\tau}_F = 10.84$ suggests that there was more competition when political donors participated in tendering process. This result aligns the discontinuities observed in figures 2.B.10. For the reform of 2016, column (8), the increase in the discretionary threshold leads to a significant higher number of offers for All sam-

ple and Non connected firms. Firms that have made donations to politicians and anonymously owned face an increase in competition, but the higher number of bids is not significant. We find that firms with personal political connections compete with fewer bids. Regarding competition we speculate that political favoritism is a channel that affects the number of bids in tendering process. For instance, contracts awarded below the discretionary threshold compete against fewer bidders as shown in figure 2.B.10. After the reforms and above the thresholds, there is an increase in the number of bidders participating in the tenders. This is expected because, above the discretionary threshold, public buyers are not allowed to use simplified procedures. Consequently, these buyers must consider other awarding procedures, such as open tendering auctions where the number of bidders is expected to be higher. As expected from economic theory, when there is more competition in a suitable context, in the procurement context there are more bids per tender, such competition leads to lower prices, which we observe in terms of a lower normalized price.

Our measure of competitiveness shows that after the reform in 2012, winning firms in Total Sample were 0.73 standard deviations above the average productivity of other firms in the same sector. This differs from the case of political donors, who were less competitive. After the reform in 2012, these firms had below-average productivity. Regarding the reform in 2016, the competitiveness remains unchanged for the total sample of contracts. Although these estimations are non-statistically significant, they indicate that after the discretion increase, there is no effect on the level of competitiveness of winning firms.

2.5 Conclusions

This paper examines the effects of changes of discretionary thresholds in public procurement and the role of political connections in the Czech Republic. We implement bunching estimations and regression discontinuity design, and exploit two reforms of the public procurement law. We show that there are adjustments in bunching patterns after the reform of 2012 that restricted discretion and after the reform of 2016 that broadened the range of discretion. Using these two reforms that decreased and subsequently increased the discretionary threshold in public procurement, we find abundant evidence for contract manipulation and some evidence for political favoritism associated with the changes in the discretionary power of procurement officers induced by the reforms. The bunching of contracts is persistent through the reforms, while it is more pronounced when the reform restricted discretion. Furthermore, we find that the final prices of the contracts awarded under restricted discretion are lower after the reform, indicating that strengthening regulations reduces the inefficiency gaps. However, we also find that granting more discretion does not affect the final prices of the contracts awarded, implying that there are no efficiency gains from relaxing regulations.

Our findings can be rationalized by the theoretical models introduced by Bandiera et al. (2009) about procurement officers' autonomy and wasteful spending associ-

ated channels, Bosio et al. (2022) about the trade off between discretion and rules in contexts of high versus low public sector capacity, Szucs (2024) on procurer officers' demand for discretion observed with the existence of discretionary thresholds, and Decarolis et al. (2025) on the two levels of agency in public procurement from congress to procurers and from procurers to firms, and Schoenherr (2019) on political connections. Our paper offers evidence of the effects of changing discretionary thresholds on market competition outcomes in public procurement. We offer insights about the intricate relationship between political favoritism, bureaucratic discretion, and inefficiency in public procurement. First of all, we start our study by focusing on analyzing the distribution of the anticipated value of contracts, or the so-called engineers estimated contract value. Therefore, the contract anticipated value not only gives us information about the running variable for the implementation of the regression discontinuity design and carries the interesting notion of demand for discretion (Szucs, 2024).

Our findings suggest that political favoritism is an important channel through which discretion is exploited. These results shed light on the trade-offs between discretion and political favoritism in public procurement and suggest that there are potentially more benefits from restricting discretion and having more oversight. The paper also highlights the adaptability of politically connected contractors to exploit loopholes in the regulations and misuse discretion via contract value manipulation.

From a policy perspective, these findings are important as they suggest that public officers adjust to the new thresholds, i.e. some of them manipulate contract values also around the new thresholds. The bunching of contracts is associated with overall higher prices of public procurement contracts. Therefore, our results carry implications for improving the transparency and accountability of public procurement processes, especially around thresholds, and reducing the scope for corruption and favoritism. Having thresholds appears to influence the reduction of factors such as favoritism, ultimately leading to improved pricing. However, the specific height of the threshold seems to be of lesser importance. Merely having a threshold in place may be the most important driver for enhancing awareness and accountability. This argues against frequent changes to thresholds, as each alteration incurs transaction costs. It also advocates against setting the threshold too low in countries or organizations with high procurement expertise. After all, if the exact height is of lesser consequence, more discretion can be granted to professionals involved in procurement.

This paper's findings also speak to the role of regulations in public procurement as a facilitator for industry policies (Kattel and Lember, 2010). Regulatory frameworks, particularly discretionary thresholds, have impacts on the interactions between governmental agencies and firms in the private sector. Therefore, the level of discretion granted to procurement officers can channel or block industrial policies. An additional policy implication relates to the European legislative framework and the discretionary thresholds of the member states. Our findings suggest that higher discretionary thresholds may be more beneficial. We agree with Decarolis et al. (2025) on the risks associated with a more uniform framework might impose. European regulatory frameworks should recognize not only the benefits of discre-

tion, but the risks of corruption that are very specific to the political economy of each country.

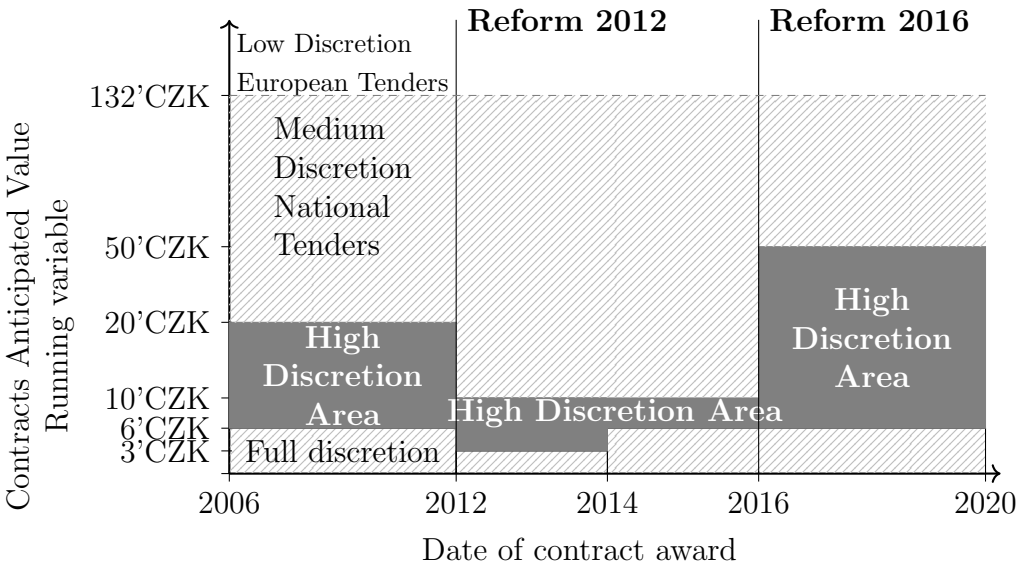
We consider that the implications of our study extend to various aspects of the organization of public procurement markets, from budget planning to the supervision of procurement officers and contractors. For instance, effective budgeting could incorporate strategies that mitigate the risks of contract value manipulation. We also consider that our findings stress the need to align the incentives of procurement officers to broader economic and social objectives. The better we understand officers' incentives, preferences for discretion, and channel of political favoritism the better it would be to facilitate the integration of public procurement into larger policy frameworks that promote fair and open competition in these markets.

Our findings are likely to be valid in the context of countries with similar levels of the economic development, the public sector capacity, and similar legal environment. This means that our findings are likely applicable, at least to some in extent, in other European Union member states in the Central Eastern Europe (for instance, Poland, Slovakia or Slovenia) and in the Southern Europe (for example, Spain and Italy). All these countries share very similar legislative framework based on the European Union directive. More generally, our findings speak to principal-agent settings affected by corrupt activities that arise as a consequence of bureaucratic discretion that can be misused. Several papers have proposed conceptual frameworks rationalizing results linking discretion, oversight, and corruption in public procurement point into this direction (Decarolis et al., 2025; Carril, 2021; Bosio et al., 2022; Bandiera et al., 2021a). These papers study markets in a number of countries across the globe and find similar patterns, which suggests more general validity of our findings.

Appendices

2.A Appendix section

Figure 2.A.1 Areas for levels of discretion during different public procurement acts. 2006 - 2021

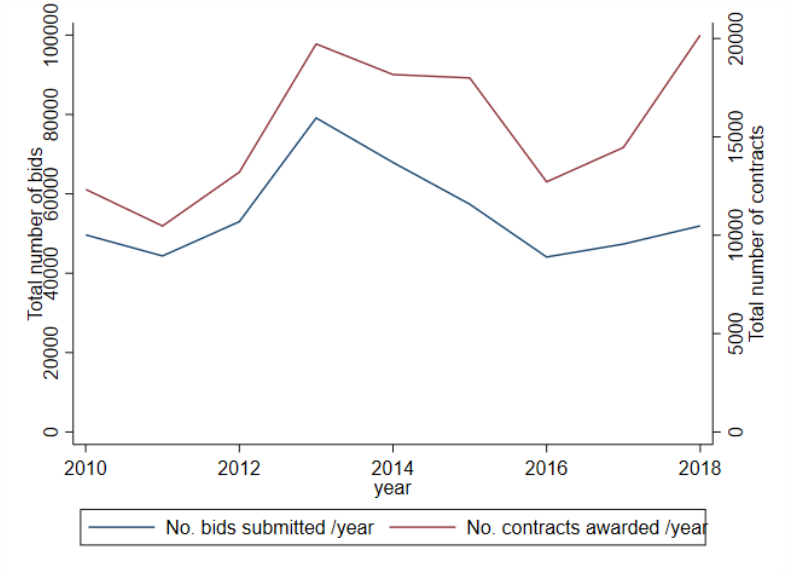


Notes: The x-axis represents years. The thick lines indicate the periods governed by different public procurement laws. In 2006 the Public Procurement Act no. 137/2006 became effective. In 2012 the Amendment no. 55/2012 Coll was enacted. In 2014 the Amendments no. 303/2013 Coll were implemented. In 2016 the Act no. 134/2016 Coll became effective. The labeled areas represent four levels of discretion. Reading them from bottom to top, the areas are as follows: Full discretion refers to contracts whose anticipated value lies under the small-value contracts; hence public procurement laws are not applicable. High level of discretion refers to contracts that are under the sub national threshold that entitles procurers to use simplified procedures. Medium discretion refers to contracts whose values lie above the sub-national threshold and but below the European threshold for public procurement; these contracts are excluded from discretionary procedures and subject to limited European oversight. Low discretion refers to contracts that exceed the European threshold and are thus tendered in the European market. The y-axis is expressed in Czech korunas, using the exchange rate from 14th July 2023: \$1 = 21.2 CZK. The values in dollars, from bottom to top, are: 141 thousand (k), 280k, 470k, 940k, and 2.36 million (M).

Table 2.A.1 Descriptive statistics - public procurement contracts Czech Republic

	Before 2012						After 2012						Before 2016						After 2016						Total Sample		
	Below 20 CZK ($940k)			Above 20 CZK ($940k)			Below 10 CZK ($470k)			Above 10 CZK ($470k)			Below 10 CZK ($470k)			Above 10 CZK ($470k)			Below 50 CZK ($2.36M)			Above 50 CZK ($2.36M)			Sample		
	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N	Mean	SD	N
Tenders characteristics																											
Num lots	1.269	1.224	3,209	1.468	2.010	1,470	1.314	1.470	3,502	2.237	3.934	2,870	1.174	0.772	2,635	2.081	3.789	5,040	1.443	2.258	7,192	1.509	1.517	460	1.589	2.598	26,378
Award M.E.A.T	0.433	0.496	3,209	0.529	0.499	1,470	0.184	0.388	3,502	0.240	0.427	2,870	0.096	0.295	2,635	0.147	0.355	5,040	0.108	0.311	7,192	0.137	0.344	460	0.202	0.402	26,378
Simplified Procedure	0.378	0.485	3,209	0.003	0.058	1,470	0.499	0.500	3,502	0.084	0.277	2,870	0.507	0.500	2,635	0.002	0.042	5,040	0.625	0.484	7,192	0.002	0.047	460	0.343	0.475	26,378
Open Procedure	0.296	0.457	3,209	0.676	0.468	1,470	0.325	0.469	3,502	0.713	0.453	2,870	0.358	0.479	2,635	0.843	0.364	5,040	0.290	0.454	7,192	0.802	0.399	460	0.484	0.500	26,378
Restricted Procedure	0.042	0.200	3,209	0.242	0.429	1,470	0.016	0.125	3,502	0.114	0.318	2,870	0.012	0.108	2,635	0.087	0.281	5,040	0.060	0.237	7,192	0.172	0.378	460	0.070	0.255	26,378
EU Funds	0.352	0.478	3,209	0.546	0.498	1,470	0.343	0.475	3,502	0.437	0.496	2,870	0.400	0.490	2,635	0.496	0.500	5,040	0.295	0.456	7,192	0.509	0.500	460	0.390	0.488	26,378
Tenders Outcomes																											
Anticipated Value	12.120	4.351	3,209	37.812	14.928	1,470	6.147	2.068	3,502	24.893	14.986	2,870	8.121	1.218	2,635	25.967	15.344	5,040	16.899	10.531	7,192	62.823	8.080	460	18.582	14.934	26,378
Final Value	9.956	5.160	3,203	31.315	15.591	1,462	4.749	9.880	3,497	18.830	17.873	2,860	7.053	13.855	2,630	20.885	17.765	5,020	14.182	9.981	7,169	52.054	18.029	450	15.086	15.707	26,291
Normalized Price	0.809	0.226	3,197	0.826	0.210	1,459	0.730	0.217	3,484	0.743	0.223	2,853	0.801	0.189	2,619	0.796	0.206	5,011	0.838	0.204	7,146	0.823	0.249	449	0.797	0.214	26,218
Log Normalized Price	-0.263	0.352	3,197	-0.235	0.332	1,459	-0.369	0.357	3,484	-0.357	0.391	2,853	-0.255	0.270	2,619	-0.285	0.423	5,011	-0.222	0.361	7,146	-0.276	0.485	449	-0.278	0.372	26,218
Number Offers	6.614	4.195	3,135	6.893	4.550	1,426	7.099	4.343	3,409	7.048	4.380	2,776	6.099	3.635	2,592	6.270	3.904	4,951	4.666	3.074	7,169	5.292	3.430	452	6.060	3.966	25,910
Std. Producitivity	0.006	0.775	1,231	0.112	0.455	589	-0.062	0.843	1,408	0.032	0.658	1,212	-0.020	0.811	976	0.038	0.661	1,934	0.167	0.133	2,553	0.138	0.156	140	0.052	0.630	10,043
Favoritism																											
Political Donor	0.186	0.389	3,209	0.210	0.408	1,470	0.196	0.397	3,502	0.210	0.408	2,870	0.185	0.388	2,635	0.209	0.406	5,040	0.199	0.399	7,192	0.189	0.392	460	0.199	0.399	26,378
Political Connection	0.068	0.252	3,209	0.079	0.270	1,470	0.083	0.276	3,502	0.071	0.256	2,870	0.104	0.305	2,635	0.078	0.269	5,040	0.096	0.295	7,192	0.089	0.285	460	0.085	0.278	26,378
Anonymously Owned	0.051	0.221	3,209	0.078	0.268	1,470	0.054	0.226	3,502	0.082	0.274	2,870	0.058	0.234	2,635	0.091	0.288	5,040	0.064	0.244	7,192	0.120	0.325	460	0.069	0.254	26,378
Donor & Connected	0.066	0.249	3,209	0.083	0.276	1,470	0.059	0.236	3,502	0.077	0.267	2,870	0.070	0.256	2,635	0.071	0.256	5,040	0.068	0.251	7,192	0.083	0.276	460	0.069	0.254	26,378
Connected & Anonymous	0.033	0.180	3,209	0.050	0.219	1,470	0.035	0.183	3,502	0.060	0.237	2,870	0.050	0.217	2,635	0.059	0.236	5,040	0.047	0.211	7,192	0.057	0.231	460	0.048	0.214	26,378
Donor & Anonymous	0.041	0.199	3,209	0.067	0.251	1,470	0.025	0.156	3,502	0.054	0.227	2,870	0.018	0.132	2,635	0.050	0.217	5,040	0.019	0.138	7,192	0.061	0.239	460	0.036	0.185	26,378
Connected & Donor & Anonymous	0.045	0.207	3,209	0.067	0.250	1,470	0.041	0.199	3,502	0.057	0.232	2,870	0.039	0.195	2,635	0.043	0.203	5,040	0.046	0.209	7,192	0.048	0.214	460	0.046	0.210	26,378
Value donations	0.185	0.268	1,086	0.203	0.293	628	0.219	0.312	1,126	0.204	0.277	1,145	0.159	0.216	823	0.183	0.253	1,875	0.171	0.219	2,386	0.151	0.180	175	0.186	0.257	9,244
Number donations	3.170	3.055	1,086	3.433	3.498	628	3.396	3.174	1,126	3.325	3.086	1,145	3.095	2.746	823	3.323	3.228	1,875	3.313	2.958	2,386	2.817	2.723	175	3.289	3.086	9,244
Number of connections	1.436	1.211	812	1.547	1.204	457	1.445	1.265	931	1.430	1.044	933	1.494	1.201	834	1.386	1.157	1,616	1.423	1.196	2,248	1.174	0.970	172	1.429	1.179	8,003

Figure 2.A.2 Total number of bids per year and total number of contracts



2.A.1 Number of firms and bids over years

Institutional changes, such as legal reforms, can have general equilibrium effects. For instance, the pool of firms interested in public procurement contracts might adjust due to the new regulatory framework. In general, we do not observe differences in the extensive margin, or big changes in the landscape of firms. Figure 2.A.2 shows the total number of bids and the total number of contracts awarded per year from 2010 to 2018. The number of contracts awarded is a proxy of the size of the procurement market and the number of bids is a proxy of the pool of firms participating in the market. This graph informs that over the period studied when there were more contracts in the market, there were also more bidders competing for these contracts. Similarly, when the market was small there were less firms participating in the market.

We present the percentage of contracts awarded to firms with political favoritism per year. table 2.A.2 shows a relatively stable proportion in the number of politically connected firms who are awarded public procurement contracts. Since the political favoritism was relatively stable during the period of analysis, our estimations of the effect of discretion in market outcomes are not confounded by the likelihood of corruption.

Table 2.A.2 Number of contracts and number of firms per year

Year	Connections		Donations		Anonymous		Total	Total
	Number	Percentage	Number	Percentage	Number	Percentage	Firms	Contracts
2006	286	0.14	340	0.17	179	0.09	2,038	5,447
2007	130	0.08	139	0.08	103	0.06	1,672	8,198
2008	101	0.08	73	0.06	71	0.05	1,322	9,191
2009	104	0.06	73	0.04	82	0.05	1,697	13,727
2010	60	0.05	56	0.05	54	0.04	1,244	12,332
2011	51	0.05	51	0.05	38	0.04	965	10,470
2012	56	0.04	49	0.04	48	0.04	1,281	13,205
2013	66	0.03	72	0.03	50	0.02	2,166	19,721
2014	44	0.03	37	0.03	42	0.03	1,435	18,166
2015	40	0.04	36	0.03	37	0.03	1,132	17,995
2016	24	0.03	22	0.03	34	0.04	828	12,713
2017	20	0.02	32	0.03	34	0.03	1,089	14,461
2018	8	0.01	45	0.03	26	0.02	1,576	20,172
2019	3	0.00	27	0.02	29	0.02	1,219	20,312
2020	3	0.00	28	0.02	32	0.03	1,138	19,577
2021	2	0.00	27	0.02	27	0.02	1,371	35,158

2.A.2 Measurement of anticipated value of contracts and final price paid

The empirical strategy of this paper uses three monetary variables of a procurement contract: anticipated value, final price, and normalized price. The anticipated value of a contract defines the level of discretion available to procurement officers. This variable measures the engineers' cost of the project or services procured, as described by Palguta and Pertold (2017, p.297) *The anticipated value should be set so as to approximate the anticipated financial obligations ensuring from the contract [...] the anticipated value does not serve as a reserve price, firms can bid a higher price.*. The anticipated value of a contract is the main variable targeted by the reforms of the discretionary thresholds in 2012 and 2016.

The second variable characteristic is the final price of a contract price, which indicates the amount paid to the winning firms including possible renegotiation. The third variable corresponds to the normalized price, which is a relative measure of the final price with respect to the anticipated value of a contract ($NormPrice = \frac{FinalPrice}{Anticipated}$). This measure has been used in the literature to proxy different aspects of the procurement process, such as competition Szucs (2024), winning rebate Coviello et al. (2018), and cost-overrun Palguta and Pertold (2017), Carril (2021) and Spenkuch et al. (2023). Procurement authorities can procure a good or service with a single contract or divide it into several contracts or lots. We conditioned our sample to include only contracts with one lot.

2.A.3 Productivity

We estimate firms' productivity following the estimation framework illustrated by (Rovigatti and Mollisi, 2018). The estimation implements the methods of Wooldridge Levinsohn-Petrin for estimating Total Factor Productivity (TFP) at the firm-level. Following (Gal, 2013) we use a subsample of firms in the Czech Republic from

the Orbis database. The resulting panel data structure allows us to implement the estimation package prodest proposed by (Rovigatti and Mollisi, 2018).

Our starting point is the commonly accepted definition of productivity as a measure that captures the efficiency by which inputs are turned into outputs. In Orbis the gross output is measured by operating revenue turnover (Gal, 2013, p.11) in table 2. We regress this operational revenue, y_{fit}, with inputs such as costs of employees w_{fit}, tangible fixed assets x_{fit}, and imputed intermediate inputs onto operational revenue turnover. We also adjust for deflators as suggested by (Gal, 2013) and (De Jaegher et al., 2025). We estimate industry-specific production functions at the NACE 2-digit industry groups similar to the estimation by (**?**). Our specification is $y_{fit} = \alpha + w_{fit}\beta + x_{fit}\gamma + v_{fit} + e_{fit}$. The TFP is computed from the estimated parameters as $TFP = y_{fit} - \hat{\alpha} + w_{fit}\hat{\beta} + x_{fit}\hat{\gamma}$

We then compute a standard score $Z_f = \frac{TFP_{fi} - \bar{TFP}_i}{\sigma_i}$. We called this standard score of productivity as competitiveness, which is our measurement of interest. As any standard score this variable is the number of standard deviations by which the productivity or TFP or a firm is above or below the mean productivity of other firms in the same industry. Firms with productivity above the average of its industry have positive standard scores, thus, are competitive high performing firms in the market. Whereas firms with productivity below the average of its industry have negative standard scores, thus are less competitive for the mean standards of their industry.

We are interested in matching the standard score lagged by one year prior to a winning award. In other words, a firm winning a contract in year t has the productivity score of the year $t-1$. In this way we exploit an information channel because a procurer who awards a contract to firm f in time t might have been informed their decision by observing the productivity of the firm in the previous period $t-1$. This enables us to test two hypotheses. First, an efficient, rational and aligned procurer who observes the productivity score Z_f would prefer to allocate the contract to a firm with positive $Z_f+ > Z_f-$. Second hypothesis is that a wasteful and misaligned procurer would prefer or be indifferent to allocate a contract to a firm with negative productivity score $Z_f- \leq Z_f+$. Notice that the mechanisms through which a procurer is informed, observes or screens this productivity score involve political favoritism as we discuss in the body of the paper.

2.B Robustness checks estimates excess mass

Table 2.B.1 Estimates excess mass below thresholds by winning firms' likelihood of corruption.

		Reform 2012		Reform 2016	
		Before	After	Before	After
All Contracts	$\hat{b}$	1.43	0.77	1.68	2.11
	$\hat{se}_b$	0.395	0.358	0.348	0.566
	$\hat{B}$	390	205	582	75
	N	7,446	5,128	5,130	1,158
Non connected	$\hat{b}$	1.44	0.56	2.37	4.00
	$\hat{se}_b$	0.524	0.313	0.996	1.526
	$\hat{B}$	170	74	338	50
	N	3,422	2,437	2,327	453
Connected	$\hat{b}$	1.84	0.68	2.00	0.20
	$\hat{se}_b$	0.971	0.609	0.392	0.927
	$\hat{B}$	28	12	65	1
	N	500	391	501	109
Donated	$\hat{b}$	1.78	0.78	1.10	3.11
	$\hat{se}_b$	0.851	0.498	1.229	2.023
	$\hat{B}$	88	41	86	13
	N	1,469	1,024	995	216
Anonymous	$\hat{b}$	1.49	0.35	0.40	-0.41
	$\hat{se}_b$	0.553	0.715	1.993	0.756
	$\hat{B}$	35	6	12	-2
	N	503	324	323	118

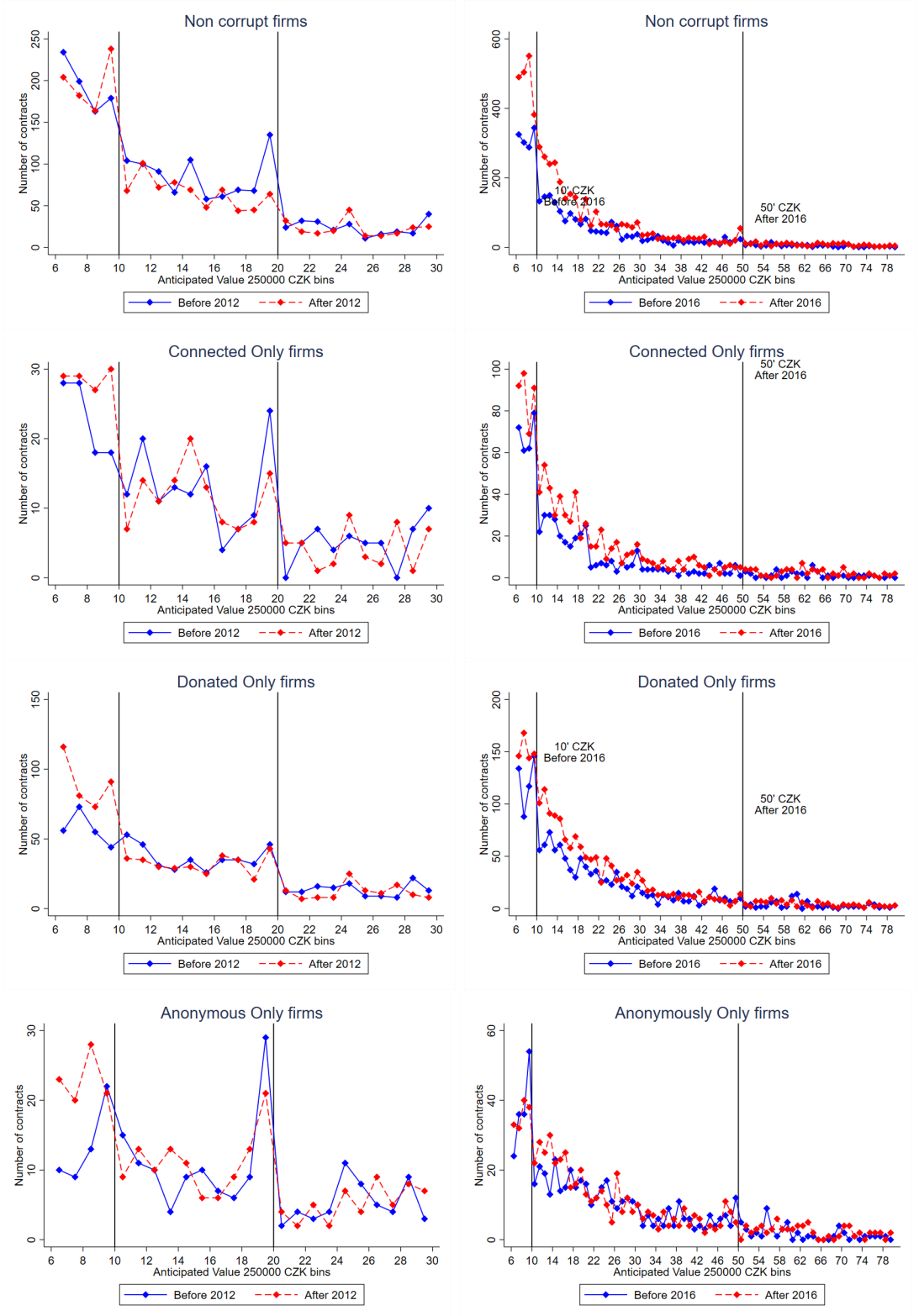
Notes: The estimations are computed following Chetty et al. (2011) and extend the results presented in Table 2.2. The excess of mass of contracts is quantified by $\hat{b}$ in parentheses are the standard errors. $\hat{B}$ corresponds to the estimated number of contracts in the bunching bins. N is the total number of bins.

Table 2.B.2 Estimates excess mass below thresholds by winning firms' joined likelihood of corruption.

		Reform 2012		Reform 2016	
		Before	After	Before	After
Connected and Donated	$\hat{b}$	1.15	1.76	-4.00	2.37
	$\hat{se}_b$	0.457	1.165	0.033	1.800
	$\hat{B}$	47	51	*n.c*	10
	N	953	580	558	144
Connected and Anonymous	$\hat{b}$	0.75	1.38	-4.00	1.34
	$\hat{se}_b$	0.316	1.445	0.000	0.906
	$\hat{B}$	23	29	*n.c*	7
	N	685	454	476	131
Donated and Anonymous	$\hat{b}$	1.23	1.15	-4.00	0.19
	$\hat{se}_b$	0.624	1.946	2.769	1.035
	$\hat{B}$	36	23	*n.c*	1
	N	702	414	350	109
Connected & Donated & Anonymous	$\hat{b}$	1.04	1.91	-4.00	1.05
	$\hat{se}_b$	0.529	6.479	0.000	1.854
	$\hat{B}$	18	20	*n.c*	2
	N	394	248	200	61

Notes: The estimations are computed following Chetty et al. (2011) and extend the results presented in Table 2.2. The excess of mass of contracts is quantified by $\hat{b}$ in parentheses are the standard errors. $\hat{B}$ corresponds to the estimated number of contracts in the bunching bins. N is the total number of bins.

Figure 2.B.1 Distribution of contracts anticipated value by winning firms' likelihood of corruption



We also estimate two alternative specifications of our bunching regressions. A first alternative specification consists in using relative binned values. The equation in 2.B.1 represents a pooled estimation of both pre-reform threshold and post-reform threshold equalized at zero. In this case, there is one single indicator variable for the intervals below the discretionary threshold. Thus we are interested in estimating β_i

$$NC_{jt} = \alpha_j + \alpha_t + \sum_{i=-AC_Y}^{0} \beta_i \mathbb{1}[Bin_j = i]\mathbb{1}[t > Y_{LM}] + \varepsilon_{jt} \tag{2.B.1}$$

The second alternative specification consists in identifying the whole area below the cutoff thresholds instead of only some bins. As shown in equation 2.B.2 this regression considers the distribution of contracts at the two thresholds after the reform. This means we include two dummies variables, one for the old limit threshold and one for the new limit threshold. This specification aims to observe whether there were differences in the distribution of contracts after the reform around the new threshold, and whether there were differences after the reform around the old threshold. In other words, we can test the hypothesis bunching occurs at the new threshold and disappears at the old threshold.

$$NC_{jt} = \alpha_j + \alpha_t + \delta_1 \mathbb{1}[j < C_Y^-]\mathbb{1}[t > R_Y] + \delta_2 \mathbb{1}[j < C_Y^+]\mathbb{1}[t > R_Y] + \varepsilon_{jt} \tag{2.B.2}$$

Table 2.B.3 Estimates excess mass below thresholds - alternative estimation in equation 2.B.1 Reform 2012

	(1) All contracts	(2) Non connected	(3) Connected Only	(4) Donated Only	(5) Anonymously Only	(6) Connected Donated	(7) Connected Anonymous	(8) Donated Anonymous	(9) Connected Donated Anonymous
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\hat{\beta}_1$	-0.084	-0.080	0.197	-0.052	-0.141	0.022	0.084	-0.124	0.178
	(0.076)	(0.107)	(0.162)	(0.119)	(0.165)	(0.139)	(0.153)	(0.144)	(0.202)
$\hat{\beta}_2$	0.345***	0.295***	0.302	0.571***	0.594***	0.315**	0.072	0.318**	0.152
	(0.075)	(0.096)	(0.193)	(0.108)	(0.163)	(0.151)	(0.136)	(0.151)	(0.156)
Constant	4.424***	3.868***	1.819***	2.665***	1.490***	2.096***	1.854***	1.818***	1.252***
	(0.033)	(0.048)	(0.096)	(0.062)	(0.071)	(0.061)	(0.064)	(0.060)	(0.086)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bins FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	293	293	250	290	254	290	273	281	252
ll	-901.8	-683.5	-357.9	-571.1	-372.8	-496.8	-455.5	-463.8	-340.6

Standard errors in parentheses
* pj0.1, ** pj0.05, *** pj0.01

Notes: Table 2.B.3 represents the estimations from Poisson Fixed Effects regression of equation 2.B.1 for the 2012 reform. Reported coefficients are not exponential. Coefficients can be interpreted as $(\exp(\gamma - 1)) * 100$ and $(\exp(\eta - 1)) * 100$

Table 2.B.4 Estimates excess mass below thresholds - alternative estimation in equation 2.B.1 Reform 2016

	(1) All contracts (1)	(2) Non connected (2)	(3) Connected Only (3)	(4) Donated Only (4)	(5) Anonymously Only (5)	(6) Connected Donated (6)	(7) Connected Anonymous (7)	(8) Donated Anonymous (8)	(9) Connected Donated Anonymous (9)
$\hat{\beta}_1$	-0.084	-0.050	-0.234**	-0.078	-0.101	-0.320***	-0.399***	-0.261	-0.495**
	(0.062)	(0.083)	(0.108)	(0.085)	(0.120)	(0.114)	(0.147)	(0.163)	(0.195)
$\hat{\beta}_2$	0.049	0.073	0.032	0.115	-0.274	-0.078	0.167	-0.065	-0.142
	(0.092)	(0.125)	(0.233)	(0.196)	(0.185)	(0.210)	(0.209)	(0.198)	(0.340)
Constant	4.471***	3.796***	2.219***	2.806***	1.716***	2.321***	1.981***	1.722***	1.437***
	(0.044)	(0.064)	(0.115)	(0.092)	(0.079)	(0.101)	(0.100)	(0.089)	(0.174)
Bins FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
year_group	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	413	413	398	407	403	409	401	404	370
ll	-1,290.5	-1,028.5	-564.8	-791.9	-600.1	-654.6	-649.4	-596.8	-459.4

Standard errors in parentheses
* $p_i0.1$, ** $p_i0.05$, *** $p_i0.01$

Notes: Table 2.B.4 represents the estimations from Poisson Fixed Effects regression of equation 2.B.1 for the 2016 reform. Reported coefficients are not exponential. Coefficients can be interpreted as $(\exp(\gamma - 1)) * 100$ and $(\exp(\eta - 1)) * 100$

Table 2.B.5 Estimates excess mass below thresholds - alternative estimation in equation 2.B.2 Reform 2012

	(1) All contracts	(2) Non connected	(3) Connected Only	(4) Donated Only	(5) Anonymously Only	(6) Anonymously Only	(7) Connected Donated	(8) Connected Anonymous	(9) Donated Anonymous
$\hat{\delta}_1$	0.485***	0.518***	0.610**	0.735***	0.040	0.040	0.485***	0.214*	0.128
	(0.072)	(0.108)	(0.301)	(0.080)	(0.176)	(0.176)	(0.081)	(0.113)	(0.235)
$\hat{\delta}_2$	0.256***	0.185*	0.589**	0.258*	0.634***	0.634***	0.342***	0.054	0.567*
	(0.074)	(0.098)	(0.232)	(0.133)	(0.181)	(0.181)	(0.089)	(0.114)	(0.293)
Constant	4.469***	3.908***	1.986***	2.779***	1.560***	1.560***	2.158***	1.899***	1.825***
	(0.018)	(0.021)	(0.035)	(0.027)	(0.034)	(0.034)	(0.031)	(0.029)	(0.032)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bins FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	293	293	250	290	254	254	290	273	281
ll	-902.6	-679.5	-357.6	-577.3	-374.8	-374.8	-496.5	-455.6	-463.8

Standard errors in parentheses
* $p_i0.1$, ** $p_i0.05$, *** $p_i0.01$

Notes: Table 2.B.5 represents the estimations from Poisson Fixed Effects regression of equation 2.B.2 for the 2012 reform. Reported coefficients are not exponential. Coefficients can be interpreted as $(\exp(\gamma - 1)) * 100$ and $(\exp(\eta - 1)) * 100$

Table 2.B.6 Estimates excess mass below thresholds - alternative estimation in equation 2.B.2 Reform 2016

	(1) All contracts (1)	(2) Non connected (2)	(3) Connected Only (3)	(4) Donated Only (4)	(5) Anonymously Only (5)	(6) Connected Donated (6)	(7) Connected Anonymous (7)	(8) Donated Anonymous (8)	(9) Connected Donated Anonymous (9)
$\hat{\delta}_1$	0.221	0.461	1.011	0.197	-0.914***	-0.241	0.497	0.182	-0.093
	(0.280)	(0.333)	(0.732)	(0.338)	(0.344)	(0.459)	(0.336)	(0.336)	(0.554)
$\hat{\delta}_2$	-0.389	-0.609	-0.493**	-0.322	0.744**	-0.270	-1.449***	-1.071	-1.479**
	(0.313)	(0.554)	(0.193)	(0.495)	(0.292)	(0.336)	(0.329)	(0.742)	(0.732)
Constant	4.480***	3.821***	2.190***	2.848***	1.594***	2.242***	2.006***	1.672***	1.309***
	(0.013)	(0.018)	(0.026)	(0.019)	(0.026)	(0.025)	(0.031)	(0.032)	(0.043)
Bins FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
year_group	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	413	413	398	407	403	409	401	404	370
ll	-1,290.9	-1,026.0	-566.3	-792.3	-599.2	-659.2	-652.1	-597.5	-462.6

Standard errors in parentheses
* $p_i0.1$, ** $p_i0.05$, *** $p_i0.01$

Notes: Table 2.B.6 represents the estimations from Poisson Fixed Effects regression of equation 2.B.2 for the 2012 reform. Reported coefficients are not exponential. Coefficients can be interpreted as $(\exp(\gamma - 1)) * 100$ and $(\exp(\eta - 1)) * 100$

2.B.1 Regression Discontinuity Design

Figure 2.B.2 Representation of the sample split for the two reforms and the fuzzy RDD design

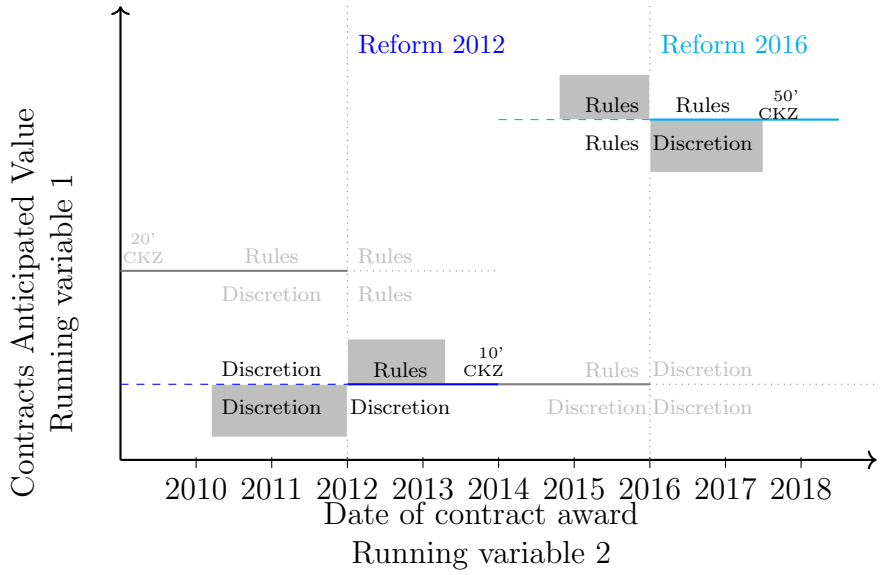
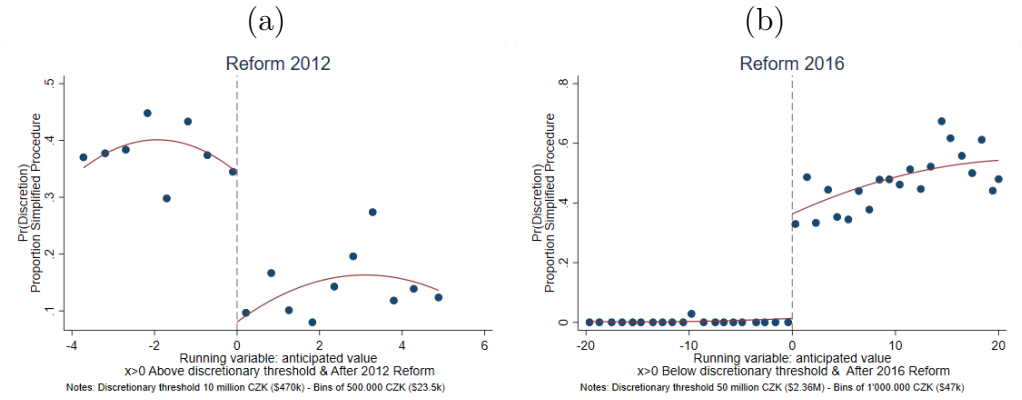


Figure 2.B.3 Intention to treat and compliance of new regulatory threshold



Notes: The Intention to treat refers to the Sharp discontinuity in treatment assignment is implicit because after the Reforms (2012 and 2016) the threshold changes were enforced. The compliance refers to the Fuzzy discontinuity since the uptake or implementation of simplified procedures does not change for all contract units. We observe a jump in the probability of treatment.

McCrary test and continuity in running variable

Figure 2.B.4 McCrary density continuity plot

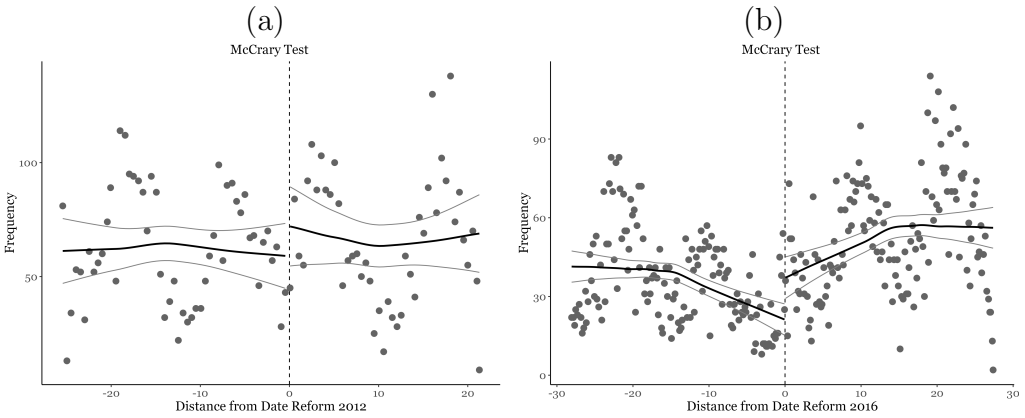


Figure 2.B.5 Contracts' date of award distribution with respect to date reform

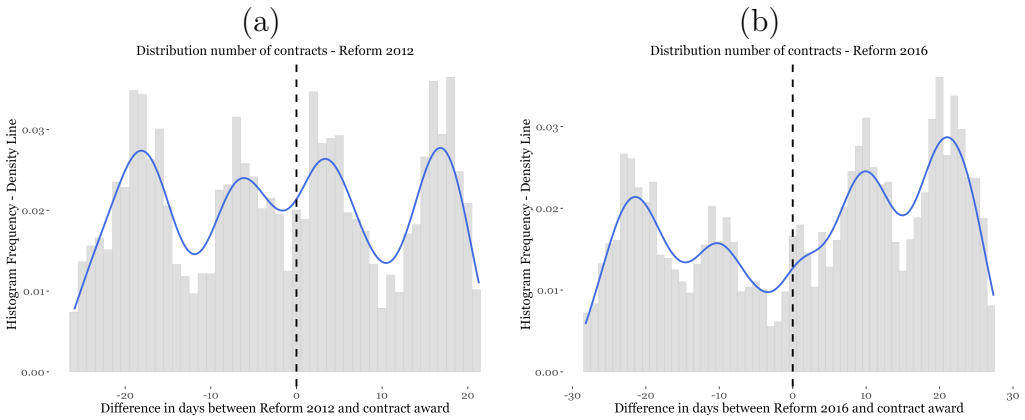


Figure 2.B.6 Contracts' date of award distribution with respect to date reform, by size of contract whether above or below

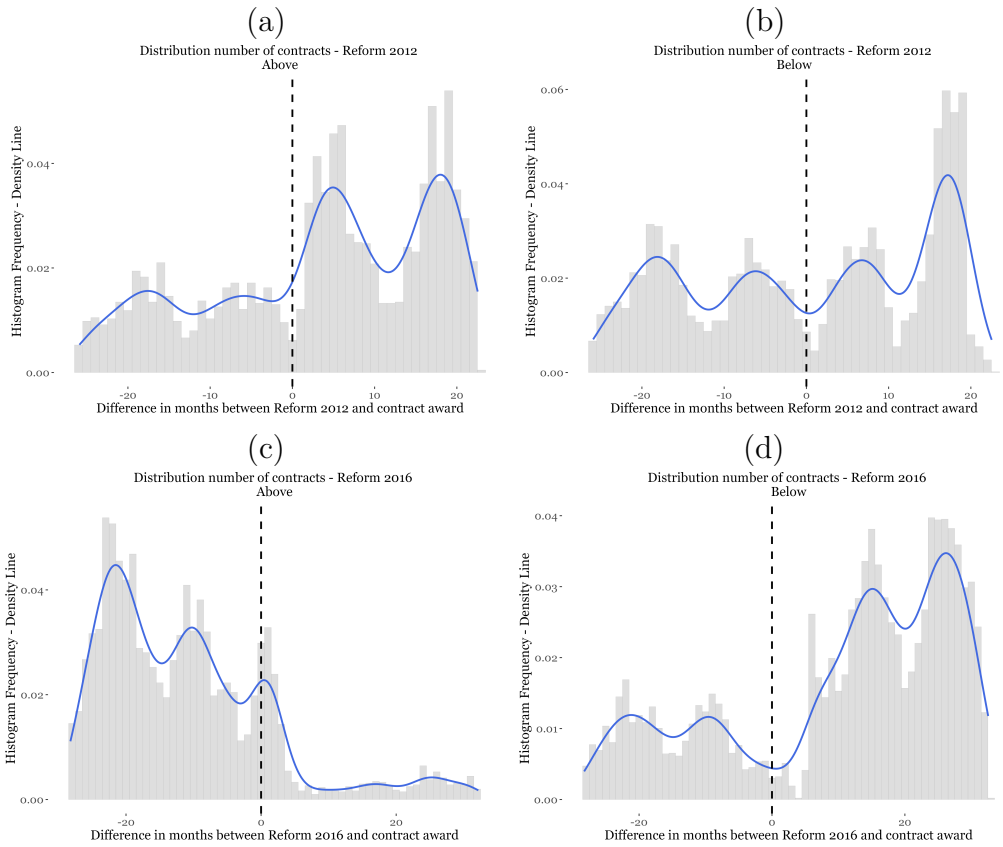
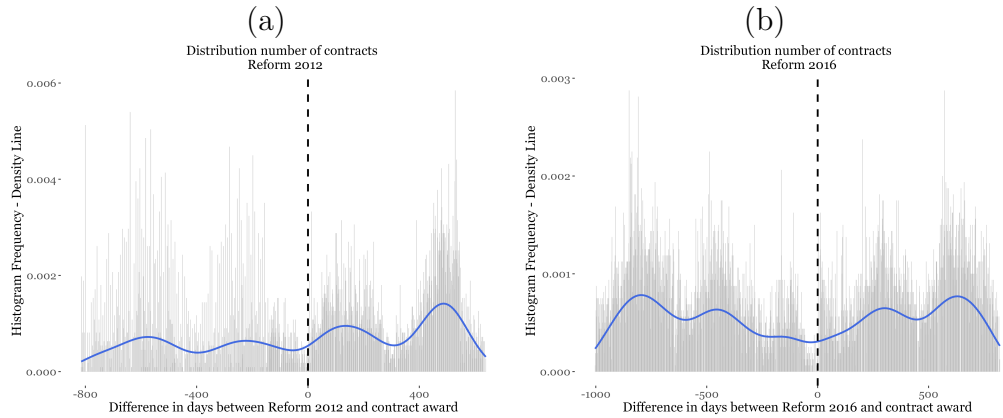


Figure 2.B.7 Contracts' date of award distribution with respect to date reform- Days



Continuity in multiscored running variable

Figure 2.B.8 Multiscored running variable Local Polynomial Density Estimation

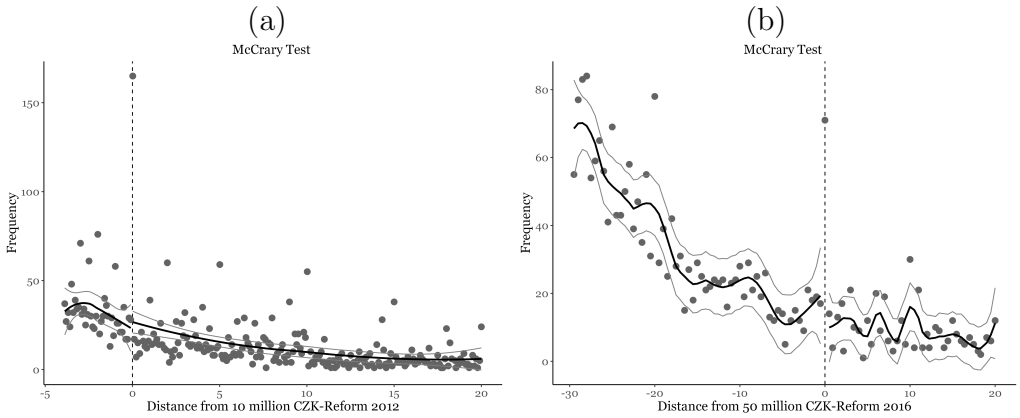


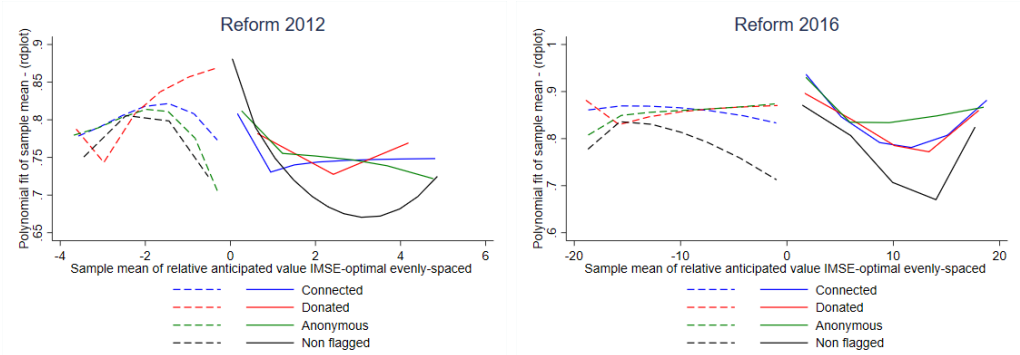
Table 2.B.7 Density discontinuity tests - McCrary manipulation testing

Running variable	θ	s.e	z	$p-value$	Bin Size	Bandwidth	Cutoff
Multi-Score 2012	-0.15	0.04	-3.52	0.00	0.10	20.00	0.00
Multi-Score 2016	-0.18	0.11	-1.58	0.11	0.50	25.00	0.00
Anticipated value 2012	-0.15	0.04	-3.52	0.00	0.10	20.00	0.00
Anticipated value 2016	-0.15	0.11	-1.36	0.17	0.50	25.00	0.00
Date Award 2012	0.30	0.05	6.23	0.00	0.50	30.00	0.00
Date Award 2016	0.22	0.10	2.20	0.03	0.20	5.00	0.00

Notes: θ: estimated log difference in heights at the cutpoint. *s.e*: the standard error of θ. z: the z statistic of the test. $p-value$: p-value of the test. Estimations based on `rdd` package from `DCdensity` function by (Dimmery, 2016).

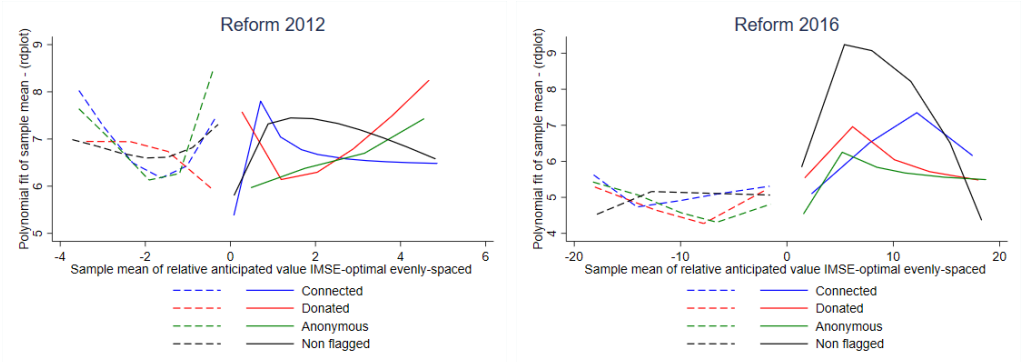
Discontinuities in outcomes with multiscored running variable

Figure 2.B.9 Price over cost ratio after the discretionary reforms by type of flagged favoritism



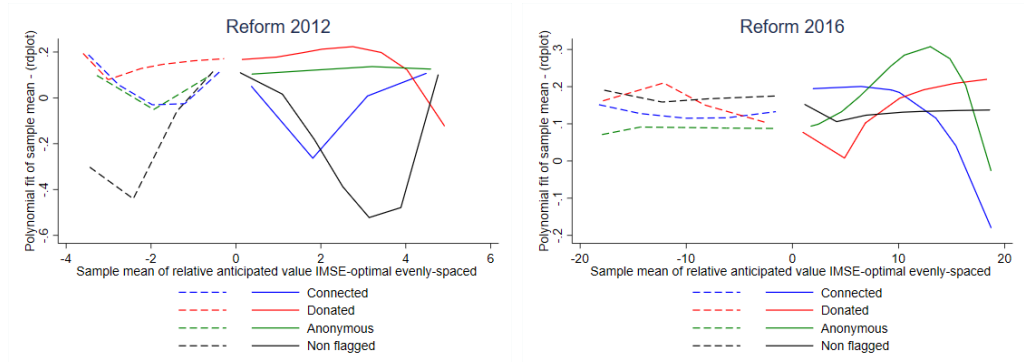
Notes: Plots based on `rdplot` package by Calonico et al. (2014). IMSE-optimal evenly-spaced method using polynomial regression.

Figure 2.B.10 Average number of bids



Notes: Plots based on `rdplot` package by Calonico et al. (2014). IMSE-optimal evenly-spaced method using polynomial regression.

Figure 2.B.11 Competitiveness (Lagged productivity)



Notes: Plots based on `rdplot` package by Calonico et al. (2014). IMSE-optimal evenly-spaced method using polynomial regression.

Robustness checks with covariates

Table 2.B.8 RDD estimations for *normalized price* - Reform 2012

	$\hat{b}$	$\hat{se}_b$	$p-v$	N	N left	N right	Bandwidth L	Bandwidth R
2016 - Fuzzy								
Favoritism Flagged	0.42	0.33	0.201	521	270	251	0.97	0.97
Non-flagged	0.23	0.50	0.645	397	211	186	1.18	1.18
Total Sample	0.77	0.42	0.068	918	481	437	1.22	1.22
Connected Firms	0.58	0.60	0.334	262	134	128	0.97	0.97
Donor Firms	-0.89	0.50	0.079	215	116	99	0.88	0.88
Anonymous Ownership	-0.34	0.81	0.674	159	82	77	0.88	0.88
2012 - Sharp								
Favoritism Flagged	0.10	0.06	0.127	521	270	251	0.97	0.97
Non-flagged	0.11	0.25	0.647	397	211	186	1.18	1.18
Total Sample	0.23	0.12	0.051	918	481	437	1.22	1.22
Connected Firms	0.14	0.09	0.126	262	134	128	0.97	0.97
Donor Firms	-0.13	0.11	0.235	215	116	99	0.88	0.88
Anonymous Ownership	0.02	0.04	0.647	159	82	77	0.88	0.88

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts were funded by European-level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshold or date of award.

Table 2.B.9 RDD estimations for *normalized price* - Reform 2016

	$\hat{b}$	$\hat{se}_b$	$p-v$	N	N left	N right	Bandwidth L	Bandwidth R
2016 - Fuzzy								
Favoritism Flagged	0.15	0.12	0.191	281	189	92	2.57	2.57
Non-flagged	0.06	0.13	0.641	177	133	44	5.11	5.11
Total Sample	0.18	0.15	0.241	458	322	136	4.20	4.20
Connected Firms	-0.48	0.25	0.060	141	105	36	2.82	2.82
Donor Firms	0.64	2.50	0.798	109	68	41	6.90	6.90
Anonymous Ownership	1.04	1.22	0.395	98	69	29	3.89	3.89
2016 - Sharp								
Favoritism Flagged	-0.09	0.05	0.092	281	189	92	2.57	2.57
Non-flagged	-0.04	0.09	0.684	177	133	44	5.11	5.11
Total Sample	-0.06	0.06	0.344	458	322	136	4.20	4.20
Connected Firms	0.91	0.48	0.059	141	105	36	2.82	2.82
Donor Firms	-0.21	0.11	0.055	109	68	41	6.90	6.90
Anonymous Ownership	0.47	0.38	0.218	98	69	29	3.89	3.89

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts was European level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshold or date of award.

Table 2.B.10 RDD estimations for *number of bids* - Reform 2012

	$\hat{b}$	$\hat{se}_b$	$p-v$	N	N left	N right	Bandwidth L	Bandwidth R
2016 - Fuzzy								
Favoritism Flagged	2.43	6.35	0.702	1,066	541	525	1.31	1.31
Non-flagged	0.67	2.56	0.794	1,164	705	459	0.87	0.87
Total Sample	0.76	2.95	0.797	2,230	1,246	984	0.91	0.91
Connected Firms	-47.05	50.70	0.353	499	247	252	0.73	0.73
Donor Firms	8.51	6.95	0.221	381	211	170	1.34	1.34
Anonymous Ownership	-11.67	8.49	0.169	384	173	211	1.11	1.11
2012 - Sharp								
Favoritism Flagged	0.46	0.94	0.625	1,066	541	525	1.31	1.31
Non-flagged	0.29	1.07	0.785	1,164	705	459	0.87	0.87
Total Sample	0.19	0.80	0.815	2,230	1,246	984	0.91	0.91
Connected Firms	3.08	1.60	0.054	499	247	252	0.73	0.73
Donor Firms	3.15	1.63	0.053	381	211	170	1.34	1.34
Anonymous Ownership	-3.50	1.50	0.020	384	173	211	1.11	1.11

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts was European level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshold or date of award.

Table 2.B.11 RDD estimations for *number of bids* - Reform 2016

	$\hat{b}$	$\hat{se}_b$	$p-v$	N	N left	N right	Bandwidth L	Bandwidth R
2016 - Fuzzy								
Favoritism Flagged	-0.30	2.21	0.894	800	556	244	6.67	6.67
Non-flagged	1.93	2.19	0.378	486	354	132	4.77	4.77
Total Sample	0.32	1.58	0.840	1,286	910	376	6.66	6.66
Connected Firms	-8.15	3.47	0.019	361	260	101	3.75	3.75
Donor Firms	-0.26	1.45	0.857	241	170	71	3.80	3.80
Anonymous Ownership	-4.69	3.91	0.231	351	236	115	5.71	5.71
2016 - Sharp								
Favoritism Flagged	0.11	0.82	0.893	800	556	244	6.67	6.67
Non-flagged	-1.01	1.24	0.416	486	354	132	4.77	4.77
Total Sample	-0.14	0.67	0.839	1,286	910	376	6.66	6.66
Connected Firms	4.22	1.40	0.003	361	260	101	3.75	3.75
Donor Firms	0.19	1.10	0.864	241	170	71	3.80	3.80
Anonymous Ownership	1.33	0.96	0.166	351	236	115	5.71	5.71

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts was European level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshold or date of award.

Table 2.B.12 RDD estimations for *productivity* - Reform 2012

	$\hat{b}$	$\hat{se}_b$	$p-v$	N	N left	N right	Bandwidth L	Bandwidth R
2016 - Fuzzy								
Favoritism Flagged	0.42	0.33	0.201	521	270	251	0.97	0.97
Non-flagged	0.23	0.50	0.645	397	211	186	1.18	1.18
Total Sample	0.77	0.42	0.068	918	481	437	1.22	1.22
Connected Firms	0.58	0.60	0.334	262	134	128	0.97	0.97
Donor Firms	-0.89	0.50	0.079	215	116	99	0.88	0.88
Anonymous Ownership	-0.34	0.81	0.674	159	82	77	0.88	0.88
2012 - Sharp								
Favoritism Flagged	0.10	0.06	0.127	521	270	251	0.97	0.97
Non-flagged	0.11	0.25	0.647	397	211	186	1.18	1.18
Total Sample	0.23	0.12	0.051	918	481	437	1.22	1.22
Connected Firms	0.14	0.09	0.126	262	134	128	0.97	0.97
Donor Firms	-0.13	0.11	0.235	215	116	99	0.88	0.88
Anonymous Ownership	0.02	0.04	0.647	159	82	77	0.88	0.88

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts was European level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshold or date of award.

Table 2.B.13 RDD estimations for *productivity* - Reform 2016

	$\hat{b}$	$\hat{se}_b$	$p-v$	N	N left	N right	Bandwidth L	Bandwidth R
2016 - Fuzzy								
Favoritism Flagged	0.15	0.12	0.191	281	189	92	2.57	2.57
Non-flagged	0.06	0.13	0.641	177	133	44	5.11	5.11
Total Sample	0.18	0.15	0.241	458	322	136	4.20	4.20
Connected Firms	-0.48	0.25	0.060	141	105	36	2.82	2.82
Donor Firms	0.64	2.50	0.798	109	68	41	6.90	6.90
Anonymous Ownership	1.04	1.22	0.395	98	69	29	3.89	3.89
2016 - Sharp								
Favoritism Flagged	-0.09	0.05	0.092	281	189	92	2.57	2.57
Non-flagged	-0.04	0.09	0.684	177	133	44	5.11	5.11
Total Sample	-0.06	0.06	0.344	458	322	136	4.20	4.20
Connected Firms	0.91	0.48	0.059	141	105	36	2.82	2.82
Donor Firms	-0.21	0.11	0.055	109	68	41	6.90	6.90
Anonymous Ownership	0.47	0.38	0.218	98	69	29	3.89	3.89

Notes: Estimations based on Sharp and Fuzzy RDD, from `rdrobust` package by Calonico et al. (2014). Covariate-adjusted fuzzy RDD estimates using local polynomial regression. We implement a uniform Kernel function. The bandwidth is selected based on optimal mean-squared error (MSE). We include covariates to control as fixed effects for type of public buyer, whether the contracts was European level funds, and a dummy for contracts located at exact cutoff value of the discretionary threshsold or date of award.

CHAPTER 3

Quality of Public Projects and Procurement Officers Value Added

Public procurement tenders are governed by complex regulations, yet procurement officers retain substantial discretion. How they exercise this discretion can influence both procurement efficiency and the quality of delivered projects. Using the universe of public procurement tenders in the Czech Republic, we estimate procurement officers' value added to project quality, measured by the need for subsequent repair contracts. We find that moving from the 10th to the 90th percentile of the value-added distribution is associated with a 4.5% reduction in expected repair costs. This corresponds to a reduction in 5-year expected repair costs of CZK 520,000 (EUR 21,000) for a median-size contract and CZK 2.1 million (EUR 86,000) for a mean-size contract. We further show that officers with legal education, and those who are more engaged with developments in public procurement legislation, tend to underperform, resulting in higher subsequent repair costs.

Keywords: Procurement, Bureaucratic performance, Project quality.
JEL Codes: H57, H83, J45, H11

3.1 Introduction

Governments spend between one-quarter and one-third of their total expenditures through public procurement tenders (OECD, 2025). Ensuring that these procurement processes deliver high-quality projects at reasonable cost is therefore essential for the effective functioning of governments. In most countries, public procurement is governed by extensive rules and restrictions that, for example, specify which selection procedures may be used for a given tender or set minimum environmental

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standards that delivered projects must meet. Despite a tight regulatory framework, procurement officers retain substantial discretion at several stages of the procurement process.

Procurement officers exercise discretion along several dimensions, including the choice of award criteria and procurement procedures, engagement in market consultations, and the design of contractual clauses. While existing evidence suggests relatively homogeneous preferences among procurement officers, at least within a country (Tukiainen et al., 2024), other individual characteristics, such as experience, knowledge, hard and soft skills, and motivation, may nonetheless generate substantial variation in officers' performance and procurement outcomes.

In this paper, we study procurement officers' value added to the quality of public procurement projects. We first use the universe of public procurement contracts from the Czech Republic to estimate the quality of individual construction contracts. The quality measure is based on the total estimated value of the procurement tender that the procuring authority must incur in subsequent years to repair or remedy deficiencies in the original project (for the same approach, see Baranek and Titl, 2024; De Jaegher et al., 2025). To illustrate, consider two contracts that each deliver 50 km of highway for EUR 10 million. Over the subsequent five years, the first project requires repairs worth EUR 500,000, whereas the second requires only EUR 200,000. Our measure then assigns a higher quality score to the latter project.

Having introduced and quantified a measure of project quality, we next estimate the value added of procurement officers to the quality of public procurement projects. Drawing on the value-added modeling literature in education (Koedel and Rockoff, 2015; McCaffrey et al., 2004; Chetty et al., 2014a,b), we quantify the extent to which individual officers contribute to the quality of delivered projects. To this end, we combine detailed information on the quality of construction contracts with data identifying the responsible procurement officers. For each public tender, the procuring authority publishes an email address for contacting the responsible officer, which we use to identify and link individual officers across tenders. This enables us to construct officer-specific measures of value added to the quality of delivered projects.

A key advantage of estimating procurement officers' value added using the quality of delivered projects is that this outcome is rarely observed contemporaneously (i.e., within the same fiscal year). As a result, it is unlikely to be incorporated into incentive schemes and is therefore less susceptible to deliberate manipulation by procurement officers (or their supervisors) seeking to meet performance targets. In this respect, our measure captures the quality of public officers' performance more accurately and improves upon contemporaneously observed indicators, such as those based on spending levels or final contract prices.

We find substantial variation in procurement officers' value added. In our primary specification, comparing officers at the 10th and 90th percentiles reveals a 4.5 point difference in the estimated value of repairs for a given project. Given that the median construction contract in our sample is worth CZK 11.5 million (EUR 470,000), replacing a 10th-percentile officer with a 90th-percentile officer would re-

duce estimated five-year repair costs by CZK 520,000 (EUR 21,000). For a mean contract value of CZK 46 million (EUR 1.9 million), the corresponding reduction in expected repair costs amount to CZK 2.1 million (EUR 86,000). While this is not a causal exercise, it illustrates the economically meaningful magnitude of variation in officers' value added.

We then link officers' value added to a unique survey dataset collected directly from procurement officers. This allows us to examine which individual characteristics of procurement officers, as well as features of the organizational structure of procurement teams, help explain the observed variation in officers' value added. We find that officers with a legal education and those who report higher levels of legal engagement in public procurement, measured by more intensive monitoring of legal and legislative developments, tend to underperform, leading to higher expected repair costs. This finding is robust across alternative definitions of our quality measures, officer value added, and regression models examining the effects of legal education and legal engagement.

In contrast, we find no evidence that other self-reported characteristics of officers are significantly correlated with their value added. In particular, neither officers' perceived hard nor soft skills help explain variation in officers' value added. This includes core competencies for procurement officers, such as knowledge of the relevant market, familiarity with procurement methods, and the ability to choose contract parameters and procurement procedures that ensure cost-effective outcomes.[1]

Our data and empirical strategy do not allow us to identify the precise mechanisms underlying the association between legal background, legal engagement, and project quality. Nonetheless, two mechanisms can help explain the observed patterns. First, procurement officers may face a trade-off between mission-oriented objectives and legal compliance, balancing project quality against procedural risk. Greater legal engagement may shift this balance toward compliance, particularly in complex projects where following the rules can limit flexibility in contract design. Second, occupational sorting and comparative advantage may matter. In the absence of formal selection or certification systems, individuals may sort into procurement roles based on their skill, and legal training may confer a comparative advantage in managing contractual risk rather than optimizing performance (Henderson et al., 2025) leading to systematic differences in value added.

This paper introduces a research design that links survey-based measures of procurement officer competences to administrative data on contract-level performance, enabling the study of bureaucratic characteristics in a value added framework. Existing procurement literature relies extensively on administrative datasets (among others, see Bajari et al., 2014; Baranek and Titl, 2024; Coviello and Mariniello, 2014; Decarolis, 2018; Titl and Geys, 2019; Titl et al., 2021), less often uses surveys with real public procurement officials (see Tukiainen et al., 2024)[2], but rarely

[1]This does not mean that these skills are unimportant or that they cannot, in fact, explain variation in officers' value added; rather, officers may not be able to accurately self-assess these skills.

[2]An extensive literature in management and supply chain research uses surveys and qualitative methods with purchasing and supply management professionals (Giunipero, 2000; Knight et al.,

combines the two (Bandiera et al., 2021a)[3]. By linking self-reported competences and characteristics to realized quality outcomes for the universe of all construction contracts in the Czech Republic, we provide large-scale evidence connecting bureaucratic characteristics to ex-post public service delivery. Although our results are based on Czech procurement officers, all public buyers in the EU operate under the same European public procurement directives. Thus, our approach is potentially scalable and replicable in other countries.

Our findings carry clear policy implications: procurement officers play a central role in determining the quality and effectiveness of public procurement. Existing policy debates and much of the literature focus primarily on procurement rules and standards, yet our results suggest that comparable attention should be paid to the individuals responsible for implementing tenders under the rules. Effective procurement policy requires both well-designed regulations and sustained investments in the selection, training, and professional development of procurement officers[4]. Our findings align with those of Bosio et al. (2022), who show that procurement practices, rather than formal rules, are positively associated with higher procurement quality. All in all, this paper findings can inform the assessment of recently developed guidelines in the public procurement field regarding skills training and professional certification (OECD, 2025b). Our results imply that buyers could benefit from trainings in how to conduct good procurement, rather than primarily focusing on knowing the rules.

Our study contributes to three strands of the literature. First, we contribute to the public procurement literature by proposing an innovative approach to measuring the quality procurement project. Measuring quality in public procurement is inherently challenging, as post-award quality assessments are rarely collected or systematically reported. Existing studies, therefore, rely on indirect measures such as cost overruns and time delays (Coviello et al., 2018; Decarolis, 2014; Best et al., 2023). We adopt a novel quality measure proposed by Baranek and Titl (2024), which is specific to the construction sector and rests on the premise that higher-quality projects require fewer and less costly repairs. Accordingly, we define project quality as the negative value of follow-up repair expenditures, expressed as a share of the initial procurement contract value. To the best of our knowledge, we are the first to estimate individual officer value added in a setting where the outcome is an objective measure of realized project quality in a complex contracting environment such as the construction sector. Our approach allows us to move

2014; Stek, 2021; Stentoft Arlbjørn and Vagn Freytag, 2012). This descriptive evidence highlights systematic differences between public and private sector procurement, suggesting that the public sector involves additional incentives and constraints such as transparency requirements, budget efficiency and political accountability (Rejeb et al., 2023).

[3]One important study is the large scale experiment conducted by Bandiera et al. (2021a), whose design randomly varied incentives and autonomy across procurement officers, and linked the experimental design to actual purchases decisions. They find that increased autonomy reduced procurement prices, but explicitly note that their results may not generalize to complex procurement environments in which the quality dimension is salient.

[4]The selection and retention of public officers tasked with administrative responsibilities have attracted increasing scholarly attention (Brierley et al., 2023).

beyond organizational level heterogeneity (Bandiera et al., 2009, 2021a; Allende et al., 2024; Bucciol et al., 2020; Decarolis et al., 2020, 2021) and beyond proxy performance indicators such as completion rates (Rasul and Rogger, 2018). We move this discussion forward by measuring completion and follow-ups of construction projects and showing that quality of procurement, and not only prices, vary systematically with respect to individual buyers. The level of granularity of our value added measure is related to the work of Best et al. (2023), who construct a non-price index of spending quality and show, using variance decomposition, that individual officers explain less variation in quality than in prices. In contrast, our findings indicate that individual officers explain economically significant variation in realized quality outcomes in complex contracts, suggesting that officer heterogeneity becomes particularly salient when procurement involves non-standardized projects and non-contractible quality considerations.

Second, our research speaks to the literature of discretion in public procurement. We provide evidence of the role of discretion and public sector employees' performance in ex-post contract procurement outcomes. While the literature has mostly focused on procurement processes performance, such as supplier selection where discretionary decisions are relevant (see Baltrunaite et al., 2021; Coviello et al., 2018; Szucs, 2024; Celis Galvez et al., 2025; Baranek and Titl, 2024). Recent evidence seems to confirm the benefits of granting these public workers more discretion in countries with high public capacity (Bosio et al., 2022; Bandiera et al., 2021a; Rasul and Rogger, 2018; Duflo et al., 2018). We contribute to this literature by showing that procurement officers contribute to long-term procurement outcomes that materialize after the contract award, where discretion is less prominent.

Third, our findings contribute to the public economics literature on bureaucratic capacity and state effectiveness (Bertrand et al., 2020; Besley et al., 2022). We provide suggestive evidence that officers' law background and their preference for legal compliance posit a trade off in achieving higher public service quality. These results are consistent with models in which motivated agents differ in mission preferences and intrinsic motivations (Tukiainen et al., 2024; Besley and Ghatak, 2005; Ashraf et al., 2020; Barfort et al., 2019; Banuri and Keefer, 2016). Moreover, our findings relate to policy discussions emphasizing the professionalization of the public procurement workforce, particularly in procurement and financial management (Saussier and Tirole, 2015; OECD, 2009). Future research could explore how training civil servants with diverse occupational and educational backgrounds affects procurement outcomes. While evidence suggests that training managers increases the use of strategic goals in decision-making (Kroll and Moynihan, 2015) and that assigning better managers to larger offices increases productivity (Fenizia, 2022), little is known about the effects of training and allocating lawyers or technical civil servants in administrative roles. By estimating individual value added in public procurement, we extend the worker effects framework of labor economics to the public sector and document that bureaucrats; similar to teachers (Chetty et al., 2014a,b; Bau and Das, 2020), managers (Bertrand and Schoar, 2003), and firm workers (Abowd et al., 1999); exhibit persistent, economically meaningful produc-

tivity differences that aggregate into state capacity.

The remainder of the paper is organized as follows. Section 3.2 presents the institutional background. Section 3.3 describes the data sources. Section 3.4 introduces the empirical approach. Section 3.5 presents the results. Section 3.6 ends with a discussion and conclusion of the results.

3.2 Institutional context

The public procurement workforce handles approximately 2 trillion Euros per year on the purchase of services, works and supplies in the European Union (EU) (European Commission, 2026b) and faces increasing demands to carry sustainable and innovative contracting (OECD, 2023). Nonetheless, only one-fifth of the member states of the EU report having a formal definition of the occupation for public procurement officers (European Commission, 2020a, page 15)[5]. Without a standard accepted definition, the occupation is associated with different roles and professions (e.g. purchaser manager, procurement lawyer). This diversity has made salient the mismatch of skills of public buyers, and the lack of technical knowledge and expertise of procurement officials (Saussier and Tirole, 2015; OECD, 2021). Diverse definitions of the occupation might disrupt the channels that procurement officers' decisions and practices take to affect procurement outcomes, especially long-term ones such as project quality and ultimately a fragmented design of targeted training programs.

In response to these institutional challenges, the European Commission launched the European Competency Framework for Public Procurement Professionals (hereafter, ProcurCompEU) (European Commission, 2026a) with the goal of providing a common reference for the competences required for the occupation. ProcurCompEU is a comprehensive framework consisting of 30 competences, classified by levels of expertise and hard and soft skills, that public officers are expected to demonstrate to perform their job effectively and achieve value for money (European Commission, 2020a, page 17). An initial assessment of the ProcurCompEU reveals a predominance of legal and procedural training to ensure compliance, which suggests a lack of training in innovation, sustainability, and quality criteria (European Commission, 2020b, page 16-17). Additionally, the framework has been piloted in countries outside the European Union (OECD, 2023) and has led to further steps for developing certification schemes for the procurement occupation (OECD, 2025b). The findings and policy actions are relevant to our research, as they indicate that similar gaps in skill development may arise in other institutional contexts.

The ProcurCompEU framework serves as a building block for this study. We construct a questionnaire (see Table 3.A.1 in the Appendix) based on the definitions of knowledge and skills included in the framework and apply it to public

[5]For OECD countries this proportion is slightly higher, one third of member countries report formally acknowledging public procurement officers as a standalone profession (OECD, 2021, page 170)

procurement officers in the Czech Republic. The Czech public procurement system is primarily governed by EU procurement directives as transposed into national law[6]. There is no uniform statutory requirement for the mandatory certification of procurement officers. However, the National Public Procurement Strategy for the period 2024–2028 (Ministry of Regional Development of the Czech Republic, 2024) identifies the professionalization of public procurement as a key priority, emphasizing the need to strengthen skills, knowledge, and awareness among procurement officers, as well as to develop a national competence framework and targeted educational programs for procurement officers[7].

Existing training initiatives remain largely focused on legal and procedural aspects. The European Commission (European Commission, 2020b) reports seminars and trainings offered by the Ministry of Regional Development and Office for the Protection of Competition, the latter organizes an annual Methodological Days on Public Procurement [8]. This institutional context makes the Czech Republic a relevant setting to study potential gaps in the development of procurement competences.

3.3 Data

3.3.1 Public procurement contracts

We use the universe of Czech public procurement contracts with estimated values above the applicable thresholds, which are approximately EUR 240,000 for construction contracts and EUR 80,000 for other contracts. By law, procurement authorities are required to disclose detailed information about all contracts with an estimated value above the aforementioned thresholds in an online system called the Gazette of Public Procurement. We use a hand-cleaned version of the dataset processed by a private company called Datlab s.r.o.

This dataset contains date information when the contract was awarded, the estimated value of the contract, the number of bidders, the winning bid, the final price of the contract, including extra payment due to cost overruns, the selection method, public procurement procedures used to award the contract, the exact identity of a buyer, and the email address of a responsible procurement officer. Such detailed information on individual public procurement contracts allows us to estimate contract quality based on the total cost of repairs and to aggregate contract characteristics at the procurement officer level, identified by their email address.

[6]Total public procurement spending in Czechia amounts to about 12 % of GDP (OECD, 2025). In 2024, this corresponded to approximately CZK 1 trillion (around EUR 41 billion). Public procurement is carried out by central government bodies, regional governments, municipalities, and other public institutions, including hospitals, schools, and other statutory public-law entities.

[7]The National Public Procurement Strategy in the Czech Republic (2024–2028) launched an online portal providing webinars and training for procurement civil servants in line with the European Commission's Competence Framework (See portal-vz.cz/vzdelavani)

[8]The methodological days organized by *The Office for the Protection of Competition* https://uohs.gov.cz.

3.3.2 Survey

We conducted a large-scale survey with procurement officers to elicit their unobserved characteristics, including self-assessed skills and knowledge, their intrinsic motivation, and whether they work in teams. The survey was designed as a battery of statements, and participants were asked to indicate the extent of their agreement on a five-point Likert scale: strongly agree, agree, neutral, disagree, or strongly disagree. The questionnaire is mostly based on the ProcurCompEU framework (see 3.2). The questions on motivation were formulated based on the World Values Survey. The original questions were prepared in Czech; their English translations are provided in the appendix (see Table 3.A.1).

To invite participants to complete the survey, we used email addresses provided in tender documentation and available in the dataset covering the universe of Czech public procurement contracts. The survey was conducted between May and July 2023. In total, 46,077 email addresses were invited to participate. A reminder invitation was sent two weeks after the initial mailing. Approximately 8% of those invited initiated the survey, and 3.1% of invitees completed it in full.

Table 3.A.2 presents the distribution of responses across fourteen survey-based measures of procurement officers' competences and motivations. Overall, the distributions are skewed toward agreement, indicating that respondents tend to assess their competences positively. For most variables, the modal category is agree often followed by strongly agree, while the shares of disagreement remain consistently low.

Within the domain of hard skills and knowledge, responses suggest a moderate self-assessed proficiency. Skills such as negotiation, procedure selection, and dispute resolution exhibit relatively high agreement rates, with over half of respondents selecting agree. In contrast, more specialized areas such as market knowledge and procurement methods display greater dispersion, with a substantial proportion of neutral responses, around one-third of the answers, suggesting more heterogeneity in perceived expertise.

The soft skills measures display the strongest positive skew. Communication, collaboration, and time efficiency all show very high shares of agreement and strong agreement, with disagreement rates close to zero. This pattern is consistent with findings in the literature that soft skills are often perceived as core professional competences in the public sector (Ashraf et al., 2020) and in our case these skills may be more uniformly distributed across procurement officers, in line with the findings of Tukiainen et al. (2024).

3.3.3 Datasets merge

We use the email addresses provided in the tender documentation to invite procurement officers to participate in the survey and subsequently to merge their survey responses with estimates of their value added to project quality. This novel approach enables procurement-officer-level analysis, but it is also subject to several caveats.

First, because the procurement dataset started several years before the survey was conducted, some email addresses obtained from the public tender database may have been deactivated or otherwise become obsolete by the time of the survey; for example, due to procurement officers changing positions, moving to different organizations, or changes in email address structures at contracting authorities. This reduced the effective universe of potential survey participants and mechanically lowered the response rate. Contracts linked to these email addresses are still used to estimate the distribution of procurement officer value added, but they cannot be included in the subsequent analysis that explores heterogeneity in value added across officers.

Second, some contracts are associated with multiple responsible procurement officers. In these cases, we link each contract to every officer involved and treat each contract–officer combination as a separate observation. As a result, contracts associated with multiple email addresses appear multiple times in the data—once for each officer—while sharing the same contract information.

From a universe of 233,090 tenders, we observe 21,537 unique email addresses, which we use to estimate the value added of procurement officers. In the administrative data, tenders are associated with one or more email addresses. Specifically, 41.6% of contracts report a single contact email address, suggesting that in many cases one procurement officer serves as the main contact point responsible for the tendering process. In addition, 54% of contracts list two email addresses, indicating that multiple procurement officers are often involved in the same contract, while contracts with more than two email addresses are relatively rare, accounting for 4.4% of the sample. After conducting the survey, we are able to link 10.7% of contracts to the survey, corresponding to 24,957 contracts and 1,074 completed questionnaires. Among the contracts linked to the survey, 7,168 contracts have one email address, while 15,439 contracts have two email addresses but only one of them completed the survey. In 743 contracts with two or more email addresses, two officers completed the survey, and in four contracts three email addresses are listed and all three officers responded. Procurement officers are associated with one or more contracts: in the administrative data, 43.9% of email addresses are linked to a single contract, 16.8% to two contracts, and the remaining 39.3% to three or more contracts. Similarly, among the officers who completed the survey, 358 (33.3% of the 1,074 respondents) are associated with a single contract, while the remaining 66.7% are associated with two or more contracts.

3.4 Empirical approach

This section describes our empirical approach to quantify the value-added of procurement officers to project quality in public procurement. We begin by introducing a novel measure of project quality based on ex post repair costs, which captures the durability and performance of completed construction projects. Building on this outcome variable, we implement two steps. First, we estimate procurement officers' value added by exploiting variation in project quality across contracts. Second, we

examine the determinants of this estimated value added by relating it to officers' self-reported skills, preferences, and characteristics.

3.4.1 Quality of the procurement projects

Measuring the quality of procurement projects is challenging. Public projects have multiple dimensions that plausibly matter for quality assessment, and what constitutes quality may vary across types of projects. For public contracts involving relatively homogeneous goods for which there is little variation in the quality of the delivered item (e.g., pharmaceuticals), other aspects of the contract are more informative indicators of quality, such as delivery time, contractual clauses, or supplier reliability. For more complex projects, cost overruns or delays, both used as quality measures in the previous literature, are likely to be more suitable proxies for project quality (Coviello et al., 2018; Decarolis, 2014).

We focus on the quality of the substance: the quality of the product delivered in the specific setting of construction contracts. In particular, we argue that the costs of subsequent repairs provide a useful and observable measure of the quality of the initial contract (Baranek and Titl, 2024; De Jaegher et al., 2025). Consider two contracts that each deliver 50 km of highway for EUR 10 million. Over the subsequent five years, however, the first project requires repairs costing EUR 500,000, whereas the second requires repairs of only EUR 200,000. We argue that the higher repair costs of the first project indicate that it was initially delivered at a lower quality. While reducing project quality to a single dimension, such as future repair costs, is informative in the context of construction contracts, its applicability to other types of public projects may be limited and may not fully capture overall project quality.

In our measure of project quality, aspects such as contractual clauses have only instrumental value, as they serve as means through which the procuring authority seeks to ensure a higher quality of the delivered project.

3.4.2 Value added to project quality

Our research design is based on value-added models, which isolate the contribution of an individual input, such as a teacher, manager, or public officer, to an outcome of interest (McCaffrey et al., 2004; Koedel and Rockoff, 2015). A large literature in education economics uses value-added models to estimate teachers' value added to student achievement (Chetty et al., 2014a,b; Bau and Das, 2020). More recently, this approach has been adapted to study bureaucrats and managers in the public sector (Bertrand et al., 2020; Limodio, 2021). We build on this literature to estimate the extent to which a procurement officer contributes to project quality. We use the novel measure of public procurement quality for each contract as the outcome variable and propose to estimate the regression in equation 3.4.1 with a set of observed contract characteristics and the fixed effects of procurement officers as explanatory variables.

$$\text{Project Quality}_{itj} = \eta_i + \psi X_{tj} + \zeta_t + \mu_{itj} \tag{3.4.1}$$

where η_i represents officers' fixed effects, X includes characteristics of the tender j that are time-varying controls, ζ_t are year fixed effects, and μ_{itj} is an error term. We are interested in the vector of η's, capturing individual officer value added we observe across multiple tenders. η is not a causal contribution of officer i to project quality, but a key input. From our specification we are able to estimate η separately from other fixed effects as long as specific procurement officers work on different tenders.

3.4.3 Officers' characteristics and their value added

We examine which self-reported skills and individual characteristics of procurement officers help explain their value added to Project Quality. We propose to estimate regressions as shown in equation 3.4.2. The right-hand-side variable in this equation is the estimated vector $\hat{\eta}_i$, which is the individual procurement officer value-added from equation 3.4.1, thus $OfficerValueAdded_i \equiv \hat{\eta}_i$. In the left-hand-side we include measures of preferences Prefer_i^F and skill Skills_i^F where the index F refers to different definitions from the European Competency Framework (see section 3.2). Finally, we include officers' characteristics X and an error term ε_i.

$$\text{Officer Value Added}_i = \alpha_0 + \beta_1 \text{Prefer}_i^F + \gamma_1 \text{Skills}_i^F + \delta_1 \text{X}_i + \varepsilon_i \tag{3.4.2}$$

Similar regressions for exploring patterns of public buyer characteristics have been estimated with organizations as the unit of analysis (Allende et al., 2024; Bandiera et al., 2009). To the best of our knowledge, we are the first to estimate how much individual officers' self-perceived preferences and skills explain their value added in the context of public procurement and administrative bureaucrats. This exercise is exploratory, but aims to document whether there are overarching patterns in procurement skills and professional background, which is the blind spot in the current literature, and can inform policy making.

3.5 Results

3.5.1 Public projects' quality

To construct the quality measure for a given contract, we sum the estimated value of all repair procurement contracts awarded within five years after the initial contract.[9] To ensure comparability across contracts of different types and sizes, we express these additional costs relative to the estimated value of the initial contract. Finally, we multiply this ratio by minus one so that the index decreases with higher repair costs and increases with higher project quality. In the illustrative example

[9]To provide more robust evidence, we replicate the measure using both shorter and longer time windows and find no qualitative differences in the results. The trade-off in choosing the window length is that a shorter period provides limited variation, since most projects have not yet required repairs, while a longer period substantially reduces the sample of contracts, as it includes only contracts awarded in the early years of the study period.

considered above, the quality measure for the first contract equals $-1 \times \frac{500{,}000}{10 \text{ million}} = -0.05$, while for the second contract it equals $-1 \times \frac{200{,}000}{10 \text{ million}} = -0.02$.

Ideally, one would observe all follow-up repairs associated with each public procurement contract. Unfortunately, no dataset we are aware of provides such information. In our setting, however, we benefit from the fact that repairs are typically implemented through new procurement contracts, which are themselves included in our dataset. To address the absence of unique identifiers linking initial contracts to their subsequent repairs, we use text-based matching algorithms to identify the original contract and its associated repair tenders. Manual validation indicates that approximately 93% of the matches are correct (Baranek and Titl, 2024). Based on this matching exercise, we then compute the quality measure as described above.

The quality of the delivered contract is a primary policy concern, as low-quality projects generate higher fiscal expenditures through unnecessary repairs and impose additional non-fiscal costs on society, including extended traffic restrictions or closures. Compared with other indicators, such as project delays or final price, our measure has the clear advantage of being contemporarily unobserved by procurement officers and their supervisors, since the quality of a delivered project only becomes apparent over time. Consequently, it cannot be incorporated into incentive schemes for the procurement officers and is therefore unlikely to be deliberately manipulated.

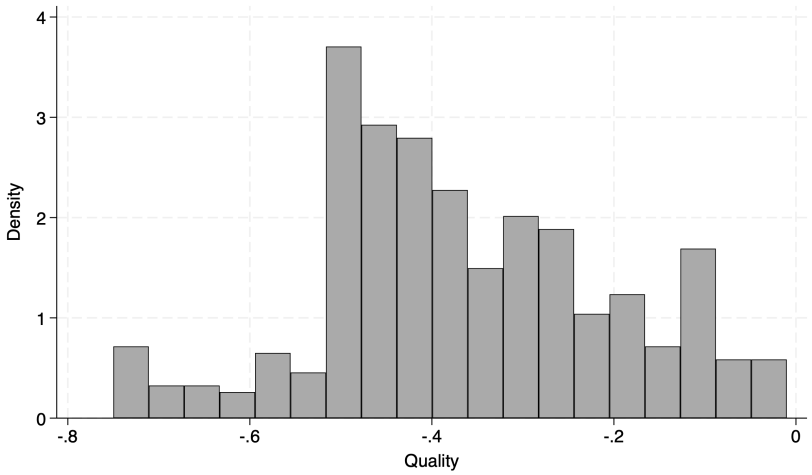
Importantly, although the construction project is ultimately delivered by the supplier, procurement officers can influence the expected quality of the delivered project through multiple channels at different stages of the procurement process. The most influential stage is likely prior to supplier selection and contract award. At this stage, officers can ensure high-quality tender documentation; clear, well-designed tender documents are more likely to attract a larger pool of bidders, thereby increasing the probability of selecting a reliable and capable supplier. Officers can then apply a selection procedure to choose the *best* offer and prepare a contract that specifies clear, enforceable quality standards. Once the contract is awarded and signed, procurement officers remain responsible for contract management and oversight. In complex projects with long implementation periods, delivery typically involves multiple stages at which officers can intervene to limit deviations and safeguard quality; for instance through monitoring, coordination, and enforcement. Finally, at the completion stage, procurement officers are responsible for formally accepting the delivered project, which provides a final opportunity to verify compliance with contractual quality requirements. Overall, there are many stages in the life cycle of a tender (e.g. from preparation of tender documentation to final delivery) at which the procurement officer can influence the quality of the project delivered.

Our primary measure of quality sums the value of repairs needed over a 5-year horizon after the initial project.[10] Only a fraction of contracts required repairs, while almost 95% of contracts require no repairs within the 5-year horizon. Among

[10]We replicate the results with a 3-year horizon in the appendix.

projects that required repairs, the mean value of repair is 37% indicating that the average repair costs were 37% of the initial project's estimated cost. In Figure 3.1, we plot the distribution of quality across procurement projects in our sample.

Figure 3.1 Public projects' quality



Notes: This figure displays the histogram of project quality. For the sake of clarity, we plot only non-zero quality scores. Quality is defined as the value of all follow-up repairs within five years after the contract award divided by the value of the initial project multiplied by minus one.

3.5.2 Value added of procurement officers

In our primary specification of equation 3.4.1, X includes selection procedure, evaluation criteria, the number of bidders, the buyer category[11], a dummy variable indicating whether the public tender was funded by the EU, size of the team, fixed effect for the year when the tender was announced, and fixed effect of the procuring organization. We refer to the value added derived from this specification as the primary measure of value added.

Some characteristics of the tender may be driven by the procurement officers themselves. For example, *better* officers may be both more likely to ensure high-quality project delivery during the monitoring stage and more willing to bear the higher costs associated with organizing an open tender procedure. In this case, controlling for the selection procedure would bias the estimated officers' value added, as it would absorb part of the variation in officer performance. We therefore re-estimate the model while controlling only for the year in which the tender was announced, procuring-organization fixed effects, and the buyer category. We refer to this specification as an alternative measure of officers' value added and present the results alongside those based on the primary measure.

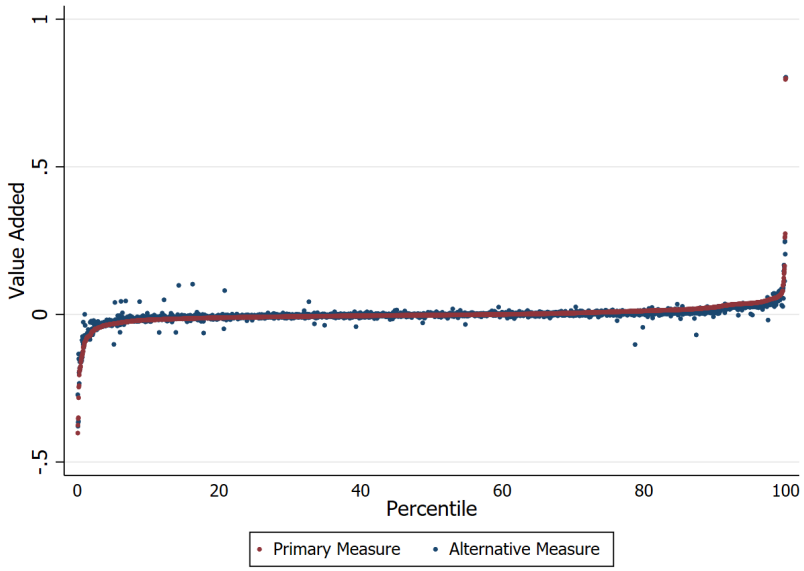
We exclude officers who managed fewer than four procurement projects. The results are qualitatively similar when using alternative cutoffs of seven and ten contracts, although, naturally, these alternative specifications use fewer observations.

[11]We split the buyers into 6 categories National authority, regional authority, public body, National agency, regional agency; European Agency and other

A procurement officer's value added can be interpreted as the average conditional contribution of a given procurement officer to the quality of the projects they managed. We plot the coefficients η, from estimating equation 3.4.1, in Figure 3.2. Coefficients from the primary measure are shown in red, while those from the alternative measure (omitting most of the control variables form regression 3.4.1) are shown in blue. Both sets of coefficients are ordered according to their values under the primary measure. A comparison of the two measures reveals only minor differences: officers with higher value added under one measure also tend to exhibit higher value added under the alternative measure. The correlation between the primary and alternative measures of procurement officers' value added is 97% (see Figure 3.A.3 in the appendix).

To demonstrate the variation in the officers' value added, consider a procurement officer at the 10th percentile whose contribution is around -2% and a procurement officer at the 90th percentile whose contribution is 2.5%. Given that the median of the estimated value of the contracts considered is CZK 11.5 million (and the mean contract value of CZK 46 million), replacing the procurement officers from the 10th to the 90th percentile implies a cost reduction in the future repairs of CZK 520,000 (EUR 21,000) for each contract of the median value and CZK 2.1 million (EUR 86,000) for the average contract.

Figure 3.2 Procurement officers' value added



Notes: This figure displays both the primary and alternative measures of procurement officers' value added. Procurement officers are ordered from those with the most negative contribution to project quality on the left—corresponding to a higher relative value of repairs—to those with the highest positive conditional contribution on the right—corresponding to a lower relative value of repairs. Replacing a procurement officer at the 10th percentile with one at the 90th percentile would reduce expected repair costs by approximately CZK 520,000 (EUR 21,000) for a median-sized tender and by about CZK 2.1 million (EUR 86,000) for an average-sized tender.

3.5.3 Procurement officers' and quality of public projects

Having documented variation in procurement officers' value added to the quality of procurement projects, we extend our analysis by examining survey-based measures of procurement officers' characteristics to help explain this variation. This exercise is exploratory in nature and does not support causal claims; nevertheless, the observed patterns are stable across various specifications and provide useful guidance for future research. Merging the value-added dataset with the survey respondents reduces the sample to 472 procurement officer–level observations.[12]

Survey participation. We first examine the characteristics of officers who participated in the survey. In particular, we estimate a regression predicting survey participation as a function of procurement officers' value added. Table 3.1 shows that the estimated coefficient is statistically insignificant across six different specifications, varying both the estimation method (LPM, probit, and logit) and the primary and the alternative measures of procurement officers' value added. Figure 3.A.5 further illustrates that officers who participated in the survey are distributed relatively evenly across the value-added distribution, with the possible exception of the top and bottom 1% of the value added. Overall, these results suggest that the sample of officers who completed the survey is representative of the value added in the population of procurement officers.

Table 3.1 Participation regression

	(1)	(2)	(3)	(4)	(5)	(6)
	Survey Particiaption (= 1)					
Value Added (primary)	-0.135	-1.551	-2.871			
	(0.105)	(1.092)	(2.079)			
Value Added (alternative)				-0.140	-1.685	-3.067
				(0.113)	(1.209)	(2.270)
Constant	0.0575***	-1.578***	-2.801***	0.0593***	-1.563***	-2.769***
	(0.004)	(0.035)	(0.073)	(0.004)	(0.033)	(0.070)
Observations	3,472	3,472	3,472	3,724	3,724	3,724
Model	LPM	Probit	Logit	LPM	Probit	Logit

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports regression results examining whether procurement officers' value added predicts survey participation. Columns (1)–(3) use the primary measure of value added, while columns (4)–(6) use the alternative measure. For each measure, we estimate linear probability Model (LPM), probit, and logit models. Across all specifications, value added has no statistically significant predictive power for procurement officers' survey participation.

Legal engagement and legal education. We next present findings focusing on two variables that relate to officers' legal background and legal exposure. In par-

[12]The 472 observations includes procurement officers who are later excluded from the analysis because they did not handle a sufficient number of procurement tenders.

ticular, we show that officers with a legal education and those who report paying closer attention to legal regulation tend to underperform in terms of the quality of procurement projects. Following equation 3.4.2, we estimate the following regression 3.5.1:

$$\begin{aligned}\text{Officer Value Added}_i = \alpha + \beta_1 \text{Legal Engagement}_i + \beta_2 \text{Legal Education}_i + \\ + \gamma_1 \text{Hard Skills \& Knowledge}_i + \gamma_2 \text{Soft Skills}_i + \gamma_3 \text{Motivation}_i + \\ + \delta_1 \text{Type of Buyer}_i + \delta_2 \text{Personal Characteristics}_i + \varepsilon_i, \end{aligned} \tag{3.5.1}$$

where Officer Value Added denotes the procurement officer's value added estimated in regression 3.4.1, β_1 and β_2 are the two parameters of our interest. The former one, β_1, captures the effect of Legal Engagement, i.e., the extent to which officers agree, on a five-point scale, with the statement that they monitor legal and legislative developments in the field of public procurement; the latter one, β_2, estimates the effect of a dummy variable indicating whether the officer holds a legal degree.

Hard skills & Knowledge (γ_1) comprise a set of self-reported variables that include officers' assessments of their negotiation skills on a five Likert scale; knowledge of the relevant market; knowledge of procurement methods; ability to choose contract parameters and procurement procedures to ensure cost-effective outcomes; ability to resolve disputes; ability to manage sustainable public procurement; and ability to identify and manage risks associated with public contracts.

Soft Skills (γ_2) comprise a set of self-reported variables capturing officers' ability to communicate effectively with suppliers, collaborate with other departments within the same organization, and manage their time efficiently.

The third set of survey variables, **Motivation** (γ_3), includes assessments of whether officers' work attitudes are driven by goal-oriented motivation, identification with institutional goals, and the perception that their job contributes to the provision of public goods.

In some specifications, we additionally include **buyer-category** (δ_1) fixed effects (national authority, regional authority, public body, national agency, regional agency, European agency, and other), and other **Personal Characteristics** (δ_2) of the officer, including experience, the number of completed training courses, professional certifications, and whether the officer serves as a specialist or a supervisor within the team.

Table 3.2 reports results from four specifications using the primary measure of value added as the outcome variable. Column (1) presents results from the simplest specification, which includes only legal engagement and legal education and no additional control variables. Column (2) adds hard skills & knowledge, soft skills, and motivation controls as specified in regression 3.5.1. Columns (3) and (4) further control for buyer category fixed effects and other personal characteristics of the officers.

Across all specifications reported in Table 3.2, the point estimates for both legal engagement and legal education are negative and statistically significant. Officers with a legal education deliver, on average and conditional on their level of legal engagement, projects that are 1.3 points lower in quality (and 1.8 points lower in

the fully saturated specification in column (4)); their projects thus exhibit relative costs that are approximately 1.3% higher than those handled by officers without a legal education.[13] For legal engagement, a one-step increase in the response scale is associated with a 0.3 point reduction in project quality.

Table 3.2 Exploring officers value added: legal engagement and education

	(1)	(2)	(3)	(4)
	Procurement Officer Value Added			
Legal Engagement	-0.003*	-0.006**	-0.006**	-0.005**
	(0.002)	(0.003)	(0.003)	(0.003)
Legal Education	-0.013**	-0.013**	-0.013*	-0.018**
	(0.006)	(0.006)	(0.007)	(0.008)
N	199	171	171	171
Skills & Motivation	No	Yes	Yes	Yes
Type of Buyer	No	No	Yes	Yes
Other Controls	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

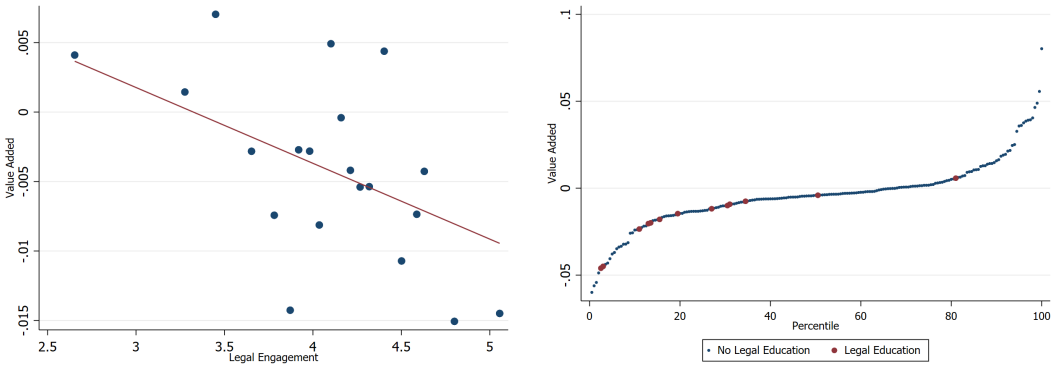
Notes: This table reports results from four specifications using the primary measure of value added as the outcome variable. Column (1) presents results from a simple regression that includes only legal engagement and education. Columns (2)–(4) report results from the specifications outlined in regression 3.5.1. Across all specifications, coefficients for both legal engagement and legal education are negative and statistically significant at least at the 10% significance level.

We replicate the estimates using the alternative measure of procurement officers' value added. The results, presented in Table 3.A.8 in the appendix, show that all coefficients of interest remain negative. The estimated effect of legal engagement is of a similar magnitude, between 0.3 and 0.4 points, and remains statistically significant, while the effect of legal education remains negative but shrinks to roughly half of its original size.

In addition, because legal engagement is measured on a five-point Likert scale that does not necessarily carry cardinal properties, we reclassify officers' responses into a binary indicator which equals to one if the officer agrees or strongly agrees with the statement, and zero otherwise. Table 3.A.6 shows that the estimated effect of legal engagement roughly doubles, yielding magnitudes comparable to those obtained under the primary specification when used the five-point scale. Officers who agree or strongly agree that they monitor legal developments deliver lower-quality projects by approximately 0.8 points compared to those who do not agree with that statement.

[13]Because our survey does not cover officers in the top and bottom 1% of the value-added distribution, any asymmetric sorting of legally educated officers into these tails would attenuate the estimated effect, implying that the true effect may be even more negative.

Figure 3.3 Role of legal engagement and legal education



(a) Legal Engagement **(b)** Legal Education

Notes: The left panel shows a negative relationship between procurement officers' value added and marginal effect of legal engagement: officers who report more intensive monitoring of legal developments in public procurement tend to have lower (more negative) value added. The relationship is shown conditional on the covariates included in Equation 3.5.1. The right panel plots value added for officers who participated in the survey, highlighting those with a legal education in red color. Officers with legal education are disproportionately concentrated in the lower part of the value-added distribution, consistent with their average negative contribution to project quality.

Figure 3.3 illustrates these effects graphically. The left panel shows a negative relationship between procurement officers' value added and their reported extent of monitoring legal developments in public procurement, conditional on all control variables included in regression 3.5.1. It reveals a clear downward-sloping relationship: the higher the level of legal engagement an officer reports, the lower their value added; this pattern is robust to the influence of outliers. The right panel of Figure 3.3 presents the value-added distribution of surveyed procurement officers, highlighting those with a legal background in red. Officers with legal education appear disproportionately concentrated on the left side of the distribution, corresponding to lower (more negative) value added.

In the appendix, we replicate the main results using an alternative measure of project quality that counts repairs over a three-year horizon. Measuring repair costs over a shorter period leads to lower overall repair costs, more frequent zeros, and reduced variation. This is reflected in the distribution of procurement officers' value added, as the gap between the 10th and 90th percentiles shrinks to 4.2 points, compared to 4.5 points in the primary specification. The established underperformance of procurement officers with legal education and higher legal engagement is replicated and remains quantitatively similar to the main results.

In the Table 3.A.7, we present the full specification of the regression 3.5.1. Columns (2-4) indicate a positive association between collaboration skills and value added. The point estimates suggest that an increase in the response scale is associated with a 0.5 point increase in project quality. This particular result aligns with previous discussions that public procurement is an interdisciplinary activity requiring effective within-office collaboration (Decarolis et al., 2020).

We find no evidence that other self-reported preferences or skills are correlated with officers' value added. In particular, neither perceived hard nor soft skills ex-

plains variation in performance. This includes core procurement competencies such as knowledge of the relevant market, familiarity with procurement methods, and the ability to select contract parameters and procedures that ensure cost-effective outcomes. This lack of evidence is interesting, given that these competences are emphasized in the Competences Framework promoted by the European Commission (European Commission, 2020a) and the OECD (OECD, 2023). At the same time, the heterogeneity in value added we document, mainly driven by officers with a legal orientation, is consistent with the broader literature showing that individual bureaucrats influence public sector performance despite differing backgrounds, preferences, motivations, and incentives (Bandiera et al., 2021a; Tukiainen et al., 2024).

3.6 Discussion and conclusion

High-quality public procurement is vital for the effective functioning of government, yet relatively little is known about how to measure such quality, how much variation exists, and what drives differences across procurement officers. We address these questions by building on a recently introduced measure of public procurement quality that derives project quality from the value of necessary repairs incurred over a given period after project delivery. Empirically, we propose a novel research approach to measuring procurement officers' contribution to public procurement quality, by combining administrative procurement data with our a unique survey of officers' self-reported competences. We document two central findings. First, procurement officers exhibit substantial heterogeneity in their contribution to project quality, even within a highly regulated environment. Second, officer background, particularly legal training, is systematically associated with lower project quality conditional on engagement.

We argue that procurement officers can affect project quality through multiple channels, ranging from the pre-award phase to contract design and monitoring during project implementation. Compared to existing alternative quality measures (Best et al., 2023; Rasul and Rogger, 2018), the proposed measure has the advantage of being less susceptible to manipulation by procurement officers seeking to satisfy contemporaneous incentive schemes, as the need for repairs is typically revealed only with a delay. We find sizeable variation in officers' contributions. The gap between officers at the 10th and 90th percentiles of the value-added distribution is 4.5 points, which corresponds to a difference of approximately CZK 520,000 (EUR 21,000) in expected repair costs for a median-size contract and roughly three times that amount for a mean-size contract. These estimations highlight that, despite regulations and rules that constrain public tenders, procurement officers play an important role in ensuring a well-functioning public procurement system. Our findings support previous literature showing that public procurement officers perceive they can influence tender outcomes through their own work (Tukiainen et al., 2024). Furthermore, the variation we find suggests that individual officers constitute a quantitatively important input into procurement outcomes.

We also find that procurement officers with a legal background are associated with lower project quality. Conditional on their level of legal engagement, these officers deliver projects that are 1.3 points lower in quality, corresponding to approximately 1.3% higher expected costs relative to officers without legal training. One interpretation is that legal training shifts the emphasis of procurement decisions toward compliance and the mitigation of legal risks, potentially at the expense of performance-oriented dimensions of contract design. In a setting where officers trade off project quality against the risk of litigation or procedural non-compliance, legally trained officers may place greater weight on limiting ex post risks, leading to more conservative procurement choices. This interpretation is consistent with recent literature showing that legal training is associated with reducing firms' litigation risks (Henderson et al., 2025) and the allocation of contractual risk (Karsten et al., 2022). This finding also speaks to ongoing institutional debates about the professionalization of procurement (Saussier and Tirole, 2015; European Commission, 2017, 2020b), where legal expertise has been noticed, our results suggest that this emphasis may come with measurable trade-offs in project quality.

Our data and empirical strategy do not allow us to identify the precise mechanisms underlying the association between legal background, legal engagement, and project performance. Nonetheless, we propose two plausible mechanisms through which procurement officers may affect project quality. A first mechanism is a trade-off between mission-oriented objectives (Besley and Ghatak, 2005) and legal compliance (Bosio et al., 2022). Procurement officers operate within an institutional environment where their role is vaguely defined, and combines elements of service delivery (e.g. direct relation with suppliers) with administrative oversight (e.g. design and monitoring of tenders) and legal accountability (e.g. procurement laws). As a result, their decision-making may reflect multiple, potentially competing objectives, such as maximizing project quality on the one hand, and ensuring compliance with legal and procedural requirements on the other. In a context where officers face monitoring from audits or legal challenges, greater legal engagement may increase the weight placed on minimizing procedural risk relative to achieving high-quality outcomes. This trade-off is likely to be salient in more complex projects, where strict adherence to rules may come at the expense of flexibility in contract design. Under this interpretation, differences in project quality reflect heterogeneity in how officers balance performance-oriented and compliance-oriented goals.

A second mechanism operates through occupational sorting and comparative advantage (Murphy et al., 1991; Roy, 1951). In the absence of formal selection procedures or specialized certification for procurement roles, individuals with different educational backgrounds may sort into procurement positions based on their skill sets. Procurement tasks require a combination of legal, managerial, and technical competencies, yet legal training may provide a comparative advantage primarily in managing contractual risk and compliance rather than in optimizing project performance. If individuals with legal backgrounds disproportionately select into procurement roles, and if their skills are more closely aligned with minimizing legal exposure than maximizing quality, this sorting process may generate systematic differences in value added across officers. More broadly, this mechanism highlights

how the institutional design of recruitment and professionalization in public procurement can shape the mapping between individual skills and project outcomes. If these two mechanisms are in place, then they would suggest that observed differences in project quality reflect both heterogeneity in officers' objectives and the allocation of individuals with different comparative advantages into procurement roles.

Some limitations of our analysis should be acknowledged. One limitation of our project quality measure is its restricted applicability beyond construction contracts. Thus, our findings should be interpreted within the context of the construction sector, which represents a complex contracting environment characterized by non-standardized projects and substantial uncertainty. Additionally, our data do not provide information on the allocation of time or responsibilities across procurement officers. In practice, the officer listed in the tender documentation may work together with other officers on different tasks. Therefore, our interpretations hinge on the assumption that the officer identified in the documentation is the primary contact and responsible officer, rather than the sole individual involved in the procurement process.

Although our results are based on Czech procurement officers, all public buyers in the EU operate under the same European public procurement directives, and given widespread concerns across the EU about overly formalistic, legally driven, and non-optimal public procurement practices (i.e. concerns that have contributed to the revision of the directives), we expect our findings to be broadly applicable across Member States. Considering the scale of public procurement in the EU, estimated by the commission at around 2.5 EUR trillion annually, even modest improvements in the value added by public procurement officers, particularly by raising lower-performing buyers toward the level of above-average performers, could have substantial financial and quality implications.

From a policy perspective, understanding what leads a procurement officer to procure high quality, is useful in order to design effective selection criteria, training programs, organizational structures, and overall rules that facilitate more efficient performance. We take a first step in this direction by examining which officer characteristics, collected through the survey, explain variation in officers' value added. The evidence we provide, even though, not causal, is consistent with mechanisms documented in related settings (Bandiera et al., 2021a): officers who place excessive emphasis on legal compliance may perform worse along economic dimensions, such as delivering cost-effective public procurement, particularly in a domain that is already heavily constrained by legal rules. Finally, our findings can inform the development and assessment of two policy guidelines: skills training and professional certification policies (OECD, 2025b) and (ii) the selection procedure for administrative bureaucrats (Dahis et al., 2025). Against the background of public procurement becoming increasingly formalistic and legally driven, our results imply that buyers should be trained more in how to conduct good procurement, that is, how to play the game well, rather than primarily focusing on knowing the rules, that is, how to act as a good referee.

Our paper also carries implications for future research. A fruitful research av-

enue would be to explore how procurement outcomes depend on both who procurement officers are (e.g. educational or preference type) and how they allocate effort across competing objectives (e.g compliance vs performance). Furthermore, it would be valuable to develop richer measures of procurement quality that extend repair costs, particularly for procurement sectors other than construction projects. In this line, future work could explore heterogeneity in officers' value added across different segments of the quality distribution or across different types of procurement sectors. Testable hypotheses to consider are that legally trained procurement officers may perform relatively better in the procurement of standardized goods with lower contractual complexity; and that officers with managerial or market-oriented skills may contribute more in complex contracting environments.

Appendices

3.A Appendix section

3.A.1 Questionnaire and survey descriptive

Table 3.A.1 Description of survey questionnaire

Legal Engagement	I monitor legal and legislative developments in the field of public procurement to help my organisation with public procurement in compliance with legislation.
Hard Skills & Knowledge	
Negotiation Skill	I conduct negotiations and solve simple problems for routine and less complex projects.
Market Knowledge	I know the markets in which I request construction work, supplies or services through public procurement very well (e.g. I know how many suppliers operate on them).
Procurement Methods	When preparing public contracts, I can effectively use a wide range of procurement methods, methods for electronic submission, types of contracts, clauses and sanctions.
Procedure Selection	I am capable of choosing the parameters and type of procurement procedure that will guarantee a cost-effective solution, taking into account the market situation and the applicable legislation
Dispute Resolution	I am capable of resolving disputes with suppliers and/or applying corrective measures in case of deviation from contractual terms.
Sustainability	I am capable of fulfilling the objectives of sustainable public procurement.
Risk management	I can identify and manage the risks associated with public contracts.
Soft Skills	
Communication	In communication with the client, I define relevant requirements, oversee their implementation and delivery, and create conditions for two-way and effective communication with the supplier.
Collaboration	I am capable of establishing and maintaining relationships across departments of my organization, with suppliers and with stakeholders.
Time Efficiency	I use my time efficiently and am capable of organizing my work in such a way as to meet deadlines.
Motivation	
Goal Orientation	In my work, I demonstrate perseverance and determination to achieve the goals of the entire organization.
Performance Orientation	Through my work, I contribute to achieving cost-effective results.
Common Goods	It is important to me that my work contributes to the common good.

Table 3.A.2 Descriptive statistics of survey measures

Variable	**Strongly Disagree** (%)	**Disagree** (%)	**Neutral** (%)	**Agree** (%)	**Strongly Agree** (%)
Legal Engagement	1.7	7.1	21.4	51.6	18.2
Hard Skills & Knowledge					
Negotiation Skill	1.6	7.3	14.8	56.9	19.3
Market Knowledge	3.3	19.9	32.7	37.4	6.8
Procurement Methods	4.5	22.6	29.1	35.3	8.4
Procedure Selection	3.0	14.2	26.4	46.9	9.6
Dispute Resolution	3.0	13.1	21.7	51.7	10.5
Sustainability	2.5	9.5	36.2	45.4	6.4
Risk Management	1.2	5.5	25.2	55.5	12.6
Soft Skills					
Communication	1.0	5.6	16.9	63.0	13.6
Collaboration	0.8	3.4	16.1	61.7	18.0
Time Efficiency	0.7	2.8	17.1	60.1	19.3
Motivation					
Goal Orientation	0.5	0.7	12.4	58.9	27.6
Performance Orientation	0.1	0.7	9.6	53.7	36.0
Common Goods	0.4	1.4	12.8	52.5	33.0

Notes: Descriptive statistics for the self-assessment measures collected in the survey.

3.A.2 Survey response selection and sample representativeness

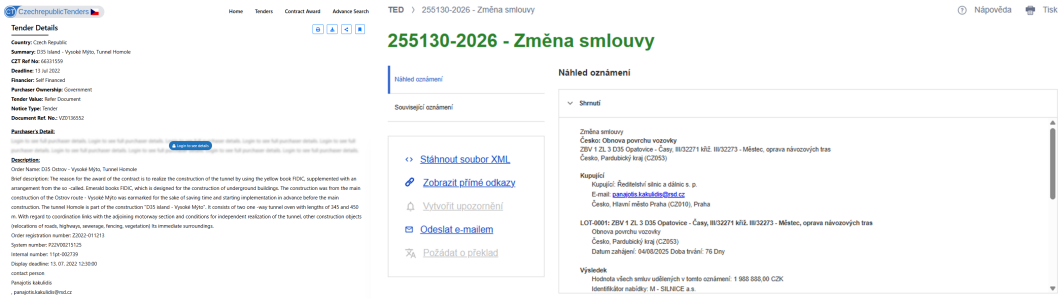
Our empirical strategy combines the survey response data with available administrative or contract-level information. This section describes the construction of the respondent and non-respondent samples, the matching procedure, and any exclusions. To start with, we extracted email addresses from the contract administrative data (We refer to this administrative data as Vestnik). After cleaning the email addresses, a total of 31408 unique email addresses were identified. We then completed this list of email addresses with an additional list of 14,669[14]. This completed a list of 46,077 unique email addresses. We then matched the list of email addresses to the survey response data, which resulted in a total of 46,077 unique email addresses to which we reached out to with our questionnaire via Qualtrics.

The email addresses extracted from the contract data correspond to real people listed in the procurement records. In most cases, these individuals are identified as the contact person for the tender. We therefore expect and assume that they are familiar with the details of the tender and with the description of the procured goods, services, or works. Figure 3.A.1 shows two examples of how our email addresses appear in the contract records.

After the survey was completed, we merged the email addresses of the respon-

[14]For the additional email addresses, we did not have a direct link to contract administrative data. Therefore, we rely only on the 31,408 email addresses that we could match to the contract administrative data to compare respondents and non-respondents on observable characteristics. Here this additional list is called Kontakty.

Figure 3.A.1 Examples of contact people identified in the contract records



dents with the email addresses in the contract data[15]. There were a total of 243,795 contracts in the administrative data, 233,090 or 95.6% of which had at least one email address. The survey was initially linked to 45,224 contracts, while 24,957 contracts were linked to completed survey responses. Therefore, the representation in terms of contracts is 10.7% for completed survey responses. The representation in terms of contracts is summarized in Table 3.A.3.

Table 3.A.3 Contract-level email and completed-survey coverage

Panel A: Email addresses coverage of contracts		
Status	Contracts	Share (%)
With email address	233,090	95.61
No email address	10,705	4.39
Total contracts	243,795	100.00

Panel B: Completed-survey coverage among contracts with email		
Status	Contracts	Share (%)
Linked to completed survey	24,957	10.71
Not linked to completed survey	208,133	89.29
Contracts with email address	233,090	100.00

In terms of outreach, the survey was sent to 46,077 email addresses, see Table 3.A.4. Of these, 3,675 respondents started the survey and 1,458 completed it. The overall response rate is therefore 7.98% for started surveys and 3.16% for completed surveys. The response rate is slightly higher for the Kontakty source (the additional list of email addresses) (9.32% started and 3.15% completed) than for the Vestnik source (7.94% started and 3.42% completed). The highest response rate is observed for the email addresses that were present in both sources (14.37% started and 5.26% completed). Overall, the rate of response of the 31,408 email addresses is 3.42%.

Another perspective on the representativeness of the survey respondents is their

[15]In principle all email addresses of officers who responded to the survey should be in the contract data; however, it is possible to forward the survey to another email address

Table 3.A.4 Survey sampling and response

Step	Count
Contract lots in Vestnik data	382,921
Contract lots with at least one valid email	371,838
Distinct valid emails in Vestnik	31,408
Emails sent	46,077
Survey started	3,675
Survey completed	1,458
Completed and found in Vestnik	1,074

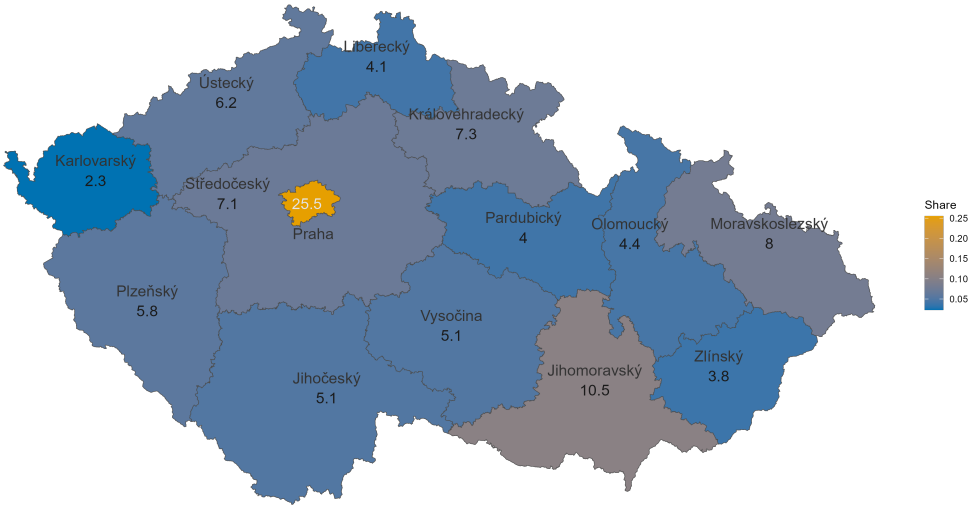
Notes: In table 3.A.4 Counts are unique contract lots or email addresses at each stage of the sampling and response process. Vestnik refers to email addresses extracted from the administrative contract records; emails sent also include additional addresses from the Kontakty source. The final row counts completed surveys that could be matched back to Vestnik and are therefore available for linked administrative comparisons.

geographic distribution. We identified the region of each respondent and use this information to assess whether responses are geographically dispersed or concentrated in a small number of regions. Although we cannot directly compare this distribution with the regional distribution of all contracting authorities in the administrative data, Figure 3.A.2 provides a useful descriptive check. The figure shows that respondents are present across all regions, suggesting that the survey is not driven by only one part of the country. Prague accounts for the largest share of respondents, at 25%, which is expected given its status as the capital and the largest region in population terms. Outside Prague, the distribution is relatively even across regions. The main exception is the Karlovy Vary region, which accounts for only 2.3% of responses. The remaining regions show no clear east-west or urban-rural clustering.

Figure 3.A.2 shows that survey respondents are geographically distributed across all Czech regions, although the distribution is not uniform. The color scale ranges from dark blue for lower respondent shares to yellow/orange for higher shares. Prague is the only region shown in bright yellow, reflecting its much larger share of respondents at 25.5%. The remaining regions are mostly shown in blue or gray-blue tones, indicating substantially lower and relatively more balanced shares. The largest regional shares outside Prague are observed in South Moravia (10.5%) and Moravia-Silesia (8.0%), while the lowest representation is in Karlovy Vary, shown in dark blue, with 2.3%

Table 3.A.5 compares procurement officers whose email addresses can be linked to the administrative contract records, distinguishing between officers who finished the survey and officers who did not. This is a balance table at the email-address level. It does not use the full outreach list, because some email addresses were added from sources that cannot be linked directly to contract-level information. The table is therefore best interpreted as a comparison of observable procurement histories for the matched part of the survey frame. Since the availability of administrative variables differs across measures, the number of observations varies by row. The largest comparisons include more than 30,000 email addresses, while the quality measures are available for smaller subsets.

Figure 3.A.2 Share of survey respondents by region



Notes: The map uses Czech regional boundaries from the `RCzechia` R package. See the `RCzechia` vignette: `https://cran.r-project.org/web/packages/RCzechia/vignettes/vignette.html`

Table 3.A.5 Balance respondents versus non-respondents

	No finished survey		Finished survey			
Variable	N	Mean	N	Mean	Diff.	p-value
Number of contracts	29,833	5.60	1,026	8.53	2.92	0.000
Tender final price (th)	30,334	214.43	1,074	444.60	230.17	0.007
Mean bids	25,907	3.72	955	3.67	-0.05	0.508
Procurer tenure (years)	19,957	1.77	751	3.41	1.64	0.000
Contracts per active year	19,957	2.44	751	2.65	0.21	0.286
Average quality	16,621	-0.01	701	-0.01	0.00	0.234
Average quality (W)	265	-0.40	13	-0.34	0.06	0.079

Notes: Table 3.A.5 reports the number of observations, group means, mean differences, and p-values for seven officer-level characteristics: number of contracts, tender final price in thousands of CZ Korunas, mean bids, procurer tenure, contracts per active year, average project quality, and average project quality winsorized to consider only projects with repairs.

The main finding is that officers who completed the survey are more active in the procurement records. Respondents are linked to an average of 8.53 contracts, compared with 5.60 contracts for officers who did not finish the survey. The difference is 2.92 contracts and is statistically significant. This indicates that survey completion is positively related to the amount of observed procurement activity. In practical terms, the completed-survey sample contains officers with a larger administrative footprint. These officers may be more engaged with procurement tasks, more experienced with tender procedures, or more likely to view the questionnaire as relevant to their work.

A related difference appears for tender final price. Officers who finished the survey are associated with contracts with an average final price of 444.60, while non-finishers are associated with an average of 214.43. The difference of 230.17 is statistically significant, with a p-value of 0.007. This suggests that respondents are more likely to be connected to larger-value procurement. The table alone cannot determine whether this comes from the type of contracting authority, the seniority of the officer, or the kinds of procurement procedures handled by respondents. However, it does show that the respondent group is not perfectly balanced with non-respondents in terms of the scale of procurement activity.

The tenure comparison points in the same direction. Officer tenure is measured at the procurement-officer level, where an officer is defined as a distinct email address observed in the contract records. For each email address, we identify the first and last linked contract using the ordering of contract dates in the Vestnik data. Tenure is then calculated as the time elapsed between the first and last contract, expressed in years. This measure captures the span of observed procurement activity, while the number of contracts and contracts per active year capture the volume and intensity of that activity. Respondents have an average observed procurement tenure of 3.41 years, compared with 1.77 years for non-respondents. The difference of 1.64 years is statistically significant and large relative to the non-respondent mean. This means that officers who completed the survey tend to have a longer observed history in the procurement data. This is consistent with the higher number of contracts among respondents. From the perspective of survey quality, this pattern has two interpretations. On one hand, more experienced respondents may provide especially informed answers about procurement practice. On the other hand, the survey may under-represent newer officers or officers who appear only occasionally in the administrative records.

In contrast, the table shows much weaker differences for measures related to tender competition and procurement intensity. The mean number of bids is almost identical across the two groups: 3.67 for respondents and 3.72 for non-respondents. The difference is -0.05 and is not statistically significant. Contracts per active year are also similar. Respondents have 2.65 contracts per active year, compared with 2.44 for non-respondents, and the difference is not statistically significant. These results suggest that respondents are linked to more contracts partly because they have longer observed tenure, not because they necessarily handle many more contracts in each active year.

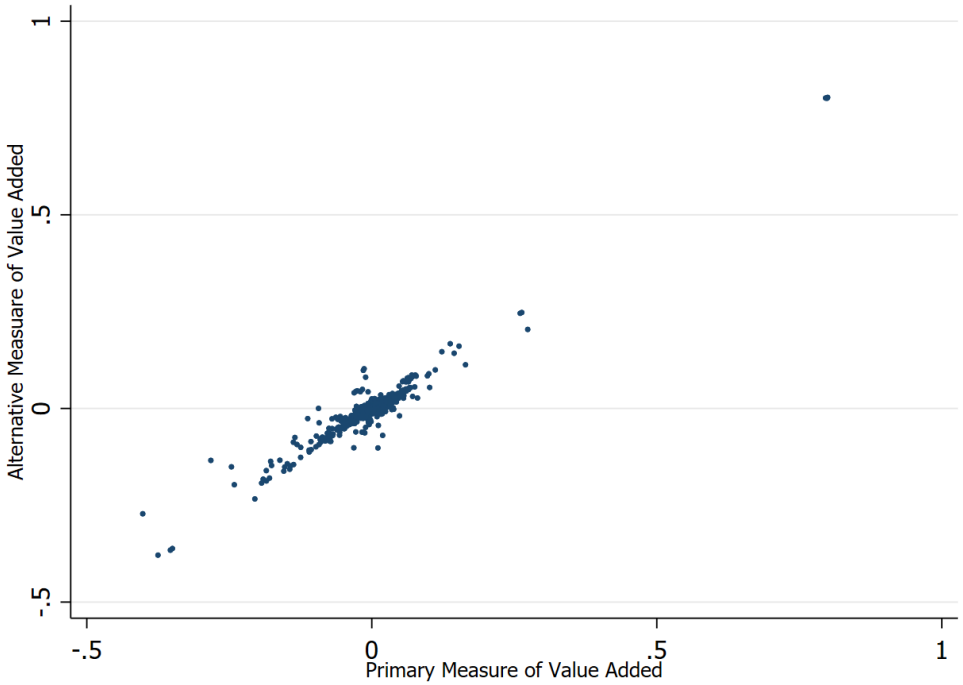
The quality variables also provide little evidence of imbalance. Mean quality is

-0.01 for both respondents and non-respondents, and the difference rounds to zero. The winsorized quality measure is slightly higher for respondents, at -0.34 compared with -0.40 for non-respondents, but this estimate is based on a much smaller sample and is only marginally significant. The balance table suggests that response selection is concentrated mainly in officer experience, number of contracts, and contract value. There is less evidence that respondents differ from non-respondents in terms of average competition, annual procurement intensity, or measured tender quality.

The survey results should therefore be interpreted as reflecting a more experienced and more administratively visible group of procurement officers, while remaining broadly comparable on several important tender-level characteristics. Nevertheless, our findings should primarily be interpreted as evidence on active procurement professionals rather than on the full population of more occasional procurement officers.

3.A.3 Primary and alternative value added measures

Figure 3.A.3 Procurement officers' value added correlation of primary and alternative measures



Notes: This figure shows a scatterplot of the primary measure of officers' value added and an alternative measure of officers' value added. The primary measure is obtained from the fully saturated regression in equation 3.4.1, while the alternative measure is estimated from a regression that controls only for the year in which the tender was announced, procuring-organization fixed effects, and buyer category. The two measures are strongly correlated (97%).

3.A.4 Additional specifications main results

Table 3.A.6 Officers' value added: legal engagement and education categorized legal engagement

	(1)	(2)	(3)	(4)
	Procurement Officer Value Added			
Legal Engagement (Agree = 1)	-0.007*	-0.009*	-0.008*	-0.008*
	(0.004)	(0.005)	(0.005)	(0.005)
Legal Education	-0.013**	-0.012**	-0.012*	-0.016**
	(0.005)	(0.006)	(0.006)	(0.007)
N	199	171	171	171
Skills & Motivation	No	Yes	Yes	Yes
Type of Buyer	No	No	Yes	Yes
Other Controls	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This Table replicates the main results using a categorized measure of legal engagement. In this specification, the legal engagement variable takes the value of 1 if officers agreed or strongly agreed with the statement that they monitor legal and legislative developments, and 0 otherwise.

Table 3.A.7 Officers' value added: full results

	(1)	(2)	(3)	(4)
	Procurement Officers' Value Added			
Legal Engagement	-0.003*	-0.006**	-0.006**	-0.005**
	(0.002)	(0.003)	(0.003)	(0.003)
Legal Education	-0.013**	-0.013**	-0.013*	-0.018**
	(0.005)	(0.006)	(0.007)	(0.008)
Negotiation Skill		0.002	0.002	0.002
		(0.002)	(0.002)	(0.002)
Market Knowledge		-0.002	-0.002	-0.002
		(0.002)	(0.002)	(0.002)
Procurement Methods		0.001	0.001	0.001
		(0.002)	(0.002)	(0.003)
Procedure Selection		-0.003	-0.004	-0.004
		(0.003)	(0.003)	(0.003)
Dispute Resolution		-0.001	-0.001	-0.000
		(0.003)	(0.003)	(0.003)
Sustainability		0.004	0.003	0.002
		(0.003)	(0.003)	(0.003)
Risk Management		0.002	0.003	0.003
		(0.003)	(0.003)	(0.003)
Communication		0.001	0.000	0.001
		(0.003)	(0.003)	(0.003)
Collaboration		0.006**	0.006**	0.005*
		(0.002)	(0.002)	(0.003)
Time Efficiency		-0.004	-0.003	-0.002
		(0.002)	(0.003)	(0.003)
Goal Orientation		0.005	0.004	0.004
		(0.003)	(0.003)	(0.003)
Common Goods		-0.003	-0.003	-0.003
		(0.002)	(0.002)	(0.002)
Performance Orientation		0.000	0.000	-0.001
		(0.003)	(0.003)	(0.003)
Estimated Cost		-0.000	-0.000	-0.000
		(0.000)	(0.000)	(0.000)
Type Buyer (=1)			-0.002	-0.001
			(0.007)	(0.007)
Type Buyer (=2)			0.007	0.003
			(0.010)	(0.011)
Type Buyer (=3)			-0.005	-0.003
			(0.006)	(0.007)
Type Buyer (=4)			-0.008	-0.008
			(0.005)	(0.005)
Type Buyer (=5)			-0.007	-0.006
			(0.009)	(0.009)
Training				-0.009
				(0.006)
Certificate				-0.003
				(0.004)
Experience				-0.000
				(0.000)
Specialist				-0.006
				(0.005)
Supervisor				-0.004
				(0.005)
Constant	0.011	-0.008	0.001	0.015
	(0.008)	(0.017)	(0.017)	(0.019)
N	199	171	171	171

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the full regression results corresponding to those presented in Table 3.2 of the regression 3.5.1.

3.A.5 Additional results: alternative value added measures

In this section, we present additional figures, tables, and findings that support and extend the results from the main text. Tables 3.A.8, and Figures 3.A.4 and 3.A.5, extend the main results by using the total amount of repairs over a five-year horizon to define procurement project quality or the alternative measure of officers' value added.

Table 3.A.8 Exploring Officers Value Added: Legal Engagement and Education

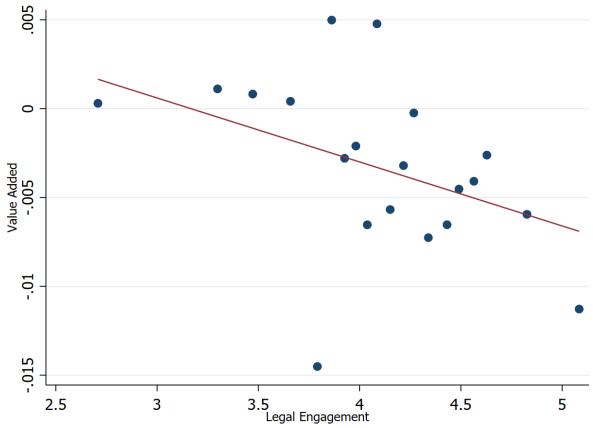
	(1)	(2)	(3)	(4)
	Officer Value Added (Alternative Measure)			
Legal Engagement	-0.004**	-0.004**	-0.004**	-0.004*
	(0.001)	(0.002)	(0.002)	(0.002)
Legal Education	-0.008**	-0.006	-0.006	-0.008
	(0.004)	(0.005)	(0.005)	(0.005)
N	220	189	189	189
Skills & Motivation	No	Yes	Yes	Yes
Type of Buyer	No	No	Yes	Yes
Other Controls	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

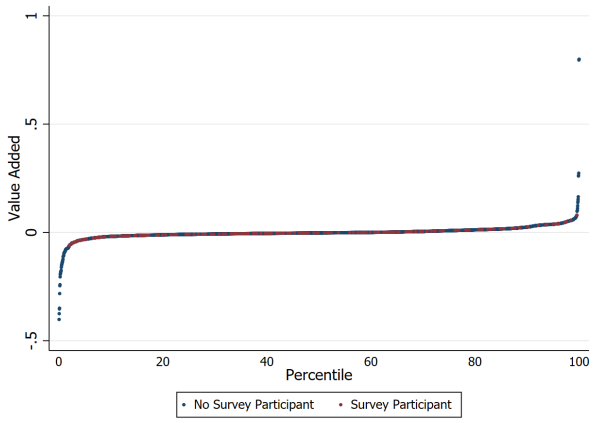
Notes: This Table replicates Table 3.2 for the alternative measure of officers' value added. Column (1) presents results from a simple regression that includes only legal engagement and legal education. Columns (2)–(4) report results from the specifications outlined in regression 3.5.1. Across all specifications, both legal engagement and education are negative.

Figure 3.A.4 Officers' value added and legal engagement



Notes: This Figure shows the conditional negative relationship between officers' value added and the self-reported frequency of monitoring legal and legislative developments. It replicates Figure 3.3a using an alternative measure of officers' value added estimated from a regression that controls only for the year in which the tender was announced, procuring-organization fixed effects, and buyer category.

Figure 3.A.5 Procurement officer value added and survey participation



Notes: This figure shows the distribution of officers' value added and highlights those who participated in the survey. The figure indicates that survey participants are well distributed across the value-added distribution, with the only notable exceptions being officers in the top and bottom 1% of the distribution.

3.A.6 Quality of delivered project based on 3-year duration measure

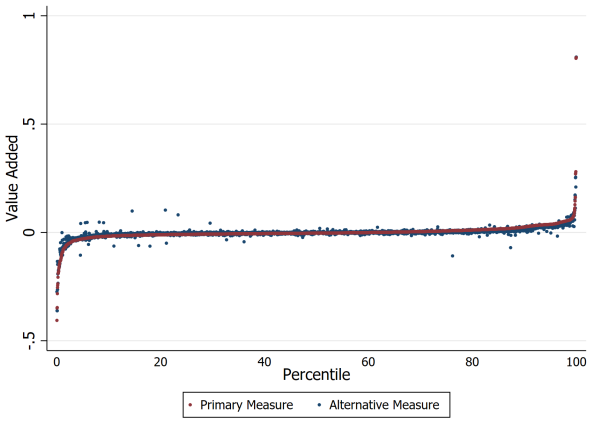
In this section, we replicate the results from the main text using a three-year horizon to sum the required repairs in the definition of project quality, instead of five years. Measuring repair needs over a shorter period, on the one hand, increases the number of observations; on the other hand, it attenuates the total value of repairs, as some repairs may not yet have occurred. This, in turn, results in less variation in the quality measure and, consequently, in procurement officers' estimated value added.

We first plot procurement officers' value added and then replicate the exploratory analyses, estimating the effects of legal education and legal engagement. Figure 3.A.6 shows the distribution of officers' value added using both the primary and the alternative measures. Both measures are constructed from the same regressions as in the main text. Compared to Figure 3.2 presented in the main text, the distribution exhibits less variation. In particular, the gap between the 10th and 90th percentiles is 4.2 points, which is 0.3 points smaller than in the main specification. This compression of the distribution is driven by the shorter period over which repair costs are aggregated.

For a median-size contract, replacing an officer at the 10th percentile with one at the 90th percentile would reduce expected three-year repair costs by CZK 480,000 (approximately EUR 20,000). The magnitude is marginally lower than in the main specification.

Table 3.A.9 replicates Table 3.2 from the main text, showing that officers with legal education and those monitoring legal and legislative development more often tend to underperform. The point estimates are negative and statistically significant for both variables in all four specifications that vary in the control variables included.

Figure 3.A.6 Procurement officers' value added 3-year quality measure



Notes: This figure shows the distribution of procurement officers' value added using the primary and alternative measures. Project quality is derived from the total value of repairs required over three years. Both measures exhibit less variation than in Figure 3.2, as fewer repairs are observed over the shorter horizon. The gap between officers at the 10th and 90th percentiles is 4.2 points. Note that the 5-year measure and the 3-year measure do not preserve officers' rankings.

Table 3.A.9 Exploring officers value added: legal engagement and education 3-year quality measure

	(1)	(2)	(3)	(4)
	Procurement Officer Value Added			
Legal Engagement	-0.004**	-0.006**	-0.006***	-0.005**
	(0.002)	(0.002)	(0.002)	(0.002)
Legal Education	-0.010**	-0.012*	-0.013**	-0.019***
	(0.005)	(0.006)	(0.006)	(0.007)
N	199	171	171	171
Skills & Motivation	No	Yes	Yes	Yes
Type of Buyer	No	No	Yes	Yes
Other Controls	No	No	No	Yes

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table replicates Table 3.2, showing that officers with legal education and those more engaged with legal developments in the area of public procurement underperform in terms of their value added. The table reports results using officers' value added derived from the three-year quality measure.

CHAPTER 4

Competition, Entry, and Firm Dynamics in Public Procurement

This chapter studies how entry into public procurement markets affects firms' financial performance. I implement an event-study design that exploits variation in entry timing by assembling a firm-year panel that combines administrative Dutch public procurement contract data with firm-level balance sheet information. I estimate dynamic post-entry changes for general entry events (first contract) and competitive entry events (first contract in an open tender with more than two competitors). General entry is associated with asset growth and higher liabilities, while liquidity declines. In contrast, gains from competitive entry are weaker and less persistent. Firm heterogeneity analysis suggests that established firms sustain improvements, capacity-constrained firms display limited persistence, and short-term participants experience only temporary gains. I discuss two mechanisms. First, short-term participants engage in aggressive underbidding consistent with winner's-curse dynamics and learning-by-doing. Second, entrants sort into tenders with fewer bidders, suggesting information frictions. The results suggest that firms' capabilities may play a role in leveraging public demand into firm growth.

Keywords: Procurement, Firm entry, Competition.
JEL Codes: H57, D22, L13, D44

4.1 Introduction

Public procurement is an important source of demand for firms. Around 16% of GDP of European Union members is spent on the purchase of services, works, and supplies (European Commission, 2026b), indicating an important link between public demand and private supply. These markets are important for firms, especially when public contracts provide access to large buyers, repeated business opportu-

nities, and reputational signals that may affect future growth. Understanding how firms respond when they first enter procurement markets is central to evaluating whether public demand translates into firm expansion. However, public procurement markets have become less competitive over the past decade, as measured by a decline in the average number of bids submitted per tender. By 2024, the average number of bids for a tender was 3.2, two fewer bids than a decade earlier (ECA, 2023). This declining trend in participation is particularly puzzling in light of a growing strand of empirical evidence documenting that firms benefit from winning public procurement contracts, such as increased productivity (Chiappinelli et al., 2025; Ferraz et al., 2015), access to credit (di Giovanni et al., 2024), and sales (Hoekman et al., 2024). Additionally, public entities and international agencies promote public procurement as a channel through which governments can achieve innovation, green growth, and societal goals (OECD, 2026; World Bank, 2026; Mazzucato et al., 2025).

This chapter examines how entry into public procurement markets is associated with firms' financial performance, and whether changes depend on how firms enter. I define entry as the first public contract awarded to a firm, and distinguish between **general entry** and **competitive entry**, where the latter is restricted to open tenders with more than two bidders. This distinction allows me to isolate differences in outcomes across procurement environments that vary in competitive intensity. Existing literature focuses on firm growth and the effects of winning public contracts (Hebous and Zimmermann, 2021; Gugler et al., 2015; Srhoj and Dragojević, 2024; Fadic, 2020; di Giovanni et al., 2024), but discusses little about the type of entry or treat it as a homogeneous event and abstracts from the competitive environment in which contracts are awarded. The auction literature provides detailed analyses of bidding behavior and entry in procurement settings primarily within first-price auction frameworks (Krasnokutskaya and Seim, 2011; De Silva et al., 2009) but focuses on equilibrium behavior rather than post-award firm performance. By distinguishing between general and competitive entry, this chapter bridges these literatures: it brings variation in competitive conditions, central to auction models, into the analysis of firm-level outcomes following procurement entry. The distinction is empirically relevant given the documented decline in the average number of bidders in public procurement markets, and it provides a framework to better understand how market structure relates to firm outcomes at entry.

Answering this question is challenging because firms do not enter procurement markets randomly. Firms that win public contracts may differ from non-suppliers in age, size, sector, capabilities, financial conditions, and expected growth even before entry. These differences make simple comparisons between suppliers and non-suppliers difficult to interpret. A credible empirical strategy must therefore construct a comparison group that approximates the counterfactual financial path entrants would have followed had they not entered the procurement market.

A second challenge is that procurement entry occurs in a staggered fashion. Firms win their first public contract in different years, while many firms never enter during the sample period. In this setting, standard two-way fixed effects event-study estimators can be sensitive to treatment-effect heterogeneity across

cohorts and over time since entry. Already-treated firms may implicitly serve as controls for newly treated firms, generating contaminated comparisons and non-transparent weights (Goodman-Bacon, 2021). This concern is especially relevant in procurement markets, where the effects of entry are likely to differ between firms entering through competitive procedures and firms entering through other procurement routes.

This chapter studies entry into Dutch public procurement markets above the European procurement thresholds between 2016 and 2024[1]. I define entry as the first time a firm is awarded a public procurement contract above these thresholds. I combine contract-level procurement data with firm-level balance sheet information and estimate dynamic effects around entry using a staggered difference-in-differences design. The main comparison group consists of matched firms that never become public suppliers during the sample period.

The empirical framework treats procurement entry as a persistent state rather than a one-time transaction. Once a firm wins its first public contract, it may accumulate procurement-specific experience, learn how to bid, gain market access, and remain engaged in contract delivery. I therefore consider firms treated after their first contract, even if they do not immediately win another one. This interpretation is consistent with models of procurement participation in which firms become potential suppliers after incurring entry and learning costs (Athey et al., 2013; De Silva and Rosa, 2024).

The main identifying assumption is conditional parallel trends. Conditional on pre-entry observables, including firm age, firm size, and sector, entrants and comparable non-entrants are assumed to have followed similar trends in financial outcomes absent entry. This assumption allows for selection into procurement based on observed characteristics and time-invariant differences, but requires that remaining differences in untreated outcome dynamics are not systematically related to the timing of entry. The estimates should therefore be interpreted as relying on a structured comparison between entrants and similar non-entrants, rather than on random assignment.

I construct the dataset from three sources. First, I use contract-level data from TenderNed, the Dutch public procurement platform, covering contracts above the European thresholds. These data identify awarded contracts, buyers, procurement procedures, contract values, number of bidders, and winning firms. Second, I merge the procurement records to firm-level financial information from Moody's Orbis, which provides balance sheet outcomes for Dutch firms over 2012–2024. Third, I use complementary bid-level data from ProRail, a large Dutch public buyer, to study bidding behavior and mechanisms.

The main outcomes capture firms' financial performance and scale. I study current ratio, solvency, shareholder funds, total assets, asset growth, liabilities, liability growth, number of employees, and employment growth. The final event-study sample contains 4,235 general entry events, of which 1,095 correspond to

[1]According to the OECD public procurement spending accounted for 20% of the Dutch GDP in 2022 (OECD, 2025), positioning the Netherlands on the right tail of this distribution.

competitive entry. Entry contracts are smaller than the average contract in the full procurement data but attract relatively similar number of bidders, suggesting that entry occurs in procurement environments where competition is meaningful.

To take the model to the data, I implement the group-time difference in differences estimator of Callaway and Sant'Anna (2021). Firms are grouped into cohorts based on the year in which they first enter the procurement market. I first estimate cohort-specific treatment effects for each entry cohort and calendar year, comparing entrants to matched never-supplier firms. I then aggregate these group-time effects by relative event time to obtain dynamic event-study estimates. This approach avoids using already-treated firms as controls and allows treatment effects to vary across cohorts and over time since entry.

I address selection into procurement in two ways. First, I construct a matched comparison group of never-supplier firms using pre-entry firm characteristics, including age, sector, and size. Second, I estimate robustness specifications using not-yet-supplier firms as the comparison group, excluding never suppliers. This alternative comparison relies only on variation in entry timing among firms that eventually become suppliers. I also examine heterogeneity by classifying entrants as short-term participants, capacity-constrained firms, and established firms, and I use bid-level data to study whether differences in bidding behavior help explain the results.

I document that entry into public procurement, when defined broadly, is associated with improvements in firms' financial indicators. In particular, I find patterns consistent with increases in asset accumulation and liabilities, which align with interpretations in the literature linking procurement participation to improved access to credit and firm expansion. However, when focusing on competitive entry, the observed gains appear less persistent. Increases in assets are concentrated around the year of entry and attenuate thereafter.

The main results show that procurement entry is associated primarily with balance-sheet adjustments. General entry is followed by a decline in liquidity, measured by the current ratio, and by a substantial increase in liabilities. Overall liabilities rise by about EUR 97,975 after general entry, with liability growth increasing by 22.3%. Competitive entry is also associated with higher liabilities, but the effects are smaller: liabilities increase by about EUR 71,887, with liability growth of 13.5%. These patterns suggest that entry is associated with greater financial obligations and weaker short-term liquidity.

The evidence on firm expansion is more mixed. General entry is associated with statistically significant asset growth of about 9.9%, while the effect on total assets in levels is positive but not statistically significant. For competitive entry, asset growth is smaller, about 2.5%, and not statistically significant. Employment effects are limited across both entry definitions. The results therefore suggest that entry into procurement markets does not uniformly generate broad-based firm growth. Instead, public procurement entry appears to reshape balance sheets, especially through higher liabilities, while sustained growth in assets and employment is concentrated among specific types of firms.

This pattern is consistent with evidence from experimental and quasi experi-

mental settings suggesting that competitive environments may limit the persistence of gains from initial contract awards (Fadic, 2020). My contribution lies in showing that, in observational data, entry through more competitive channels is systematically associated with weaker and less sustained financial improvements.

A potential concern is that these patterns may reflect selection on observable or unobservable firm characteristics. To further investigate heterogeneity in post-entry trajectories. Firms typically remain active a couple of years following entry, as contracts take on average 36 months to complete. This institutional feature implies that post-entry differences are not driven by short-run market exit, but rather by heterogeneity in firms' ability to obtain additional contracts over time. I classify firms based on their subsequent participation patterns: established firms, which consistently win contracts after entry; capacity constrained firms, which participate intermittently; and short-term participants, who do not win additional contracts beyond entry. This classification reveals systematic differences.

Established firms exhibit positive changes in assets following entry, although these effects are attenuated in competitive entry settings. Capacity-constrained firms display similar, albeit weaker, patterns. In contrast, short-term participants experience only temporary improvements under general entry, which are not evident under competitive entry. These findings suggest that observable firm characteristics, such as size and capacity, play an important role in shaping post-entry outcomes. At the same time, the attenuation of effects under competitive entry is observed across all firm types, indicating that competitive entry environments may pose challenges even for relatively stronger firms, potentially due to higher adjustment, transaction, or information costs.

Finally, I explore mechanisms that may rationalize these patterns. I present evidence consistent with entrants engaging in aggressive bidding strategies, including underbidding, which may expose them to outcomes resembling a winner's curse, particularly among short-term participants. At the same time, the patterns for established firms are consistent with a process of learning-by-doing following entry. I also document that short-term participants tend to enter contracts with lower observed levels of competition. Observable contract characteristics do not provide clear evidence that these contracts are systematically less attractive, more complex, or riskier. This suggests that unobserved contract attributes or informational advantages of incumbents may play a role in shaping entry outcomes.

Overall, the chapter contributes to the literature by highlighting the importance of distinguishing between entry environments and by documenting that entry through more competitive procurement channels is associated with weaker and less persistent financial improvements, even after accounting for firm heterogeneity. These findings suggest that entry alone is insufficient: firms' ability to bid, execute contracts, and remain active is central to whether public demand supports growth.

This chapter also carries policy implications. This chapter highlights that in public procurement markets, entry and permanence are central to the competition outcomes. The implications of these results are informative to public entities and international agencies that promote public procurement as a channel through which governments can achieve innovation, green growth, and societal goals (OECD, 2026;

World Bank, 2026). Furthermore, studying competition in public procurement markets provides evidence to enlighten salient discussions on European competitiveness (Draghi, 2024).

This chapter contributes to three strands of literature. First, I speak to a growing literature that exploits public procurement as a source of exogenous demand shocks to study firms' growth, labor demand, and financial outcomes. The literature shows that winning a public procurement contract positively affects firms' performance (Hebous and Zimmermann, 2021; Hoekman et al., 2024; di Giovanni et al., 2024; Pantea and Laurentsyeva, 2026; Czarnitzki et al., 2020). However, there is substantial heterogeneity in persistence. While some studies find only short-run effects on employment (Gugler et al., 2015; Srhoj and Dragojević, 2024; Fadic, 2020), others document long-run growth (Ferraz et al., 2015; di Giovanni et al., 2024; Lee, 2021). However, the existing literature has paid little attention to the competitive mechanism firms face when winning. My chapter contributes by focusing on how competitive mechanisms in public procurement markets may shape these growth trajectories, an aspect absent in this strand of literature.

Second, the firm heterogeneity exploration contributes to a larger literature that examines the role of demand shocks in determining firms' dynamics (Foster et al., 2016, 2008) and how firms approach new markets (Alfaro-Urena et al., 2022; Amiti et al., 2024). In the context of public procurement, a few papers have studied heterogeneity as such Hebous and Zimmermann (2021); constrained versus constraints, Ferraz et al. (2015) by small versus big; and Hoekman et al. (2024) by regular versus irregular status. My findings move the empirical literature forward by showing that entry into a public procurement market improves firm outcomes on average, but these gains are not driven by competitive entry and are concentrated among firms that remain active and have the capacity to scale.

Third, my findings that small firms underbid and participate in tenders with fewer competitors speak to the literature that studies the trade-off between price competition in public procurement (Carril et al., 2026; De Silva and Rosa, 2024; Jääskeläinen et al., 2026; Titl, 2025). My chapter contributes by documenting that competitive pressures influence participation and bidding behavior, and that these, in turn, determine the allocation and impact of demand shocks.

The rest of the paper is presented as follows. Section 4.2 describes the institutional setting and data sources. Section 4.3 presents the empirical framework and identification assumptions. Section 4.4 presents the main results, including the staggered entry patterns, the event-study estimates for general and competitive entry, and the firm heterogeneity analysis. Section 4.5 explores the mechanisms underlying the main findings, focusing on underbidding behavior and sorting into less competitive procurement environments. Section 4.6 discusses limitations and directions for future research. Section 4.7 concludes.

4.2 Institutional setting

Public procurement is a key market for the Dutch economy. The Netherlands ranks among the largest public buyers in the OECD, with procurement accounting for 20.9% of GDP compared to an OECD average of 12.9% in 2021 (OECD, 2025). The institutional setting is characterized by a decentralized governance structure in which provinces, municipalities, and regional water authorities have substantial autonomy over their procurement decisions[2].

The regulatory framework of the procurement market is defined by the Public Procurement Act 2012 *Aanbestedingswet 2012* that establishes rules aimed at ensuring fair competition, transparency, and objectivity. This framework applies to contracts above and below the European procurement thresholds. An important amendment occurred on 1 July 2016, when the European Union procurement directives 2014/24/EU and 2014/25/EU were transposed to the Dutch Act. These directives introduced two changes for contracts above the European thresholds: (i) mandatory electronic tendering and (ii) use of the most economically advantageous tender criteria, which is an umbrella term for lowest price, best price-quality ratio, and lowest cost criteria.

This paper focuses on Dutch public procurement above EU thresholds over the period 2016-2024[3]. Since the transposition of the EU procurement directives in 2016, this segment operates within a stable and harmonized regulatory framework. In this context, 2016 represents a shift in the entry margin by increasing the probability that potential suppliers become aware of tender opportunities, increasing expected returns to participation. By focusing on this segment, the analysis can capture the dynamics of entry and competition in a setting where information frictions are reduced and participation is likely to emerge.

4.2.1 Data sources

I construct a panel dataset of Dutch firms with balance sheet information and the year of entry in the European public procurement market. I use two primary data sources to build this dataset. First, TenderNed, which includes tenders above the European procurement thresholds between 2016 and 2024. This dataset provides a contract-level panel that links awarded tenders with winning firms. Second, I use firm-level financial information from Orbis Moody's, a global database of company financials covering 2012 to 2024. Additionally, I use a complementary dataset from

[2]The decentralized public bodies consist of: 12 provinces, 355 municipalities, and 21 regional water authorities. Other semi-public organizations, such as educational organizations, are also considered contracting authorities when purchasing works, goods, and services (EZK, 2021)

[3]The thresholds for public contracts are defined by type of procurement; for instance, the threshold for public works contracts is EUR 5,350,000, for public supply and service contracts is EUR 214,000, and for public supply and service contracts awarded by central government authorities is EUR 428,000. Section 2, article 4 of the European Directives 2014/24/EU define the thresholds with value net of value-added tax Single Market Europa Web. The thresholds described here are remained unchanged for the period covered in the analysis. I use the threshold amount to impute missing contract values (see appendix 4.A.3)

ProRail, a large public buyer, to study bid-level behavior.

Procurement data and definition of entry. The main source of procurement data is TenderNed, a Dutch national platform that publishes public procurement contracts above the European thresholds[4]. The data cover the period 2016–2024 and provide contract-level information on procurement procedures, including the identity of the winning firm, the number of bidders, the procurement method, and contract characteristics. In the appendix 4.A, I provide a detailed description of the data cleaning process, which includes recovering misreported information on the price and number of bids, as well as text similarity checks to ensure accurate matching of winning firms to their financial data in Orbis.

I use these data to define firm entry into the public procurement market. A firm enters when it is first observed winning a contract above the European thresholds. I distinguish between two definitions of entry. **General entry** is defined as the first year in which a firm wins any contract above the threshold. **Competitive entry** restricts attention to contracts awarded through open procedures with at least two bidders, capturing entry into contested procurement markets. General entry reflects access to public procurement, and competitive entry isolates entry into contracts where competition forces are more likely to shape outcomes.

Firm-level data and panel construction. To study firm outcomes around entry, I merge the procurement data with firm-level financial information from Orbis (Moody's Analytics). Orbis provides annual balance sheet data for a large population of Dutch firms over the period 2012–2024. The merged dataset is structured as a firm-year panel. This allows me to track firms both before and after entry into procurement markets, as well as to compare entrants to firms that never supply to the government. The availability of a large firm-level database is important in this context, as it enables the identification of a comparison group of non-supplier firms.

For each firm-year, I construct a set of nine financial outcomes capturing firm performance and scale, including measures of assets, liabilities, and employment, as well as their growth rates. These variables allow me to study how entry into procurement markets is associated with changes in firm dynamics. The raw dataset includes roughly 1.3 million firms per year in the Netherlands. Of these, around 10,000 firms are matched to the procurement data and form the main analysis sample, where I measure financial performance and firm characteristics including total assets, liabilities, shareholder funds, and employment, as well as their growth rates.

The estimation sample is constructed by linking the original matched panel dataset to the samples effectively used in the event studies estimations. For each

[4]TenderNed publishes procurement tender notices and tender awards above the EU thresholds for the Netherlands and provides comprehensive coverage of large public contracts. Existing estimates indicate that these procedures account for about 29.2% of Dutch public procurement. In 2023, EUR 33.9 billion in tenders out of EUR 116.2 billion of estimated total (Team Significant Synergy, 2024).

specification, outcome variable, and estimation type (never-treated and not-yet-treated), I retrieve the subset of observations retained in the estimation procedure and create indicator variables that flag whether a given firm-year observation is included in the corresponding estimation sample. This is done for all specifications (general and competitive events samples, as well as firms heterogeneity subsamples) and for all outcome variables. The resulting dataset combines these indicators with the original panel, making it possible to track sample composition across specifications and ensure consistency in the empirical analysis.

Complementary bid-level data. To complement the contract-level analysis, I use a separate sample of bidding data from a large Dutch public buyer. This dataset contains information on submitted bids, participating firms, and procurement outcomes over the period 2012–2024. The bid-level data provide a more granular view of firm behavior, allowing me to analyze participation decisions and competitive dynamics directly. While this dataset covers a subset of the procurement market, it is particularly useful for understanding the mechanisms underlying entry and competition.

Table 4.1 Descriptive statistics

	Panel A: General Entry Events				Panel B: Competitive General Entry Events					
Variable	Mean	Median	Std.Dev.	N	Mean	Median	Std.Dev.	N	Diff.	p-value
Contracts characteristics										
Number of bids	4.230	3.000	4.270	4,121	4.860	4.000	3.740	1,095	-0.640	0.000
Duration in months	36	24	42	1,040	35	24	31	432	1	0.450
Contract value	1,161	750	3,555	4,102	1,144	509	3,713	1,073	17	0.890
Services	0.720	1.000	0.450	4,235	0.690	1.000	0.460	1,095	0.030	0.090
Supplies	0.220	0.000	0.410	4,235	0.220	0.000	0.410	1,095	0.000	0.960
Works	0.060	0.000	0.240	4,235	0.090	0.000	0.290	1,095	-0.030	0.000
Firms' balance sheet information										
Current ratio	2.5	1.5	5.0	2,751	2.5	1.5	5.6	836	0.0	0.957
Solvency ratio	37.4	38.2	31.8	2,706	35.6	36.3	31.1	821	1.8	0.153
Shareholder funds	520,057	263,243	778,368	2,761	529,019	267,356	781,466	842	-8,962	0.771
Assets	1,444,288	822,338	1,669,443	2,763	1,560,260	918,630	1,787,774	837	-115,973	0.095
Log assets	13.4	13.6	1.6	2,792	13.5	13.7	1.7	849	0.0	0.449
Employees	18.6	13.0	18.4	1,736	20.0	13.5	19.8	532	-1.4	0.145
Log employees	2.4	2.6	1.2	1,736	2.5	2.6	1.1	532	-0.1	0.141
Liabilities	774,315	440,876	862,105	2,668	821,810	511,367	848,691	802	-47,494	0.166
Log liabilities	12.8	13.0	1.5	2,662	13.0	13.1	1.4	801	-0.2	0.004
Subsample bids information										
Relative bid					1.08	1.02	0.35	6,697		
Number of bids					3.90	4.00	1.56	6,716		
Duration in months					16.40	13.10	12.67	4,972		
Lowest Price					0.85	1.00	0.35	6,716		

Notes: The table is divided in three parts. The top part corresponds to characteristics of contracts including number of bids, duration of the contract, imputed contract value, and type of tender. The sample of contracts is restricted to entry events, Panel A for general events and Panel B for competitive events. For reference, table 4.A.1 shows the descriptive of all tenders. The middle part presents statistics of financial data in baseline 2016. The bottom part presents statistics of the subsample of bids from ProRail, including the relative bid, the number of bidders, and the type of procurement procedure.

Descriptive statistics. Table 4.1 reports descriptive statistics for entry events. The sample is restricted to contracts that constitute the entry event for firms in the estimation sample, excluding subsequent contracts and contracts awarded to other firms. The sample comprises 4,235 entry events of which 1,095 correspond to competitive entry. The average contract value at entry is EUR 1.16 million,

with a standard deviation of EUR 3.5 million, indicating substantial variation in contract size. The average number of bidders for entry contracts is around 4.2 in general entry events, with competitive entry contracts having a higher average number of bidders of 4.9. Contract duration averages approximately three years and does not differ meaningfully between general and competitive entry. In terms of composition, the majority of entry contracts are for services (60%), followed by works (30%) and supplies (10%). Compared to the full sample of contracts (Table 4.A.1), entry contracts are smaller on average (EUR 1.16 million versus EUR 1.8 million) but exhibit a similar distribution across contract types and procurement methods. However, entry contracts attract more bidders, suggesting that firms face relatively stronger competitive pressure when entering the procurement market.

Table 4.1 presents summary statistics for the main financial variables used in the analysis. Firms entering public procurement markets are relatively large and financially stable. For instance, the average firm has assets of about EUR 1.6 million and employs around 20 workers. Differences between general entry and competitive entry are small, although firms entering competitive procedures tend to be slightly larger and more leveraged.

4.3 Empirical framework

To estimate changes in firm financial performance post-market entry, I adopt an event study framework. I define entry as the first time a firm is awarded a public procurement contract above the European public procurement thresholds, which marks the firm's initial participation as a supplier in the public procurement market[5]. This design aligns with similar approaches in the procurement literature that study the effects of demand shocks on firms' growth and financial performance (Hebous and Zimmermann, 2021; Pantea and Laurentsyeva, 2026; Lee, 2021) and in the literature of market entry (Alfaro-Urena et al., 2022; Amiti et al., 2024). Firm entry into the public procurement market occurs in a staggered fashion. This means that not all firms supply to the government and that those who are suppliers do not become suppliers in the same year.

A canonical specification is given by equation 4.3.1[6], where $D_{i,t-j}$ is an indicator variable for event time j, with $j = 0$ denoting the year of entry (i.e., the first contract awarded).

$$y_{it} = \sum_{\substack{j=-m \\ j\neq -1}}^{n} \gamma_j \cdot D_{i,t-j} + \alpha_i + \delta_t + \epsilon_{it} \tag{4.3.1}$$

[5]One limitation is that entry is defined based on the available data window. Firms that entered the public procurement market before the start of the sample period, but are only observed later may be misclassified as entrants. The estimates should be interpreted as reflecting changes around observed entry events rather than true first-time participation.

[6]The specification in equation 4.3.1 represents the Two-Way Fixed Effects model following notation from Miller (2023), which is the same discussed in Freyaldenhoven et al. (2021), Sun and Abraham (2021), and Goodman-Bacon (2021), among others.

Using terminology from Difference-in-Differences, the treatment indicator D is defined as a firm's first win as a supplier to the government. This is a discrete event: once a firm wins its first contract, it is considered treated in all subsequent periods[7]. The specification includes firm fixed effects α_i that control for time-invariant firm characteristics, time fixed effects δ_t to absorb common yearly economic shocks, and sets $\gamma_{-1} = 0$ as a restriction condition. The parameter γ_j captures the changes in outcomes after entry relative to the year before entering.

Identification in this framework exploits the staggered timing of entry. Each year, a new cohort of firms enters the procurement market, while other firms have not yet entered or never enter during the sample period. However, when treatment effects are heterogeneous across cohorts or over time since entry, the standard two-way fixed-effects estimator may produce biased estimates. In particular, already-treated firms may serve as controls for newly treated firms, leading to contamination and negative weighting of treatment effects (Goodman-Bacon, 2021). Unfortunately, the more treatment effect heterogeneity, i.e. variation in treatment timing or multiple time periods, the more sensitivity to weighting schemes. These issues have been extensively discussed in the recent Difference-in-Differences literature with different estimators proposed. For instance, Sun and Abraham (2021) and Wooldridge (2025) suggest regression-based approaches; Borusyak et al. (2024) develop an imputation method of untreated potential outcomes, Dube et al. (2025) suggest a stacked two-way effects regression, and (Callaway and Sant'Anna, 2021), who propose estimations at the group level and is robust to covariates.

To address these concerns, I follow Callaway and Sant'Anna (2021) in two steps. First, I estimate group-time average treatment effects. Second, I aggregate these effects by relative event time. In the first step, firms are grouped into cohorts according to their year of entry into the public procurement market, denoted by g. For each entry cohort g and calendar year t, the group-time average treatment effect is defined as

$$\widehat{\gamma}(g,t) = \mathbb{E}\left[Y_t(g) - Y_t(0) \mid G = g\right] \tag{4.3.2}$$

where $Y_t(g)$ is the potential outcome at time t for firms first treated in year g, $Y_t(0)$ is the untreated potential outcome, and $G = g$ denotes membership in entry cohort g. Equation 4.3.2 is the cohort-specific treatment effect in calendar time. Since $Y_t(0)$ is not observed for treated firms after entry, identification requires a comparison group that approximates the untreated outcome path for firms in cohort g.

I use firms that never become suppliers in the public procurement market as the comparison group. These never-supplier firms are drawn from the pre-matched sample described in Section 4.C, ensuring that they are sufficiently comparable to entrant firms. Let $C_i = 1$ indicate that firm i never enters the public procurement market during the sample period and that is similar to entrants. I estimate equation 4.3.2 using the doubly robust estimator of Callaway and Sant'Anna (2021), conditioning on a covariate vector X_i that includes sector and pre-entry firm size. Under no anticipation, the estimator is

[7] I do not use a continuous measure of treatment intensity (e.g., procurement revenues), as these variables are poorly recorded in the data (see the appendix).

$$\widehat{\gamma}^{CS}(g,t) = \mathbb{E}_n\left[\left(\widehat{w}_g^{entry} - \widehat{w}_g^{never}\right)\left(Y_{it} - Y_{i,g-1} - \widehat{m}_{g,t}^{never}(X_i)\right)\right] \tag{4.3.3}$$

where the treatment and comparison weights are

$$\widehat{w}_g^{entry} = \frac{\mathbf{1}\{G_i = g\}}{\mathbb{E}_n[\mathbf{1}\{G_i = g\}]}, \qquad \widehat{w}_g^{never} = \frac{\dfrac{\widehat{p}_g(X_i)C_i}{1-\widehat{p}_g(X_i)}}{\mathbb{E}_n\left[\dfrac{\widehat{p}_g(X_i)C_i}{1-\widehat{p}_g(X_i)}\right]}. \tag{4.3.4}$$

In equation 4.3.4, $\widehat{p}_g(X_i)$ is the estimated generalized propensity score for belonging to entry cohort g, rather than to the never supplier group, conditional on sector and pre-entry firm size. The term $\widehat{m}_{g,t}^{nev}(X_i)$ in equation 4.3.3 is the estimated outcome-regression component for firms that are never suppliers, defined as the conditional expectation of $Y_{it} - Y_{i,g-1}$ given X_i among never supplier firms.

In the second step, I aggregate the estimated group-time effects by relative event time $e = t - g$. This produces event-study estimates that summarize how financial performance changes e years after entry into the procurement market. Following Callaway and Sant'Anna (2021), the event-time aggregation is

$$\widehat{\gamma}_e^{CS} \equiv \widehat{\theta}_{es}(e) = \sum_{g\in\mathcal{G}} \mathbf{1}\{g + e \leq T\}\widehat{P}(G = g \mid G + e \leq T)\widehat{\mathrm{ATT}}_{dr}^{nev}(g, g+e) \tag{4.3.5}$$

Equation 4.3.5 averages the group-time effects in equation 4.3.3 across all cohorts observed e years after entry, weighting cohorts by their relative size among the cohorts observed at that event time. The resulting parameter, $\widehat{\gamma}_e^{CS}$, is the Callaway-Sant'Anna analog of the event-study coefficient γ_j in equation 4.3.1. It traces the dynamic effect of becoming a public procurement supplier while allowing entry effects to vary across cohorts and over time since entry.

For robustness, I also estimate specifications that use not-yet-supplier firms as the comparison group. In these exercises, the structure of equation 4.3.3 remains largely the same, but the comparison group is restricted to firms that have not yet entered the public procurement market by calendar year t. The not-yet-supplier robustness specifications are estimated without covariates. These estimations exclude never-supplier firms, so identification comes from variation in treatment timing among firms that eventually become suppliers. This comparison group is informative because not-yet-supplier firms may share similar economic incentives, information, and capacity with treated firms, but enter the procurement market in a different year due to variation in timing.

4.3.1 Assumptions

My approach relies on the conditional parallel trends assumption. In the setting of staggered entry, the interpretation of $\widehat{\gamma}^{CS}(g,t)$ depends on whether, after controlling for observed covariates, firms entering the procurement market at different times, or

never entering, provide a plausible comparison group. The covariates include firm age, a size indicator (small vs. non-small), and sector fixed effects. Conditional on these characteristics, the identifying assumption is that, absent entry into the procurement market, treated and comparison firms would have followed similar trends in financial outcomes. Parallel trends can be compatible with selection into treatment when selection is based on pre-treatment information or time-invariant unobservables, as in imperfect foresight or selection on fixed effects (Ghanem et al., 2022). To make this assumption more plausible, I account for pre-entry invariant observables by implementing propensity score matching (Sekhon, 2011) and constructing a comparison group of firms with similar observable characteristics. Once these observable differences are accounted for, the remaining identifying restriction is that the evolution of untreated potential outcomes is not systematically related to the pre-treatment determinants of entry. Under these conditions, the estimates should be interpreted as relying on a structured comparison between entrants and similar non-entrants.

Staggered treatment adoption implies that once a firm participates in public procurement, it remains treated, as treatment reflects the accumulation of experience, capabilities, and market access rather than a one-time transaction. Firms that do not win a contract after they enter are still considered treated, since they continue to acquire bidding experience and may remain engaged in the delivery of previously awarded contracts, which on average take 36 months (see Table 4.1). This interpretation is consistent with the literature, which defines participation in procurement markets as a persistent state of potential bidders rather than a transitory event (e.g., Athey et al. (2013); De Silva and Rosa (2024))[8].

Furthermore, the timing of first contract awards reflects both firms' endogenous participation decisions and the outcomes of competitive procurement processes. While firms choose whether and when to participate in tenders, the exact timing of winning the first contract depends on factors such as procurement cycles, competing bids, and contract-specific conditions, which are not fully predictable at the firm level. This generates variation in the timing of entry that is plausibly unrelated to short-run fluctuations in firm performance, conditional on observables and fixed effects. In this sense, winning the first contract can be interpreted as a demand shock to the firm, reflecting the allocation of public contracts through competitive procedures (Hebous and Zimmermann, 2021). Similar empirical strategies have been used to study firms' entry into new markets under comparable identifying assumptions (e.g., Alfaro-Urena et al. (2022); Amiti et al. (2024); Chiappinelli et al. (2025); Lee (2021)).

[8]Appendix 4.D provides suggestive evidence that the first contract is more strongly associated with subsequent financial outcomes than the second contract, supporting the use of the first contract won by a firm to define entry into the public procurement market.

4.4 Results

In this section, I present the results of the event studies estimations. I begin by describing the staggered entry of firms into the public procurement market, then show the main results for general and competitive entry, and finally present results by firm heterogeneity.

4.4.1 Staggered entry

The market has a staggered adoption of entry across firms and time. Entry is firm specific and timing varies across firms. Table 4.1 presents the staggered entry of firms in the public procurement market and the number of contracts they were competing for each year. In total, there were 56,141 contracts awarded from 2016 to 2024, out of which 29,306 were awarded through competitive bidding. I define the year of the first observed contract as the year of entry. This is the first contract awarded to a firm above European public procurement thresholds. The data contain a total of 4,235 events rolled out from 2017 to 2024. I define competitive mechanisms as the first time a firm wins a contract via an open procedure with at least two bidders. This corresponds to 1,095 events in the final sample over the period 2016 to 2024.

The staggered timing facilitates two comparison groups, firms that never supply to the government during the sample period, and firms that have not-yet-entered by a given point in time. The preferred comparison group consists of never-supplier firms. Conditional on being similar in observable characteristics (see matching description in appendix 4.C), these never-supplier firms inform what would have happened had a supplier firm never entered the procurement market. The strategy compares a firm's financial outcomes before and after its own entry, using matched never-suppliers to absorb common trends. A potential concern is that the decision to bid for government contracts is driven by unobservable firm characteristics. As a robustness check, I also use not-yet-supplier firms as a comparison group. By definition, the size of the not-yet-supplier comparison group declines over time as more firms enter the procurement market.

The results in this section are based on the firms reported in the Staggered columns under General Entry and Competitive Entry in Table 4.1. These columns report the cumulative number of firms whose first European public procurement contract is observed by each year. For example, in 2017, 859 firms entered the procurement market under the general entry definition, which includes firms entering after submitting bids, participating in negotiations, or through other selection procedures. Of these firms, 234 entered through competitive entry, defined as winning a first public procurement contract in a procedure with at least two submitted bids. Similarly, by 2024, 4,235 firms had entered under the general entry definition, of which 1,095 had entered through competitive procedures. In annual terms, 246 firms entered for the first time in 2024, including 49 firms whose first contract was

Table 4.1 Market staggered entry

	Awarded	General Entry		Competitive Entry	
Year	Contracts *	Events	Staggered	Events	Staggered
2016	1,835				
2017	4,698	859	859	234	234
2018	5,618	629	1,488	196	430
2019	5,971	528	2,016	131	561
2020	6,205	505	2,521	155	716
2021	6,876	437	2,958	106	822
2022	8,363	511	3,469	115	937
2023	8,397	520	3,989	109	1,046
2024	8,178	246	4,235	49	1,095
Total	56,141	4,235		1,095	

Notes: This table presents the staggered entry of firms in the public procurement market. * The column Awarded Contracts refers to contracts with no lot division. Table 4.C.1 in the Appendix presents additional columns with all contracts. The Staggered columns for general entry and competitive entry report the cumulative number of firms that have entered the market by each year over the sample period.

awarded through a competitive procedure.

4.4.2 Main results: general and competitive entry

I discuss the main results starting from three financial variables: assets growth, current ratio, and liabilities. The Figures 4.1 shows the Panel A presents results for 4,235 general entry events, and Panel B for 1,095 competitive entry events. Event time 0 corresponds to the entry year. Table 4.2 presents the point estimates and standard errors. The event study plots for the other variables are presented in the Appendix in Figures 4.E.1 and Figures 4.E.2 and in Table 4.E.1.

I begin by examining assets growth subpanels (a) and (b) in figure 4.1. General entry into public procurement is associated with a positive and statistically significant increase in assets growth. In the year of entry, assets grow by approximately 8%, rising to around 11% in subsequent years. Firms that enter through competitive mechanisms experience a similar initial increase in assets of about 9% in the first year. However, this effect is not persistent and fades over time, becoming statistically insignificant one year after entry. Consistent patterns emerge in levels. The value of assets increases by EUR 55,254 relative to never suppliers, although this result is only statistically significant at the 15% level. For firms entering through competition, the results indicate that they experience an increase in assets of EUR 52,867 the first year. However, this change is not sustained. In the subsequent years, asset levels decline relative to both pre-entry values and firms that never supply to the government.

Next, I turn to the current ratio, a balance sheet measure of liquidity. General entry into public procurement is associated with a decline in firms' liquidity relative to never suppliers. The current ratio decreases by 0.241 points (on a 0–100 scale) in the year of entry. This decline persists over time and becomes more pronounced, reaching a reduction of approximately 1.5 points five years after entry. For competitive entry, the current ratio remains unchanged in the first two years

after entry. However, starting in the third year, the current ratio declines by 0.691 points, reaching a reduction of 1.02 points by the fifth year.

Entry into public procurement is also associated with an increase in liabilities. For general entry, liabilities increase by EUR 40,741 in the year following entry and reach EUR 149,864 five years after entry. Entry through competitive procedures is also associated with an increase in firms' liabilities with a rise by EUR 42,439 in the first year after entry and EUR 114,893 three years after entry.

The estimations regarding number of employees and employment growth are inconclusive on how employment responds to entry into public procurement, (see Table 4.E.1 columns (3) and (4)). For both general and competitive entry, the estimated effects on the number of employees are statistically insignificant, and although the point estimates are positive, their magnitude is economically negligible. The results for employment growth are statistically insignificant in the first four years. However, in the fifth year, general entry is associated with a decline in employment growth of 12.9%. For firms entering through competitive procedures, a decline is observed earlier, in the fourth year, amounting to a reduction of 10.6%, although this effect is less precisely estimated.

These results are suggestive of firm expansion. The results are consistent with previous literature showing that winning procurement contracts facilitates access to credit (di Giovanni et al., 2024). Regarding assets growth, these results suggest that while entry is associated with firm expansion, only general entry leads to sustained asset growth, whereas competitive entry generates short-run increases that are not maintained over time. These results are similar to those of Fadic (2020) who finds that firms that win in randomized contest their revenues and assets revert back to pre-shock levels.

The findings on employment are consistent with the mixed evidence in the literature on public procurement (Gugler et al., 2015; Srhoj and Dragojević, 2024). Several mechanisms may explain these results. First, general equilibrium effects may dampen net employment gains, as increased activity among winning firms may be offset by reduced activity among competitors (Ferraz et al., 2015; Srhoj and Dragojević, 2024). Second, firms may rely on subcontracting, particularly in labor-intensive contracts, limiting the extent to which employment expands within the firm.

Are changes in financial performance persistent? As a benchmark, I also aggregate the estimates of $\widehat{\gamma}^{CS}(g,t)$ across post-entry years to obtain the overall effect. The goal is to add to the discussion of whether entering a procurement market has persistent effects (Gugler et al., 2015). The estimates provide a measure of whether entry into public procurement markets generates persistent changes in firms' financial performance.

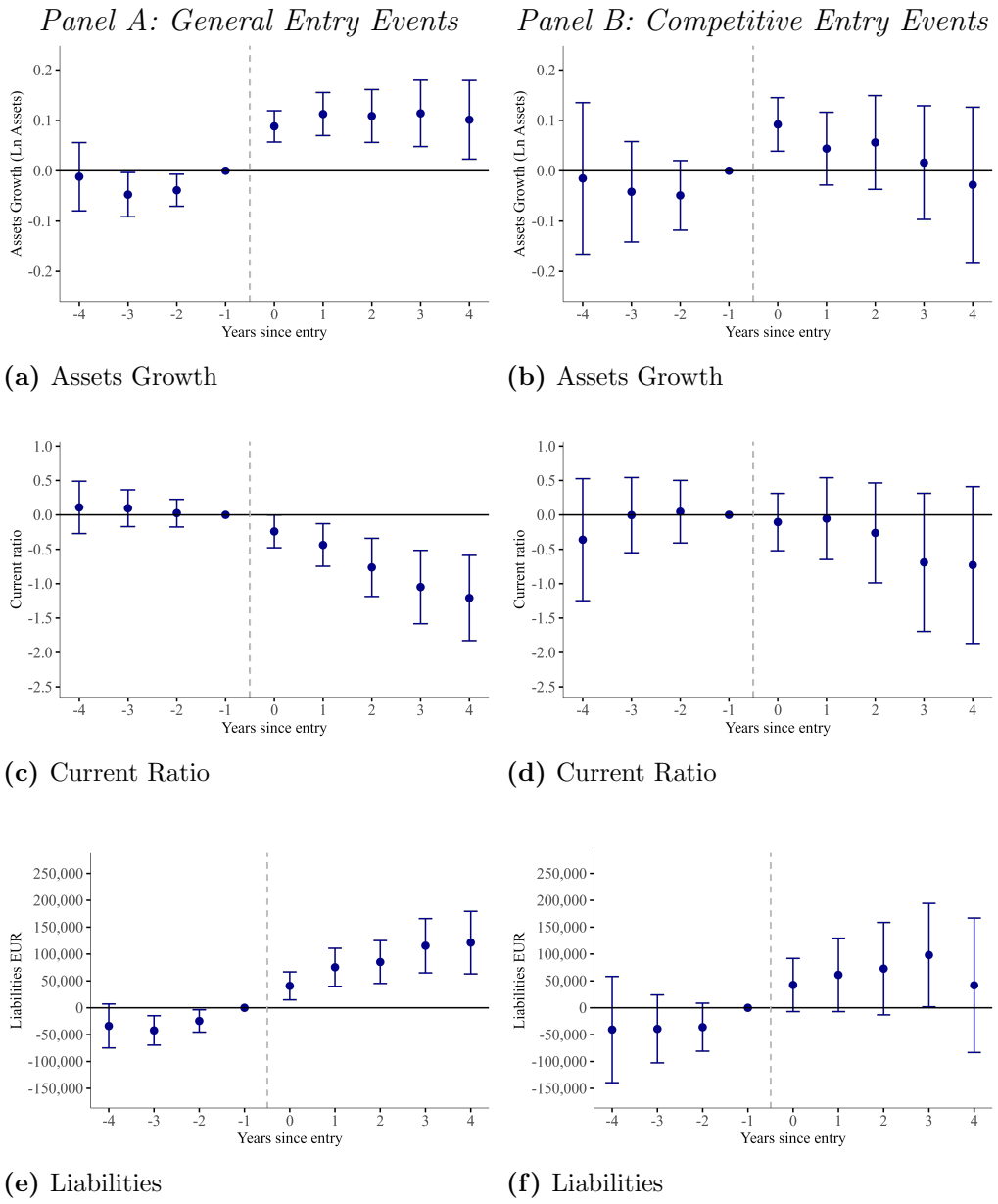
The overall estimates indicate that general entry into the public procurement market is associated with asset growth relative to matched never suppliers (see Table 4.E.2). The change in total assets in levels is positive but not statistically significant, with an estimated increase of EUR 28,065. In contrast, the log spec-

ification shows a statistically significant increase of 9.9%, suggesting that general entry is associated with growth in firms' asset base. For competitive entry, the evidence on asset growth is weaker. The estimated effect on total assets is small and statistically insignificant, and the log asset estimate is also not statistically different from zero. Thus, unlike general entry, competitive entry does not appear to be associated with an overall increase in assets relative to never suppliers.

The results for liabilities show a different pattern. Liabilities increase significantly under both general and competitive entry, although the estimated change is larger for general entry. General entry is associated with an increase in liabilities of approximately EUR 97,975 and liability growth of 22.3%, while competitive entry is associated with an increase of EUR 71,887 and liability growth of 13.5%. These results suggest that entry into the procurement market is accompanied by greater financial obligations, with stronger effects among firms entering under the broader general-entry definition.

Overall, these results suggest that entry into public procurement is associated primarily with balance-sheet adjustments rather than broad-based firm expansion. Firms appear to take on more liabilities and experience weaker liquidity after entry, while evidence of sustained growth in assets and employment is more limited, especially among firms entering through competitive procedures.

Figure 4.1 Event Study Plots Financial Performance



(a) Assets Growth **(b)** Assets Growth

(c) Current Ratio **(d)** Current Ratio

(e) Liabilities **(f)** Liabilities

Notes: Figure 4.1 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. Sub-panels (a) and (b) display estimates for capital asset growth (Columns 1 and 6 of Table 4.2); (c) and (d) for the current ratio (Columns 3 and 8); and (e) and (f) for liabilities (Columns 5 and 10). The event study plots for the other variables are presented in the Appendix in Figures 4.E.1 and Figures 4.E.2 and in Table 4.E.1.

Table 4.2 Event study estimates financial performance

	Panel A: General Entry Event					Panel B: Competitive Entry Event				
Event	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
5 years before entry	-0.017	22,353.825	0.042	-0.173*	-63,614.960*	-0.006	27,548.095	-0.458	-0.125	-54,092.621
	(0.040)	(38,011.255)	(0.240)	(0.039)	(22,052.895)	(0.085)	(80,598.633)	(0.517)	(0.088)	(39,651.157)
4 years before entry	-0.012	-18,894.554	0.109	-0.140*	-33,837.325	-0.015	-57,875.838	-0.361	-0.118	-40,624.794
	(0.025)	(27,493.495)	(0.142)	(0.029)	(14,984.239)	(0.055)	(52,554.406)	(0.339)	(0.052)	(35,346.500)
3 years before entry	-0.047*	-60,658.226*	0.097	-0.120*	-42,143.847*	-0.042	-78,318.391	-0.004	-0.094*	-39,311.854
	(0.016)	(17,595.090)	(0.100)	(0.018)	(10,041.671)	(0.037)	(33,976.448)	(0.209)	(0.033)	(22,601.316)
2 years before entry	-0.039*	-62,693.319*	0.025	-0.066*	-24,506.024*	-0.049	-61,256.044	0.046	-0.063	-36,101.234
	(0.012)	(12,432.310)	(0.075)	(0.013)	(7,626.737)	(0.025)	(22,475.746)	(0.174)	(0.025)	(15,970.613)
Year of entry	0.088*	14,952.516	-0.241*	0.113*	40,741.166*	0.092*	52,867.741	-0.104	0.134*	42,439.420
	(0.011)	(14,686.759)	(0.089)	(0.014)	(9,520.227)	(0.020)	(30,556.783)	(0.159)	(0.026)	(17,678.862)
1 year after entry	0.113*	-1,288.540	-0.437*	0.181*	75,335.956*	0.044	-46,056.163	-0.054	0.109*	61,217.398
	(0.016)	(22,275.253)	(0.116)	(0.019)	(12,948.732)	(0.027)	(41,717.855)	(0.227)	(0.036)	(24,396.528)
2 years after entry	0.109*	-493.401	-0.763*	0.221*	85,184.149*	0.056	-48,694.382	-0.262	0.151*	72,822.632
	(0.019)	(27,745.709)	(0.158)	(0.024)	(14,586.659)	(0.034)	(58,118.651)	(0.278)	(0.048)	(30,730.141)
3 years after entry	0.114*	52,254.883	-1.049*	0.267*	115,446.096*	0.016	-32,439.528	-0.691	0.161	98,080.252*
	(0.024)	(34,879.467)	(0.199)	(0.032)	(18,491.265)	(0.042)	(63,809.280)	(0.384)	(0.060)	(34,403.672)
4 years after entry	0.101*	56,550.350	-1.208*	0.249*	121,279.188*	-0.028	27,100.160	-0.729	0.079	41,869.886
	(0.028)	(43,990.882)	(0.232)	(0.037)	(21,317.577)	(0.057)	(90,660.685)	(0.436)	(0.074)	(44,740.563)
5 years after entry	0.070	46,411.588	-1.542*	0.310*	149,864.290*	-0.028	59,421.018	-1.092	0.175	114,893.540
	(0.036)	(50,483.995)	(0.275)	(0.043)	(26,807.631)	(0.064)	(96,323.112)	(0.532)	(0.082)	(54,100.662)
Observations	13,445	13,445	13,258	13,205	13,245	3,348	3,348	3,305	3,263	3,278
Cohorts	8	8	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13	13	13

Notes: This table presents the estimations of event studies using Callaway and Sant'Anna estimator. Comparison group is sample of matched never suppliers. Standard errors in parenthesis clustered at firm level. Cohort defined by year of entry in the public procurement market; the first year in which the firm was awarded a contract by a public entity.

Alternative comparison: not-yet-suppliers So far, the results are based on estimations that use matched never-suppliers as the comparison group. However, using firms that never enter the public procurement market raises concerns about their comparability with first time entrants, as these firms may differ systematically in both observable and unobservable characteristics. To address this concern, I estimate equation 4.3.3 and compute the aggregated γ_j coefficients as in equation 4.3.5, but now using firms that have not yet entered the procurement market as the comparison group. In the terminology of the difference-in-differences literature, these firms are not yet treated. This comparison group is informative, as it captures firms that may eventually enter the market and therefore provides another comparable group. This will provide insight into the role of selection into procurement. Figures 4.F.1 present the results from both specifications: estimations using never-suppliers (in blue, with solid confidence intervals) and estimations using not-yet-suppliers (in red, with dotted confidence intervals). For expositional clarity, I focus on asset growth, while the full set of results is reported in Tables 4.F.1 and 4.F.2. The results are qualitatively consistent across the two specifications for both asset growth and liability growth. However, point estimates differ in magnitude, and in some cases lose statistical significance when using not yet suppliers as the comparison group. This may be driven by the smaller effective sample size in this specification. For general entry, the estimates using not-yet-suppliers differ from the estimates using never suppliers in both magnitude and direction for the current ratio, equity, and employment growth. These differences suggest that selection into public procurement markets may play an important role in shaping firm outcomes. For firms entering through competitive mechanisms, the results remain largely stable across specifications. Finally, I find no evidence of differential pre-trends in either specification, supporting the validity of the conditional parallel trends assumption.

4.4.3 Firms heterogeneity

I study firm heterogeneity by classifying entrants into three types and re-estimating the event studies for each, with financial outcomes measured relative to matched never-suppliers. The three types are defined as follows. Short-term participants are firms observed only in the year of entry. Capacity-constrained firms are active upon entry and in subsequent years, but participate intermittently, exhibiting gaps over time. Established firms enter the market and remain continuously active thereafter, with no observable interruptions. This typology is motivated by the concern that entry may be correlated with observed and unobserved firm characteristics (Krasnokutskaya, 2011; De Silva et al., 2009), which could lead to overestimated effects due to non-random selection and unobserved heterogeneity. The aim is not to compare the three groups as if they were randomly assigned, but to decompose the average post-entry effect by procurement trajectory and document how financial outcomes differ across firms with different degrees of market engagement.

The three groups differ along observable dimensions, consistent with the view that the typology captures meaningful variation in firm characteristics (see Table

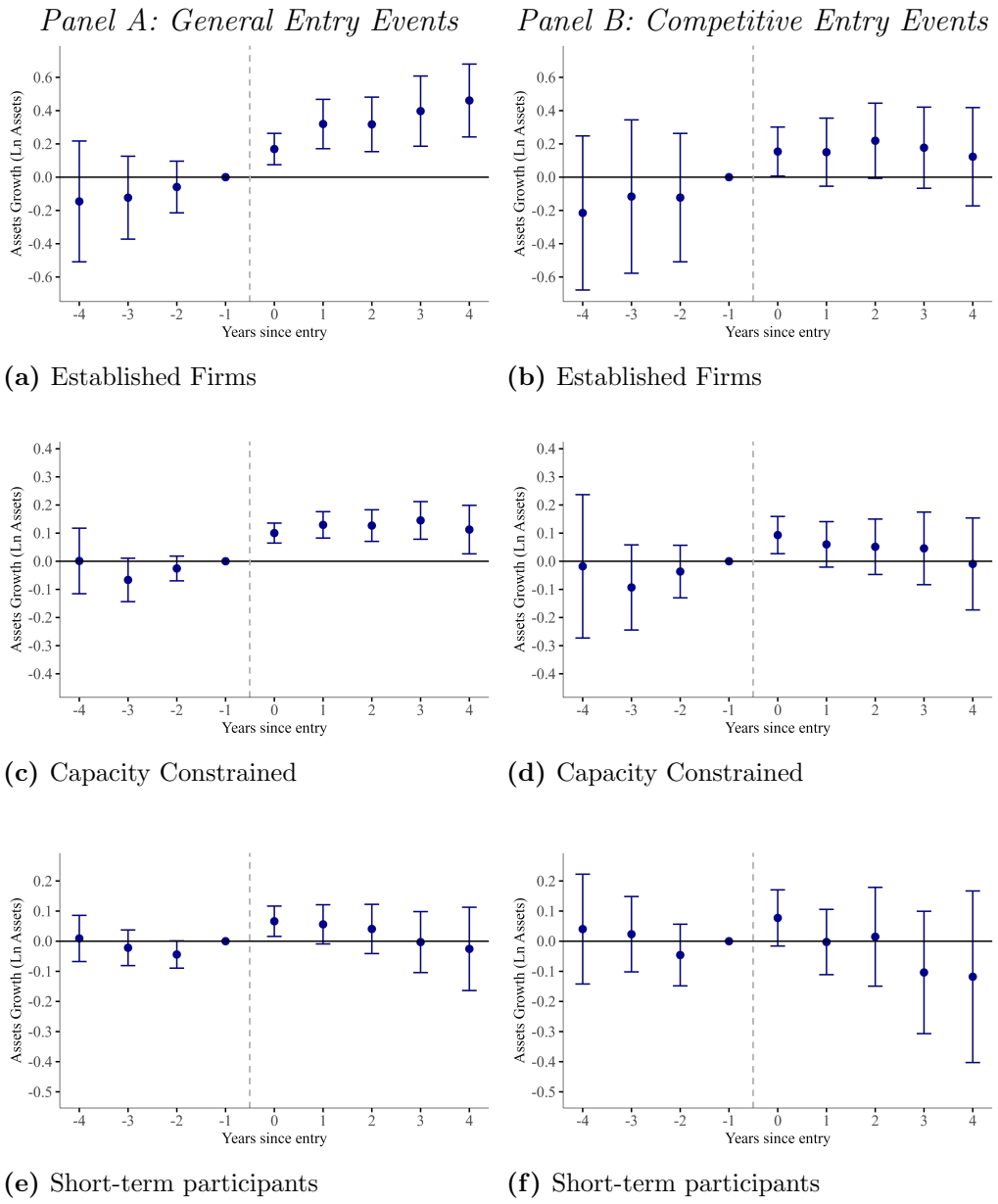
4.E.3). Established firms are medium-to-large enterprises of approximately 17 years since incorporation; upon entry they are awarded on average more than ten contracts, and they continue to scale their procurement activity over time. Capacity-constrained firms are smaller, participate intermittently, and display patterns consistent with binding resource or production constraints. Short-term participants are the smallest firms, typically awarded a single contract upon entry. All three types are heavily concentrated in the services sector (around 80 to 85%). Established firms win substantially more tenders and are awarded higher-value contracts on average, while contract duration is broadly similar across groups, approximately 36 months.

The heterogeneity analysis reveals that both the magnitude and persistence of post-entry gains are associated with firm type (see Figure 4.2). Established firms exhibit the strongest response, with asset growth of about 16% following general entry that remains persistent over time, while the 15% increase under competitive entry dissipates. Capacity-constrained firms also experience positive effects, though more moderate: asset growth reaches around 10% under general entry and remains relatively stable, whereas the 9% increase associated with competitive entry fades over time. In contrast, short-term firms display the weakest and least durable effects, with a 6% increase under general entry that disappears within two years, and a 7% gain under competitive entry that similarly dissipates. The results in other financial variables indicate a broadly consistent pattern (see Appendix 4.E.1). General entry is associated with increases in assets and liabilities for established and capacity-constrained firms, alongside positive and significant liability growth for all firm types, but a decline in liquidity (current ratio) for the former two groups. In contrast, competitive entry yields largely insignificant effects, with the exception of positive liability growth for established and capacity-constrained firms. Short-term firms show no significant changes in assets or liabilities under either entry mode. Across all specifications, shareholder funds and solvency ratios remain unaffected. Employment effects are also limited: the number of employees does not change significantly for any firm type, while employee growth is generally unaffected except for a delayed decline among short-term firms, which becomes statistically significant only four years after entry.

A potential concern with this typology is that the classification is partly based on firms' subsequent procurement participation: short-term participants are defined by not winning further contracts after entry, so weaker financial performance might appear mechanically related to the group definition. I argue this concern is limited for two reasons. First, the outcome of interest is firm-level financial performance beyond the procurement market, not subsequent procurement success; weaker post-entry financial outcomes are not mechanically implied by failing to win another public contract, and may instead reflect adjustment costs, crowding out of other activities, or genuine capacity constraints. Second, entry into public procurement can be interpreted as an intention-to-treat event: winning a first government contract may confer a reputational signal or access to new markets, but firms differ in their ability or willingness to exploit this opportunity. From this perspective, the three groups correspond to different degrees of post-entry treatment uptake; es-

tablished firms are high-uptake, capacity-constrained firms are intermittent-uptake, and short-term participants are low-uptake. The estimates should therefore be read not as causal effects of firm type, but as descriptive evidence on how financial trajectories differ across firms with varying levels of post-entry procurement engagement.

Figure 4.2 Event Study plots financial performance: assets growth



(a) Established Firms **(b)** Established Firms

(c) Capacity Constrained **(d)** Capacity Constrained

(e) Short-term participants **(f)** Short-term participants

Notes: Figure 4.2 plots the estimated $\hat{\gamma}$ event-study coefficients from the regression given in equation 4.3.3. Panel A presents the event study plots for general entry and Panel B for competitive entry. The dependent variables are: capital assets growth panels a and b, current ratio panels c and d, and liabilities panels e and f. Estimations based on Callaway and Sant'Anna estimator presented in table 4.2. In the appendix, figures 4.F.2 show the estimations using not-yet suppliers; the results are qualitatively similar.

4.5 Mechanisms

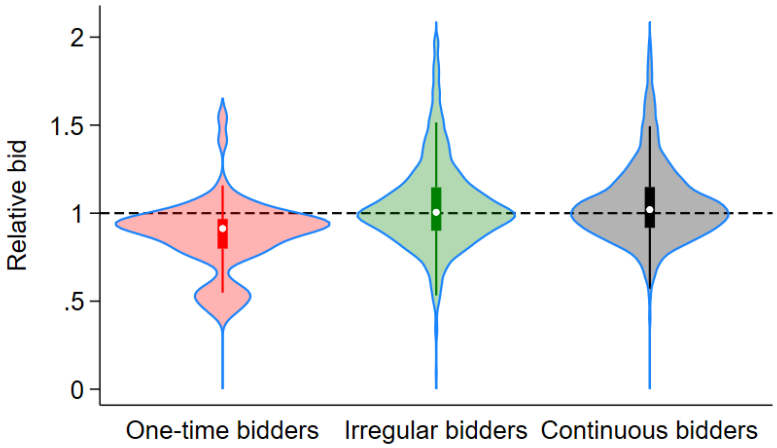
This section explores the mechanisms that rationalize why entry into public procurement markets does not uniformly translate into improved firm financial performance, particularly when entry occurs through competitive procedures. The empirical results also indicate that short-term participants exhibit short-run gains and weaker financial outcomes, while more established firms benefit persistently. I discuss two mechanisms that can account for these patterns: underbidding and selection into less competitive procurement environments.

4.5.1 Underbidding

A first mechanism relates to differences in bidding behavior across firms. I find that short-term entrants submit more aggressive bids relative to established firms. This pattern is consistent with a winner's curse mechanism, particularly observed in common value auctions. Previous literature shows that public procurement markets exhibit this pattern, specifically that firms with less experience or limited information systematically underestimate the true costs of contract execution (De Silva and Rosa, 2024; Jääskeläinen et al., 2026).

To explore pricing mechanisms, I use the subsample of auctioned tenders from one public buyer from 2012 to 2024. Figure 4.1 shows the density distribution of relative bids across firms, using violin plots over the range from 0 to 2 on the y-axis. The width of each violin reflects the density of observations at a given value, such that wider regions indicate a higher concentration of bids. Firms are grouped according to their bidding frequency. One-time bidders are observed only once, irregular bidders participate intermittently, and continuous bidders are observed consistently every year. The dashed horizontal line at 1 represents a benchmark corresponding to the engineer's cost estimate. Each violin contains a boxplot indicating the median and interquartile range of the distribution. An interesting feature of this plot is to check whether the median lies below 1, which would suggest that, on average, firms in that group submit bids below the engineer's estimate.

Figure 4.1 Distribution of relative bids by bidder type



Notes: Distribution elaborated based on ProRail data.

The visual evidence indicates that one time bidders consistently submit bids below the engineer's estimate. To quantify these differences, I estimate a reduced form regression of bid functions as in equation 4.5.1, following the structural models developed by Bajari and Ye (2003) and De Silva et al. (2003). The relative bid of firm i is denoted as rb_i and it is calculated as the ratio between the amount bid by firm i on contract c and the engineer's cost estimate for the contract, $\frac{Bid_i}{\text{Est.cost}_c}$.

The specification in equation 4.5.1 includes dummy variables for one time bidders and irregular bidders, leaving continuous firms as a baseline. In the appendix, I present an alternative version of this equation and use the definitions of short-term participants, capacity constrained firms, and established firms used in the previous firms' heterogeneity analysis (see table 4.F.3). To control for firm characteristics, I include a vector X with firms' age and status of certification, as indicated by ProRail. The buyer has in place a certification program to guarantee safety some projects need it and some not, this is a relevant control as it may be associated with the type of firms that bid and the type of projects. Additionally, I include type of project fixed effect to control for sector specific effects ω_p and I control for year effect τ_t.

$$rb_{ic} = \alpha_0 + \alpha_1 \text{OneTime} + \alpha_2 \text{Irregular} + \beta_1 \text{X}_i + \omega_p + \tau_t + \varepsilon_{ic} \tag{4.5.1}$$

Table 4.1 presents the results of estimating 4.5.1. Column (1) includes only the dummies for one-time bidders and irregular bidders and does not include any fixed-effects. The estimated α_1 indicates that one-time bidders bid approximately 17.5 percentage points lower relative to the engineer's estimate. Column (2) clusters standard errors at the firm level and does not include additional controls, even though the standard error increases the estimation remains similar. Column (3) includes fixed effects at contract level and includes the set of controls. The results now indicate that conditional on auction and firm characteristics, one-time bidders underbid by 35.3%. This suggests compositional differences across contracts that mask the extent of aggressive bidding by one-time bidders. Column (4) saturates

the regression with additional fixed effects and the estimation of α_1 remains significantly at 35.3%. The inclusion of contract fixed effects ensures that comparisons are made across firms bidding for the same contract.

Table 4.1 Relative bid reduced-form regression

	(1)	(2)	(3)	(4)
One Time Bidder	-0.172***	-0.172***	-0.353***	-0.353***
	(-4.83)	(-5.30)	(-5.82)	(-5.82)
Irregular Bidder	0.00167	0.00167	0.00983	0.0101
	(0.21)	(0.19)	(0.71)	(0.73)
Firm age			0.000301	0.000301
			(1.00)	(1.00)
Certified Firm			0.0103	0.0103
			(0.77)	(0.77)
Constant	1.069***	1.069***	1.056***	1.056***
	(264.23)	(162.87)	(78.16)	(77.96)
Observations	6672	6672	6609	6599
R^2	0.00353	0.00353	0.412	0.411
R^2adj	0.00323	0.00323	0.157	0.156
FE Year	No	No	No	Yes
FE Project	No	No	No	Yes
FE Contract	No	No	Yes	No
Type of Cluster	None	Firm	Firm	Firm

Notes: Estimations of equation 4.5.1. t statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Overall, the descriptive evidence suggests that one-time bidders and short-term participants systematically underbid relative to more regular bidder firms, a pattern consistent with a winner's curse mechanism and learning-by-doing. This behavior helps explain the main findings on entry, which indicate that new-entrants experience only short-run benefits, whereas established firms bid better and exhibit persistent and significant improvements in financial performance.

4.5.2 Number of competitors

A second mechanism concerns the type of procurement environments in which different firms participate. The number of bidders is a widely used indicator of competition in empirical studies. A higher number of participants typically increases the likelihood of attracting more efficient firms in the market and desincentivizes collusive behavior (Dimitri et al., 2006b, page 267). I test this mechanism by estimating a reduced form of the models developed by Bajari and Tadelis (2001); De Silva and Rosa (2024) as shown in equation 4.5.2. The specification consist of the number of bidders per contract N_c as dependent variable in a poisson regression with dummy variables indicating whether the contract was an entry event by including dummy indicator for general entry event and a dummy for competitive entry event. Then

I include dummies and the type of firm awarded the contract, short-term participants and capacity constrained leaving established firms as reference group. I also include controls for tender characteristics, as well as fixed effects for industry, year, and buyer, to account for systematic differences across procurement environments.

$$\begin{aligned} N_c =& \beta_0 + \sum \beta_j \text{Entry Event} + \sum \beta_g \text{TypeFirm} \\ &+ \sum \delta_{jk} \text{Entry Event} * \text{TypeFirm} + \alpha_3 \text{Z}_c + \mu_s + \psi_t + \phi_b \end{aligned} \tag{4.5.2}$$

At first the estimation of β_j indicates that entry events are associated with a higher number of bidders, Column (1) of Table 4.2, which is consistent with the simple descriptive statistics. However, once I include controls and fixed effects, the estimation of β_j indicates that general entry events are associated with a lower number of bidders. Particularly, when including short-term participants interacted with general entry events, the estimation of δ_{jk} indicates that contracts to which short-term firms enter are associated with a lower number of bidders. Conditional on contract characteristics and fixed effects, the expected number of bidders is 8.13%[9] lower for contracts to which short-term firms enter compared to contracts where established firms entered, Column (5) of Table 4.2. When including capacity constrained firms interacted with general entry events, the estimation of δ_{jk} indicates that contracts to which capacity constrained firms enter are associated with a lower number of bidders. Conditional of contract characteristics and fixed effects, the expected number of bidders is 9.74% lower for contracts to which capacity constrained firms enter compared to contracts where established firms entered.

This suggests that new entrants tend to enter in less competitive procurement environments. This pattern is consistent with the idea that less established firms may strategically target tenders with fewer competitors, which may be perceived as less risky or more accessible given their limited experience and information.

[9]Interpretation of coefficient based on $100 * (exp(-0.0848) - 1)$

Table 4.2 Number of bidders, entry events and type of firms

	Number of Bids				
	(1)	(2)	(3)	(4)	(5)
Variables					
General entry event	0.0446***	-0.0471***	-0.0415***	-0.0352***	0.0309
	(0.0098)	(0.0104)	(0.0135)	(0.0135)	(0.0247)
Competitive entry event	0.1853***	0.2592***	0.2511***	0.3060***	0.2415***
	(0.0168)	(0.0181)	(0.0230)	(0.0478)	(0.0519)
Short-term participant			0.0009	-0.0142	-0.0126
			(0.0245)	(0.0247)	(0.0248)
General × Short-term participant			-0.0142	-0.0186	-0.0848**
			(0.0316)	(0.0317)	(0.0379)
Competitive × Short-term participant			0.0202	-0.0353	0.0291
			(0.0364)	(0.0555)	(0.0591)
Capacity Constrained				-0.0254***	-0.0223***
				(0.0054)	(0.0055)
Competitive × Capacity Constrained				-0.0634	0.0248
				(0.0505)	(0.0577)
General × Capacity Constrained					-0.0915***
					(0.0290)
Log Contract Value	0.0034**	0.0106***	0.0106***	0.0104***	0.0104***
	(0.0017)	(0.0023)	(0.0023)	(0.0023)	(0.0023)
Log contract length	-0.0786***	-0.0192***	-0.0192***	-0.0189***	-0.0189***
	(0.0052)	(0.0070)	(0.0070)	(0.0070)	(0.0070)
Constant	1.839***				
	(0.0435)				
Year FE	No	Yes	Yes	Yes	Yes
Authority FE	No	Yes	Yes	Yes	Yes
Industry FE	No	Yes	Yes	Yes	Yes
Observations	43,595	43,233	43,233	43,233	43,233
Pseudo R^2	0.00209	0.13652	0.13652	0.13663	0.13667

Notes: Estimations based on a Poisson pseudo maximum likelihood regression. Interpretation based on $100 * (exp(\beta) - 1)$ transformation. Poisson fixed-effects models estimated using `fepois`. All specifications include year, contracting authority, and industry fixed effects. Standard-errors in parentheses. ****: 0.01, **: 0.05, *: 0.1.*

These findings may also indicate that contracts have different levels of attractiveness and profitability, which may influence the type of firms that choose to participate. The estimations in Table 4.2 suggest that more expensive contracts attract more bidders, while longer duration contracts are associated with fewer bidders. However, differences in other observable contract characteristics are not strongly associated with number of bidders. Table 4.F.4 in the appendix shows the descriptive statistics associated to whether the contract is part of the World Trade Organization agreement, has extensions, is an urgent procedure, includes outsourcing, is classified as a SME tender, and the type of information regarding final price, estimated cost, and type of award criterion. These characteristics are not strongly associated with the type of firms that enter, which suggests that the observed differences in number of bidders may be driven by unobservable contract characteristics.

This mechanism helps reconcile two empirical findings. On the one hand, short-

term participants are able to enter and win contracts. On the other hand, they do not experience sustained improvements in financial performance. If these firms systematically enter less competitive or less profitable tenders of the public procurement market, the contracts they win may not generate the same benefits as those obtained by more established participants. In summary, I find that short-term entrants tend to win contracts in auctions with fewer competitors. Recent studies discuss informational channels (Carril et al., 2026) and market structure (Jääskeläinen et al., 2026) as channels for similar findings. Tenders with fewer competitors may appear to have lower barriers to entry, allowing short-term participants to secure contracts despite incomplete information and their limited experience. However, these environments may also be associated with less attractive projects. For instance, contracts with fewer bidders may involve higher uncertainty, lower expected profitability, or greater complexity, deterring more established firms from participating.

4.6 Discussion

While this chapter documents significant heterogeneity in financial outcomes following entry into public procurement markets, several important avenues merit future investigation. Throughout the chapter I relied on a conditional parallel trends assumption that can be compatible with selection into treatment when this is based on pre-treatment information and time-invariant unobservables (Ghanem et al., 2022). A key limitation is that entry decisions are likely subject to dynamic selection, where firms choose to enter based on time-varying expectations about profitability that are not fully captured by observable firm characteristics or fixed effects. Unobservable dimensions such as managerial quality, technological capability, or market-specific information asymmetries may drive both the decision to enter procurement markets and subsequent financial trajectories. A fruitful avenue of research is the exploration of dynamic selection mechanisms more explicitly, for instance, by implementing Heckman and Navarro (2007) a methodological approach that bridges dynamic discrete choice and dynamic treatment effects. Decomposing bias due to dynamic selection will better inform which type of firms benefit the most from supplying to government and how market design of public procurement can be improved.

Beyond market design, future research should address the role of unobservable contract characteristics and the market structure of public procurement. Detailed information on contract complexity, technical specifications, relationship-specific investments, or strategic importance to incumbent suppliers is often unavailable in administrative data. Such unobserved heterogeneity may explain why even matched entrants experience divergent trajectories. Furthermore, the aggressive bidding strategies observed among short-term participants, particularly in less competitive procurement environments, raise questions about whether incumbents can maintain positions through tacit coordination or information advantages that effectively deter entry. Future work incorporating qualitative evidence on procure-

ment practices, interview data with firms and procurement officials, would improve our understanding of whether limited persistence reflects firms' inability to execute public contracts profitably, or whether it instead reflects market structures and practices that limit genuine competition and reduce incentives for sustained participation among new entrants.

Finally, while the contract data are relatively complete one limitation is worth noting. With some exceptions, contracts below EU procurement thresholds are not observed. My definition of entry captures the first observed participation in contracts above these thresholds, which may not coincide with a firm's initial entry into national public procurement markets. Consequently, I interpret my results as characterizing entry into large-scale procurement markets, rather than initial participation. This distinction is economically meaningful because contracts above EU thresholds represent a new market with likely higher fixed costs and complex bidding requirements. Entry at this margin reflects firms becoming first-time suppliers in more demanding segments of the public procurement market, where selection and competition may operate differently than in smaller contracts. The results should be interpreted as evidence on the specific margin of entry, while entry into smaller contracts or individual auctions remains an important avenue for future research.

4.7 Conclusions

This chapter provides new evidence on how entry into public procurement markets shapes firms' financial performance and highlights the importance of distinguishing entry mechanisms. While entry through general procurement procedures is associated with asset growth, entry through competitive bidding does not generate persistent financial improvements. This distinction is empirically important given the documented decline in bidder participation and highlights a critical gap: entry alone is insufficient to guarantee firm growth. The heterogeneity analysis reveals that gains depend substantially on firm type. Established firms obtain larger and more sustained improvements, while capacity-constrained firms exhibit weaker persistence, and short-term participants experience only temporary gains. Notably, short-term participants bid more aggressively and systematically enter less competitive procurement environments, a pattern consistent with a winner's curse mechanism and learning-by-doing effects.

The results indicate that public procurement functions more as a selection mechanism than as a driver of firm growth. Entry exposes firms to competitive pressures that not all can manage effectively, particularly when bidding strategies are aggressive and experience is limited. Many entrants fail to translate initial contract wins into sustained financial performance, while a smaller cohort of established firms consolidates their position over time. This dynamic explains both the concentration of procurement benefits among established suppliers and the declining participation in public tenders. It suggests that policies aimed solely at lowering barriers to entry are insufficient to sustain competition.

Appendices

4.A Details public procurement data

This section provides details on the main and complementary datasets used in this chapter.

4.A.1 Public procurement data extracted from the Dutch government's tendering platform

The main dataset is the public procurement data extracted from the Dutch government's tendering platform *TenderNed*. The data available in TenderNed are the main complete source of public contracts in the Netherlands. TenderNed is part of the Netherlands Enterprise Agency *Rijksdienst voor Ondernemend Nederland (RVO)* which belongs to the Ministry of Economic Affairs *Ministerie van Economische Zaken (EZ)*. While TenderNed offers a centralized platform for the announcements, contracting authorities can also use other tendering platforms for conducting the tender. The most used ones are platforms owned by Mercell. TenderNed imports these notices into their database. In fact, 55% of the tenders available in TenderNed are flagged as imported from other platforms. I extracted the datasets from TenderNed during January and February 2025. The dataset contains about 320,000 tender announcements over 2016-2024, some are contract notices and some are contract award notices. I filter the data to analyze 68,531 awarded contracts over this period, of which 56,141 are not divided into lots. Some of the relevant variables for the research design are:

- Name and chamber of commerce number of the firm awarded the contract
- Date of publication of the contract
- Value of the contract awarded
- Type of procedure used to select the supplier
- Award criteria
- Number of bidders per contract
- Name of the public buyer

While TenderNed is a reliable source of information for public procurement in the Netherlands, the available datasets pose some important limitations and misreporting issues. The most relevant ones for this study are that the name and identification number of the firm that was awarded a contract are not always available, the value of the contract is missing in several cases, the estimated value of the contract is missing as well in several cases. According to TenderNed, about 72% of the available data corresponds to the information found in the original tender

documents (TenderNed, 2025a)[10]. This implies that the missing information is not directly recoverable from scraping the original announcement and documentation. Considering this limitation, I implement a text similarity check to recover names and identification of the winning firms, and adopt an imputation strategy to recover contract awarded value.

4.A.2 Firm names recovering text similarity

The identity of the winning firm is required to link the procurement data to external sources of information. However, in the raw data, 6,916 tenders (9.4%) did not report the name of the winning firm, and 34,484 tenders (47%) reported a firm name but no identifying number. The tenders without any winner name (9.4%) contained no further information from which a firm identity could be recovered[11]; these cases are excluded from further identification efforts. The remaining 47% of tenders provided a firm name but lacked a firm identifier. Given that identifiers are crucial to enable linking with financial datasets, I implement four steps to recover them whenever possible.

First, I search for additional information on firm names and firm identifiers from alternative TenderNed sources, including the public API[12]. Second, I construct a consolidated pool of firm names and identifiers derived from both the API and the initial data extraction. In this step, I standardize and clean names by removing punctuation, special characters, and common particles. Third, I fill in missing identifiers by grouping entries with the same name. This is, I group the data by names and I propagate the identifier to the missing ones. This happens when a firm is awarded two different tenders, in one tender, the public buyer published all the information of the tender and in the other tender the buyer only published the name of the firm without the firm identifier. Fourth, I use a complete reference list of company names, each linked to at least one legal identifier [13], and apply

[10]Recently, TenderNed has put into place several strategies to improve the quality of the data. In September 2025, TenderNed announced that they had implemented a data extraction tool from tender documents. According to their preliminary findings about 72% of the tender information is found in both original dataset and the scrapped documents. This report also implies that the documents are not as complete as they could or should be. This suggestion implies that data used in this chapter underestimates some tender characteristics such as contract duration.

[11]An initial exploration of the website and pdfs downloaded from it revealed that misreporting and missing information take place from the original sources. Therefore, I discarded a web-scraping alternative. I performed this exploration before TenderNed published their results from a similar strategy (TenderNed, 2025b)

[12]TenderNed has an API available where information is available for queries. The query relies on publication identifier, not tender identifier. In TenderNed, each publication is linked to one contract, and its lots when the contract is divided into lots, but each contract can have more than one publication. In fact, 45% of tenders have two associated publications: one contract notice and one contract award notice. I use both types of publications to recover the missing company names or company identifier to use in the pool of names and for feeding the text similarity algorithm. Ultimately, in my sample, I use tenders awarded to only one company and with no division into lots.

[13]According to the Dutch Chamber of Commerce, every legal entity has a Legal Entity Partnership Information Number (RSIN), every business has a single Chamber of Commerce number

text-similarity algorithms to match firm names in the procurement data to this reference list and recover missing identifiers. Through this process, I recover 21,773 firm identifiers that are suitable for linking with additional data sources. This result allows me to effectively use 13,695 firms in the empirical strategy. At the time of performing this procedure, I did not have access to additional data from Statistics Netherlands (CBS), whose researchers merged tendered data with their microdata in 2023 (Bos and Sleijpen-Snoek, 2023; Jansen et al., 2023). Their studies cover the period 2017-2021. In contrast, here, I have aimed to provide a slightly longer historical scope, covering 2016 to 2024. Future collaboration could further extend and validate this linkage effort.

4.A.3 Final contract price and income from public procurement

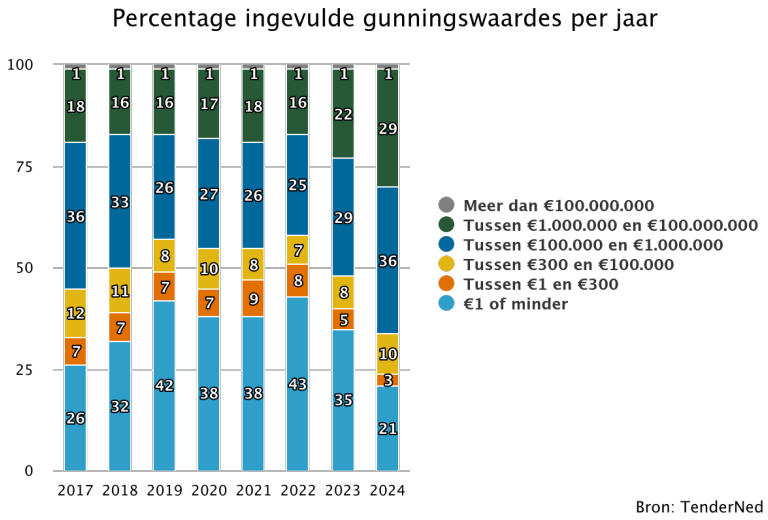
Public procurement contracts that exceed the European thresholds are expected to report the estimated value of the contract, and when publishing a contract award notice, this shall also indicate the final price or range of prices (maximum/minimum) paid (European Directive, 2014, page 218). The final price is the total amount the public entity pays to the winning firm. However, the data suffer from a large misreporting issue with an average of 34% of tenders published with an award value of 1 Euro or less (see figure 4.A.1 TenderNed (2025a)). The 1 Euro tenders phenomenon is less evident in tenders from central government institutions, such as ministries and central government organizations. However, this misreporting is a common feature of the data.

Besides the 1 Euro phenomenon, one third of the raw data have missing information in the final value. To address this issue, I construct an imputed measure of contract value that follows a transparent hierarchy of information sources in accordance to the requirements suggested in the European Directive 2014. The guiding principle is to use the most reliable monetary information first, and only rely on more indirect proxies when necessary.

I exploit the fact that tenders in TenderNed have four fields with monetary information: the estimated value of the contract, the final price of the tender, and the lowest bid received and the highest bid received in the competition; and that tenders may appear multiple times because they are associated with different publication types, such as the contract notice, the contract award notice, corrections, rectification, and pre-announcements. These multiple publications often contain complementary monetary fields. The imputation proceeds as follows. First, I start with the final contract value of the tenders from the award publication. Second, when the final value is not reported, but the estimated contract value is available in the award publication, I assign the estimated value to the missing final value. The estimated value is a good proxy because it '*... includes all expected revenues for*

in the Trade Register (KvK-nummer), some firms have an establishment number (Vestigingsnnummer), and lastly, some firms whose owner acts as a sole proprietorship, or director of a private limited company, or chairman of an association, then the firm would be assigned the Personal Registration Number of the owner(Kamer van Koophandel, 2024). Some firms can also be identified by their European VAT number.

Figure 4.A.1 Misreporting of contract values of public procurement contracts



Notes: Graph extracted from TenderNed.nl

the best bidder resulting from winning the contract. This includes payments from the client to the contractor, any commissions, bonuses, design fees, all third-party funds, interest payable, and so on.' Pianoo (2025). Third, if neither final nor estimated values are available in the award notice, I look for these variables in other publication types for the same tender. Here I make use of the assumption that the estimated value was published in the contract notice but the final price was not published in the contract award notice, or that the final price was published in a rectification notice. Fourth, some tenders are divided into lots. The European Directive 2014 indicates that 'contract notices shall include the estimated total order of magnitude of contract(s) and this information shall be provided for each lot' (page 212). In that case, I add up the final value of each individual lot to create the tender final value. When the final value is missing, I add up the estimated value of each lot and use this sum to impute the final value of the tender. As a third step, I impute remaining missing values by adding up the values of the rectification publications, if any. Nonetheless, for my main analysis, I limit my sample to tenders with only one lot, and only one winner. Fifth, some award publications do not report the final price perhaps due to the ambiguity in the European Directive as it states that 'that award notices shall provide the value of the successful tender (tenders) **or** the highest tender and lowest tender taken into consideration for the contract award or awards' (Part D). Thus, I make use of these two sources of monetary information to impute the remaining missing values. Following suggestive evidence that approximately 59% of the tenders in the Netherlands are awarded to the lowest bidder(Schotanus et al., 2022), I compute an estimation that assigns 60% of probability for the tender to be the lowest bid and 40% probability to be the average between the highest and the lowest bid, assuming that in the other cases, the

allocation mechanism is a scoring auction[14]. Finally, for European single-lot tenders with missing or implausibly low estimated values, I assign standard EU procurement threshold values based on the type of contract. In particular, I replace such values with EUR 5,538,000 for works contracts, EUR 216,000 for supplies, and EUR 750,000 for services. This step ensures consistency with the regulatory framework, as tenders above these thresholds are subject to European procurement rules, and prevents distortions arising from missing or misreported estimated values.

4.A.4 Additional description of procurement contracts

Table 4.A.1 shows descriptive statistics for the contract-level data without restricting the sample to events and event of firms in the estimation sample. The average contract has a value of EUR 1.8 million. The average number of bidders per tender is around 3.8, indicating moderate competition in the market. The distribution of contract types is relatively balanced between works, services, and supplies.

Table 4.A.1 Descriptive statistics for all contracts

	Panel A: All Contracts				Panel B: Competitive Contracts					
Variable	Mean	Median	Std.Dev.	N	Mean	Median	Std.Dev.	N	Diff.	p-value
Number of bids	3.9	3.0	4.0	54,582	3.7	3.0	3.3	18,691	0.2	0.0000
Duration in months	38	30	40	13,381	40	36	44	6,826	-2	0.0100
Contract value	1,800	750	5,798	54,476	1,568	698	4,796	18,347	231	0.0000
Services	0.7	1.0	0.5	56,141	0.7	1.0	0.5	18,759	0.0	0.3200
Supplies	0.2	0.0	0.4	56,141	0.2	0.0	0.4	18,759	0.0	0.0000
Works	0.1	0.0	0.3	56,141	0.1	0.0	0.3	18,759	0.0	0.0000

Notes: This table presents characteristics of contracts including number of bids, duration of the contract, imputed contract value, and type of tender. Unlike table 4.1 this table shows all contracts without restricting entry events, Panel A for all contracts and Panel B for competitive contracts.

According to the sectoral distribution of contracts, shown in Table 4.A.2, in general entry events the majority of contracts are awarded in business services, which account for 30.8% of contracts, followed by construction with 20%, and IT services with 13.7%. In competitive entry events, construction accounts for 23% of contracts, followed by IT services with 19%, and business services with 18%.

4.B Details firm's balance sheet information

The Orbis global database, from Bureau van Dijk (BvD) a Moody's Analytics company, is a cross country database with firm level balance sheet information. The dataset offers good information for building a nationally representative panel (Kalemli-Ozcan et al., 2015), identifying firms with state participation (Dall-Olio et al., 2022), and estimating Total Factor Productivity at the firm and national levels (Gal, 2013). However, Orbis has an unbalanced yearly coverage (Bajgar et al., 2020), I find that in the Netherlands 13% of the firms have detailed financial and investment information, 28% have limited financial reports, and 49% do not have

[14] $0.6(\text{LowestBid}) + 0.4(\text{HighestBid})$

Table 4.A.2 Sectoral distribution of contracts

	All Contracts		General Entry		Competitive Entry	
Variable	N	Percent	N	Percent	N	Percent
Agriculture, Electricity	3,274	5.83	170	4.01	72	6.58
Business services	16,246	28.94	1,304	30.79	197	17.99
Construction	11,164	19.89	849	20.05	254	23.20
Education, Health, RD	2,774	4.94	267	6.30	74	6.76
Financial and Insurance Activities	1,469	2.62	50	1.18	12	1.10
IT Services, Communication	5,635	10.04	582	13.74	207	18.90
Manufacturing	9,618	17.13	576	13.60	150	13.70
Other Service Activities	1,300	2.32	129	3.05	28	2.56
Public Administration and Defence	1,058	1.88	68	1.61	23	2.10
Water Supply, Sewerage, Waste Management	3,255	5.80	212	5.01	72	6.58
Wholesale and Retail Trade	348	0.62	28	0.66	6	0.55
Total	56,141	100.00	4,235	100.00	1,095	100.00

Notes: This table provides the distribution of contracts awarded in the Netherlands above European thresholds across sectors defined by CPV codes for all contracts, for contracts awarded in general entry, and for contracts awarded in competitive entry.

information[15]. I extracted information for nine financial variables from the balance sheet, as well as firms characteristics. Some of the key variables are the following:

- Balance sheet information: assets, liabilities, equity, and revenue.
- Number of employees
- Firms' year of incorporation
- Firms' sector NACE

A challenge arises from the structure of the financial data. In particular, firms do not report financial statements every year, resulting in an unbalanced panel. Consequently, for a subset of firms, the time series of financial outcomes contains gaps that complicate the identification of dynamic effects around the event of interest. For some financial variables. such as profit and sales, the number of usable firm-year observations falls below what would be ideal to obtain precise point estimates and tighter confidence intervals. Therefore, I focus on financial variables such as assets, shareholder funds, liabilities, current ratio, and number of employees to guarantee a stronger empirical power of the event study design.

4.C Matching algorithm

I begin by constructing baseline cohort samples using the general definition of entry. Specifically, for each entry year $g \in \{2017, \ldots, 2024\}$, I retain firms that first enter procurement in year g, and firms that never participate, which serve as controls. I exclude firms with non-treated procurement activity under this definition and

[15]From the total data, 66.5% have *Limited Financials*. According to Moodys this implies 'The information available for such companies is often based on rounded figures, or class level, officially available (sometimes also collected from other directories or web sites). In most cases only the number of employees and the Operating Revenue are available.'

define treatment based on the indicator of entry event. All variables are measured in the year prior to entry $(g-1)$, aiming for a pre-treatment baseline. Each cohort dataset contains one observation per firm and includes sector, firm's size, age, and baseline outcomes of the balance sheet information. I do this separately for general entry events and competitive entry events.

I estimate propensity scores separately for each cohort using a logistic regression model of treatment on time-invariant firm characteristics. Specifically, for each entry year g, I regress the treatment indicator on firm age, sector fixed effects, and size categories, using only variables measured in the baseline year $(g-1)$. This procedure is implemented for both definitions of entry. First, I estimate propensity scores for the broad general entry definition, where treatment corresponds to any procurement participation. Second, I repeat the estimation using the more restrictive definition of entry into competitive procurement.

After estimating the propensity scores, I construct matching samples by keeping the observations used in the logistic regression. For each cohort, I align the baseline outcome, the treatment indicator, and the estimated propensity score. I then remove any remaining observations with missing values to ensure consistency across variables. This procedure yields clean vectors and corresponding indices that are used in the subsequent matching analysis.

I implement matching based on the estimated propensity score. I follow and use the package developed by Sekhon (2011). Defining the propensity score as $e(X_i) = \Pr(T_i = 1 \mid X_i)$ from the logistic regression, the matching is performed by pairing each treated firm with the control firm whose estimated propensity score is closest, thereby conditioning on a one-dimensional summary of the covariates. Formally, for each treated unit i, I select the control unit j such that, for each treated firm i with $T_i = 1$, I select the control firm j with $T_j = 0$ whose propensity score is closest to that of firm i, $j(i) = \arg\min_{j:T_j=0} |\hat{e}(X_i) - \hat{e}(X_j)|$. In practice I implement a one nearest neighbor $(M = 1)$ with ties randomly broken.

This approach follows the standard propensity score matching framework, which ensures that, under unconfoundedness, treated and control units are comparable in terms of observed characteristics. I aggregate these cohort-specific matched datasets into a single matched sample for each definition of entry: general and competitive, ensuring that each firm appears only once where appropriate. Finally, I merge the matched firm identifiers back into the full firm-level panel. This step recovers the complete longitudinal history of the matched treated and control firms, rather than only their baseline observations. I verify that the matched firm list does not contain duplicate identifiers before constructing the final panel. The resulting matched panel is then used to estimate the event-study specifications.

4.C.1 All contracts and complete staggered entry

Table 4.C.1 reports the annual number of public procurement contracts awarded above the European procurement thresholds. The column All includes 68,531 awarded

contracts between 2016 and 2024,[16] while the column 1 Lot includes 56,141 contracts awarded without lot division. The table provides a benchmark for the number of available entry events under the general-entry and competitive-entry definitions, and for the number of events retained in the final sample after matching entrants to never-supplier firms and estimating the event-study specifications.

Table 4.C.1 Sample restrictions and staggered events

	Awarded Contracts		General Entry				Competitive Entry			
Year	All	1 Lot	Events	Staggered	Events In Sample	Staggered In Sample	Events	Staggered	Events In Sample	Staggered In Sample
2016	2,355	1,835								
2017	5,813	4,698	2,618	2,618	859	859	712	712	234	234
2018	6,951	5,618	1,939	4,557	629	1,488	613	1,325	196	430
2019	7,293	5,971	1,771	6,328	528	2,016	429	1,754	131	561
2020	7,724	6,205	1,646	7,974	505	2,521	457	2,211	155	716
2021	8,507	6,876	1,524	9,498	437	2,958	330	2,541	106	822
2022	10,061	8,363	1,801	11,299	511	3,469	334	2,875	115	937
2023	10,000	8,397	1,624	12,923	520	3,989	308	3,183	109	1,046
2024	9,827	8,178	1,788	14,711	246	4,235	179	3,362	49	1,095
Total	68,531	56,141	14,711		4,235		3,362		1,095	5,841

Notes: This table presents the staggered entry of firms in the public procurement market. The columns awarded contracts present all awarded contracts as observed in the TenderNed platform. The column 1 lot indicates a sample restriction of keeping tenders without lot division. The columns events indicate the observed first time contracts of firms and in sample restricts the analysis to firms observed in the financial panel data.

[16] Public information from TenderNed reports a total of 78,950 awarded contracts from 2017. See Aanbestedingsmonitor 2017–2025.

4.D First contracts test

The empirical analysis of this chapter relies on identifying the first contract won by a firm to define entry into the public procurement market. I provide suggestive evidence indicating that the first contract is more important than subsequent contracts. As shown in equation 4.D.1, I test whether the first contract is significantly associated with financial outcomes of entrants, captured by β_1. The test is conditional on year τ_t, sector ψ_s, buyer ϕ_B, and firm ζ_i fixed effects.

$$y_{it} = \beta_1 \text{FirstContract}_{it} + \beta_2 \text{SecondContract}_{it} + \psi_s + \tau_t + \phi_B + \zeta_i + \varepsilon_{it} \quad (4.D.1)$$

I test the hypothesis that the effect of the first contract is statistically different from zero: $H_0 : \beta_1 = 0 \; ; H_1 : \beta_1 \neq 0$. Rejecting $H_0 : \beta_1 = 0$ suggests that the first contract is more strongly associated with financial outcomes than later contracts, evidence consistent with the approach of using the first contract won as entry in the market that affects financial outcomes. The regression test also includes an indicator variable for the second contract to test the statistical significance of β_2, with the hypothesis that $H_0 : \beta_2 = 0 \; ; H_1 : \beta_2 \neq 0$. In this case, failing to reject the null hypothesis, $H_0 : \beta_2 = 0$, makes a stronger case for the importance of the first contract.

Table 4.D.1 Test relevance of the first contract

Model:	Assets Growth (1)	Liabilities EUR (2)	Current Ratio (3)	Assets EUR (4)	Liabilities Growth (5)	Solvency (6)	Shareholder EUR (7)	Employee (8)	Employee Growth (9)
Variables									
First Contract	-0.0030	113,345.7***	-0.1833***	175,145.1***	0.0138	-1.861***	22,851.2	-0.4599	-0.0387***
	(0.0125)	(16,549.7)	(0.0659)	(27,311.8)	(0.0161)	(0.4644)	(15,200.7)	(0.3302)	(0.0140)
Second Contract	-0.0106	24,585.5*	-0.0472	31,384.5	-0.0209	-0.2856	7,965.6	-0.4067	-0.0146
	(0.0105)	(13,788.8)	(0.0553)	(22,828.9)	(0.0134)	(0.3897)	(12,713.3)	(0.2741)	(0.0117)
FE									
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CPV 2-digits	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Buyer	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	16,539	14,204	16,434	15,339	14,187	16,298	15,620	11,598	11,598
Pseudo R^2	0.83426	0.06510	0.27950	0.07035	0.73244	0.18139	0.06588	0.26503	0.88838

Notes: This table presents OLS estimations of equation 4.D.1. Standard errors in parentheses. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The results of the test are shown in Table 4.D.1. For these estimations, I use only the pooled sample of firms with public procurement contracts. The first row shows the estimates of β_1 for the nine financial variables of interest. The statistical significance displayed by the p-value stars indicates that the first contract is statistically significant in five financial outcomes. In contrast, the coefficient of the second contract indicator is not statistically different from zero. Therefore, these results provide suggestive evidence indicating that the first contract is more strongly associated with financial outcomes than subsequent contracts.

4.E Estimation tables

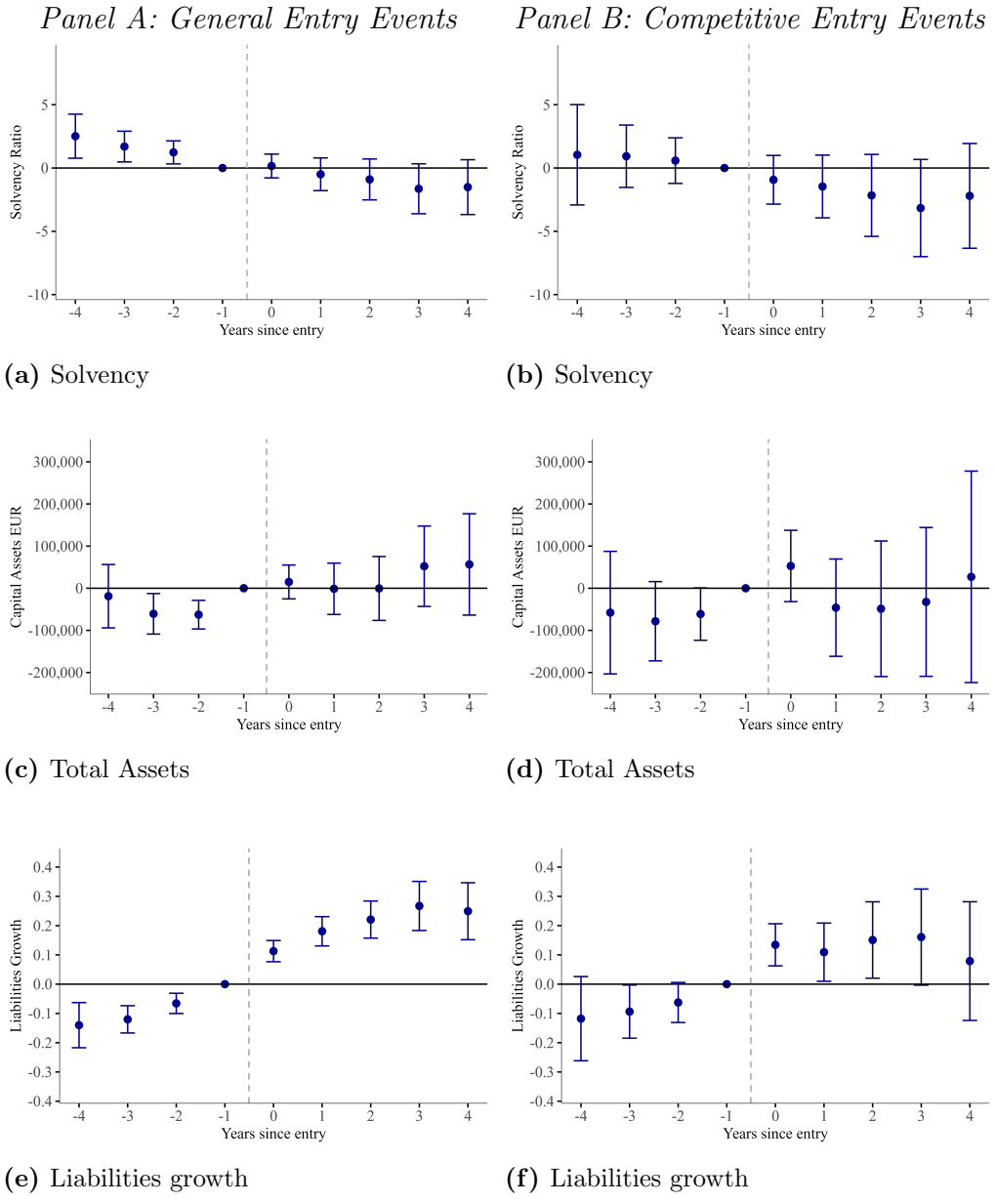
This section presents the second part of estimation tables for the main event study estimations for general entry event and competitive entry event in table 4.E.1. This second part corresponds to the remaining dependent variables: solvency, shareholder funds, number of employees, and employment growth.

Table 4.E.1 Event Study estimates financial performance (part 2)

	Panel A: General Entry Event				Panel B: Competitive Entry Event			
Event	Solvency	Shareholder	Employee	Ln Employee	Solvency	Shareholder	Employee	Ln Employee
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
5 years before entry	2.353	84,500.282*	-0.980	-0.002	3.232	96,372.139	-0.369	0.088
	(0.908)	(19,032.007)	(0.456)	(0.032)	(1.990)	(40,862.276)	(0.971)	(0.065)
4 years before entry	2.513*	70,744.342*	-0.416	0.041	1.044	48,102.051	-0.143	0.056
	(0.623)	(13,883.595)	(0.369)	(0.028)	(1.411)	(29,997.880)	(0.740)	(0.051)
3 years before entry	1.694*	33,263.133*	-0.615	0.027	0.931	19,325.002	0.094	0.084
	(0.434)	(8,507.272)	(0.272)	(0.020)	(0.877)	(17,671.290)	(0.515)	(0.034)
2 years before entry	1.234*	23,784.965*	-0.360	0.018	0.584	20,869.953	0.353	0.047
	(0.326)	(6,783.719)	(0.230)	(0.015)	(0.641)	(11,784.432)	(0.406)	(0.030)
Year of entry	0.160	18,340.010	0.114	0.020	-0.931	18,034.212	0.362	0.032
	(0.337)	(7,299.621)	(0.239)	(0.017)	(0.685)	(15,755.654)	(0.473)	(0.029)
1 year after entry	-0.494	19,447.391	0.319	-0.001	-1.459	-9,072.905	0.335	-0.014
	(0.463)	(11,480.974)	(0.324)	(0.019)	(0.882)	(22,767.946)	(0.586)	(0.035)
2 years after entry	-0.901	-1,031.977	0.324	-0.025	-2.164	-37,861.180	0.793	0.001
	(0.579)	(13,972.311)	(0.367)	(0.025)	(1.153)	(25,660.320)	(0.745)	(0.044)
3 years after entry	-1.640	-5,646.098	0.714	-0.024	-3.162	-34,020.320	0.806	-0.019
	(0.707)	(18,959.244)	(0.449)	(0.030)	(1.367)	(37,756.529)	(0.870)	(0.055)
4 years after entry	-1.510	11,136.842	0.193	-0.089	-2.201	-19,394.085	0.346	-0.106
	(0.776)	(24,637.740)	(0.558)	(0.035)	(1.474)	(47,556.658)	(1.047)	(0.070)
5 years after entry	-2.519	-16,287.526	0.028	-0.129*	-4.876*	-83,626.498	1.403	-0.069
	(0.922)	(28,673.184)	(0.704)	(0.042)	(1.641)	(52,686.095)	(1.347)	(0.080)
Observations	13,345	13,351	10,296	10,296	3,325	3,311	2,596	2,596
Cohorts	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13

Notes: This table presents the estimations of event studies using Callaway and Sant'Anna estimator. Panel A presents the estimations for general entry events and Panel B for competitive entry events. Cohorts are defined by year of entry in the public procurement market: the first year in which the firm was awarded a contract by a public entity. The comparison group corresponds to matched never suppliers. Standard errors are presented in parenthesis and clustered at firm level.

Figure 4.E.1 Event Study plots financial performance - part 2



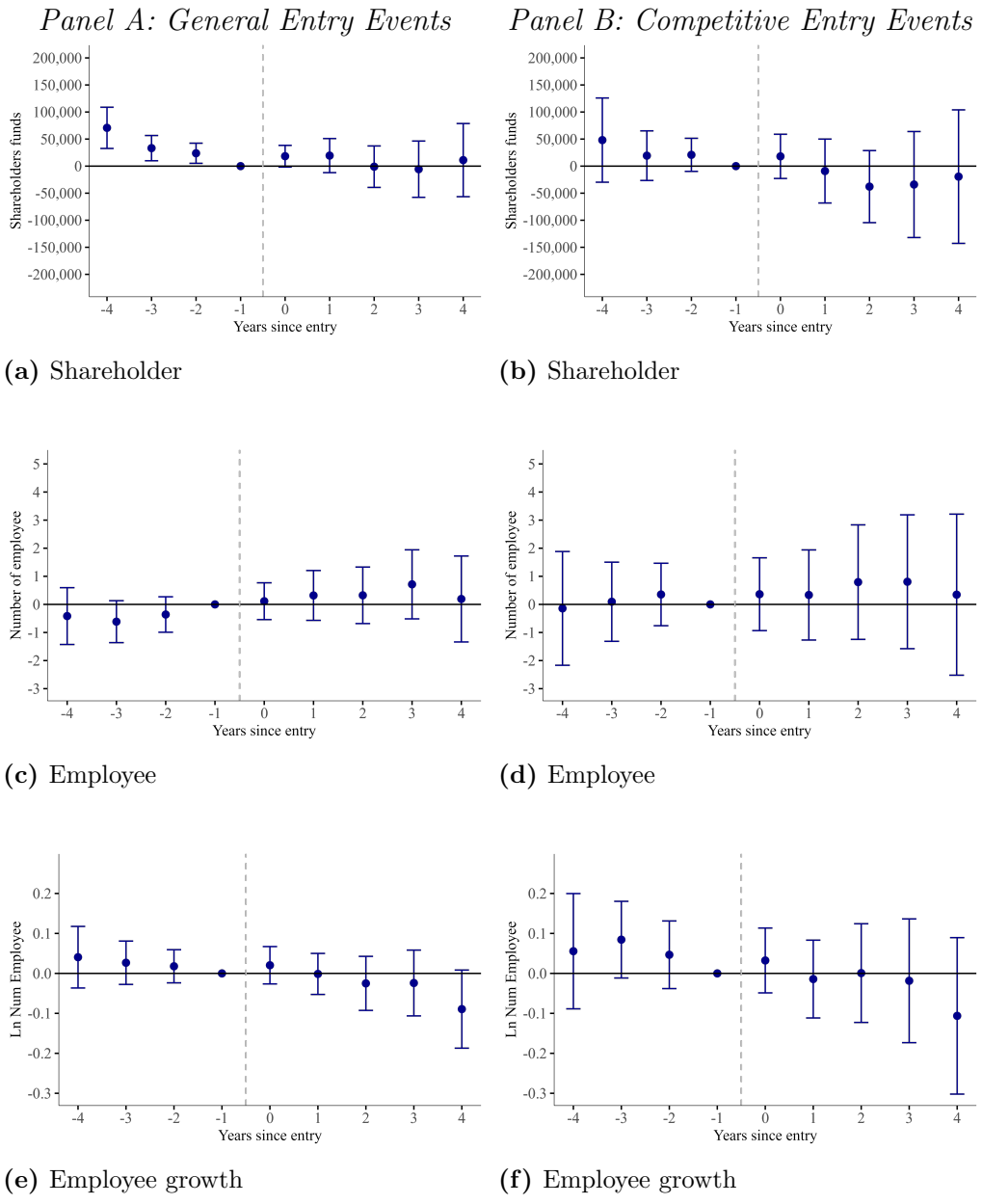
(a) Solvency **(b)** Solvency

(c) Total Assets **(d)** Total Assets

(e) Liabilities growth **(f)** Liabilities growth

Notes: Figure 4.E.1 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. Sub-panels (a) and (b) display estimates for solvency (Columns 1 and 5 of Table 4.2); (c) and (d) for total assets (Columns 2 and 6 of Table 4.2); (e) and (f) for Liabilities growth (Columns 4 and 9 of Table 4.2).

Figure 4.E.2 Event Study plots financial performance - part 3



(a) Shareholder **(b)** Shareholder

(c) Employee **(d)** Employee

(e) Employee growth **(f)** Employee growth

Notes: Figure 4.E.1 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. Sub-panels (a) and (b) display estimates for shareholders funds (Columns 2 and 6 of Table 4.E.1); (c) and (d) for number of employees (Columns 3 and 7 of Table 4.E.1); and (e) and (f) for employee growth (Columns 4 and 8 of Table 4.E.1).

As a benchmark, I also aggregate the estimates of $\widehat{\gamma}^{CS}(g,t)$ across post-entry years to obtain the overall change. The Callaway-Sant'Anna estimator first estimates cohort-specific treatment effects for each group of firms that enters the procurement market in year g and each calendar year t, comparing their outcome changes to an untreated comparison group such as never-supplier firms. These group-time effects are then aggregated by relative event time $e = t - g$ to obtain dynamic event-study estimates that allow treatment effects to vary across cohorts and over time since entry. The goal is to add to the discussion of whether entering a procurement market has persistent effects.

Table 4.E.2 Overall coefficients

	Current ratio	Solvency	Shareholders	Total Assets	Ln Assets	Employee	Ln Employee	Liabilities	Ln Liabilities
General Entry									
Overall ATT	-0.873*	-1.151	4,326.441	28,064.566	0.099*	0.282	-0.041	97,975.141*	0.223*
Std. Error	(0.154)	(0.491)	(13,425.796)	(25,709.455)	(0.017)	(0.331)	(0.022)	(13,079.157)	(0.022)
Observations	13258	13345	13351	13445	13445	10296	10296	13245	13205
Cohorts	8	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13	13
Competitive Entry									
Overall ATT	-0.489	-2.466	-27,656.796	2,033.141	0.025	0.674	-0.029	71,887.188*	0.135*
Std. Error	(0.260)	(0.913)	(25,235.989)	(51,266.068)	(0.032)	(0.705)	(0.043)	(23,704.665)	(0.046)
Observations	3305	3325	3311	3348	3348	2596	2596	3278	3263
Cohorts	8	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13	13

Notes: The table presents the aggregation of the event studies estimations using the Callaway and Sant'Anna estimator. The Section on the top presents the estimations for general entry events and Section on the bottom for competitive entry events.

4.E.1 Firm heterogeneity

In this section, I present in more detail the heterogeneity estimations by firm type. I define three types of firms: (i) short-term participants that are observed as active only in the year of entry; (ii) capacity constrained firms that are active in the procurement market in the year of entry and in subsequent years, but whose participation is intermittent, exhibiting gaps over time; and (iii) established firms that enter the market and remain continuously active in subsequent years, with no observable interruptions in participation[17].

Table 4.E.3 Firm Heterogeneity Characteristics

	Short-term participants				Capacity constrained				Established firms			
Variable	Mean	Median	Std.Dev.	N	Mean	Median	Std.Dev.	N	Mean	Median	Std.Dev.	N
General Entry												
Firms characteristics												
Small	0.61	1.00	0.49	3,675	0.42	0.00	0.49	2,629	0.37	0.00	0.48	454
Medium	0.37	0.00	0.48	3,675	0.54	1.00	0.50	2,629	0.56	1.00	0.50	454
Large	0.02	0.00	0.14	3,675	0.04	0.00	0.20	2,629	0.07	0.00	0.25	454
Firm age (years)	18.46	14.00	13.95	3,675	22.46	18.00	16.26	2,629	16.95	12.00	13.41	454
Sector construction	0.08	0.00	0.27	3,675	0.09	0.00	0.29	2,629	0.08	0.00	0.27	454
Sector manufacturing	0.04	0.00	0.20	3,675	0.04	0.00	0.19	2,629	0.05	0.00	0.22	454
Sector other activities	0.03	0.00	0.16	3,675	0.02	0.00	0.15	2,629	0.03	0.00	0.18	454
Sector services	0.85	1.00	0.36	3,675	0.85	1.00	0.36	2,629	0.84	1.00	0.37	454
Contracts characteristics												
Number of tenders	1.12	1.00	0.45	3,675	4.83	3.00	5.11	2,629	14.39	8.00	19.03	454
Number of lowest price	0.23	0.00	0.44	3,675	1.01	0.00	1.57	2,629	2.53	1.00	4.38	454
Contract duration (months)	39.06	28.37	52.73	523	37.38	30.58	27.30	943	35.44	30.37	24.55	210
Contract value (Thousand Euro)	1 233.99	750.00	3 766.03	1,977	1 350.84	697.96	2 602.17	2,038	1 359.64	691.76	2 821.91	393
Competitive Entry												
Firms characteristics												
Small	0.57	1.00	0.50	824	0.40	0.00	0.49	732	0.26	0.00	0.44	119
Medium	0.42	0.00	0.49	824	0.55	1.00	0.50	732	0.66	1.00	0.48	119
Large	0.02	0.00	0.13	824	0.05	0.00	0.21	732	0.08	0.00	0.28	119
Firm age (years)	19.69	16.00	14.61	824	23.12	19.00	16.20	732	19.06	15.00	14.42	119
Sector construction	0.10	0.00	0.31	824	0.12	0.00	0.33	732	0.08	0.00	0.27	119
Sector manufacturing	0.04	0.00	0.20	824	0.03	0.00	0.17	732	0.08	0.00	0.28	119
Sector other activities	0.04	0.00	0.19	824	0.03	0.00	0.17	732	0.04	0.00	0.20	119
Sector services	0.81	1.00	0.39	824	0.82	1.00	0.39	732	0.80	1.00	0.40	119
Contracts characteristics												
Number of tenders	1.10	1.00	0.39	824	4.77	3.00	4.21	732	11.84	9.00	11.89	119
Number of lowest price	0.75	1.00	0.51	824	1.97	2.00	1.95	732	4.80	3.00	5.98	119
Contract duration (months)	36.95	34.00	30.59	212	37.63	30.96	30.45	359	40.24	33.95	31.63	77
Contract value (Thousand Euro)	1 036.18	585.60	2 472.98	508	1 339.21	666.29	2 626.68	639	1 333.50	750.00	2 142.92	114

Notes: The table reports summary statistics for firm characteristics across three groups of firms: short-term participants, capacity constrained firms, and established firms. The top section reports firms in general entry events. The bottom section reports firms in competitive entry events.

Table 4.E.3 presents descriptive statistics for the firms and for the contracts awarded to them. The top section of the table reports summary statistics for firms in general entry events and the bottom section for firms in competitive entry events. In terms of size, short-term participants are predominantly small firms, whereas capacity-constrained firms are more evenly split between small and medium-sized firms, and established firms are primarily medium-sized with a somewhat larger share of large firms. In terms of age, capacity-constrained firms are the oldest on average, followed by short-term participants, while established firms are somewhat younger, though differences across groups are moderate. All three types are heavily

[17]The analysis focuses on firms that enter the market; therefore, I do not consider incumbents or firms whose year of entry is not observed in the data. For reference, these incumbents are large firms that have been incorporated for 25 years on average, and are awarded around 45 contracts between 2016 and 2024.

concentrated in the services sector (around 80 to 85%), with construction, manufacturing, and other activities representing small and broadly comparable shares.

In terms of the characteristics of the contracts they are awarded, established firms win more tenders than the other groups, averaging 14, followed by capacity-constrained firms, with an average of 5, while short-term participants win one tender on average. Contract duration is broadly similar across groups, three years on average, suggesting that differences in firm outcomes are unlikely to reflect differences in contract length. The average value of the contracts is higher for capacity constrained and established firms, although there is substantial within-group dispersion. These patterns are mirrored under competitive entry, with relative differences across firm types remaining consistent.

Ideally, I would describe firms' sales, output, and revenue sources, the data provide limited coverage. Nonetheless, I calculate the public procurement revenue as a share of total sales revenue using a small subsample of firms with observed turnover. The goal is to understand to what extent entrant firms rely on supplying to the government for their operations. Table 4.E.4 presents the average share of public procurement revenue in total firm revenue and tests the differences in means with respect to incumbents. I find that the shares are on average 11% for established firms, 8.4% for capacity-constrained firms, and 5.2% for short-term participants. The differences with respect to incumbents are statistically significant at the 1% level for all three groups, suggesting that public procurement revenue constitutes a smaller share of total turnover for entrants compared to incumbents. This supports the idea that entrant firms are not primarily dependent on procurement revenue.

Table 4.E.4 Share of procurement revenue in total firm revenue

Type of firm	Mean	Median	Std.Dev.	N	Diff	P-value
Established firms	10.8%	2.6%	18.0%	322	-2.8%	0.0278
Capacity constrained	7.4%	1.7%	14.8%	1,566	-6.2%	0.0000
Short-term participants	5.2%	1.1%	10.3%	200	-8.4%	0.0000
Incumbents	13.6%	5.6%	19.1%	597		

Notes: The table reports summary statistics for the share of procurement revenue in total operating turnover revenue. The table shows the three types of firms: short-term participants, capacity constrained firms, and established firms; as well as incumbents. The column $Diff$ shows the differences in means between each type and incumbents.

The estimations by firm heterogeneity reveal a broadly consistent pattern in financial outcomes. Figures in 4.E.3 show that established firms exhibit increases in total assets (Column (2) of Table 4.E.5), whereas entry through competitive channels does not generate statistically significant changes in asset holdings (Column (7) of Table 4.E.5). For capacity constrained and short-term firms (Columns (2) and (7) of Table 4.E.7 and Columns (2) and (7) of Table 4.E.9, respectively), I find no statistically significant changes in asset levels.

With respect to liquidity, measured by the current ratio, general entry is associated with a decline for both established and capacity constrained firms, while no statistically significant effect is observed for short-term firms. Under competitive entry, the estimates indicate no significant changes in the current ratio across any

firm type (see Figures 4.E.4). I find increases in liabilities for established and capacity constrained firms following general entry. Although the point estimates for liabilities under competitive entry are positive, they are not statistically significant. Short-term firms do not exhibit significant changes in liabilities under either form of entry (see Figures 4.E.5). Examining liability growth, the results indicate positive and statistically significant effects for all three firm types under general entry. In contrast, under competitive entry, liability growth is positive and significant for established and capacity constrained firms, but not for short-term participants (see Figures 4.E.6). Shareholder funds remain statistically unchanged across all firm types and entry modes. While the estimates for short-term participants are negative, they are not statistically distinguishable from zero (see Figures 4.E.7). The solvency ratio does not exhibit any statistically significant changes for any firm type under either form of entry (see Figures 4.E.8). Turning to employment outcomes, no significant effects are detected in the number of employees for any firm type (see Figures 4.E.9). Similarly, I find no statistically significant effects on employee growth for established and capacity constrained firms (see Figures 4.E.10), with the exception of short-term participants, for whom the estimates suggest a decline in employee growth; although they are only statistically significant in the fourth year after entry.

Table 4.E.5 Event study estimates financial performance established participants - part 1

	Panel A: General Entry Event					Panel B: Competitive Entry Event				
Event	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
5 years before entry	0.002	199,376.411	0.405	-0.837*	-267,712.794*	0.254	550,428.091	-0.276	-0.197	-251,006.298*
	(0.193)	(273,701.178)	(0.312)	(0.298)	(63,717.195)	(0.182)	(314,424.818)	(0.615)	(0.150)	(88,871.955)
4 years before entry	-0.146	-40,577.334	0.309	-0.368	-102,131.593	-0.215	-121,191.558	0.129	-0.282	-194,406.097
	(0.131)	(131,996.406)	(0.227)	(0.186)	(103,234.033)	(0.181)	(261,286.171)	(0.441)	(0.167)	(179,505.675)
3 years before entry	-0.123	-97,237.470	0.431	-0.099	-42,672.333	-0.117	-167,729.962	0.399	-0.067	-70,999.884
	(0.090)	(83,467.897)	(0.239)	(0.067)	(44,094.970)	(0.180)	(166,622.315)	(0.314)	(0.096)	(91,527.857)
2 years before entry	-0.059	-73,538.350	-0.113	-0.039	4,677.833	-0.123	-137,180.912	0.083	-0.080	-98,988.569
	(0.056)	(43,788.363)	(0.168)	(0.056)	(37,298.874)	(0.151)	(100,186.071)	(0.202)	(0.078)	(70,045.081)
Year of entry	0.169*	58,750.476	-0.373	0.257*	146,564.646*	0.154*	20,272.449	0.098	0.248*	139,001.884
	(0.034)	(55,951.696)	(0.170)	(0.042)	(35,949.396)	(0.058)	(119,999.309)	(0.209)	(0.062)	(67,567.796)
1 year after entry	0.320*	132,284.192	-0.521	0.391*	160,364.172*	0.150	31,454.983	-0.162	0.304*	134,704.330
	(0.054)	(91,424.018)	(0.259)	(0.069)	(55,430.387)	(0.080)	(160,343.391)	(0.335)	(0.101)	(98,042.689)
2 years after entry	0.317*	149,767.676	-0.714*	0.405*	184,004.684*	0.219	-9,793.463	-0.515	0.410*	260,536.246
	(0.059)	(111,414.562)	(0.278)	(0.075)	(67,939.785)	(0.088)	(211,407.638)	(0.311)	(0.117)	(117,401.862)
3 years after entry	0.397*	501,629.077*	-1.357*	0.500*	319,049.865*	0.177	104,280.482	-0.759	0.394*	320,020.407
	(0.076)	(158,564.461)	(0.334)	(0.097)	(82,361.841)	(0.095)	(234,862.588)	(0.392)	(0.140)	(121,172.855)
4 years after entry	0.461*	540,478.174*	-1.901*	0.581*	433,949.582*	0.123	43,575.548	-1.230*	0.376	186,236.786
	(0.079)	(194,273.720)	(0.347)	(0.114)	(101,255.967)	(0.116)	(288,288.719)	(0.420)	(0.159)	(161,713.505)
5 years after entry	0.416*	544,863.902	-1.913*	0.618*	540,208.567*	0.056	193,784.718	-1.324*	0.299	371,429.906
	(0.094)	(212,989.729)	(0.473)	(0.168)	(129,131.549)	(0.132)	(442,993.654)	(0.473)	(0.190)	(249,968.737)
Observations	7,141	7,141	6,959	6,968	7,002	1,792	1,792	1,750	1,728	1,742
Cohorts	7	7	7	7	7	7	7	7	7	7
Years	13	13	13	13	13	13	13	13	13	13

Notes: This table reports event-study estimates using the Callaway and Sant'Anna estimator for established firms for the first set of financial outcomes. The comparison group consists of matched never suppliers. Standard errors, in parenthesis, are clustered at firm level. Cohorts are defined by the year of entry into the public procurement market; the first year in which a firm is awarded a procurement contract. Panel A presents the estimations for general entry, while Panel B reports results for competitive entry. The outcomes are organized as follows: asset growth (Columns 1 and 6), total assets (Columns 2 and 7), current ratio (Columns 3 and 8), liabilities growth (Columns 4 and 9), and total liabilities (Columns 5 and 10).

Table 4.E.6 Event Study estimates financial performance established participants - part 2

	Panel A: General Entry Event				Panel B: Competitive Entry Event			
Event	Solvency	Shareholder	Employee	Ln Employee	Solvency	Shareholder	Employee	Ln Employee
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
5 years before entry	1.841	7,346.612	1.184	0.182	-5.685	-289,504.533*	-0.605	0.187
	(4.663)	(76,757.707)	(1.929)	(0.117)	(4.174)	(105,790.503)	(2.804)	(0.146)
4 years before entry	4.879	-604.971	-1.393	0.055	3.405	-162,520.056	-0.404	0.058
	(2.832)	(55,791.436)	(1.483)	(0.111)	(4.499)	(133,697.512)	(2.983)	(0.176)
3 years before entry	2.473	28,901.230	0.865	0.086	5.099	28,100.487	2.572	0.150
	(1.681)	(35,145.851)	(1.174)	(0.074)	(2.645)	(62,322.936)	(2.481)	(0.127)
2 years before entry	1.549	32,056.944	0.685	0.102	3.948	132.448	0.599	0.050
	(1.177)	(25,600.163)	(0.905)	(0.060)	(2.006)	(34,194.433)	(1.737)	(0.108)
Year of entry	-1.644	44,178.628	0.745	0.101	-2.163	41,458.318	0.968	0.125
	(1.110)	(24,639.352)	(0.851)	(0.056)	(1.988)	(58,385.098)	(1.446)	(0.094)
1 year after entry	-1.666	88,984.779	2.264	0.064	-8.005	-96,336.460	5.726	0.265
	(1.648)	(43,366.401)	(1.327)	(0.068)	(3.623)	(78,656.235)	(2.707)	(0.106)
2 years after entry	-1.552	21,852.758	-0.750	-0.095	-5.843	-26,924.710	0.561	0.074
	(1.898)	(55,216.668)	(1.413)	(0.074)	(3.185)	(108,301.161)	(3.063)	(0.120)
3 years after entry	-2.609	62,974.102	1.673	-0.012	-0.862	79,963.173	-0.219	0.006
	(1.949)	(72,511.078)	(1.668)	(0.082)	(3.190)	(127,787.783)	(2.611)	(0.118)
4 years after entry	-4.399	123,890.751	2.922	-0.045	1.251	233,545.829	1.485	-0.043
	(2.343)	(91,579.342)	(1.740)	(0.088)	(3.630)	(151,220.961)	(3.353)	(0.138)
5 years after entry	-5.272	150,894.740	6.584*	0.066	-7.063	-85,644.353	5.406	0.099
	(2.572)	(101,204.977)	(2.257)	(0.097)	(4.492)	(204,303.878)	(4.106)	(0.202)
Observations	7,068	7,061	4,721	4,721	1,773	1,760	1,218	1,218
Cohorts	7	7	7	7	7	7	7	7
Years	13	13	13	13	13	13	13	13

Notes: This table reports event-study estimates using the Callaway and Sant'Anna estimator for established firms for the second set of financial outcomes. The comparison group consists of matched never suppliers. Standard errors, in parenthesis, are clustered at firm level. Cohorts are defined by the year of entry into the public procurement market; the first year in which a firm is awarded a procurement contract. Panel A presents the estimations for general entry, while Panel B reports results for competitive entry. The outcomes are organized as follows: solvency (Columns 1 and 5), shareholder funds (Columns 2 and 6), number of employees (Columns 3 and 7), and employment growth (Columns 4 and 8).

Table 4.E.7 Event Study estimates financial performance capacity constrained firms - part 1

	Panel A: General Entry Event					Panel B: Competitive Entry Event				
Event	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
5 years before entry	-0.046	32,848.606	-0.616	-0.149	-77,258.269	-0.029	-16,882.783	-1.653	-0.085	-70,563.249
	(0.074)	(83,832.268)	(0.344)	(0.074)	(39,803.455)	(0.152)	(138,883.581)	(0.755)	(0.133)	(68,781.587)
4 years before entry	0.001	-6,786.845	-0.173	-0.136	-18,824.517	-0.018	-39,274.902	-0.795	-0.103	-5,122.794
	(0.044)	(49,427.116)	(0.222)	(0.051)	(29,872.854)	(0.094)	(101,563.857)	(0.498)	(0.095)	(62,926.909)
3 years before entry	-0.066	-98,004.003*	-0.179	-0.110*	-37,969.380	-0.093	-131,629.836	-0.322	-0.087	-31,469.824
	(0.029)	(27,449.137)	(0.136)	(0.029)	(18,036.849)	(0.056)	(51,495.196)	(0.325)	(0.053)	(33,646.568)
2 years before entry	-0.026	-73,112.556*	-0.042	-0.060*	-26,399.938	-0.037	-66,305.135	-0.146	-0.059	-37,533.597
	(0.016)	(17,908.975)	(0.092)	(0.019)	(11,812.882)	(0.034)	(30,473.031)	(0.248)	(0.032)	(22,646.713)
Year of entry	0.100*	31,874.975	-0.231	0.136*	52,292.701*	0.093*	57,973.555	-0.109	0.138*	41,121.902
	(0.013)	(21,572.604)	(0.105)	(0.017)	(13,118.667)	(0.024)	(32,076.596)	(0.242)	(0.032)	(23,727.134)
1 year after entry	0.129*	40,073.961	-0.531*	0.198*	100,016.811*	0.060	-560.760	-0.291	0.106	62,518.617
	(0.018)	(27,703.417)	(0.120)	(0.022)	(17,314.839)	(0.030)	(52,074.551)	(0.261)	(0.044)	(31,320.910)
2 years after entry	0.127*	44,709.955	-0.736*	0.235*	97,552.049*	0.051	32,448.216	-0.556	0.142*	58,469.269
	(0.021)	(35,180.694)	(0.161)	(0.026)	(17,689.067)	(0.036)	(68,016.372)	(0.306)	(0.049)	(37,770.788)
3 years after entry	0.145*	94,117.076	-1.167*	0.289*	134,614.384*	0.046	16,201.296	-1.149*	0.199*	108,455.059
	(0.025)	(42,500.047)	(0.192)	(0.033)	(21,573.760)	(0.048)	(83,161.390)	(0.372)	(0.062)	(40,924.479)
4 years after entry	0.113*	82,214.225	-1.309*	0.260*	126,407.834*	-0.010	64,676.959	-0.894	0.108	37,879.819
	(0.032)	(53,254.306)	(0.237)	(0.044)	(26,689.253)	(0.060)	(94,531.012)	(0.466)	(0.077)	(50,914.332)
5 years after entry	0.079	57,363.953	-1.668*	0.326*	162,635.930*	-0.015	57,989.144	-1.588*	0.238	145,192.618
	(0.036)	(61,905.661)	(0.269)	(0.046)	(33,758.104)	(0.068)	(115,060.597)	(0.492)	(0.089)	(62,055.394)
Observations	9,316	9,316	9,133	9,119	9,156	2,405	2,405	2,362	2,331	2,346
Cohorts	6	6	6	6	6	6	6	6	6	6
Years	13	13	13	13	13	13	13	13	13	13

Notes: This table reports event-study estimates using the Callaway and Sant'Anna estimator for capacity constrained firms for the first set of financial outcomes. The comparison group consists of matched never suppliers. Standard errors, in parenthesis, are clustered at firm level. Cohorts are defined by the year of entry into the public procurement market; the first year in which a firm is awarded a procurement contract. Panel A presents the estimations for general entry, while Panel B reports results for competitive entry. The outcomes are organized as follows: asset growth (Columns 1 and 6), total assets (Columns 2 and 7), current ratio (Columns 3 and 8), liabilities growth (Columns 4 and 9), and total liabilities (Columns 5 and 10).

Table 4.E.8 Event Study estimates financial performance capacity constrained firms - part 2

	Panel A: General Entry Event				Panel B: Competitive Entry Event			
Event	Solvency	Shareholder	Employee	Ln Employee	Solvency	Shareholder	Employee	Ln Employee
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
5 years before entry	3.547	89,759.364	-2.591*	-0.112	5.062	154,097.956	-0.877	0.106
	(1.633)	(37,487.261)	(0.770)	(0.057)	(3.240)	(71,231.725)	(1.715)	(0.116)
4 years before entry	2.661	62,325.235	-0.370	0.030	-0.625	82,447.501	0.448	0.102
	(1.142)	(25,465.060)	(0.702)	(0.045)	(2.451)	(53,913.997)	(1.416)	(0.088)
3 years before entry	1.298	9,080.571	-0.729	0.045	0.122	4,097.337	0.227	0.128
	(0.690)	(13,590.385)	(0.409)	(0.028)	(1.335)	(23,650.769)	(0.831)	(0.050)
2 years before entry	0.901	10,944.279	-0.316	0.040	-0.440	7,625.560	0.729	0.084
	(0.483)	(9,899.044)	(0.352)	(0.021)	(0.863)	(15,743.170)	(0.672)	(0.044)
Year of entry	-0.343	16,886.473	0.806	0.052	-0.469	26,057.491	1.117	0.051
	(0.433)	(10,489.678)	(0.341)	(0.021)	(0.931)	(17,055.969)	(0.757)	(0.040)
1 year after entry	-0.818	31,278.546	1.108	0.032	-0.563	23,741.940	1.190	0.009
	(0.577)	(15,356.225)	(0.430)	(0.022)	(1.006)	(28,752.299)	(0.753)	(0.049)
2 years after entry	-0.918	27,093.052	1.282	0.031	-2.062	-7,255.527	1.713	0.068
	(0.586)	(17,280.186)	(0.476)	(0.028)	(1.266)	(31,288.993)	(0.869)	(0.048)
3 years after entry	-1.923	20,197.885	1.273	0.001	-4.529*	-15,029.162	1.703	0.035
	(0.741)	(23,245.713)	(0.550)	(0.030)	(1.373)	(42,935.009)	(0.936)	(0.055)
4 years after entry	-1.949	18,996.021	0.647	-0.059	-3.267	-27,713.867	0.660	-0.075
	(0.889)	(27,696.140)	(0.602)	(0.040)	(1.664)	(50,992.013)	(1.213)	(0.075)
5 years after entry	-3.403*	-21,801.072	-0.072	-0.139*	-6.030*	-106,529.733	1.448	-0.066
	(1.017)	(32,681.981)	(0.728)	(0.046)	(1.836)	(54,389.172)	(1.538)	(0.090)
Observations	9,239	9,231	6,766	6,766	2,385	2,371	1,782	1,782
Cohorts	6	6	6	6	6	6	6	6
Years	13	13	13	13	13	13	13	13

Notes: This table reports event-study estimates using the Callaway and Sant'Anna estimator for capacity constrained firms for the second set of financial outcomes. The comparison group consists of matched never suppliers. Standard errors, in parenthesis, are clustered at firm level. Cohorts are defined by the year of entry into the public procurement market; the first year in which a firm is awarded a procurement contract. Panel A presents the estimations for general entry, while Panel B reports results for competitive entry. The outcomes are organized as follows: solvency (Columns 1 and 5), shareholder funds (Columns 2 and 6), number of employees (Columns 3 and 7), and employment growth (Columns 4 and 8).

Table 4.E.9 Event Study estimates financial performance short-term participants - part 1

	Panel A: General Entry Event					Panel B: Competitive Entry Event				
Event	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
5 years before entry	0.026	38,315.037	0.376	-0.120*	-31,536.717	0.042	79,883.957	0.359	-0.079	3,467.807
	(0.035)	(32,996.950)	(0.221)	(0.038)	(19,447.749)	(0.078)	(67,112.788)	(0.427)	(0.079)	(39,599.083)
4 years before entry	0.009	-2,909.338	0.257	-0.105*	-25,340.759	0.040	-13,335.328	-0.062	-0.078	-20,771.386
	(0.029)	(28,290.632)	(0.166)	(0.030)	(15,623.183)	(0.067)	(58,048.611)	(0.249)	(0.057)	(32,636.508)
3 years before entry	-0.022	-24,326.089	0.228	-0.121*	-40,584.344*	0.023	741.672	0.188	-0.089	-30,437.932
	(0.022)	(20,624.920)	(0.151)	(0.023)	(12,506.229)	(0.046)	(43,314.960)	(0.246)	(0.048)	(25,470.390)
2 years before entry	-0.044	-53,938.817*	0.078	-0.070*	-25,745.946	-0.046	-45,578.711	0.204	-0.060	-23,692.761
	(0.017)	(16,448.953)	(0.118)	(0.017)	(9,690.678)	(0.038)	(33,145.718)	(0.260)	(0.038)	(22,059.484)
Year of entry	0.066*	-5,704.321	-0.223	0.075*	17,927.912	0.077	50,178.608	-0.125	0.113*	29,565.437
	(0.019)	(24,296.102)	(0.137)	(0.023)	(13,435.554)	(0.034)	(53,848.969)	(0.196)	(0.042)	(30,034.445)
1 year after entry	0.056	-63,797.137	-0.320	0.127*	37,695.626	-0.003	-113,672.611	0.220	0.076	46,706.147
	(0.024)	(30,646.871)	(0.195)	(0.028)	(16,178.517)	(0.040)	(60,808.486)	(0.352)	(0.060)	(36,141.476)
2 years after entry	0.041	-86,279.789	-0.755*	0.160*	51,457.180	0.015	-183,681.914	0.236	0.099	55,271.512
	(0.031)	(45,573.363)	(0.228)	(0.040)	(24,300.056)	(0.061)	(93,315.940)	(0.475)	(0.090)	(49,780.179)
3 years after entry	-0.003	-98,716.478	-0.752	0.179*	48,883.342	-0.104	-169,631.637	0.137	0.023	22,571.078
	(0.038)	(57,694.787)	(0.312)	(0.046)	(25,832.997)	(0.075)	(110,534.190)	(0.608)	(0.089)	(54,099.750)
4 years after entry	-0.026	-103,085.700	-0.808	0.148	49,612.478	-0.118	-67,114.852	-0.297	-0.032	19,846.715
	(0.052)	(74,643.674)	(0.349)	(0.060)	(36,592.019)	(0.105)	(158,370.858)	(0.628)	(0.142)	(70,945.187)
5 years after entry	-0.094	-114,066.588	-1.059	0.178	36,256.854	-0.169	-44,625.528	0.276	-0.037	-26,264.303
	(0.066)	(87,155.892)	(0.445)	(0.077)	(44,261.034)	(0.128)	(174,483.511)	(1.030)	(0.150)	(84,862.145)
Observations	10,362	10,362	10,176	10,156	10,193	2,497	2,497	2,455	2,424	2,438
Cohorts	8	8	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13	13	13

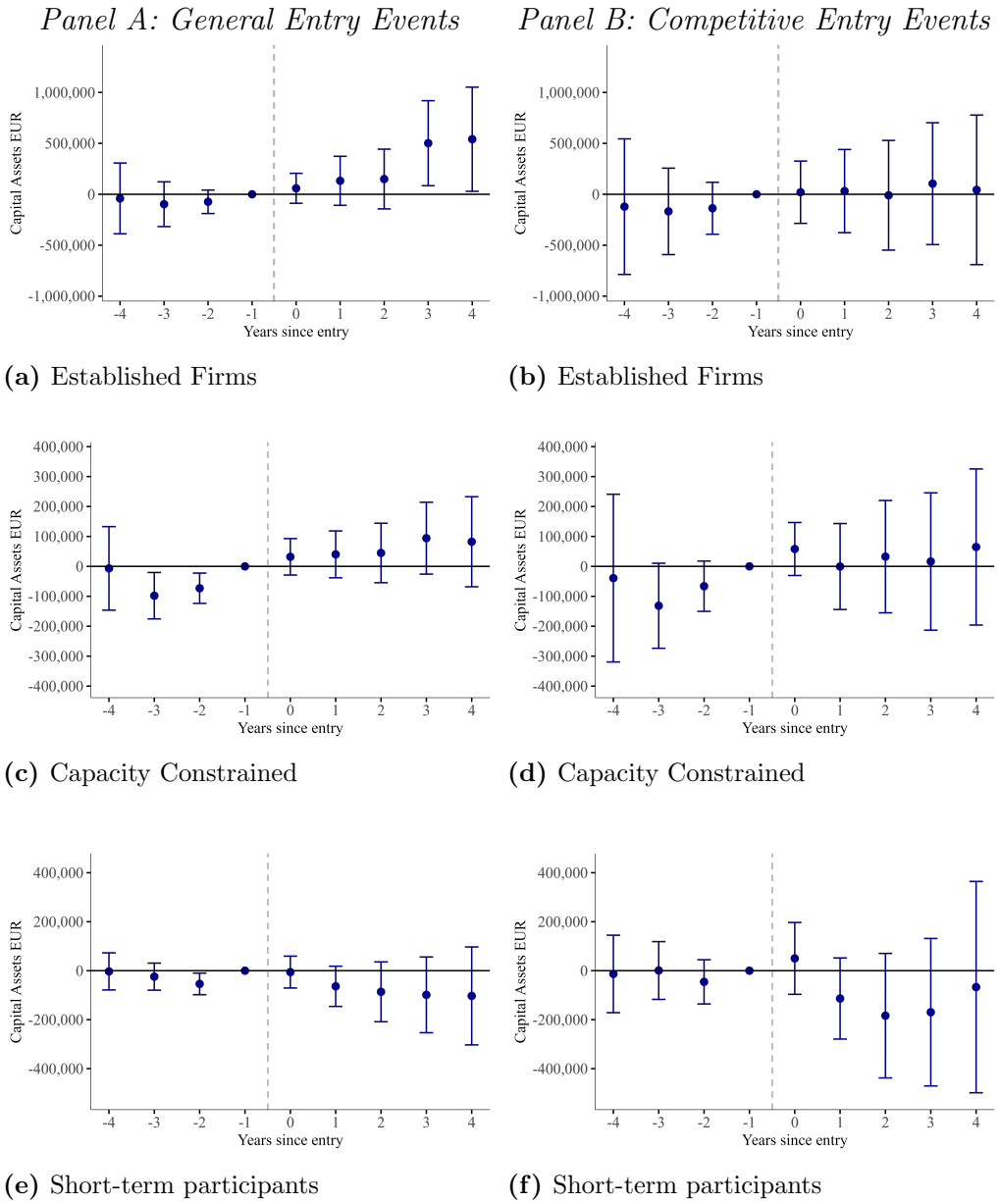
Notes: This table reports event-study estimates using the Callaway and Sant'Anna estimator for short-term participants for the first set of financial outcomes. The comparison group consists of matched never suppliers. Standard errors, in parenthesis, are clustered at firm level. Cohorts are defined by the year of entry into the public procurement market; the first year in which a firm is awarded a procurement contract. Panel A presents the estimations for general entry, while Panel B reports results for competitive entry. The outcomes are organized as follows: asset growth (Columns 1 and 6), total assets (Columns 2 and 7), current ratio (Columns 3 and 8), liabilities growth (Columns 4 and 9), and total liabilities (Columns 5 and 10).

Table 4.E.10 Event Study estimates financial performance short-term participants - part 2

	Panel A: General Entry Event				Panel B: Competitive Entry Event			
Event	Solvency	Shareholder	Employee	Ln Employee	Solvency	Shareholder	Employee	Ln Employee
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
5 years before entry	1.813	98,883.578*	0.164	0.071	2.464	93,630.261	0.581	0.087
	(0.960)	(17,051.118)	(0.502)	(0.036)	(1.972)	(34,580.289)	(0.873)	(0.074)
4 years before entry	2.004	85,771.229*	-0.004	0.071	1.669	55,331.730	0.134	0.052
	(0.745)	(15,406.191)	(0.421)	(0.030)	(1.479)	(30,767.007)	(0.766)	(0.059)
3 years before entry	1.738*	52,065.069*	-0.506	0.018	0.876	36,601.722	0.167	0.061
	(0.603)	(12,644.420)	(0.342)	(0.026)	(1.239)	(24,548.211)	(0.659)	(0.052)
2 years before entry	1.377*	31,433.404*	-0.448	-0.005	1.030	35,520.995	0.061	0.016
	(0.494)	(8,900.925)	(0.293)	(0.021)	(1.083)	(17,680.482)	(0.623)	(0.045)
Year of entry	0.797	15,320.746	-0.605	-0.020	-1.135	5,377.786	-0.470	-0.000
	(0.534)	(11,523.056)	(0.386)	(0.025)	(1.096)	(24,813.883)	(0.646)	(0.042)
1 year after entry	0.045	-4,752.661	-1.052	-0.059	-1.216	-34,417.414	-1.830	-0.104
	(0.739)	(15,794.635)	(0.452)	(0.029)	(1.503)	(37,604.964)	(0.783)	(0.056)
2 years after entry	-0.704	-43,557.742	-1.102	-0.108	-1.399	-90,667.999	-0.948	-0.130
	(0.865)	(21,253.507)	(0.607)	(0.039)	(1.787)	(49,589.563)	(1.220)	(0.078)
3 years after entry	-0.941	-62,717.980	-0.999	-0.105	-1.144	-98,148.299	-1.713	-0.183
	(1.144)	(29,439.952)	(0.788)	(0.042)	(2.327)	(51,064.802)	(1.163)	(0.087)
4 years after entry	-0.025	-37,796.233	-2.279	-0.220*	-0.450	-64,021.333	-1.648	-0.247
	(1.333)	(36,615.232)	(0.920)	(0.063)	(2.794)	(75,612.452)	(1.482)	(0.117)
5 years after entry	0.149	-62,011.315	-2.614	-0.224*	-1.137	-68,101.655	-1.519	-0.234
	(1.642)	(43,858.739)	(1.257)	(0.070)	(3.081)	(100,733.500)	(2.289)	(0.151)
Observations	10,266	10,273	7,429	7,429	2,475	2,462	1,810	1,810
Cohorts	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13

Notes: This table reports event-study estimates using the Callaway and Sant'Anna estimator for short-term participants for the second set of financial outcomes. The comparison group consists of matched never suppliers. Standard errors, in parenthesis, are clustered at firm level. Cohorts are defined by the year of entry into the public procurement market; the first year in which a firm is awarded a procurement contract. Panel A presents the estimations for general entry, while Panel B reports results for competitive entry. The outcomes are organized as follows: solvency (Columns 1 and 5), shareholder funds (Columns 2 and 6), number of employees (Columns 3 and 7), and employment growth (Columns 4 and 8).

Figure 4.E.3 Event Study plots financial performance: assets thousand Euro



(a) Established Firms

(b) Established Firms

(c) Capacity Constrained

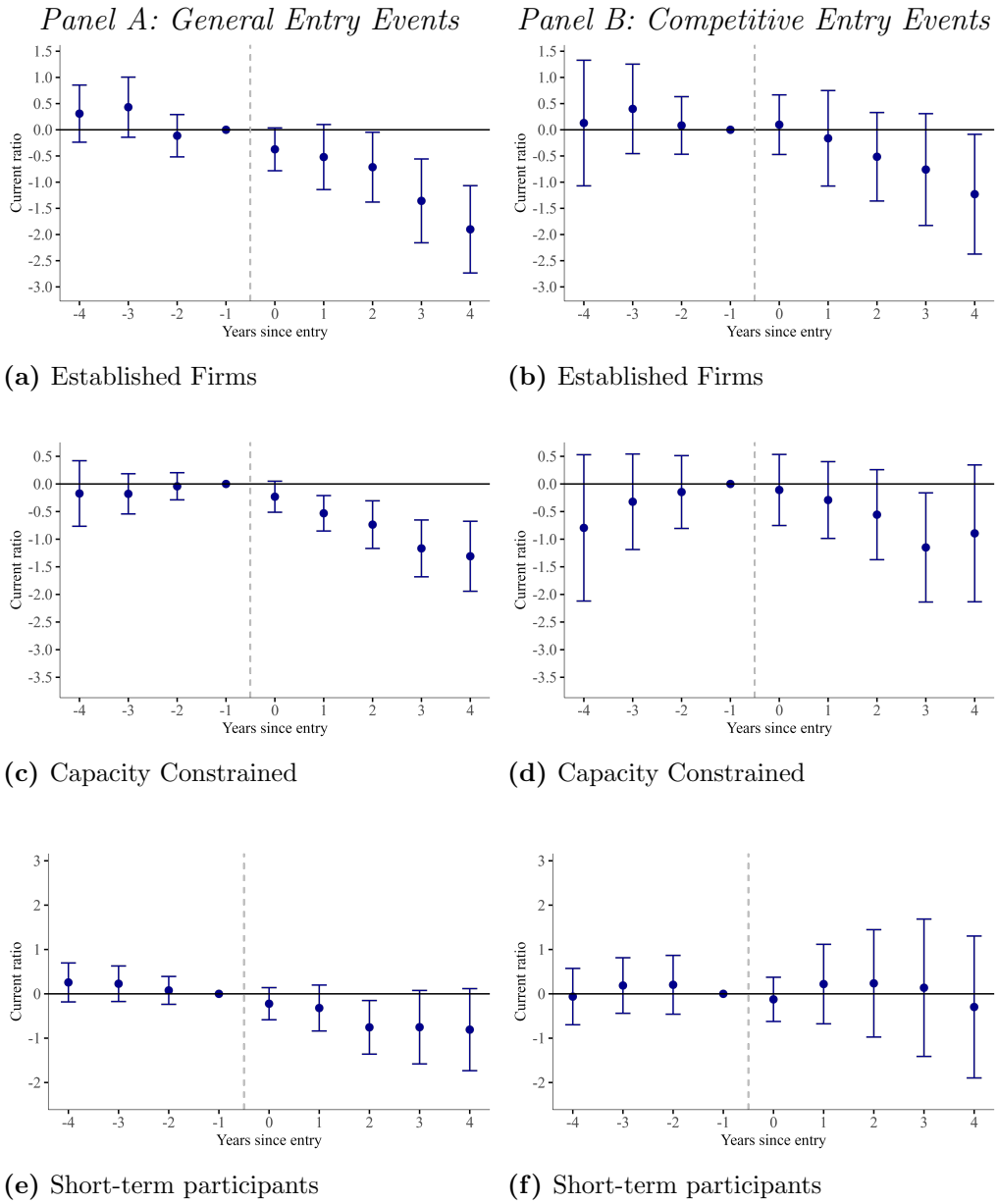
(d) Capacity Constrained

(e) Short-term participants

(f) Short-term participants

Notes: Figure 4.E.3 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is capital assets (in thousands of euros). Sub-panels (a) and (b) correspond to established firms (Columns 2 and 7 of Table 4.E.5); (c) and (d) to capacity-constrained firms (Columns 2 and 7 of Table 4.E.7); and (e) and (f) to short-term participants (Columns 2 and 7 of Table 4.E.9).

Figure 4.E.4 Event Study plots financial performance: current ratio



(a) Established Firms

(b) Established Firms

(c) Capacity Constrained

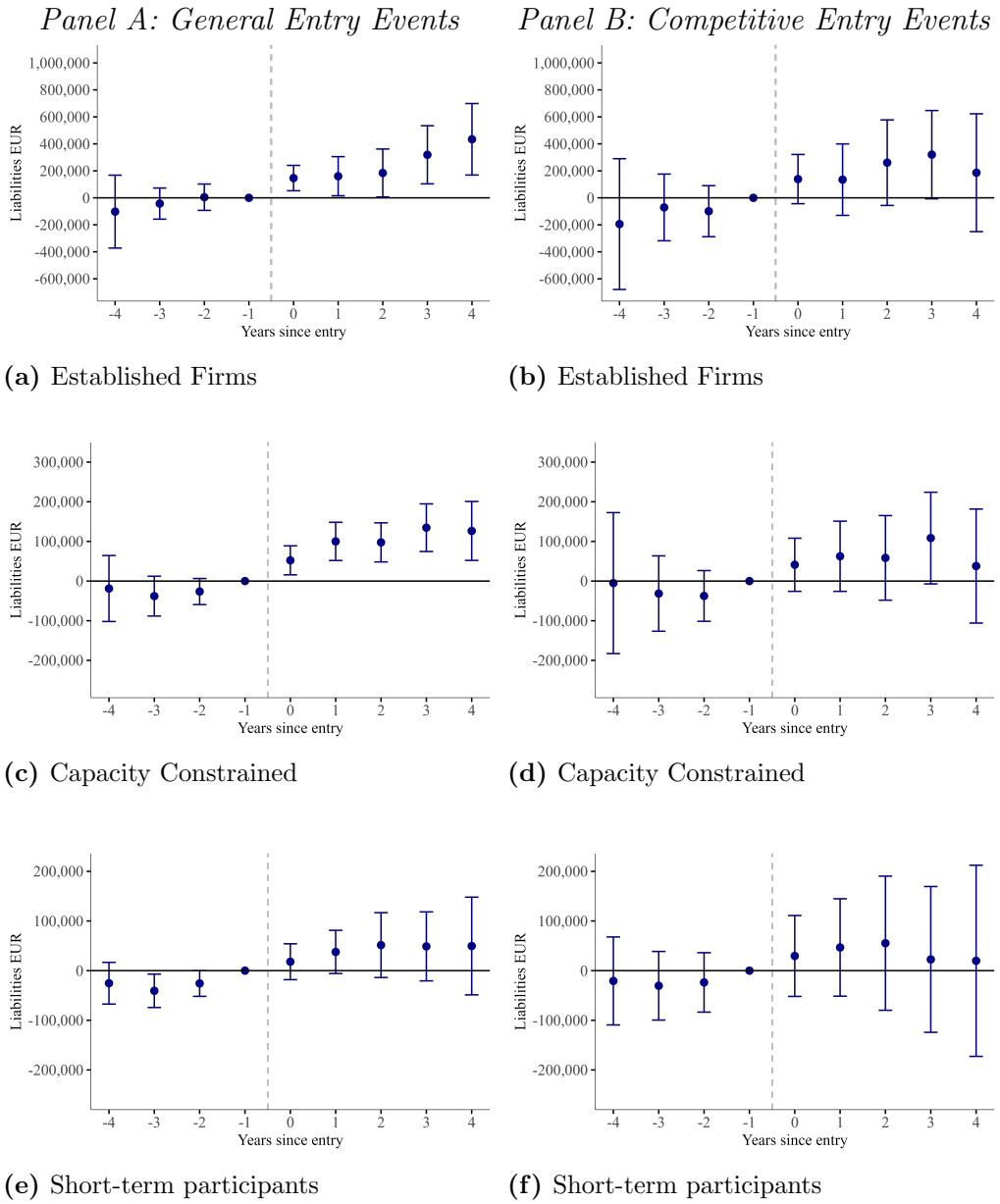
(d) Capacity Constrained

(e) Short-term participants

(f) Short-term participants

Notes: Figure 4.E.4 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is current ratio. Sub-panels (a) and (b) correspond to established firms (Columns 3 and 8 of Table 4.E.5); (c) and (d) to capacity-constrained firms (Columns 3 and 8 of Table 4.E.7); and (e) and (f) to short-term participants (Columns 3 and 8 of Table 4.E.9).

Figure 4.E.5 Event Study plots financial performance: liabilities thousand Euro



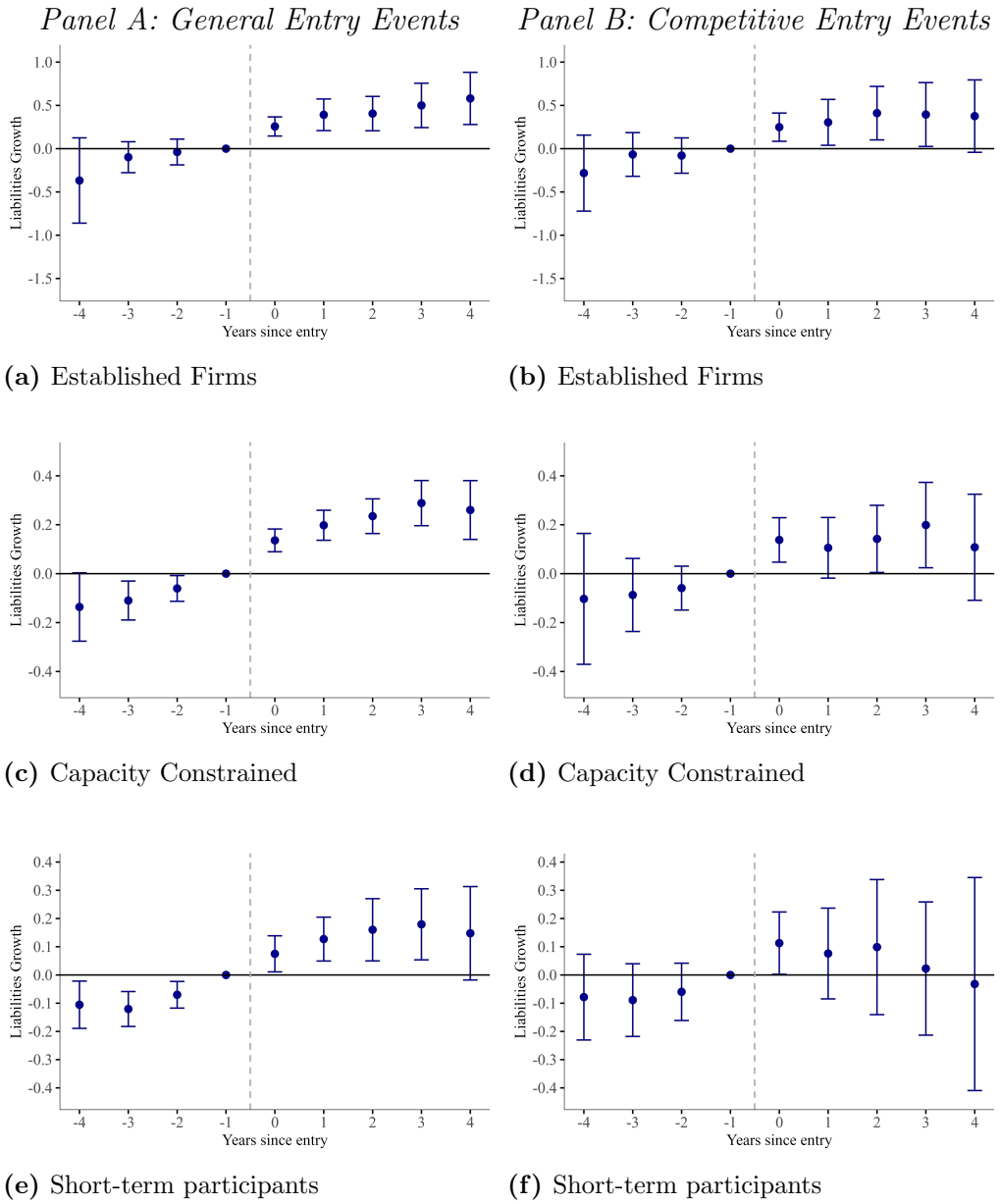
(a) Established Firms **(b)** Established Firms

(c) Capacity Constrained **(d)** Capacity Constrained

(e) Short-term participants **(f)** Short-term participants

Notes: Figure 4.E.5 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is liabilities in thousands of Euro. Sub-panels (a) and (b) correspond to established firms (Columns 5 and 10 of Table 4.E.5); (c) and (d) to capacity-constrained firms (Columns 5 and 10 of Table 4.E.7); and (e) and (f) to short-term participants (Columns 5 and 10 of Table 4.E.9).

Figure 4.E.6 Event Study plots financial performance: liabilities growth



(a) Established Firms

(b) Established Firms

(c) Capacity Constrained

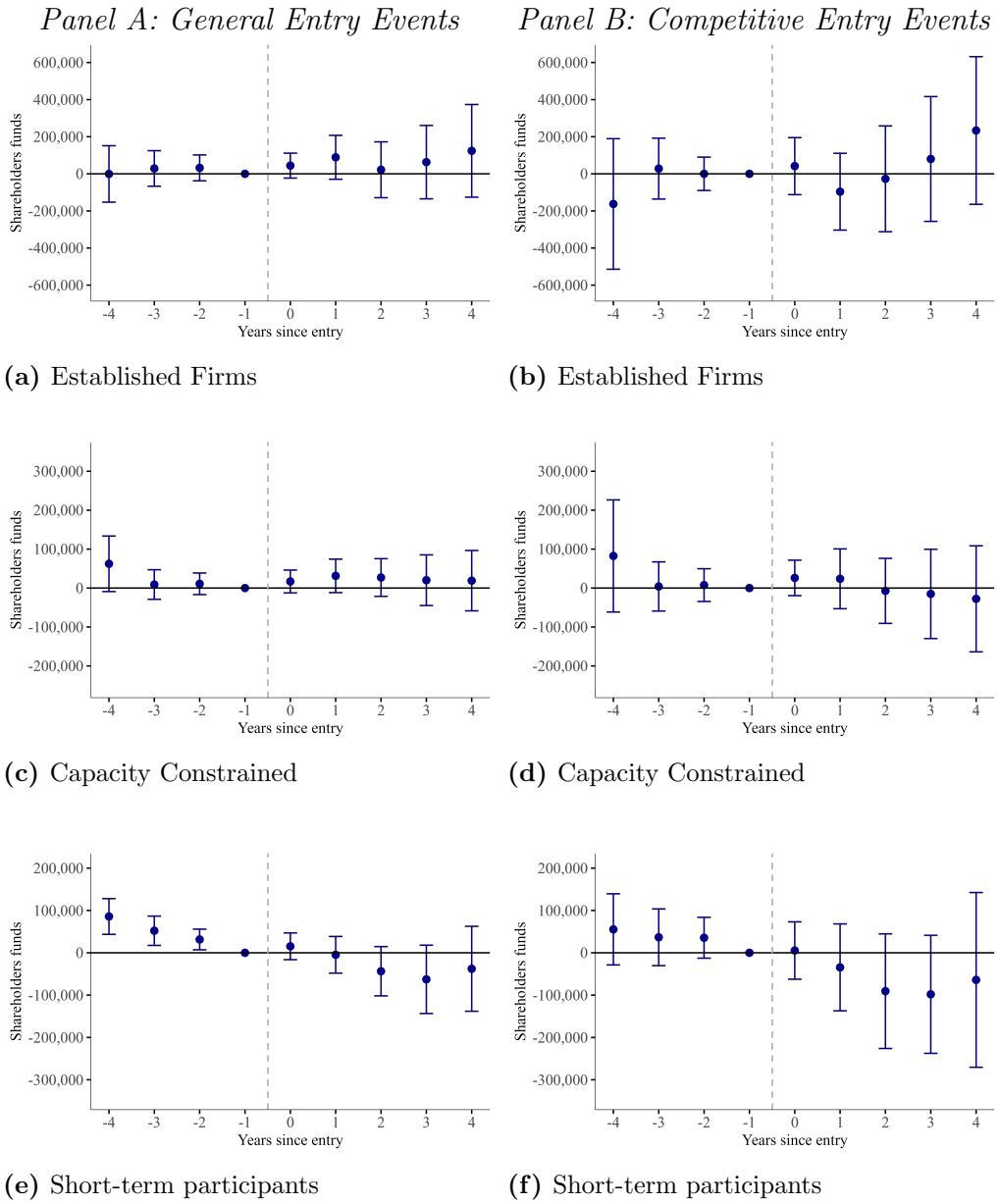
(d) Capacity Constrained

(e) Short-term participants

(f) Short-term participants

Notes: Figure 4.E.6 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is liabilities growth. Sub-panels (a) and (b) correspond to established firms (Columns 4 and 9 of Table 4.E.5); (c) and (d) to capacity-constrained firms (Columns 4 and 9 of Table 4.E.7); and (e) and (f) to short-term participants (Columns 4 and 9 of Table 4.E.9).

Figure 4.E.7 Event Study plots financial performance: shareholders funds



(a) Established Firms

(b) Established Firms

(c) Capacity Constrained

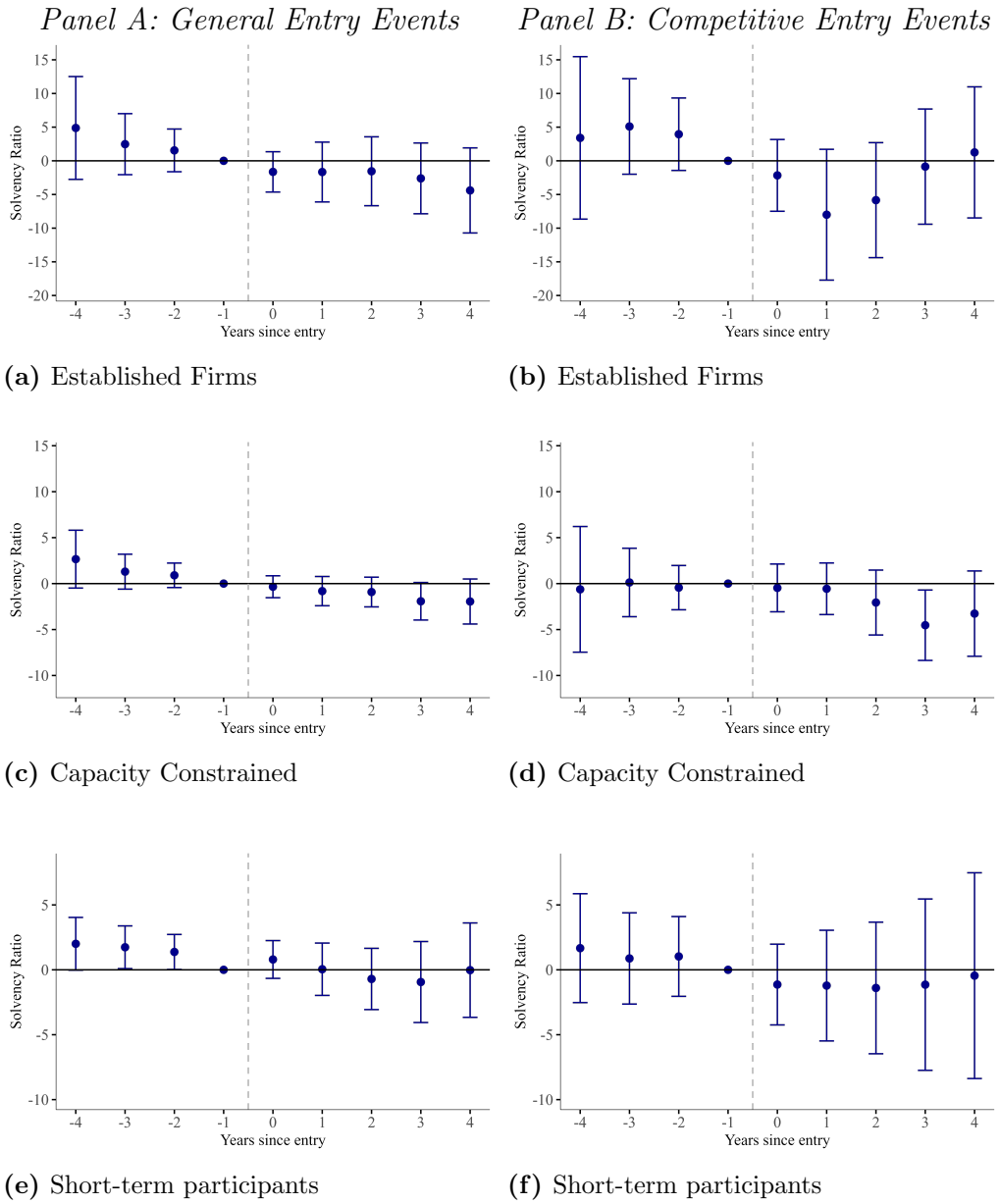
(d) Capacity Constrained

(e) Short-term participants

(f) Short-term participants

Notes: Figure 4.E.7 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is liabilities growth. Sub-panels (a) and (b) correspond to established firms (Columns 2 and 6 of Table 4.E.6); (c) and (d) to capacity-constrained firms (Columns 2 and 6 of Table 4.E.8); and (e) and (f) to short-term participants (Columns 2 and 6 of Table 4.E.10).

Figure 4.E.8 Event Study plots financial performance: solvency



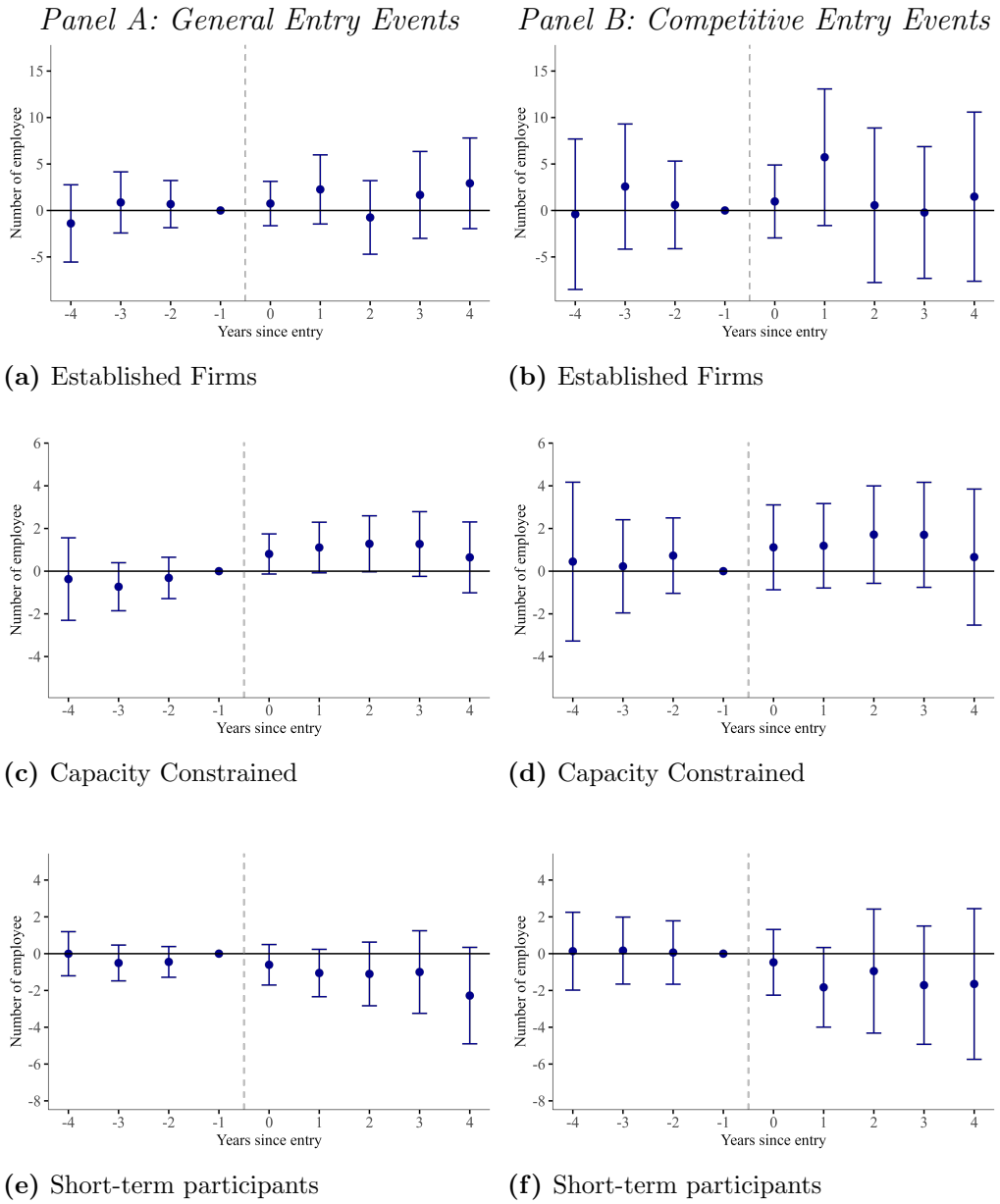
(a) Established Firms **(b)** Established Firms

(c) Capacity Constrained **(d)** Capacity Constrained

(e) Short-term participants **(f)** Short-term participants

Notes: Figure 4.E.8 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is liabilities growth. Sub-panels (a) and (b) correspond to established firms (Columns 1 and 5 of Table 4.E.6); (c) and (d) to capacity-constrained firms (Columns 1 and 5 of Table 4.E.8); and (e) and (f) to short-term participants (Columns 1 and 5 of Table 4.E.10).

Figure 4.E.9 Event Study plots financial performance: number of employees



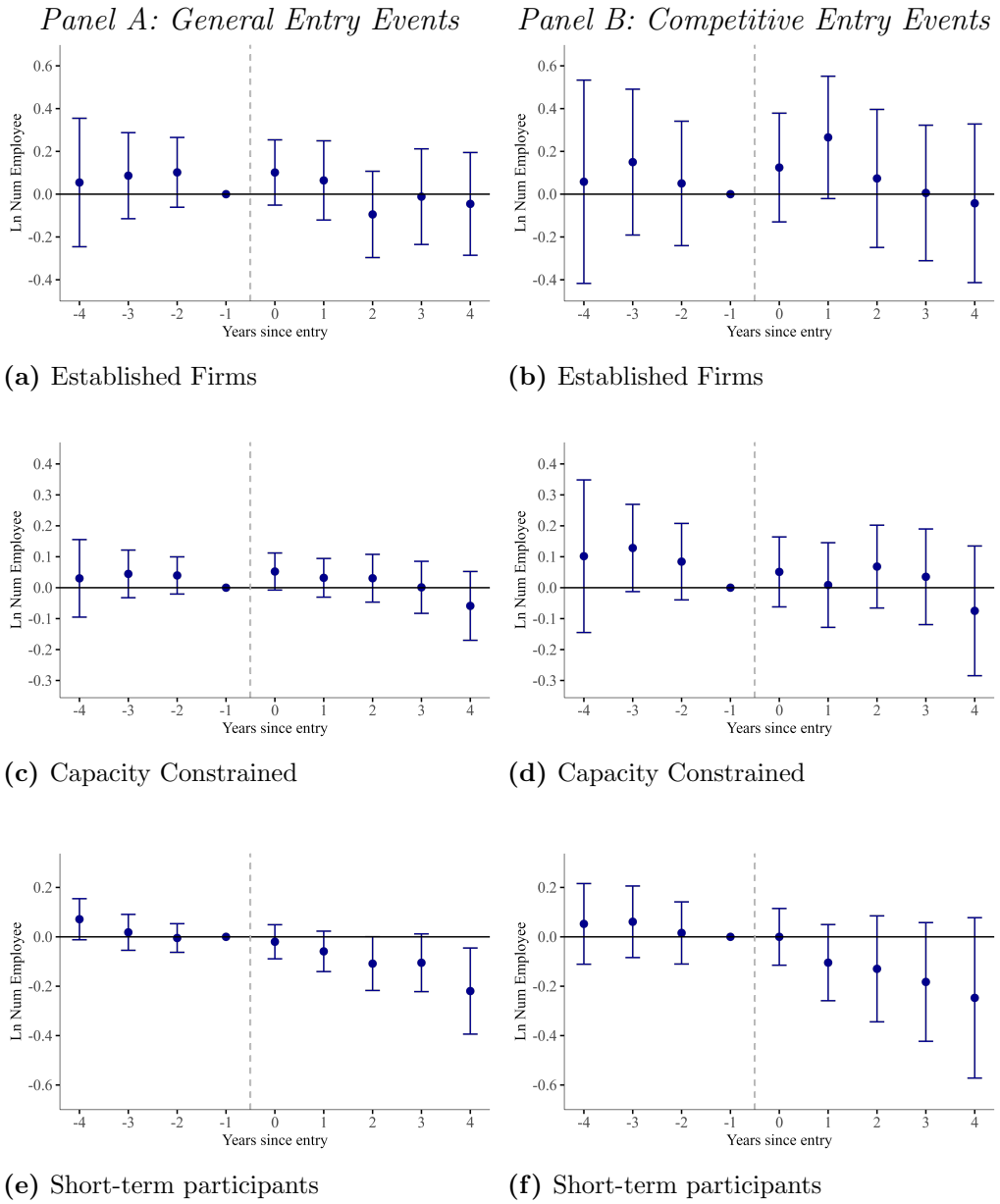
(a) Established Firms **(b)** Established Firms

(c) Capacity Constrained **(d)** Capacity Constrained

(e) Short-term participants **(f)** Short-term participants

Notes: Figure 4.E.9 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is liabilities growth. Sub-panels (a) and (b) correspond to established firms (Columns 3 and 7 of Table 4.E.6); (c) and (d) to capacity-constrained firms (Columns 3 and 7 of Table 4.E.8); and (e) and (f) to short-term participants (Columns 3 and 7 of Table 4.E.10).

Figure 4.E.10 Event Study plots financial performance: employee growth



(a) Established Firms **(b)** Established Firms

(c) Capacity Constrained **(d)** Capacity Constrained

(e) Short-term participants **(f)** Short-term participants

Notes: Figure 4.E.10 plots the estimated event-study coefficients, $\hat{\gamma}$, from equation 4.3.3. Panel A presents results for general entry, while Panel B reports results for competitive entry. The dependent variable is liabilities growth. Sub-panels (a) and (b) correspond to established firms (Columns 4 and 8 of Table 4.E.6); (c) and (d) to capacity-constrained firms (Columns 4 and 8 of Table 4.E.8); and (e) and (f) to short-term participants (Columns 4 and 8 of Table 4.E.10).

4.F Robustness checks

This section presents the event study plots and regression tables of the estimations using not-yet-supplier firms as comparison groups. Table 4.F.1 presents the estimations for assets growth, capital assets, current ratio, liabilities growth, and liabilities. Table 4.F.2 presents the estimations for solvency, shareholder funds, number of employees, and employment growth.

Table 4.F.1 Event Study estimates financial performance not yet suppliers (part 1)

	Panel A: General Entry Event					Panel B: Competitive Entry Event				
Event	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities	Ln Assets	Assets	Current Ratio	Ln Liabilities	Liabilities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
5 years before entry	-0.017	22,353.825	0.042	-0.173*	-63,614.960*	-0.006	27,548.095	-0.458	-0.125	-54,092.621
	(0.040)	(38,011.255)	(0.240)	(0.039)	(22,052.895)	(0.085)	(80,598.633)	(0.517)	(0.088)	(39,651.157)
4 years before entry	-0.012	-18,894.554	0.109	-0.140*	-33,837.325	-0.015	-57,875.838	-0.361	-0.118	-40,624.794
	(0.025)	(27,493.495)	(0.142)	(0.029)	(14,984.239)	(0.055)	(52,554.406)	(0.339)	(0.052)	(35,346.500)
3 years before entry	-0.047*	-60,658.226*	0.097	-0.120*	-42,143.847*	-0.042	-78,318.391	-0.004	-0.094*	-39,311.854
	(0.016)	(17,595.090)	(0.100)	(0.018)	(10,041.671)	(0.037)	(33,976.448)	(0.209)	(0.033)	(22,601.316)
2 years before entry	-0.039*	-62,693.319*	0.025	-0.066*	-24,506.024*	-0.049	-61,256.044	0.046	-0.063	-36,101.234
	(0.012)	(12,432.310)	(0.075)	(0.013)	(7,626.737)	(0.025)	(22,475.746)	(0.174)	(0.025)	(15,970.613)
Year of entry	0.088*	14,952.516	-0.241*	0.113*	40,741.166*	0.092*	52,867.741	-0.104	0.134*	42,439.420
	(0.011)	(14,686.759)	(0.089)	(0.014)	(9,520.227)	(0.020)	(30,556.783)	(0.159)	(0.026)	(17,678.862)
1 year after entry	0.113*	-1,288.540	-0.437*	0.181*	75,335.956*	0.044	-46,056.163	-0.054	0.109*	61,217.398
	(0.016)	(22,275.253)	(0.116)	(0.019)	(12,948.732)	(0.027)	(41,717.855)	(0.227)	(0.036)	(24,396.528)
2 years after entry	0.109*	-493.401	-0.763*	0.221*	85,184.149*	0.056	-48,694.382	-0.262	0.151*	72,822.632
	(0.019)	(27,745.709)	(0.158)	(0.024)	(14,586.659)	(0.034)	(58,118.651)	(0.278)	(0.048)	(30,730.141)
3 years after entry	0.114*	52,254.883	-1.049*	0.267*	115,446.096*	0.016	-32,439.528	-0.691	0.161	98,080.252*
	(0.024)	(34,879.467)	(0.199)	(0.032)	(18,491.265)	(0.042)	(63,809.280)	(0.384)	(0.060)	(34,403.672)
4 years after entry	0.101*	56,550.350	-1.208*	0.249*	121,279.188*	-0.028	27,100.160	-0.729	0.079	41,869.886
	(0.028)	(43,990.882)	(0.232)	(0.037)	(21,317.577)	(0.057)	(90,660.685)	(0.436)	(0.074)	(44,740.563)
5 years after entry	0.070	46,411.588	-1.542*	0.310*	149,864.290*	-0.028	59,421.018	-1.092	0.175	114,893.540
	(0.036)	(50,483.995)	(0.275)	(0.043)	(26,807.631)	(0.064)	(96,323.112)	(0.532)	(0.082)	(54,100.662)
Observations	13,445	13,445	13,258	13,205	13,245	3,348	3,348	3,305	3,263	3,278
Cohorts	8	8	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13	13	13

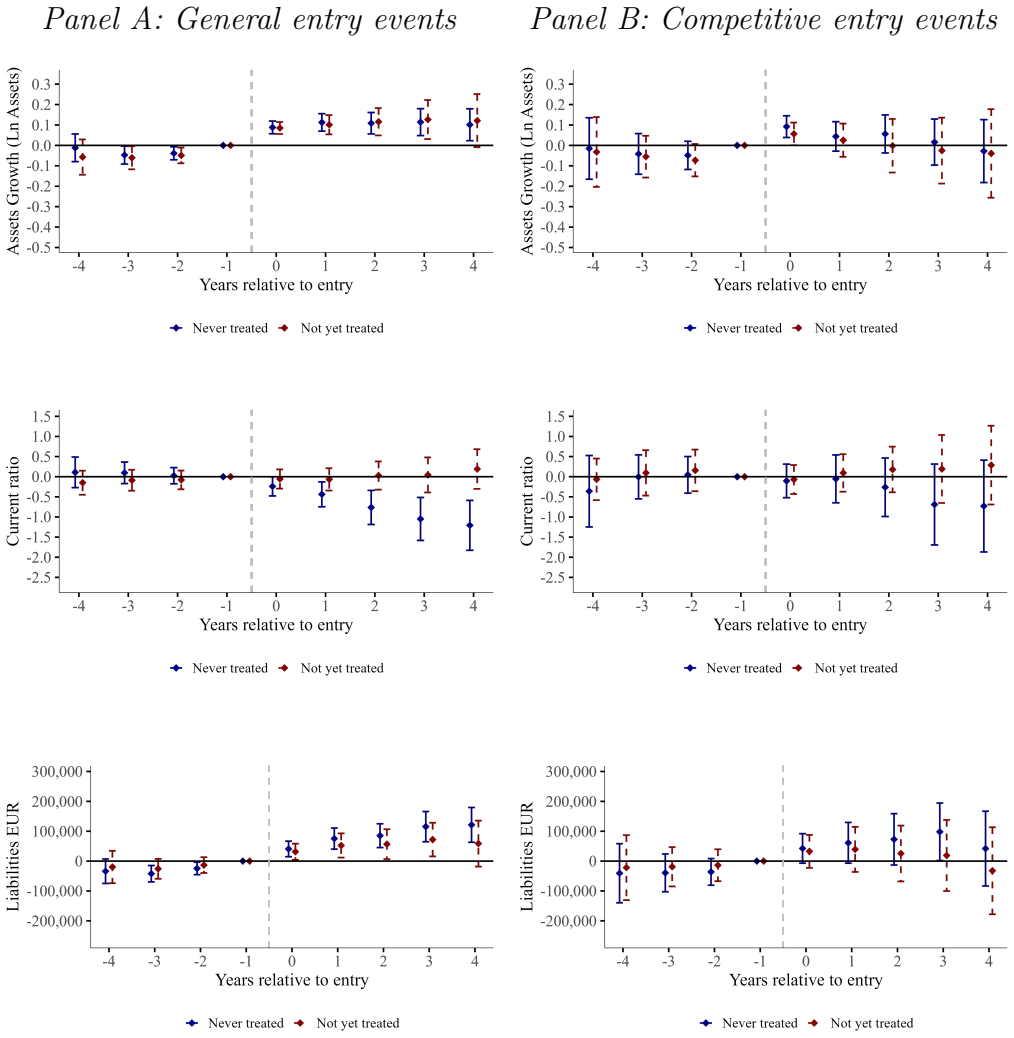
Notes: This table presents the estimations of event studies using Callaway and Sant'Anna estimator. Panel A presents the estimations for general entry events and Panel B for competitive entry events. Cohorts are defined by year of entry in the public procurement market: the first year in which the firm was awarded a contract by a public entity. The comparison group corresponds to not yet suppliers, never suppliers are excluded. Standard errors are presented in parenthesis and clustered at firm level.

Table 4.F.2 Event Study estimates financial performance not yet suppliers (part 2)

	Panel A: General Entry Event				Panel B: Competitive Entry Event			
Event	Solvency	Shareholder	Employee	Ln Employee	Solvency	Shareholder	Employee	Ln Employee
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
5 years before entry	2.353	84,500.282*	-0.980	-0.002	3.232	96,372.139	-0.369	0.088
	(0.908)	(19,032.007)	(0.456)	(0.032)	(1.990)	(40,862.276)	(0.971)	(0.065)
4 years before entry	2.513*	70,744.342*	-0.416	0.041	1.044	48,102.051	-0.143	0.056
	(0.623)	(13,883.595)	(0.369)	(0.028)	(1.411)	(29,997.880)	(0.740)	(0.051)
3 years before entry	1.694*	33,263.133*	-0.615	0.027	0.931	19,325.002	0.094	0.084
	(0.434)	(8,507.272)	(0.272)	(0.020)	(0.877)	(17,671.290)	(0.515)	(0.034)
2 years before entry	1.234*	23,784.965*	-0.360	0.018	0.584	20,869.953	0.353	0.047
	(0.326)	(6,783.719)	(0.230)	(0.015)	(0.641)	(11,784.432)	(0.406)	(0.030)
Year of entry	0.160	18,340.010	0.114	0.020	-0.931	18,034.212	0.362	0.032
	(0.337)	(7,299.621)	(0.239)	(0.017)	(0.685)	(15,755.654)	(0.473)	(0.029)
1 year after entry	-0.494	19,447.391	0.319	-0.001	-1.459	-9,072.905	0.335	-0.014
	(0.463)	(11,480.974)	(0.324)	(0.019)	(0.882)	(22,767.946)	(0.586)	(0.035)
2 years after entry	-0.901	-1,031.977	0.324	-0.025	-2.164	-37,861.180	0.793	0.001
	(0.579)	(13,972.311)	(0.367)	(0.025)	(1.153)	(25,660.320)	(0.745)	(0.044)
3 years after entry	-1.640	-5,646.098	0.714	-0.024	-3.162	-34,020.320	0.806	-0.019
	(0.707)	(18,959.244)	(0.449)	(0.030)	(1.367)	(37,756.529)	(0.870)	(0.055)
4 years after entry	-1.510	11,136.842	0.193	-0.089	-2.201	-19,394.085	0.346	-0.106
	(0.776)	(24,637.740)	(0.558)	(0.035)	(1.474)	(47,556.658)	(1.047)	(0.070)
5 years after entry	-2.519	-16,287.526	0.028	-0.129*	-4.876*	-83,626.498	1.403	-0.069
	(0.922)	(28,673.184)	(0.704)	(0.042)	(1.641)	(52,686.095)	(1.347)	(0.080)
Observations	13,345	13,351	10,296	10,296	3,325	3,311	2,596	2,596
Cohorts	8	8	8	8	8	8	8	8
Years	13	13	13	13	13	13	13	13

Notes: This table presents the estimations of event studies using Callaway and Sant'Anna estimator. Panel A presents the estimations for general entry events and Panel B for competitive entry events. Cohorts are defined by year of entry in the public procurement market: the first year in which the firm was awarded a contract by a public entity. The comparison group corresponds to not yet suppliers, never suppliers are excluded. Standard errors are presented in parenthesis and clustered at firm level.

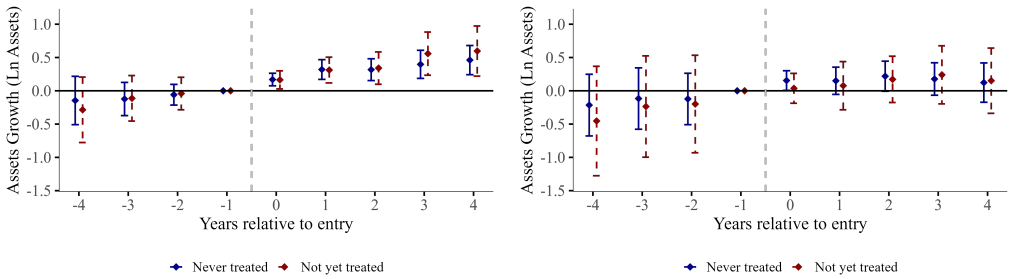
Figure 4.F.1 Event Study plots financial performance



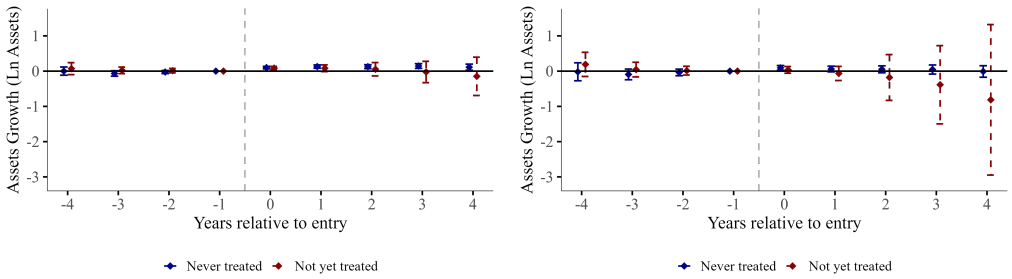
Notes: Figure 4.F.1 plots the estimated $\hat{\gamma}$ event-study coefficients from the regression given in equation 4.3.3 in the blue markers and solid lines. The red markers and dashed lines plot the estimated $\hat{\gamma}$ event-study coefficients comparing entrants with not yet suppliers while excluding explicitly never suppliers. Panel A presents the event study plots for general entry and Panel B for competitive entry. The dependent variables are: capital assets growth panels a and b, current ratio panels c and d, and liabilities panels e and f. Estimations based on Callaway and Sant'Anna estimator presented in table 4.2.

Figure 4.F.2 Event Study plots financial performance assets growth by firms heterogeneity

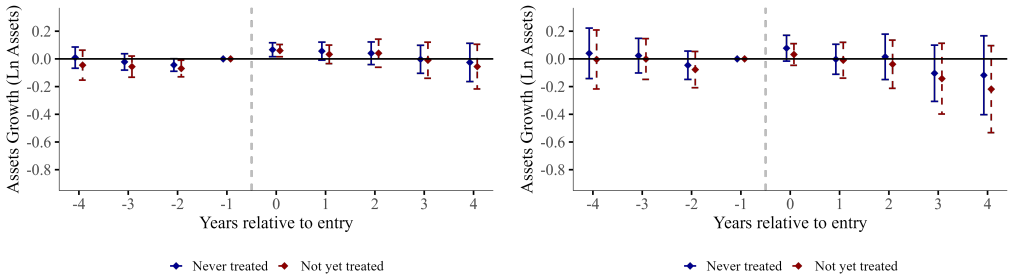
Panel A: General entry *Panel B: Competitive entry*



(a) Established Firms **(b)** Established Firms



(c) Capacity Constrained **(d)** Capacity Constrained



(e) Short-term participants **(f)** Short-term participants

4.F.1 Additional regressions from mechanisms

Table 4.F.3 Relative bid reduced-form regression - alternative specification

	(1)	(2)	(3)	(4)
Short-term participant	-0.0456***	-0.0456**	-0.0530**	-0.0579**
	(-2.82)	(-2.34)	(-2.58)	(-2.30)
Capacity constrained	0.00244	0.00244	-0.00278	-0.0179
	(0.33)	(0.26)	(-0.30)	(-1.58)
Lowest Price (FPA)			0.0190**	0.0193*
			(2.02)	(1.95)
Firm age at auction			0.000203	0.000217
			(0.75)	(0.85)
Certified Firm			-0.0156	-0.000527
			(-1.51)	(-0.05)
Constant	1.069***	1.069***	1.055***	1.058***
	(190.97)	(263.92)	(84.31)	(73.12)
Observations	6672	6672	6663	6652
R^2	0.00138	0.00138	0.00291	0.0276
R^2adj	0.00108	0.00108	0.00217	0.0214
Year	No	No	No	Yes
Project	No	No	No	Yes
VCE	None	Firm Cluster	Firm Cluster	Firm Cluster

Notes: The specification in equation 4.5.1 includes dummy variables for short-term participants and capacity constrained firms, leaving established firms as a baseline. The main difference from the estimations from table 4.1 is that the firms here could have bid several times before their first win. t statistics in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4.F.4 Contracts characteristics

	Panel A: General			Panel B: Competitive					Panel C: Entry Events					Panel B: Entry Competitive				
Variable	Mean	Std.Dev	N	Mean	Std.Dev	N	Diff	p-value	Mean	Std.Dev	N	Diff	p-value	Mean	Std.Dev	N	Diff	p-value
WTO Agreement	0.40	0.49	14,711	0.42	0.49	3,362	-0.01	0.1167	0.44	0.50	4,235	-0.04	0.0000	0.53	0.50	1,095	-0.12	0.0000
Extensions	0.01	0.12	14,711	0.03	0.16	3,362	-0.01	0.0000	0.01	0.11	4,235	0.00	0.1486	0.02	0.13	1,095	0.01	0.0502
Electronic Auction	0.04	0.19	14,711	0.00	0.03	3,362	0.04	0.0000	0.01	0.07	4,235	0.03	0.0000	0.00	0.00	1,095	0.00	0.0833
Urgent Procedure	0.01	0.07	14,711	0.01	0.08	3,362	-0.00	0.3641	0.01	0.08	4,235	-0.00	0.6021	0.01	0.10	1,095	-0.00	0.3365
Outsourcing	0.01	0.09	14,112	0.01	0.10	3,362	-0.00	0.3228	0.01	0.09	4,122	0.00	0.9143	0.01	0.09	1,095	0.00	0.7535
SME tender	0.80	0.40	14,112	0.76	0.43	3,362	0.04	0.0000	0.86	0.35	4,122	-0.06	0.0000	0.87	0.34	1,095	-0.10	0.0000
Final Price Reported	0.79	0.41	14,711	0.99	0.07	3,362	-0.20	0.0000	0.82	0.39	4,235	-0.02	0.0004	1.00	0.06	1,095	-0.00	0.5229
Estimated Cost Reported	0.33	0.47	14,711	0.37	0.48	3,362	-0.04	0.0000	0.35	0.48	4,235	-0.02	0.0519	0.37	0.48	1,095	-0.00	0.9932
Final < Cost	0.25	0.43	14,711	0.32	0.47	3,362	-0.07	0.0000	0.29	0.46	4,235	-0.05	0.0000	0.33	0.47	1,095	-0.01	0.5312
Final = Cost	0.11	0.31	14,711	0.12	0.32	3,362	-0.01	0.1158	0.11	0.31	4,235	0.00	0.7299	0.12	0.33	1,095	-0.00	0.8267
Final > Cost	0.12	0.32	14,711	0.18	0.38	3,362	-0.06	0.0000	0.11	0.31	4,235	0.01	0.0736	0.18	0.38	1,095	0.00	0.9429
Missing Prices	0.52	0.50	14,711	0.38	0.49	3,362	0.14	0.0000	0.49	0.50	4,235	0.03	0.0001	0.37	0.48	1,095	0.01	0.4850
Quality Price Ratio	0.35	0.48	14,711	0.43	0.49	3,362	-0.08	0.0000	0.44	0.50	4,235	-0.09	0.0000	0.54	0.50	1,095	-0.11	0.0000
Lowest Price	0.03	0.16	14,711	0.05	0.22	3,362	-0.02	0.0000	0.04	0.19	4,235	-0.01	0.0005	0.07	0.26	1,095	-0.02	0.0150
Award Missing	0.62	0.48	14,711	0.52	0.50	3,362	0.10	0.0000	0.52	0.50	4,235	0.10	0.0000	0.39	0.49	1,095	0.14	0.0000

Notes: This table

CHAPTER 5

Discussion

What factors shape the efficiency and effectiveness of public procurement markets? This dissertation focuses on three drivers shaping the efficiency and effectiveness of public procurement markets: regulated discretion, public buyers' skills and buyers' preferences, firms' entry and bidding strategies (see Figure 5.1). These drivers operate through three main channels: political favoritism, public sector capacity, and market structure. Political favoritism shapes whether bureaucratic discretion is used to improve procurement outcomes or to manipulate contract allocation. Public capacity matters because procurement officers' skills, preferences, and professional backgrounds influence procurement decisions and ultimately contract outcomes. Market structure affects procurement performance because firms' entry and bidding decisions determine competitive intensity, supplier viability, and the allocation of contracts.

This dissertation contains three essays on the empirical economics of public procurement, an emerging applied area within economic science. The research lies at the intersection of political economy, public economics, and industrial organization and aims to provide empirical evidence on open questions: (i) to examine how discretionary thresholds distort the allocation of public resources, affect the efficiency of procurement markets, and impact political favoritism in contract awards; (ii) to study the value-added of individual procurement officers' characteristics on the quality of public construction tenders; and (iii) to study firms' financial performance upon entry into the public procurement market. Overall, this dissertation contributes to the understanding of public procurement markets by showing how institutional design, procurement officers' decision-making, and firm dynamics shape efficiency, quality, and competition.

In what follows, I briefly summarize and discuss channels for effective and efficient public procurement, limitations and external validity, future research, and policy implications.

5.1 Summary and main findings

Chapter 2 studies two law reforms in the Czech Republic with opposite objectives: the first restricted discretion, while the second allowed greater discretion. We implement bunching estimations and a regression discontinuity design using the threshold limits defined before and after the reforms. We find that the final prices of contracts awarded under restricted discretion are lower after the reform, suggesting that stronger regulations reduce inefficiency gaps. However, we find that final prices do not change significantly after the reform that increased discretion, suggesting that relaxing regulations does not immediately lead to efficiency gains or losses. We also find evidence of political favoritism as an important mechanism through which discretion is misused, leading to contract value manipulation and bunching.

The evidence from the two reforms shows that tightening regulations reduces price inefficiencies, whereas relaxing regulations does not. This asymmetry suggests that more room for discretion is not per se efficiency-enhancing, but neither does it inherently generate inefficiency. These results hint toward the importance of administrative capacity as a third factor in the commonly discussed rules-discretion trade-off to make it a *competence-rules-discretion* trade-off. Consider a public buyer facing budgeting decisions. When the public buyer has strong procurement skills and knowledge (i.e. the competence to make informed procurement choices), efficient behavior does not depend on external constraints, and the public buyer will allocate its budget efficiently. In contrast, when the public buyer has weak skills and knowledge, external constraints, such as budget limits, become more necessary to prevent inefficient spending. In the latter case, relaxing constraints without improving skills and knowledge is likely to lead to worse outcomes. The role of administrative capacity of public buyers in determining the efficiency effects of discretion, helps motivate the turn in Chapter 3 to explore public buyers characteristics. It focuses on public buyers, operating within an unreformed institutional context, who pursue quality in the delivery of public tenders, and whose skills and competences ultimately determine their value added to procurement outcomes.

Chapter 3 studies the importance of individual bureaucrats in determining the quality of public procurement projects. We provide a novel approach to measuring procurement officers' contributions to public procurement quality by combining administrative procurement data with our unique survey of self-reported competences. Our findings indicate that officers add value to the likelihood that a project will not require costly maintenance in the years following its delivery. We find suggestive evidence that heterogeneity in procurement officers' legal backgrounds and preferences may play an important role in shaping procurement outcomes. These effects highlight that, despite regulations and rules that constrain public tenders, procurement officers play an important role in determining the quality and effectiveness of public procurement.

While administrative capacity is central to procurement outcomes, firms are strategic actors whose participation and bidding decisions shape market outcomes. Through favoritism, information asymmetries, and incumbency advantages, some

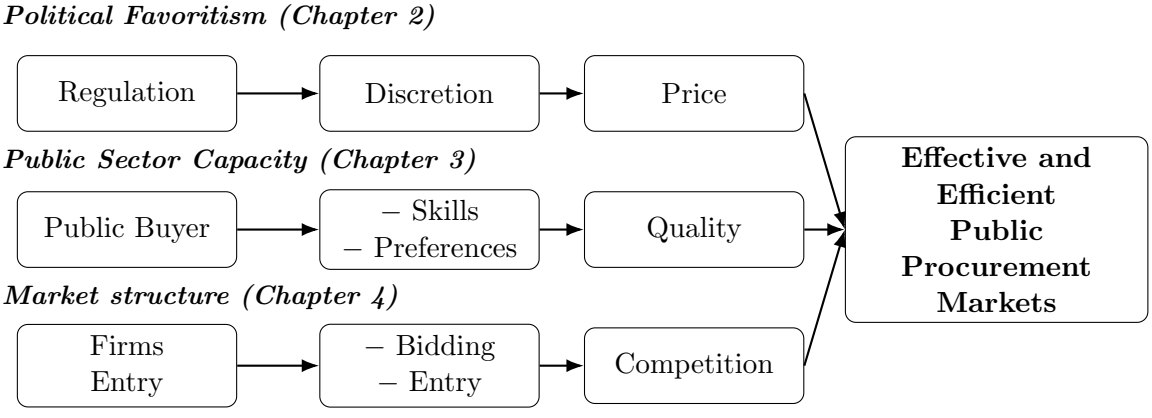
firms consolidate their positions in procurement markets. This raises two questions: which firms enter procurement markets, and which firms capture procurement rents. Focusing on competitive procedures and abstracting from discretionary pre-selection, Chapter 4 studies changes in firms' financial performance after entry into public procurement markets.

In Chapter 4, I study the changes in firm financial performance upon entry into the public procurement market. I define entry as the first time a firm wins a procurement contract and distinguish between two margins: general entry, corresponding to the first contract win, and competitive entry, defined as the first win in an open tender with at least two competing bidders. The results show that general entry is associated with sustained increases in firm assets and financial leverage. Additionally, I find that entry through competitive procurement leads to only temporary gains in assets, which dissipate within three years, alongside a persistent increase in financial liabilities. This divergence suggests that the conditions under which firms enter the procurement market play a central role in shaping their financial performance. I also find that short-term participants tend to bid more aggressively and participate in tenders with fewer competitors. These patterns are consistent with strategic underbidding, which may contribute to their weaker financial outcomes. Thus, the competitive context of the market serves as a channel that determines the effects of demand shocks created by public procurement contracts.

5.2 Channels for effective and efficient public procurement

Based on the results and insights presented in this dissertation, I discuss three channels that mediate the efficiency and effectiveness of public procurement markets: (i) political favoritism, (ii) public capacity, and (iii) market structure. Figure 5.1 illustrates how the chapters relate to these channels.

Figure 5.1 Channels for effective and efficient public procurement



First, the findings from Chapter 2 suggest the existence of a political favoritism channel (top part of Figure 5.1), where political favoritism enables the misuse of bureaucratic discretion in public procurement and conditions its efficiency. I con-

ceptualize this channel as operating through three mechanisms: (i) the trade-off between discretion and rules, (ii) individual and institutional capacity, and (iii) information asymmetries between public buyers and private firms. While the first mechanism is examined empirically in this dissertation, the latter two are discussed as complementary explorations, but remain untested. To begin with, the trade-off between discretion and rules has been widely discussed in public procurement literature (Titl and Geys, 2019; Carril, 2021; Szucs, 2024; Bosio et al., 2022). The premise is that institutional contexts with strong state capacity benefit from greater bureaucratic discretion, while contexts characterized by weaker public sector capacity or corruption risks tend to rely on stricter regulations to safeguard societal goals. Consistent with this view, our findings show that restricting discretion reduces contract manipulation and improves efficiency on average, although these gains are attenuated in the presence of political favoritism. In contrast, increasing discretion leads to a reallocation of contract manipulation toward the new thresholds, without corresponding improvements in efficiency.

Two other mechanisms within the political favoritism channel are not empirically tested, but relevant to mention in the context of this dissertation. The effective use of discretion depends on the experience, workforce skills, and institutional capacity of procurement officers and procurement organizations. Limited knowledge or weak institutional support may lead to an ineffective use of discretion, even in the absence of intentional distortions. While this mechanism is not directly examined in Chapter 2, Chapter 3 provides related empirical evidence on the role of public buyers' skills. Finally, other mechanisms related to information asymmetries between buyers and firms and their interaction with favoritism may also shape procurement efficiency (Schoenherr, 2019). These channels are not directly tested in Chapter 2.

Second, the findings from Chapter 3 suggest the importance of a public capacity channel (middle part of Figure 5.1), specifically, that the public administration workforce contributes meaningfully to the effectiveness of the public procurement market (Best et al., 2023), particularly project quality. The empirical evidence provided in this dissertation suggests that this channel might operate through the heterogeneity in public buyers' characteristics (Tukiainen et al., 2024). Although the underlying mechanisms remain descriptive, I introduce here two. The first mechanism reflects a trade-off between their mission-oriented (Besley and Ghatak, 2005) and legal compliance goals. This trade-off arises from the ambiguous institutional positioning of procurement officers, who are responsible for delivering value for money while guaranteeing compliance with regulatory frameworks. Therefore, officers may differ in the extent to which they prioritize project quality versus legal compliance. In complex procurement projects, procurement officers might exhibit legal compliance preference as legal risks are more salient. The second mechanism through which public buyers contribute to the effectiveness of the market is occupational and skills sorting. In the absence of formal selection procedures or certifications, individuals with heterogeneous backgrounds may self-select into different roles within the public administration, including public procurement departments. This institutional setting shapes both the allocation of skills within the

public sector and their retention. Consequently, variation in skills and professional backgrounds may translate into differences in procurement performance.

Finally, the findings from Chapter 4 suggest the relevance of the market structure channel (bottom part of Figure 5.1), particularly, that firms' entry and participation decisions shape the degree of competition and, ultimately, procurement efficiency (Carril et al., 2026). I examine this channel under the premise of holding fixed (ceteris paribus) institutional design (Chapter 2) and public buyers' behavior (Chapter 3). I suggest two mechanisms. The first mechanism relates to the competitive nature of entry. When firms enter through competitive bidding, short-term participants or overly optimistic firms may underbid due to imperfect information about project costs or execution risks. If this mechanism is at play, it can lead to ex post allocative inefficiency, as contracts would be awarded to firms that are unable to deliver at the estimated cost. The second mechanism operates through strategic underbidding as an entry strategy (De Silva et al., 2003). Firms may bid aggressively to secure contracts and establish a market presence; however, that can generate persistent financial strain, leading to underinvestment, lower quality, or weaker buyer-supplier relationships. As a result, entry dynamics may lead to a shortage of supplier financial viability and ultimately affect contract performance. While Chapter 4 provides empirical evidence on these mechanisms, an open question remains what is *the nature of the firm that contracts with the government.*

These channels can be understood as distinct margins through which the effectiveness of public procurement markets can be achieved and sustained. This framework is neither exhaustive nor exclusive. Other mechanisms, not explicitly analyzed in this dissertation, may also influence procurement outcomes. These include, for example, Green Public Procurement (Nicolas et al., 2025), the organizational design of procurement functions (Schotanus, 2023), market engagement and innovation policies (Berkhout et al., 2026), and the adoption of AI-driven tools (Longobucco and Ferwerda, 2025). Each of these can be interpreted as affecting either the objectives, constraints, or information environment of key actors (as mentioned in introduction section), thereby influencing equilibrium outcomes in public procurement markets.

5.3 Limitations and external validity

The three empirical research papers developed are subject to limitations. First, while I address three channels of efficiency in public procurement markets (see Figure 5.1), other factors may also play an important role. My analysis abstracts from these additional factors and therefore does not capture how they may interact with the channels studied here. Under the assumption that additional drivers improve market efficiency, the results I discuss will remain in the same direction. However, it is an open question to what extent different channels interact, reinforce, or offset each other in determining procurement efficiency.

Second, the external validity of the results depends on institutional context. My analysis focuses on two European countries, whose public procurement markets are

governed by the same European directives. The results are likely to inform other contexts within the European Union characterized by competitive markets and formal accountability mechanisms. However, in contexts where public procurement regulations are nascent, the underlying mechanisms may operate differently, limiting the applicability of the results. In settings with weaker enforcement, limited transparency, or less developed regulatory frameworks, the underlying mechanisms, particularly those related to discretion and favoritism, may operate differently, limiting the applicability of the results.

5.4 Future research

Future research could build on the findings of this dissertation in several ways.

One fruitful avenue for future studies would be to develop empirical tests to differentiate contract value manipulation generated by active versus passive waste (Bandiera et al., 2009) in complex public procurement environments. Future work could explore heterogeneity in officers' value-added across different segments along the distribution of active and passive waste. For instance, testable hypotheses include whether passive waste is primarily driven by risk aversion (e.g., legal compliance) or by lack of competence (e.g., insufficient training), and whether officers with managerial or market-oriented skills contribute more to efficiency (e.g., less waste) in more complex contracting environments.

Future work could explore the *public procurement wedge*, defined by Decarolis and Srhoj (2025) as the efficiency loss of realized social value relative to a benchmark. Considering that measuring quality in public procurement is a challenging task (De Jaegher et al., 2025; Baranek and Titl, 2024), it would be valuable to explore alternative measures of procurement quality and allocative efficiency that extend beyond procurement outcomes and repair costs, particularly in sectors where complex public procurement is carried out. For instance, the quality or efficiency loss could be assessed using indicators such as contract renegotiations and cost overruns as proxies for reliability and failure-free operation. Along this line of studying the procurement wedge, future research could provide insights about whether and how politically connected firms experience persistent productivity gains or profit advantages from preferential access to public procurement contracts, and to what extent favoritism generates moral hazard or adverse selection in procurement markets (Carril et al., 2026). Addressing these questions could shed light on the selection and participation of firms in public procurement markets, as well as their effects on the political economy of this part of the public administration.

An overdue avenue of research is to study the nature of firms contracting with the government, i.e. who contracts with the government and why (di Giovanni et al., 2024; Carril et al., 2026). Future work could explore the sources of entry frictions, quantify participation costs of firms, and other market incentives. Documenting stylized facts and causality can provide insights into the determinants to achieve long-run competitiveness of public procurement markets.

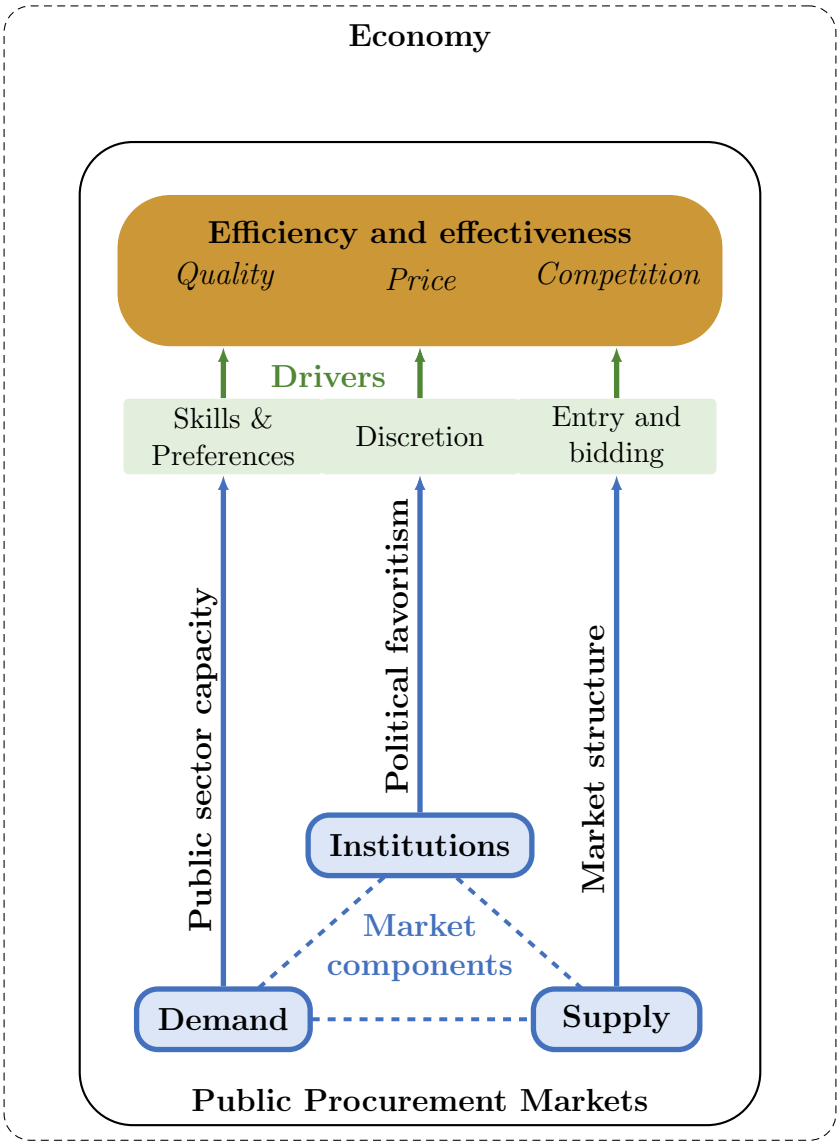
5.5 Policy implications

The results of this dissertation are informative for public policy regarding the design of discretionary thresholds and the broader role of rules and discretion in channeling policies for public procurement markets. First, the existence and level of discretionary thresholds could be determined efficiently after a thorough assessment of corruption risks and objective definitions of state capacity. Based on our results, one policy implication is that setting discretionary thresholds too low in countries with high state capacity or procurement expertise can be detrimental for achieving efficiency. For example, studying the European context, our findings suggest that higher discretionary thresholds may be more beneficial, since the European Union and its single-market framework, as institutions, have relatively strong state capacity. Nonetheless, corruption poses specific risks to each country's political economy. Identifying an optimal interval for discretionary thresholds may reduce incentives to strategically manipulate contract values while preserving sufficient discretion for procurement officers to design efficient contracts. At the same time, frequent changes to thresholds incur transaction costs for both public buyers and suppliers and may introduce uncertainty into public procurement markets. More broadly, these findings highlight that public procurement regulation can influence the effectiveness of industrial policies (Mazzucato, 2021). Regulatory frameworks, particularly discretionary thresholds, affect interactions between government agencies and private-sector firms. When procurement officers have sufficient discretion and are competent, mission-oriented professionals, they can exercise it to support policy objectives. For example, by awarding contracts to small, sustainable, and innovative firms, social entrepreneurs, or by limiting risks by favoring reliable firms or firms that are less risky from a national security perspective.

The policy guidelines and frameworks developed by the OECD and the European Commission regarding public procurement can benefit from the insights of this dissertation. Particularly on the discussion of the public procurement occupation, selection of officers, and skills training. Policymakers should systematically identify and prioritize the educational and professional backgrounds most suited to managing public resources through contracting. This implies establishing formal selection criteria and merit-based recruitment processes for procurement roles, particularly for positions that involve discretionary authority. In parallel, procurement systems should invest in targeted training and certification programs to strengthen the skills of public buyers. Overall, a more structured approach to the selection and development of procurement officials is necessary to improve the public capacity that channels decisions toward efficient public procurement.

Finally, from a policy perspective, improving participation conditions for new entrants by reducing administrative barriers, increasing transparency, or improving access to procurement information may help sustain competition in public procurement markets. Encouraging persistent participation could expand the pool of active suppliers and mitigate the decline in bidding rates observed in recent years.

Figure 5.1 Economics of Public Procurement: Drivers of Efficiency and Effectiveness



Notes: Figure 5.1 summarizes this dissertation. It positions public procurement markets as part of the wider economy and distinguishes three market components: demand, supply, and institutions, shown at the bottom of the figure. The functioning of these components is shaped by three channels examined in this dissertation: public sector capacity, political favoritism, and market structure, represented by the vertical solid lines. The empirical evidence presented in this dissertation suggest that some of the key drivers shaping how these channels affect procurement outcomes include public buyers' skills and preferences, the regulation of discretion, and firms' entry and bidding decisions (middle shaded rectangle). These drivers affect procurement outcomes: quality, price, and competition; which together determine the efficiency and effectiveness of public procurement (top shaded rectangle).

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Nederlandstalige Samenvatting

Dit proefschrift analyseert met empirische economische methoden hoe verschillende factoren de efficiëntie en effectiviteit van publieke inkoop beïnvloeden. Publieke inkoop is het proces waarbij overheden goederen, diensten en werken inkopen bij ondernemingen. Deze activiteiten vormen een substantieel deel van de economische bedrijvigheid en vertegenwoordigen een aanzienlijk aandeel van het bruto binnenlands product in OECD- en EU-landen.

Overheidsmarkten spelen een belangrijke rol in de verdeling van publieke middelen, het functioneren van de overheid en de interactie tussen overheden en ondernemingen. Ondanks dit belang kampen deze markten met structurele uitdagingen, zoals afnemende concurrentie, risico's op corruptie en favoritisme en wisselende deskundigheid. Inzicht in de drijfveren van efficiëntie en effectiviteit van aanbestedingen is daarom belangrijk voor zowel de economische prestaties als het publieke vertrouwen in de overheid.

Het proefschrift gaat in op drie relaties. Ten eerste beïnvloedt het institutionele ontwerp, en in het bijzonder de regulering van de beslisruimte van overheden (discretie), de efficiëntie en effectiviteit van publieke inkoop. Ten tweede worden de uitkomsten van aanbestedingen niet alleen bepaald door formele regels, maar ook door de juridische voorkeur van inkopers. Ten derde beïnvloedt structurele of juist incidentele deelname aan aanbestedingen door ondernemingen hun financiële prestaties. Deze relaties vormen de basis voor een analytisch kader waarin drie componenten centraal staan: instituties, inkopers en ondernemingen.

De empirische analyse is gebaseerd op data uit Tsjechië en Nederland. Hoewel beide landen gebonden zijn aan dezelfde Europese aanbestedingsrichtlijnen, verschillen zij in hun nationale institutionele context. Hierdoor bieden de analyses inzicht in mechanismen die zowel contextspecifiek als breder relevant zijn voor publieke inkoop binnen de EU. Het onderzoek combineert aanbestedingsdata en financiële bedrijfsgegevens met surveydata. In de verschillende hoofdstukken worden diverse econometrische methoden toegepast waaronder *regression discontinuity designs*, *bunching-analyses* en *difference-in-differences*.

Discretie en politiek favoritisme

Het tweede hoofdstuk van dit proefschrift gaat in op de relatie tussen de keu-

zevrijheid van de inkoper en politiek favoritisme. Discretie biedt de flexibiliteit die nodig is voor complexe opdrachten, maar creëert ook ruimte voor misbruik, zoals het bevoordelen van politiek verbonden ondernemingen. Op basis van hervormingen in Tsjechië wordt aangetoond dat overheden strategisch handelen om strikte regels te omzeilen, bijvoorbeeld door contractwaarden net onder wettelijke drempelwaarden te houden, waardoor gebruik kan worden gemaakt van een lichter aanbestedingsregime. De analyse laat zien dat het beperken van deze vrijheid leidt tot lagere inkoopprijzen en dat efficiëntiewinsten vooral ten goede komen aan ondernemingen zonder politieke connecties.

De rol van de inkoper

Het derde hoofdstuk richt de aandacht op de menselijke factor en de kwaliteit van aanbestede projecten. Omdat kwaliteit vaak moeilijk direct meetbaar is, wordt in dit onderzoek een innovatieve maatstaf gebruikt: de waarde van noodzakelijke herstelcontracten na projectvoltooiing. De resultaten tonen aan dat individuele inkopers een belangrijke rol spelen; hun persoonlijke toegevoegde waarde verklaart significante verschillen in projectkwaliteit. Opvallend is dat inkopers met een juridische achtergrond vaker geassocieerd worden met een lagere projectkwaliteit, wat kan duiden op een spanningsveld tussen strikte juridische naleving en een effectieve praktische uitvoering.

Toetreding en concurrentie door ondernemingen

Het vierde hoofdstuk analyseert de effecten van deelname aan aanbestedingen op de financiële prestaties van ondernemingen in Nederland. Hoewel het winnen van een eerste groot overheidscontract vaak leidt tot schaalvergroting (toename van activa en passiva), zijn de effecten afhankelijk van de ervaring van de onderneming. Minder ervaren ondernemingen in aanbesteden neigen tot agressief bieden, wat kan leiden tot de *winner's curse*: het winnen van een contract tegen te lage marges door onderschatting van de kosten. De meest duurzame positieve effecten worden gerealiseerd door ondernemingen die structureel actief blijven op de overheidsmarkt.

Conclusie en implicaties

Dit proefschrift identificeert drie kanalen voor aanbestedingsefficiëntie: het beheersen van politiek favoritisme, het versterken van inkoopbekwaamheid en het waarborgen van een gezonde marktstructuur. Deze kanalen zijn onderling afhankelijk; zo hangt de effectiviteit van regelgeving samen met de bekwaamheid van de inkoper. De bevindingen onderstrepen dat een verbetering van het publieke inkoopsysteem meer vereist dan alleen goed ontworpen regels. Het vraagt ook om investeringen in menselijk kapitaal, zoals de professionalisering van inkooppersoneel en beleid dat een gezonde marktstructuur waarborgt.

Curriculum Vitae

Marly was born in Manizales, Colombia. She obtained a bachelor's degree in Economics and a master's degree in Economics from Universidad de Manizales. She worked as a researcher at the Centro de Estudios Económicos, Cafeteros y Empresariales (CRECE) and at Universidad de Manizales. She later moved to Bogotá, where she worked as a policy analyst at the National Planning Department (DNP). In 2019, she moved to the Netherlands to pursue the Research Master's in Economics at Utrecht University. In 2021, she started her PhD at Utrecht University School of Economics. During her doctoral studies, she was a visiting researcher at Northeastern University in Boston in 2023 and at Universitat Pompeu Fabra in 2025.

At U.S.E., she contributed to the academic and social life of the school through several activities. She participated in the Econometrics Reading Group, helped organize informal thesis defenses for research master's students, and represented U.S.E. at the Econometric Game in 2024. She was also a member of the teachers' football team.

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